

Sightline Intelligence

Command, Control, and Script API

v3.11



[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) ■ [Camera Configuration](#)

Sightline Intelligence Command, Control, and Script API

Introduction

This document describes two distinct but related methods for controlling Sightline Intelligence video stabilization and tracking systems:

- A packet-based command and control protocol, comprised of a set of messages, suitable for use over RS-232 and Ethernet IP connections. This interface is intended for command and control by a remote computer system, for example, a PC running SightLine's Panel+ application.
 - All messages have:
 - a header comprised of a special signature, a message ID, and a length
 - a payload made up of a data structure
 - a terminating checksum byte to ensure data integrity.
- A function-based API that can be accessed through a Lua script engine built into SightLine Applications systems. The new script interface allows customization of I/O responses and enables the development of complex autonomous behaviors. Protocol messages map directly to script functions and data structures.

The packet-based messaging protocol outlined in this document is valid for this release and any prior release. Changes between revisions are noted throughout the document. Lua script support is new in the 2.23 release.

Both methods of control utilize similar data formats and concept of operations. The remainder of this document describes details specific to the two control methods (see [Command and Control Protocol](#) and [Script API](#)). For detailed descriptions of messages and functions, see [Class List](#)

Common Features of Protocol and API

See [Message Identifiers](#) section for a list of data structures in this API, or see the corresponding list of [Lua Script Functions](#).

Protocol messages and API data structures have a 1-to-1 relationship. For example, the message defined by [SLASetStabilizationParameters_t](#) defines a corresponding data structure that may be used by the API function `SLASetStabilizationParameters` in a Lua script. Except where specified, the terms "messages" and "data structures" are used interchangeably in the remainder of this document.

Bit and Byte Order

Messages are defined as a list of parameters. The basic data types and sizes used for each parameter are:

Type	Length in bytes	Description
u8	1	Unsigned 8-bit integer
s8	1	Signed two's-complement 8-bit integer
u16	2	Unsigned 16 bit integer
s16	2	Signed two's-complement short
u24	3	Unsigned 24-bit integer
u32	4	Unsigned 32-bit integer
s32	4	Signed two's-complement 32-bit integer
u64	8	Unsigned 64-bit integer
s64	8	Signed two's-complement 64-bit integer
u16be	2	Unsigned 16-bit integer (in MSB-LSB order)
u32be	4	Unsigned 32-bit integer (in MSB-LSB order)

All multi-byte fields are encoded with the Least Significant Byte (LSB) first, followed by MSB.

Some parameters are defined as bit values within a larger basic type. In these cases, all bits are right aligned. For example, a parameter of type *u8* with value *0xC9* (hexadecimal notation) has an upper nibble with value 'C' and a lower nibble with value '9'. The upper nibble, 'C', has the binary representation '1100' and the lower nibble '9' is '1001' in binary. Thus a parameter of type *u8* with value *0xC9* has the following bit pattern:

7	6	5	4	3	2	1	0
1	1	0	0	1	0	0	1

Variable Naming Convention

- Variable names that end in a number indicate that the value will be scaled by that many bits.

Strings and Arrays

The variable length types **SVPEndString_t** and **SVPLenString_t** are two different ways that names and other text strings are represented. For protocol messages, the length of **SVPEndString_t** is determined by the length of the packet (with the string length filling in 0 to 255 bytes needed to bring the packet length to the value specified in the message header). For API data structures, an array of 255 bytes is used to store a null-terminated string – some of the 255 bytes may be unused depending on the actual length of the string. **SVPLenString_t** uses a *u8*, **SVPLenString_t::len**, followed immediately by a text string. For protocol messages, only the number of bytes specified by **SVPLenString_t::len** are part of the message. For API data structures, space is reserved for the maximum length string, but only the appropriate number of bytes is utilized.

Two kinds of arrays are also referenced in the data structure documentation. Fixed-length arrays of a sub-struct, currently used only for **SLASetUserPalette_t** and **SLACurrentUserPalette_t**, indicates a repetition of the struct. For example, **SLASetUserPalette_t::palette** contains 256 y-u-v triplets. Variable length arrays similarly indicate a repetition of the struct. However, the actual number of repetitions is controlled by the *u8* immediately preceding the array. See, for example, **SLATrackingBoxPixelStats_t::stat** which contains up to 255 sets of statistics elements with the actual number present represented by the preceding value **SLATrackingBoxPixelStats_t::numTracks**.

There are two further data structures [SVPLenData_t](#) and [SVPSignedData_t](#) which are very similar to [SVPLenString_t](#). The first byte is the number of bytes to follow. The only difference is that [SVPLenData_t](#) is intended for an array of u8's and the [SVPSignedData_t](#) is intended for an array of s8's.

Time Codes

Throughout the document certain packets may refer to time codes. Each time code shall be a 64-bit unsigned integer representing the micro-second time at which this frame was captured. The time code is based on the internal high resolution clock. This clock is initialized to 1970-01-01 00:00:00.0 on start-up. The clock can be synchronize with a Universal Time Code via [SLASetMetadataValues_t::utcTime](#).

The UTC time is not preserved during a power cycle to the SightLine hardware.

Coordinate Systems

Image coordinates are referenced as row and column coordinates, with the origin in the upper left corner of the frame. Increasing column values are to the right, and increasing row values are downward in the frame. Unless otherwise identified, a video frame is 640 pixels wide and 480 pixels high.

Application Bits

System features are controlled through application bits (AppBits) in a license file unique to each hardware platform. See [SLAVersionNumber_t](#).

Command and Control Protocol

Individual getter commands have been replaced with a single generic getter command of [SLAGetParameters_t](#). This command takes the corresponding setter command's message ID as an input. For example, [SLAVersionNumber_t](#) can be returned by using either [SLAGetVersionNumber_t](#) or the new way using [SLAGetParameters_t](#) with [SLAGetParameters_t::id](#) set to 0x00. The complete message would look like: 0x51, 0xAC, 0x03, 0x28, 0x00, 0x73.

The objective in making this change is to reduce the number of new message IDs required in the future. Currently, when a new feature is implemented, the following new commands are implemented: a setter, getter, and a result reply. With 2.17 and future releases, only a new setter and result reply will be needed. Most of the getter functions do not require any additional parameters. A single generic getter has been implemented, which uses the setter command ID as its parameter.

Sample Code

For development of C/C++ command and control applications, SightLine provides sample code for writing applications that implement our protocol. This is primarily available in [slfip.h](#) and [slfip.cpp](#) and can be downloaded as part of the Example Code sample. This code can be used as a starting point for writing your own application and is also useful for conforming to any changes to the protocol that may take place over time. SightLine Command and Control Example Code and Resources can be downloaded from the command and control page at <https://knowledge.sightlineintelligence.com/>.

Command Packets

Command Packet Header

Every command packet sent over a serial or Ethernet connection begins with a pair of signature header bytes (0x51, 0xAC)

and a length field. The value specified in the length field is the number of bytes that follow with the packet up to and including the checksum. The length field can be 1 byte or 2 bytes long depending on the length of the command packet. If the command packet length is greater than 127 bytes, then the length field occupies 2 bytes.

The following tables specify how the length field is encoded:

Normal packets (Length<128 bytes):

Header1	Header2	Length	Message ID	Message-dependent	Checksum
0x51	0xAC	len	id	data_bytes...	cs

Extended length packets (Length>=128 bytes):

Header1	Header2	Length Low	Length High	Message ID	Message-dependent	Checksum
0x51	0xAC	xx	yy	id	data_bytes...	cs

xx: Lower 7 bits of the length. Bit 7 must be set to 1.

yy: Upper bits of the length.

To obtain the length from xx and yy, here is a code snippet in C language:

```
Length = (yy << 7) | (xx & ~0x80);
```

For example, 128 bytes is encoded as xx: 0x80, yy: 0x01.

NOTE: two byte length field could be used for packets whose length is less than 128.

Checksum is calculated for the data highlighted in bold above (Message ID + Message-dependent data bytes).

Example of packet with length = 3

Header1	Header2	Length	Message ID	Mode	Checksum
0x51	0xAC	0x03	0x01	0x02	0xBC

NOTE: Command Packets are not the same as UDP packets. A UDP packet sent to the Ethernet command port can contain one or more Command Packets.

Length is the length of the rest of the packet including Checksum.

Message ID

Every command packet has a Message ID (previously known as the Type or Type ID). This is a 1 byte field. The message ID can also be used to query for state using [SLAGetParameters_t](#).

Message-dependent Data

Each packet may have 0 or more bytes of payload data. This data is defined for each message.

Checksum Calculation

Checksums are calculated over the bytes following the length field, up to but not including the checksum field. To ensure proper packet framing, if the checksum fails, the bytes following the faulty signature header (0x51, 0xAC) are scanned for

the signature header bytes again.

NOTE: The checksum is also necessary when communicating over Ethernet with SightLine hardware.

The checksum can be calculated using the following table and pseudo code:

```
const uint8 crc8_Table[ ] =
{
    0,  94, 188, 226,  97,  63, 221, 131, 194, 156, 126,  32, 163, 253,  31,  65,
  157, 195,  33, 127, 252, 162,  64,  30,  95,  1, 227, 189,  62,  96, 130, 220,
   35, 125, 159, 193,  66,  28, 254, 160, 225, 191,  93,   3, 128, 222,  60,  98,
  190, 224,   2,  92, 223, 129,  99,  61, 124,  34, 192, 158,  29,  67, 161, 255,
   70,  24, 250, 164,  39, 121, 155, 197, 132, 218,  56, 102, 229, 187,  89,   7,
  219, 133, 103,  57, 186, 228,   6,  88,  25,  71, 165, 251, 120,  38, 196, 154,
  101,  59, 217, 135,   4,  90, 184, 230, 167, 249,  27,  69, 198, 152, 122,  36,
  248, 166,  68,  26, 153, 199,  37, 123,  58, 100, 134, 216,  91,   5, 231, 185,
  140, 210,  48, 110, 237, 179,   81,  15,  78,  16, 242, 172,  47, 113, 147, 205,
   17,  79, 173, 243, 112,  46, 204, 146, 211, 141, 111,  49, 178, 236,  14,  80,
  175, 241,  19,  77, 206, 144, 114,  44, 109,  51, 209, 143,  12,  82, 176, 238,
   50, 108, 142, 208,  83,  13, 239, 177, 240, 174,  76,  18, 145, 207,  45, 115,
  202, 148, 118,  40, 171, 245,  23,  73,   8,  86, 180, 234, 105,  55, 213, 139,
   87,   9, 235, 181,   54, 104, 138, 212, 149, 203,  41, 119, 244, 170,  72,  22,
  233, 183,  85,  11, 136, 214,  52, 106,  43, 117, 151, 201,  74,  20, 246, 168,
  116,  42, 200, 150,  21,  75, 169, 247, 182, 232,  10,  84, 215, 137, 107,  53
};

crc = 0x01;

for ( each byte_Value between length and checksum fields)
{
    crc = crc8_Table[ crc ^ byte_Value ] ;
}
```

Checksum Calculation Example

Data	Description	Action	CRC
0x51	Header 1	Ignored	0x01
0xAC	Header 2	Ignored	0x01
0x02	Length	Ignored	0x01
0x07	Message ID	Used	0x01 ^ 0x07 = 0x06
Look up result	-	-	Table[6] = 221 (0xDD)

CRC should equal 221.

Communication

Serial Ports

Command and Control is available on most serial ports. Serial port parameters are configured to 57600 Baud, 8 data bits, 1 stop bit, no parity, and no handshake by default. See [SLASetPortConfiguration_t](#) for information on additional modes supported. Refer to the specific hardware ICD to determine if serial port is 3.3V TTL or RS-232C level signals.

Ethernet

Command and control is also available over Ethernet.

SVP1 / listenPort1

All commands can be sent as a UDP packet to the IP address of the system on port 14001. Each packet may contain one or more commands. All replies will be sent to the IP address of the sender on port 14002. Examples of this are Panel Plus, ground control stations, autopilots, or other software that needs primary control over the system. These ports can be changed using [**SLASetNetworkParameters_t**](#).

The SightLine system listens on port 14001 for incoming commands by default. It always sends responses to port 14002 of the last sender's IP address. When a message is received, the sender's IP address is recorded and it is used globally as the destination when sending responses or telemetry data. This method assumes a single client/server relationship. When there are multiple clients that need to change settings on the server, this method doesn't work well: the response may not be returned to the correct sender depending on the timing.

**** Unsolicited messages **** One of the design considerations for having a dedicated port such as 14002 on the client is allow the SightLine system to send out unsolicited message. Examples of unsolicited messages include the User Warning message ([**SLAUserWarningMessage_t**](#)), Telemetry packets ([**Telemetry**](#)), etc. The SightLine system sends out the User Warnings to port 14002 of the last IP address. This allows for the development of a client that passively listens for messages and takes action.

SVP2 / listenPort2

Additional ports (listenPort2) were introduced for to support more [**User Programs**](#). For these additional ports, the response is always sent back to the original sender. So multiple clients can talk to the same system at the same time and they get their responses correctly. listenPort2 is supposed to be used by user programs and is also used in the Upgrade Utility, RTSP, etc. This is based on the typical UDP client/server model.

This additional port is available for onboard user programs, which needs to communicate to VideoTrack. Examples such as a lens control application that may need to receive focus data in order to drive a lens connected to one of the serial ports. One or more commands can be sent as a UDP packets to VideoTrack on port **14003**. All replies are sent back to the client on the same socket that received the command packet. Multiple clients can communicate with VideoTrack simultaneously on this port. Replies are sent to the client that sent the request.

Note

There can be implications for ARM applications that receive commands from SLACommandPassThrough (0x3D). It may not be clear which client should receive that data.

Multiple Telemetry Destinations

The SightLine protocol allows an additional 4 clients to register to receive only telemetry data. The idea being that integrators might have one controller (the autopilot) using the port pair (14001 & 14002), but other software clients are still interested in telemetry (focus, tracking results, etc.) A client sends its IP address and a port number to the OEM ([**SLASetTelemetryDestination_t**](#)), the pair is saved in a list, and then the OEM sends data forever to everyone in that list. This allows for a more explicit setup for unsolicited messages, rather than the implicit method defined by 14002.

NOTE: each client receives the same tracking output. The system cannot be configured to send focus stats to one client and landing position to a different client.

IP Address Assignment

The static IP address of the SightLine hardware can be set using [**SLASetNetworkParameters_t**](#). If a static IP address has not been set, the SightLine hardware will attempt to obtain an IP address using DHCP. If DHCP fails, the system will use a

specific link local (RFC 3927) type address 169.254.1.180.

IP Address Assignment Flowchart

	If Static...	Else try DHCP...	If DHCP Fails try...
IP Address	User Defined	DHCP Defined	169.254.1.*{180,181,182}*
Subnet Mask	User Defined	DHCP Defined	255.255.0.0
Gateway	User Defined	DHCP Defined	NOT DEFINED

NOTE: Use this table as guidance for setting the IP address of your PC so that it is on the same subnet as the SightLine hardware.

Network Port Numbers

The range of valid port numbers is 1 through 65536. 0 is reserved and is either invalid or indicates no change, depending on the specific message. Before assigning a number, make sure the port is not already in use. Multiple sockets using the same port can result in errors.

Network Ports commonly used by SightLine

Port	Description
14001	Inbound commands on SightLine HARDWARE
14002	Input reply port on PC
51000	SLDISCOVER Listen Port
5004	Default port for RTP-MJPEG
15004	Default port for MPEG2-TS H.264
21	FTP port
23	SSH port
52000	Watchdog timer port for diagnostics information
14003	Inbound commands on SightLine HARDWARE from internal ARM programs
14004	RTSP server to VideoTrack command port
45001	TCP port number for upgrade.
65100	Acquarius protocol
554	RTSP listen port
8008	RTSP to VideoTrack UDP video (Net0)
8010	RTSP to VideoTrack UDP video (Net1)

SightLine Hardware Types

See also [SLAVersionNumber_t](#).

Hardware Type	Description	ID
---------------	-------------	----

2000	2000-OEM	0
1500	1500-OEM	7
Upgrade	Upgrade Server	10
3000	3000-OEM Revision B	12
3000C	3000-OEM Revision C	14
4000	4000-OEM	18
1750	1750-OEM	28

Host Name

The default system host name takes the form SLA<[Hardware Type](#)>_<MAC>, where MAC is last 3 octets of the MAC address. Example: SLA1500_fe01cd

Start Up Behavior

On start up, the SightLine hardware will send the [SLAVersionNumber_t](#) packet over Serial Port 0 (see [SLASetPortConfiguration_t](#)) and Ethernet port 14001 after the 3rd frame is acquired. At this point the system is ready to receive commands.

Camera Index

Throughout the API, many commands and replies use **cameraIndex** to indicate which camera the message applies to.

- In cases where **cameraIndex** is not specified (older protocols, etc.), the command applies to the camera index specified by [SLACommandCamera_t](#).
- The maximum number of camera inputs can be requested with [SLACurrentConfiguration_t](#).
- Other use cases include 255 to indicate all cameras are affected and are noted with the command.
- For telemetry or other replies, camera index indicates values are for that camera.

Software Index

Platform	cameraIndex
4000	{0,1,2,3}
1750	{0,1,2,3}

Hardware Cross reference

Board	Camera Index 0	Camera Index 1	Camera Index 2	Camera Index 3
4000	J6 Digital Input Adapter	J9 MIPI B	USB 0	USB 1
1750	J6 Digital Input Adapter	J8 MIPI B	USB 0	USB 1

Note

Refer to the Hardware ICD for additional information about video ports and capabilities.

Network Display ID

Throughout the API, many commands and replies use dispID. This refers to a Network Display ID more commonly called Net0, Net1, and Net2. Video may be sent to one or both locations. see [SLASetEthernetDisplayParameters_t](#).

Configured for dual/tri channel using [Application Bits](#). If not present, then command may apply to all Network Displays. Some exceptions apply (e.g. [SLASetSDRecordingParameters_t](#), and [SLACurrentSDCardRecordingStatus_t](#))

dispID	Network Display ID
0x0002	Net0
0x0080	Net1
0x0010	Net2
0x0092	Net0 & Net1 & Net2

Script API

Each data structure named SLA<name>_t has a corresponding Lua script function named SLA<name> that takes the following parameters:

1. void *context – System supplied script handle
2. SLA<name>_t *v – Input structure reference
3. SVPOut_t *out – Output structure reference
4. u16 nElements – Number of elements in input structure

In the example below, the function that performs a system reset is SLAResetAllParameters, which references [SLAResetAllParameters_t](#) as the second parameter. SVPOut_t provides a convenient mechanism to access the return value of the function.

NOTE: nElements should be an unchanging value for any software release, but may change if additional elements are added to messages in future releases.

```
typedef struct {
    union {
        SLAGetVersionNumber_t GetVersionNumber;
        SLAResetAllParameters_t ResetAllParameters;
        SLASetStabilizationParameters_t SetStabilizationParameters;
        SLAGetStabilizationParameters_t GetStabilizationParameters;
        SLAResetStabilizationParameters_t ResetStabilizationParameters;
        //
        // all other data structure types included here
        //
        SLAStreamingControl_t StreamingControl;
    };
} SLAUnion;

typedef struct SVPOut_t {
    s32 type;
    u32 outLen;
    SLAUnion *out;
    bool isFipEx;
} SVPOut_t;
```

The example below illustrates how to call a function and correctly pass context, out, and nElements. Since functions may return unexpected types, verify out.type before accessing out.out.

```
-- Global frame count, incremented in SLPostAnalyze
local ffi = require("ffi")
local framecount = 0
```

```
function SLPostAnalyze( _vtstate, cameraIndex )
    framecount = framecount + 1

    if framecount == 2 then
        local gv = ffi.new("SLAGetVersionNumber_t")
        local out = ffi.new("SVPOut_t")
        local rv = ffi.new("SLAUnion");
        out.out = rv;

        ffi.C.SLAGetVersionNumber( _vtstate, gv, out, ffi.sizeof(gv))
        if out.type == ffi.C.VersionNumber then
            print("Version is ", rv.VersionNumber.swMajor, ".", rv.VersionNumber.swMinor, ".",
                rv.VersionNumber.swRelease)
        end
    end
end
```

For a complete list of functions in the API, see [Lua Script Functions](#).

NOTE: Append **ULL** to 64 bit integer values. Eg. setTime.utcTime = 0x54deab2bd7500**ULL**

See [EAN-SLA-Script-Development.pdf](#) for instructions on how to configure, setup, and use Lua scripts on SightLine Applications systems.

Camera Support

See [Camera Configuration](#) for the complete list of supported cameras.

Sightline Intelligence

Command, Control, and Script

API v3.11

Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

 SLAAdvancedCaptureParameters_t	Set advanced analog video decoding parameters for the video decoder
 SLAAncillaryTextMetadata_t	Inject text into the KLV stream
 SLAAppendedMetadata_t	Append user-specified binary metadata to KLV stream embedded in network video stream
 SLABlendAlign_t	Set the alignment set for a dual camera setup
 SLACameraCalibration_t	Set camera calibration parameters to fix distortion
 SLACameraCapabilities_t	Return a message with the camera capabilities
 SLACameraSwitch_t	Deprecated , supported on 1500 and 3000 only
 SLAClassifierParameters_t	Turn on/off the classifier
 SLACommandCamera_t	Set which camera channel receives camera specific parameter settings
 SLACommandPassThrough_t	Output data payload to the port specified
 SLACoordinateReportingMode_t	Set reporting rates and telemetry types to send for telemetry packets
 SLACreateDeviceOverlay_t	Send a command to generate and compile a devicetree overlay for a device
 SLACurrentAcquisitionParameters_t	Sent in response to SLAGetParameters_t() – see SLASetAcquisitionParameters_t for additional notes and details on the camera types
 SLACurrentADCPParameters_t	Current ADC parameters – format identical to SLASetADCPParameters_t
 SLACurrentAdvancedDetectionParameters_t	Return the current advanced detection parameters – format identical to SLASetAdvancedDetectionParameters_t
 SLACurrentBlendParameters_t	Current blend parameters – see SLASetBlendParameters_t
 SLACurrentConfiguration_t	Current camera and display configuration
 SLACurrentDetectionParameters_t	Current detection parameters – format identical to SLASetDetectionParameters_t
 SLACurrentDetectionRegionOfInterestParameters_t	Current detection region of interest parameters – format identical to SLASetDetectionRegionOfInterestParameters_t
 SLACurrentDigitalCameraParameters_t	Return parameters of currently selected digital camera – similar to SLASetDigitalCameraParameters_t
	

C <u>SLACurrentDisplayParameters_t</u>	Current display parameters – format identical to <u>SLASetDisplayParameters_t</u>
C <u>SLACurrentEthernetDisplayParameters_t</u>	Current Ethernet display parameters – format identical to <u>SLASetEthernetDisplayParameters_t</u>
C <u>SLACurrentEthernetVideoParameters_t</u>	Current Ethernet video parameters – format identical to <u>SLASetEthernetVideoParameters_t</u>
C <u>SLACurrentH264Parameters_t</u>	Current H264 parameters – format identical to <u>SLASetH264Parameters_t</u>
C <u>SLACurrentImageSize_t</u>	Current image size parameters
C <u>SLACurrentLensParameters_t</u>	Deprecated , use <u>SLALensParameters_t</u>
C <u>SLACurrentLensStatus_t</u>	Current lens focus and zoom position
C <u>SLACurrentMetadataFrameValues_t</u>	Current metadata frame values – format nearly identical to <u>SLASetMetadataFrameValues_t</u>
C <u>SLACurrentMetadataRate_t</u>	Current metadata rate – format similar to <u>SLASetMetadataRate_t</u>
C <u>SLACurrentMetadataValues_t</u>	Current metadata values – format nearly identical to <u>SLASetMetadataValues_t</u>
C <u>SLACurrentMultipleAlignment_t</u>	Deprecated , use <u>SLABlendAlign_t</u>
C <u>SLACurrentNetworkList_t</u>	List of all network interfaces controller names
C <u>SLACurrentNetworkParameters_t</u>	Current network parameters – format identical to <u>SLASetNetworkParameters_t</u>
C <u>SLACurrentOverlayMode_t</u>	Current overlay generation mode – format identical to <u>SLASetOverlayMode_t</u>
C <u>SLACurrentOverlayObjectParameters_t</u>	Properties of a particular graphic overlay specified by its Object ID
C <u>SLACurrentOverlayObjectsIds_t</u>	A list of used object ids as a 255 long bit array
C <u>SLACurrentPortConfiguration_t</u>	Current I/O port configuration parameters – format identical to <u>SLASetPortConfiguration_t</u>
C <u>SLACurrentRegistrationParameters_t</u>	Deprecated , use <u>SLARegistrationParameters_t</u>
C <u>SLACurrentSDCardDirectoryInfo_t</u>	A list of file names in the current recording directory
C <u>SLACurrentSDCardRecordingStatus_t</u>	Current SD card recording status – sent in response to <u>SLASetSDRecordingParameters_t</u> command
C <u>SLACurrentSnapShot_t</u>	Current snapshot state
C <u>SLACurrentStabilizationBias_t</u>	Deprecated , use <u>SLAStabilizationBias_t</u>
C <u>SLACurrentStabilizationParameters_t</u>	Describes the current stabilization mode
C <u>SLACurrentSystemValue_t</u>	Current SystemValue – format identical to <u>SLASetSystemValue_t</u>
C <u>SLACurrentTrackingParameters_t</u>	Current tracking parameters – format identical to <u>SLASetTrackingParameters_t</u>
C <u>SLACurrentUserPalette_t</u>	Return the current user palette in YUV values – format identical to <u>SLASetUserPalette_t</u>
C <u>SLACurrentVideoEnhancementParameters_t</u>	Current video enhancement parameters – format identical to <u>SLASetVideoEnhancementParameters_t</u>
C <u>SLACurrentVideoModeParameters_t</u>	Deprecated , use <u>SLAVideoDisplay_t</u> and <u>SLAMultiDisplay_t</u>
C <u>SLACurrentVideoParameters_t</u>	Current video parameters – format identical to <u>SLASetVideoParameters_t</u>
C <u>SLACursorOnTarget_t</u>	Cursor on Target (CoT) CoT packet contents are converted from KLV Metadata inputs
C <u>SLACustomAIDetect_t</u>	Custom AI Object Detector Model

C <u>SLACustomAutoFocusParameters_t</u>	Set Custom Autofocus Parameters
C <u>SLACustomClassifier_t</u>	Custom Classifier
C <u>SLACustomResponse_t</u>	A custom response that a customer can send from an overlay DLL
C <u>SLADeadPixel_t</u>	Modify Dead Pixel List Manually
C <u>SLADeadPixelStats_t</u>	Statistics from Dead Pixel Replacement calibration process
C <u>SLADecoderParameters_t</u>	Configure format of Onboard decoder
C <u>SLADesignateSelectedTrackPrimary_t</u>	Command the system to designate the selected track as primary
C <u>SLADetailedTimingMessage_t</u>	Packet is sent from the SLA hardware to report DSP and ARM detailed timing measurements
C <u>SLADigitalVideoParserParameters_t</u>	Set parameters for the digital video SightLine protocol parser
C <u>SLADirectoryStatisticsReply_t</u>	Sent in response to a <u>SLASetSDRecordingParameters_t</u>
C <u>SLADisplayAngle_t</u>	Set display rotation angle settings for a single camera or for all cameras See also <u>SLASetDisplayParameters_t</u>
C <u>SLADoDetectSnapShot_t</u>	Execute a track/detection snapshot to the MicroSD card or an external FTP server
C <u>SLADoSnapShot_t</u>	Execute an image snapshot
C <u>SLADrawObject_t</u>	Deprecated , use <u>SLADrawOverlay_t</u>
C <u>SLADrawOverlay_t</u>	Draw user specified overlay graphics on the screen
C <u>SLAExternalProgram_t</u>	Configure external programs – Set names of external applications to be run on the SLA hardware
C <u>SLAFocusParameters_t</u>	Focus parameters – control the focus measurement
C <u>SLAFocusStats_t</u>	Periodic report of focus and brightness statistics
C <u>SLAFourAlignPoints_t</u>	Used to assist in the alignment of two cameras for blending purposes
C <u>SLAFrameIndex_t</u>	User controlled frame index applied to specified camera index
C <u>SLAGetAcquisitionParameters_t</u>	Return a <u>SLACurrentAcquisitionParameters_t</u> message
C <u>SLAGetADCPParameters_t</u>	Return a <u>SLACurrentADCPParameters_t</u> message
C <u>SLAGetBlendParameters_t</u>	Return a <u>SLACurrentBlendParameters_t</u> message
C <u>SLAGetDetectionParameters_t</u>	Return a <u>SLACurrentDetectionParameters_t</u> message
C <u>SLAGetDisplayParameters_t</u>	Return a <u>SLACurrentDisplayParameters_t</u> message
C <u>SLAGetEthernetDisplayParameters_t</u>	Return a <u>SLACurrentEthernetDisplayParameters_t</u> message
C <u>SLAGetEthernetVideoParameters_t</u>	Return a <u>SLACurrentEthernetVideoParameters_t</u> message
C <u>SLAGetH264Parameters_t</u>	Return a <u>SLACurrentH264Parameters_t</u> message
C <u>SLAGetHardwareID_t</u>	Return a <u>SLAGetHardwareID_t</u> message
C <u>SLAGetImageSize_t</u>	Return a <u>SLACurrentImageSize_t</u> message
C <u>SLAGetNetworkList_t</u>	Return a <u>SLACurrentNetworkList_t</u> message
C <u>SLAGetNetworkParameters_t</u>	Return a <u>SLACurrentNetworkParameters_t</u> message
C <u>SLAGetOverlayMode_t</u>	Return a SLACurrentOverlay_t message
C <u>SLAGetParameters_t</u>	Request return of a specified message ID

C <u>SLAGetPortConfiguration_t</u>	Return a <u>SLACurrentPortConfiguration_t</u> message
C <u>SLAGetRegistrationParameters_t</u>	Deprecated , use <u>SLARegistrationParameters_t</u>
C <u>SLAGetSnapShot_t</u>	Return a <u>SLACurrentSnapShot_t</u> message
C <u>SLAGetStabilizationParameters_t</u>	Return a <u>SLACurrentStabilizationParameters_t</u> message
C <u>SLAGetTrackingParameters_t</u>	Return a <u>SLACurrentTrackingParameters_t</u> message
C <u>SLAGetVersionNumber_t</u>	Used to query for system information such as firmware version, temperature, app bits, and more
C <u>SLAGetVideoEnhancementParameters_t</u>	Return a <u>SLACurrentVideoEnhancementParameters_t</u> message
C <u>SLAGetVideoMode_t</u>	Deprecated , use <u>SLAVideoDisplay_t</u> and <u>SLAMultiDisplay_t</u>
C <u>SLAGetVideoParameters_t</u>	Return a <u>SLACurrentVideoParameters_t</u> message
C <u>SLAGPIO_t</u>	Write to and read from a GPIO
C <u>SLAI2CCommand_t</u>	Write or read I2C bus
C <u>SLAKlvClassFilters_t</u>	Set active Classification filters on KLV data
C <u>SLALandingAid_t</u>	Control landing aid detection
C <u>SLALandingPosition_t</u>	Periodic report of landing aid location, orientation, distance and match score
C <u>SLALensCommand_t</u>	Perform a lens control command
C <u>SLALensParameters_t</u>	Lens parameters - control the lens
C <u>SLALensRanges_t</u>	Lens focus and zoom ranges
C <u>SLALogoParameters_t</u>	Used to set the opacity of the logo (watermark) and its placement relative to the bottom right corner of the output frame
C <u>SLAMetadataStaticValues_t</u>	Sets new KLV metadata values
C <u>SLAModifyTrackIndex_t</u>	Modify a particular track by its index (stop or designate as primary)
C <u>SLAModifyTracking_t</u>	NOTE: Newer functionality in <u>SLAStartTracking_t</u>
C <u>SLAMultiDisplay_t</u>	Configure multi-camera display options such as Picture-In-Picture
C <u>SLANoise3D_t</u>	Statistics from 3D Noise calculation
C <u>SLANucParameters_t</u>	Set Non Uniformity Correction (NUC) and Dead Pixel Removal (DPR) parameters
C <u>SLANudgeTrackingCoordinate_t</u>	Adjust the primary track's coordinates by adding a nudge in pixel coordinate space to the current tracking coordinates
C <u>SLAReadWriteNuc_t</u>	Used to save or load NUC/DPR tables as well as set a table to be applied on startup
C <u>SLARegistrationParameters_t</u>	Parameters used in frame to frame image registration matching
C <u>SLAResetAllParameters_t</u>	Performs one of several different system resets
C <u>SLAResetStabilizationParameters_t</u>	Reset the internal motion smoothing filters that control video stabilization
C <u>SLASaveParameters_t</u>	Commits current parameters to flash
C <u>SLASendToBTS_t</u>	Start sending data to BTS from SLA
C <u>SLASendTraceStr_t</u>	Log commands and telemetry in human-readable format
C <u>SLASendTraceStr_t</u>	

C <u>SLASetAcquisitionParameters_t</u>	Configure video input for digital camera interfaces
C <u>SLASetADCPParameters_t</u>	Set parameters of the video analog-to-digital converter for analog inputs and color parameters for color digital cameras
C <u>SLASetAdvancedDetectionParameters_t</u>	Perform advanced control of the MTI parameters
C <u>SLASetBlendParameters_t</u>	Blend an EO(visible) and an IR camera with video scaling and shifting to align frames
C <u>SLASetDetectionParameters_t</u>	Set the parameters for the detection algorithms
C <u>SLASetDetectionRegionOfInterestParameters_t</u>	Sets the parameters for Moving Target Indication (MTI) Region of Interest(ROI)
C <u>SLASetDigitalCameraParameters_t</u>	Dynamically adjust a 10 to 16 bit digital camera input to an 8 bit image
C <u>SLASetDisplayAdjustments_t</u>	Set the parameters for fine-tuned adjustments of the display
C <u>SLASetDisplayParameters_t</u>	Set new display parameter settings
C <u>SLASetEthernetDisplayParameters_t</u>	Configure output format of Ethernet Video
C <u>SLASetEthernetVideoParameters_t</u>	Affects the quality, size and frame rate of the individual video frames sent over Ethernet
C <u>SLASetH264Parameters_t</u>	Modify H264, H265 and MPEG4 encoding behavior
C <u>SLASetKlvData_t</u>	The KLV blob data is constructed by user to be sent with MPEG2-TS stream (see <u>SLASetEthernetDisplayParameters_t</u>)
C <u>SLASetLensMode_t</u>	Perform camera lens control
C <u>SLASetLensParameters_t</u>	Deprecated , use <u>SLALensParameters_t</u>
C <u>SLASetMetadataFrameValues_t</u>	Sets new KLV metadata frame data values
C <u>SLASetMetadataRate_t</u>	Deprecated , use <u>SLATagDataRate_t</u>
C <u>SLASetMetadataValues_t</u>	Set new KLV metadata data values
C <u>SLASetMultipleAlignment_t</u>	Deprecated , use <u>SLABlendAlign_t</u>
C <u>SLASetNetworkParameters_t</u>	Configure the network settings for the device
C <u>SLASetOverlayMode_t</u>	Control display of overlay graphics
C <u>SLASetPortConfiguration_t</u>	Configure one of the communication ports for a specific protocol or mode
C <u>SLASetRegistrationParameters_t</u>	Deprecated , use <u>SLARegistrationParameters_t</u>
C <u>SLASetSDRecordingParameters_t</u>	Modify settings for on board video recording
C <u>SLASetSnapshot_t</u>	Set up parameters for image snapshot
C <u>SLASetStabilizationBias_t</u>	Deprecated , use <u>SLAStabilizationBias_t</u>
C <u>SLASetStabilizationParameters_t</u>	Turn on or turn off stabilization and control re-centering rate for output video
C <u>SLASetSystemValue_t</u>	Configure Special System Operational Values or Modes
C <u>SLASetTelemetryDestination_t</u>	Configure the Ethernet IP address of clients for telemetry
C <u>SLASetTrackingParameters_t</u>	Set parameters used by tracking module
C <u>SLASetUserPalette_t</u>	Set the Y, U, V values for the user false color palette
C <u>SLASetVideoEnhancementParameters_t</u>	Set up enhancement of the display video
C <u>SLASetVideoMode_t</u>	Deprecated , use <u>SLAVideoDisplay_t</u> and <u>SLAMultiDisplay_t</u>

C <u>SLASetVideoParameters_t</u>	Set parameters used by video input module
C <u>SLASetVMTI_t</u>	Allows targets from outside SightLine to be injected into the MPEG2-TS KLV stream
C <u>SLAShiftSelectedTrack_t</u>	Command the system to shift the selected track to the next track
C <u>SLAStabilizationBias_t</u>	Adjust the stabilization solution by adding a constant bias in pixel coordinate space to the current coordinates each frame
C <u>SLAStartTracking_t</u>	Command the system to start a track, stop a track, designate a track as primary, nudge a track, etc
C <u>SLAStopSelectedTrack_t</u>	Command the system to stop the currently selected track
C <u>SLAStopTracking_t</u>	Turn off all tracks
C <u>SLAStreamingControl_t</u>	Control to Start and Stop Network Streaming
C <u>SLASystemStatusMessage_t</u>	Packet is sent from the SLA hardware to report diagnostic information such as CPU load and temperature, or the performance of software tools
C <u>SLASystemStatusMode_t</u>	Enable / disable System Status Messages (<u>SLASystemStatusMessage_t</u>) and Detailed Timing Messages (<u>SLADetailedTimingMessage_t</u>) sent from the SLA hardware at 1 Hz
C <u>SLATagData_t</u>	Set the value of any MISB Tag and subtag
C <u>SLATagDataRate_t</u>	Set the output frequency of a KLV Tag or a range of tags
C <u>SLATagSourceSelector_t</u>	Select the source for a particular klv tag or a range of klv tags
C <u>SLATrackingBoxPixelStats_t</u>	Periodic report of pixel statistics within track boxes
C <u>SLATrackingMultiClass_t</u>	Periodic report of all tracked multi classifier information
C <u>SLATrackingPosition_t</u>	Periodic report of primary track's position, measured scene translation, rotation and scale, and stabilization display offset and rotation of a camera
C <u>SLATrackingPositions_t</u>	Periodic report of all tracked positions
C <u>SLATrackingPositionsExtended_t</u>	Periodic report of all tracked positions, with classifier information
C <u>SLATrackTrails_t</u>	Tracking Trail Information
C <u>SLAUsb3VisionFeature_t</u>	Send commands to an attached USB3Vision camera or query feature value
C <u>SLAUsbWebcamFeature_t</u>	Send commands to an attached V4L2-Controlled USB WebCamera, or query feature value
C <u>SLAUserFont_t</u>	Set user font filename
C <u>SLAUserWarningLevel_t</u>	Enable / disable User Warning Messages (<u>SLAUserWarningMessage_t</u>) sent from SLA hardware
C <u>SLAUserWarningMessage_t</u>	Message sent from the SLA hardware to notify users of any issues or system status which may not be obtained any other way
C <u>SLAVersionNumber_t</u>	Reports software and hardware version numbers
C <u>SLAVideoDisplay_t</u>	Configure which cameras and resolutions to display
C <u>SLAVMTIChips_t</u>	Configure KLV VMTI Chips
C <u>SLAVMTIFields_t</u>	Set active VMTI fields and Ontology Series Update Rate
C <u>SVPEndData_t</u>	Data block without specified length (determined using command length)

C <u>SVPEndString_t</u>	Text string without specified length
C <u>SVPLenData_t</u>	Data block with specified length
C <u>SVPLenString_t</u>	Simple data structures used to pass data
C <u>SVPSignedData_t</u>	Block of signed ints of specified size

Sightline Intelligence

Command, Control, and Script

API v3.11

Message Identifiers

List of SightLine command & reply message ID (type) constants.

Message ID	Name
0x00	SLAGetVersionNumber_t
0x01	SLAResetAllParameters_t
0x02	SLASetStabilizationParameters_t
0x03	SLAGetStabilizationParameters_t
0x04	SLAResetStabilizationParameters_t
0x05	SLAModifyTracking_t
0x06	SLASetOverlayMode_t
0x07	SLAGetOverlayMode_t
0x08	SLAStartTracking_t
0x09	SLAStopTracking_t
0x0A	SLANudgeTrackingCoordinate_t
0x0B	SLACoordinateReportingMode_t
0x0C	SLASetTrackingParameters_t
0x0D	SLAGetTrackingParameters_t
0x0E	SLASetRegistrationParameters_t
0x0F	SLAGetRegistrationParameters_t
0x10	SLASetVideoParameters_t
0x11	SLAGetVideoParameters_t
0x12	SLASetStabilizationBias_t
0x13	SLASetMetadataValues_t
0x14	SLAMetadataStaticValues_t
0x15	SLASetMetadataFrameValues_t
0x16	SLASetDisplayParameters_t
0x17	SLAModifyTrackIndex_t

0x18	<u>SLASetADCParameters_t</u>
0x19	<u>SLAGetADCParameters_t</u>
0x1A	<u>SLASetEthernetVideoParameters_t</u>
0x1B	<u>SLAGetEthernetVideoParameters_t</u>
0x1C	<u>SLASetNetworkParameters_t</u>
0x1D	<u>SLAGetNetworkParameters_t</u>
0x1E	<u>SLASetSDRecordingParameters_t</u>
0x1F	<u>SLASetVideoMode_t</u>
0x20	<u>SLAGetVideoMode_t</u>
0x21	<u>SLASetVideoEnhancementParameters_t</u>
0x22	<u>SLAGetVideoEnhancementParameters_t</u>
0x23	<u>SLASetH264Parameters_t</u>
0x24	<u>SLAGetH264Parameters_t</u>
0x25	<u>SLASaveParameters_t</u>
0x28	<u>SLAGetParameters_t</u>
0x29	<u>SLASetEthernetDisplayParameters_t</u>
0x2A	<u>SLASetDisplayAdjustments_t</u>
0x2D	<u>SLASetDetectionParameters_t</u>
0x2E	<u>SLAGetDetectionParameters_t</u>
0x2F	<u>SLASetBlendParameters_t</u>
0x30	<u>SLAGetBlendParameters_t</u>
0x31	<u>SLAGetImageSize_t</u>
0x32	<u>SLADesignateSelectedTrackPrimary_t</u>
0x33	<u>SLAShiftSelectedTrack_t</u>
0x35	<u>SLANucParameters_t</u>
0x36	<u>SLAReadWriteNuc_t</u>
0x37	<u>SLASetAcquisitionParameters_t</u>
0x38	<u>SLAGetAcquisitionParameters_t</u>
0x39	<u>SLAGetEthernetDisplayParameters_t</u>
0x3A	<u>SLAGetDisplayParameters_t</u>
0x3B	<u>SLADrawObject_t</u>
0x3C	<u>SLAStopSelectedTrack_t</u>
0x3D	<u>SLACommandPassThrough_t</u>
0x3E	<u>SLASetPortConfiguration_t</u>
0x3F	<u>SLAGetPortConfiguration_t</u>
0x40	<u>SLAVersionNumber_t</u>

0x41	<u>SLACurrentStabilizationParameters_t</u>
0x42	<u>SLACurrentOverlayMode_t</u>
0x43	<u>SLATrackingPosition_t</u>
0x44	<u>SLACurrentTrackingParameters_t</u>
0x45	<u>SLACurrentRegistrationParameters_t</u>
0x46	<u>SLACurrentVideoParameters_t</u>
0x47	<u>SLACurrentADCPParameters_t</u>
0x48	<u>SLACurrentEthernetVideoParameters_t</u>
0x49	<u>SLACurrentNetworkParameters_t</u>
0x4A	<u>SLACurrentVideoEnhancementParameters_t</u>
0x4B	<u>SLACurrentVideoModeParameters_t</u>
0x4D	<u>SLACurrentBlendParameters_t</u>
0x4E	<u>SLACurrentImageSize_t</u>
0x4F	<u>SLACurrentAcquisitionParameters_t</u>
0x50	<u>SLAGetHardwareID_t</u>
0x51	<u>SLATrackingPositions_t</u>
0x52	<u>SLACurrentEthernetDisplayParameters_t</u>
0x53	<u>SLACurrentPortConfiguration_t</u>
0x54	<u>SLACurrentDetectionParameters_t</u>
0x55	<u>SLAFocusStats_t</u>
0x56	<u>SLACurrentH264Parameters_t</u>
0x57	<u>SLACurrentDisplayParameters_t</u>
0x58	<u>SLACurrentSDCardRecordingStatus_t</u>
0x59	<u>SLACurrentSDCardDirectoryInfo_t</u>
0x5A	<u>SLASendTraceStr_t</u>
0x5B	<u>SLACommandCamera_t</u>
0x5C	<u>SLADisplayAngle_t</u>
0x5D	<u>SLACurrentSnapShot_t</u>
0x5E	<u>SLASetSnapShot_t</u>
0x5F	<u>SLAGetSnapShot_t</u>
0x60	<u>SLADoSnapShot_t</u>
0x61	<u>SLASetKlvData_t</u>
0x62	<u>SLASetMetadataRate_t</u>
0x63	SLASetSystemType_t
0x64	<u>SLASetTelemetryDestination_t</u>
0x65	SLACurrentSystemType_t

0x66	<u>SLAGetNetworkList_t</u>
0x67	<u>SLACurrentNetworkList_t</u>
0x68	<u>SLACurrentOverlayObjectsIds_t</u>
0x6B	<u>SLACurrentOverlayObjectParameters_t</u>
0x6C	<u>SLASetLensMode_t</u>
0x6D	<u>SLACurrentLensStatus_t</u>
0x6E	<u>SLASetLensParameters_t</u>
0x6F	<u>SLACurrentLensParameters_t</u>
0x70	<u>SLASetDigitalCameraParameters_t</u>
0x71	<u>SLACurrentDigitalCameraParameters_t</u>
0x72	<u>SLASetUserPalette_t</u>
0x73	<u>SLACurrentUserPalette_t</u>
0x74	<u>SLASetMultipleAlignment_t</u>
0x75	<u>SLACurrentMultipleAlignment_t</u>
0x76	<u>SLASetAdvancedDetectionParameters_t</u>
0x77	<u>SLACurrentAdvancedDetectionParameters_t</u>
0x78	<u>SLATrackingBoxPixelStats_t</u>
0x79	<u>SLADirectoryStatisticsReply_t</u>
0x7A	<u>SLACurrentStabilizationBias_t</u>
0x7B	<u>SLAAdvancedCaptureParameters_t</u>
0x7C	<u>SLASetDetectionRegionOfInterestParameters_t</u>
0x7D	<u>SLACurrentDetectionRegionOfInterestParameters_t</u>
0x7F	<u>SLAUserWarningLevel_t</u>
0x80	<u>SLASystemStatusMode_t</u>
0x81	<u>SLALandingAid_t</u>
0x82	<u>SLACameraSwitch_t</u>
0x83	<u>SLALandingPosition_t</u>
0x84	<u>SLASetVMTI_t</u>
0x86	<u>SLAUserWarningMessage_t</u>
0x87	<u>SLASystemStatusMessage_t</u>
0x88	<u>SLADetailedTimingMessage_t</u>
0x89	<u>SLAAppendedMetadata_t</u>
0x8A	<u>SLAFrameIndex_t</u>
0x8B	<u>SLACurrentMetadataValues_t</u>
0x8C	<u>SLACurrentMetadataFrameValues_t</u>
0x8D	<u>SLACurrentMetadataRate_t</u>

0x8E	<u>SLACurrentConfiguration_t</u>
0x8F	<u>SLAExternalProgram_t</u>
0x90	<u>SLAStreamingControl_t</u>
0x91	<u>SLADigitalVideoParserParameters_t</u>
0x92	<u>SLASetSystemValue_t</u>
0x93	<u>SLACurrentSystemValue_t</u>
0x94	<u>SLAI2CCommand_t</u>
0x95	<u>SLAFourAlignPoints_t</u>
0x96	<u>SLATagData_t</u>
0x97	<u>SLATagDataRate_t</u>
0x98	<u>SLATagSourceSelector_t</u>
0x99	<u>SLADecoderParameters_t</u>
0x9B	<u>SLALogoParameters_t</u>
0x9C	<u>SLADrawOverlay_t</u>
0x9D	<u>SLATrackTrails_t</u>
0x9E	<u>SLARegistrationParameters_t</u>
0x9F	<u>SLAStabilizationBias_t</u>
0xA0	<u>SLATrackingPositionsExtended_t</u>
0xA1	<u>SLADeadPixelStats_t</u>
0xA2	SLAInternalCommand_t
0xA3	SLAInternalResponse_t
0xA4	<u>SLAVideoDisplay_t</u>
0xA5	<u>SLAMultiDisplay_t</u>
0xA6	<u>SLAUsb3VisionFeature_t</u>
0xA7	<u>SLACustomClassifier_t</u>
0xA8	<u>SLADeadPixel_t</u>
0xA9	<u>SLAClassifierParameters_t</u>
0xAA	SLASendScript_t
0xAB	<u>SLADoDetectSnapShot_t</u>
0xAC	<u>SLAAncillaryTextMetadata_t</u>
0xAD	<u>SLAVMTIChips_t</u>
0xAE	<u>SLAUserFont_t</u>
0xAF	<u>SLANoise3D_t</u>
0xB0	<u>SLACursorOnTarget_t</u>
0xB1	<u>SLALensParameters_t</u>
0xB2	<u>SLALensCommand_t</u>

0xB3	<u>SLAFocusParameters_t</u>
0xB4	<u>SLALensRanges_t</u>
0xB5	<u>SLACustomAutoFocusParameters_t</u>
0xB6	<u>SLAGPIO_t</u>
0xB7	<u>SLAUsbWebcamFeature_t</u>
0xB8	<u>SLACreateDeviceOverlay_t</u>
0xB9	<u>SLABlendAlign_t</u>
0xBA	<u>SLACustomAIDetect_t</u>
0xBB	<u>SLACameraCapabilities_t</u>
0xBC	<u>SLACustomResponse_t</u>
0xBD	<u>SLATrackingMultiClass_t</u>
0xBE	<u>SLASendToBTS_t</u>
0xBF	<u>SLAVMTIFields_t</u>
0xC0	<u>SLACameraCalibration_t</u>
0xC1	<u>SLAKlvClassFilters_t</u>
0xC2	SLACameraParameterFile_t

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

Lua Script Functions

Classes

struct	SLAExternalProgram_t
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Functions

u32	SLAGetVersionNumber (void *context, SLAGetVersionNumber_t *v, struct SVPOut_t *out, u16 nElements)
u32	SLAResetAllParameters (void *context, SLAResetAllParameters_t *v, struct SVPOut_t *out, u16 nElements)
u32	SLASetStabilizationParameters (void *context, SLASetStabilizationParameters_t *v, struct SVPOut_t *out, u16 nElements)
u32	SLAGetStabilizationParameters (void *context, SLAGetStabilizationParameters_t *v, struct SVPOut_t *out, u16 nElements)
u32	SLAResetStabilizationParameters (void *context, SLAResetStabilizationParameters_t *v, struct SVPOut_t *out, u16 nElements)
u32	SLAModifyTracking (void *context, SLAModifyTracking_t *v, struct SVPOut_t *out, u16 nElements)
u32	SLASetOverlayMode (void *context, SLASetOverlayMode_t *v, struct SVPOut_t *out, u16 nElements)
u32	SLAGetOverlayMode (void *context, SLAGetOverlayMode_t *v, struct SVPOut_t *out, u16 nElements)
u32	SLAStartTracking (void *context, SLAStartTracking_t *v, struct SVPOut_t *out, u16 nElements)
u32	SLAStopTracking (void *context, SLAStopTracking_t *v, struct SVPOut_t *out, u16 nElements)
u32	SLANudgeTrackingCoordinate (void *context, SLANudgeTrackingCoordinate_t *v, struct SVPOut_t *out, u16 nElements)
u32	SLACoordinateReportingMode (void *context, SLACoordinateReportingMode_t *v, struct SVPOut_t *out, u16 nElements)
u32	SLASetTrackingParameters (void *context, SLASetTrackingParameters_t *v, struct SVPOut_t *out, u16 nElements)
u32	SLAGetTrackingParameters (void *context, SLAGetTrackingParameters_t *v, struct SVPOut_t *out, u16 nElements)
u32	SLASetRegistrationParameters (void *context, SLASetRegistrationParameters_t *v, struct SVPOut_t *out, u16 nElements)
u32	SLAGetRegistrationParameters (void *context, SLAGetRegistrationParameters_t *v, struct SVPOut_t *out, u16 nElements)
u32	SLASetVideoParameters (void *context, SLASetVideoParameters_t *v, struct SVPOut_t *out, u16 nElements)

u32	<u>SLAGetVideoParameters</u>	(void *context, <u>SLAGetVideoParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASetStabilizationBias</u>	(void *context, <u>SLASetStabilizationBias_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASetMetadataValues</u>	(void *context, <u>SLASetMetadataValues_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAMetadataStaticValues</u>	(void *context, <u>SLAMetadataStaticValues_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASetMetadataFrameValues</u>	(void *context, <u>SLASetMetadataFrameValues_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASetDisplayParameters</u>	(void *context, <u>SLASetDisplayParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAModifyTrackIndex</u>	(void *context, <u>SLAModifyTrackIndex_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASetADCPParameters</u>	(void *context, <u>SLASetADCPParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAGetADCPParameters</u>	(void *context, <u>SLAGetADCPParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASetEthernetVideoParameters</u>	(void *context, <u>SLASetEthernetVideoParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAGetEthernetVideoParameters</u>	(void *context, <u>SLAGetEthernetVideoParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASetNetworkParameters</u>	(void *context, <u>SLASetNetworkParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAGetNetworkParameters</u>	(void *context, <u>SLAGetNetworkParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASetSDRecordingParameters</u>	(void *context, <u>SLASetSDRecordingParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASetVideoMode</u>	(void *context, <u>SLASetVideoMode_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAGetVideoMode</u>	(void *context, <u>SLAGetVideoMode_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASetVideoEnhancementParameters</u>	(void *context, <u>SLASetVideoEnhancementParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAGetVideoEnhancementParameters</u>	(void *context, <u>SLAGetVideoEnhancementParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASetH264Parameters</u>	(void *context, <u>SLASetH264Parameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAGetH264Parameters</u>	(void *context, <u>SLAGetH264Parameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASaveParameters</u>	(void *context, <u>SLASaveParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAGetParameters</u>	(void *context, <u>SLAGetParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASetEthernetDisplayParameters</u>	(void *context, <u>SLASetEthernetDisplayParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASetDisplayAdjustments</u>	(void *context, <u>SLASetDisplayAdjustments_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASetDetectionParameters</u>	(void *context, <u>SLASetDetectionParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAGetDetectionParameters</u>	(void *context, <u>SLAGetDetectionParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASetBlendParameters</u>	(void *context, <u>SLASetBlendParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAGetBlendParameters</u>	(void *context, <u>SLAGetBlendParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAGetImageSize</u>	(void *context, <u>SLAGetImageSize_t</u> *v, struct SVPOut_t *out, u16 nElements)

u32	<u>SLADesignateSelectedTrackPrimary</u> (void *context, <u>SLADesignateSelectedTrackPrimary_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAShiftSelectedTrack</u> (void *context, <u>SLAShiftSelectedTrack_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLANucParameters</u> (void *context, <u>SLANucParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAReadWriteNuc</u> (void *context, <u>SLAReadWriteNuc_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASetAcquisitionParameters</u> (void *context, <u>SLASetAcquisitionParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAGetAcquisitionParameters</u> (void *context, <u>SLAGetAcquisitionParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAGetEthernetDisplayParameters</u> (void *context, <u>SLAGetEthernetDisplayParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAGetDisplayParameters</u> (void *context, <u>SLAGetDisplayParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLADrawObject</u> (void *context, <u>SLADrawObject_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAStopSelectedTrack</u> (void *context, <u>SLAStopSelectedTrack_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACommandPassThrough</u> (void *context, <u>SLACommandPassThrough_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASetPortConfiguration</u> (void *context, <u>SLASetPortConfiguration_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAGetPortConfiguration</u> (void *context, <u>SLAGetPortConfiguration_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAVersionNumber</u> (void *context, <u>SLAVersionNumber_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentStabilizationParameters</u> (void *context, <u>SLACurrentStabilizationParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentOverlayMode</u> (void *context, <u>SLACurrentOverlayMode_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLATrackingPosition</u> (void *context, <u>SLATrackingPosition_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentTrackingParameters</u> (void *context, <u>SLACurrentTrackingParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentRegistrationParameters</u> (void *context, <u>SLACurrentRegistrationParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentVideoParameters</u> (void *context, <u>SLACurrentVideoParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentADCParameters</u> (void *context, <u>SLACurrentADCParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentEthernetVideoParameters</u> (void *context, <u>SLACurrentEthernetVideoParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentNetworkParameters</u> (void *context, <u>SLACurrentNetworkParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentVideoEnhancementParameters</u> (void *context, <u>SLACurrentVideoEnhancementParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentVideoModeParameters</u> (void *context, <u>SLACurrentVideoModeParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentBlendParameters</u> (void *context, <u>SLACurrentBlendParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentImageSize</u> (void *context, <u>SLACurrentImageSize_t</u> *v, struct SVPOut_t *out, u16 nElements)

u32	<u>SLACurrentAcquisitionParameters</u> (void *context, <u>SLACurrentAcquisitionParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAGetHardwareID</u> (void *context, <u>SLAGetHardwareID_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLATrackingPositions</u> (void *context, <u>SLATrackingPositions_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentEthernetDisplayParameters</u> (void *context, <u>SLACurrentEthernetDisplayParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentPortConfiguration</u> (void *context, <u>SLACurrentPortConfiguration_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentDetectionParameters</u> (void *context, <u>SLACurrentDetectionParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAFocusStats</u> (void *context, <u>SLAFocusStats_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentH264Parameters</u> (void *context, <u>SLACurrentH264Parameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentDisplayParameters</u> (void *context, <u>SLACurrentDisplayParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentSDCardRecordingStatus</u> (void *context, <u>SLACurrentSDCardRecordingStatus_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentSDCardDirectoryInfo</u> (void *context, <u>SLACurrentSDCardDirectoryInfo_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASendTraceStr</u> (void *context, <u>SLASendTraceStr_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACommandCamera</u> (void *context, <u>SLACommandCamera_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLADisplayAngle</u> (void *context, <u>SLADisplayAngle_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentSnapShot</u> (void *context, <u>SLACurrentSnapShot_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASetSnapShot</u> (void *context, <u>SLASetSnapShot_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAGetSnapShot</u> (void *context, <u>SLAGetSnapShot_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLADoSnapShot</u> (void *context, <u>SLADoSnapShot_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASetKlvData</u> (void *context, <u>SLASetKlvData_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASetMetadataRate</u> (void *context, <u>SLASetMetadataRate_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASetTelemetryDestination</u> (void *context, <u>SLASetTelemetryDestination_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAGetNetworkList</u> (void *context, <u>SLAGetNetworkList_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentNetworkList</u> (void *context, <u>SLACurrentNetworkList_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentOverlayObjectsIds</u> (void *context, <u>SLACurrentOverlayObjectsIds_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentOverlayObjectParameters</u> (void *context, <u>SLACurrentOverlayObjectParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASetLensMode</u> (void *context, <u>SLASetLensMode_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentLensStatus</u> (void *context, <u>SLACurrentLensStatus_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASetLensParameters</u> (void *context, <u>SLASetLensParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentLensParameters</u> (void *context, <u>SLACurrentLensParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASetDigitalCameraParameters</u> (void *context, <u>SLASetDigitalCameraParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)

	u16 nElements)
u32	<u>SLACurrentDigitalCameraParameters</u> (void *context, <u>SLACurrentDigitalCameraParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASetUserPalette</u> (void *context, <u>SLASetUserPalette_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentUserPalette</u> (void *context, <u>SLACurrentUserPalette_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASetMultipleAlignment</u> (void *context, <u>SLASetMultipleAlignment_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentMultipleAlignment</u> (void *context, <u>SLACurrentMultipleAlignment_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASetAdvancedDetectionParameters</u> (void *context, <u>SLASetAdvancedDetectionParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentAdvancedDetectionParameters</u> (void *context, <u>SLACurrentAdvancedDetectionParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLATrackingBoxPixelStats</u> (void *context, <u>SLATrackingBoxPixelStats_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLADirectoryStatisticsReply</u> (void *context, <u>SLADirectoryStatisticsReply_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentStabilizationBias</u> (void *context, <u>SLACurrentStabilizationBias_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAAdvancedCaptureParameters</u> (void *context, <u>SLAAdvancedCaptureParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASetDetectionRegionOfInterestParameters</u> (void *context, <u>SLASetDetectionRegionOfInterestParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentDetectionRegionOfInterestParameters</u> (void *context, <u>SLACurrentDetectionRegionOfInterestParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAUserWarningLevel</u> (void *context, <u>SLAUserWarningLevel_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASystemStatusMode</u> (void *context, <u>SLASystemStatusMode_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLALandingAid</u> (void *context, <u>SLALandingAid_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACameraSwitch</u> (void *context, <u>SLACameraSwitch_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLALandingPosition</u> (void *context, <u>SLALandingPosition_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASetVMTI</u> (void *context, <u>SLASetVMTI_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAUserWarningMessage</u> (void *context, <u>SLAUserWarningMessage_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASystemStatusMessage</u> (void *context, <u>SLASystemStatusMessage_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLADetailedTimingMessage</u> (void *context, <u>SLADetailedTimingMessage_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAAppendedMetadata</u> (void *context, <u>SLAAppendedMetadata_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAFrameIndex</u> (void *context, <u>SLAFrameIndex_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentMetadataValues</u> (void *context, <u>SLACurrentMetadataValues_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentMetadataFrameValues</u> (void *context, <u>SLACurrentMetadataFrameValues_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentMetadataRate</u> (void *context, <u>SLACurrentMetadataRate_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentConfiguration</u> (void *context, <u>SLACurrentConfiguration_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAExternalProgram</u> (void *context, <u>SLAExternalProgram_t</u> *v, struct SVPOut_t *out, u16 nElements)

u32	<u>SLAStreamingControl</u> (void *context, <u>SLAStreamingControl_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLADigitalVideoParserParameters</u> (void *context, <u>SLADigitalVideoParserParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASetSystemValue</u> (void *context, <u>SLASetSystemValue_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACurrentSystemValue</u> (void *context, <u>SLACurrentSystemValue_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAI2CCCommand</u> (void *context, <u>SLAI2CCCommand_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAFourAlignPoints</u> (void *context, <u>SLAFourAlignPoints_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLATagData</u> (void *context, <u>SLATagData_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLATagDataRate</u> (void *context, <u>SLATagDataRate_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLATagSourceSelector</u> (void *context, <u>SLATagSourceSelector_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLADecoderParameters</u> (void *context, <u>SLADecoderParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLALogoParameters</u> (void *context, <u>SLALogoParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLADrawOverlay</u> (void *context, <u>SLADrawOverlay_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLATrackTrails</u> (void *context, <u>SLATrackTrails_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLARegistrationParameters</u> (void *context, <u>SLARegistrationParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAStabilizationBias</u> (void *context, <u>SLAStabilizationBias_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLATrackingPositionsExtended</u> (void *context, <u>SLATrackingPositionsExtended_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLADeadPixelStats</u> (void *context, <u>SLADeadPixelStats_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAVideoDisplay</u> (void *context, <u>SLAVideoDisplay_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAMultiDisplay</u> (void *context, <u>SLAMultiDisplay_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAUsb3VisionFeature</u> (void *context, <u>SLAUsb3VisionFeature_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACustomClassifier</u> (void *context, <u>SLACustomClassifier_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLADeadPixel</u> (void *context, <u>SLADeadPixel_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAClassifierParameters</u> (void *context, <u>SLAClassifierParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLADoDetectSnapShot</u> (void *context, <u>SLADoDetectSnapShot_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAAncillaryTextMetadata</u> (void *context, <u>SLAAncillaryTextMetadata_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAVMTIChips</u> (void *context, <u>SLAVMTIChips_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAUserFont</u> (void *context, <u>SLAUserFont_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLANoise3D</u> (void *context, <u>SLANoise3D_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACursorOnTarget</u> (void *context, <u>SLACursorOnTarget_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLALensParameters</u> (void *context, <u>SLALensParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLALensCommand</u> (void *context, <u>SLALensCommand_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAFocusParameters</u> (void *context, <u>SLAFocusParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLALensRanges</u> (void *context, <u>SLALensRanges_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACustomAutoFocusParameters</u> (void *context, <u>SLACustomAutoFocusParameters_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAGPIO</u> (void *context, <u>SLAGPIO_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAUsbWebcamFeature</u> (void *context, <u>SLAUsbWebcamFeature_t</u> *v, struct SVPOut_t *out, u16 nElements)

u32	<u>SLACreateDeviceOverlay</u>	(void *context, <u>SLACreateDeviceOverlay_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLABlendAlign</u>	(void *context, <u>SLABlendAlign_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACustomAIDetect</u>	(void *context, <u>SLACustomAIDetect_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACameraCapabilities</u>	(void *context, <u>SLACameraCapabilities_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACustomResponse</u>	(void *context, <u>SLACustomResponse_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLATrackingMultiClass</u>	(void *context, <u>SLATrackingMultiClass_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLASendToBTS</u>	(void *context, <u>SLASendToBTS_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAVMTIFields</u>	(void *context, <u>SLAVMTIFields_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLACameraCalibration</u>	(void *context, <u>SLACameraCalibration_t</u> *v, struct SVPOut_t *out, u16 nElements)
u32	<u>SLAKlvClassFilters</u>	(void *context, <u>SLAKlvClassFilters_t</u> *v, struct SVPOut_t *out, u16 nElements)

Detailed Description

Lua script interface for system control.

Function Documentation

[_SLAGetVersionNumber\(\)](#)

```

u32 SLAGetVersionNumber ( void *context,
                           SLAGetVersionNumber\_t *v,
                           struct SVPOut_t *out,
                           u16 nElements
                         )

```

Used to query for system information such as firmware version, temperature, app bits, and more.

[_SLAResetAllParameters\(\)](#)

```

u32 SLAResetAllParameters ( void *context,
                             SLAResetAllParameters\_t *v,
                             struct SVPOut_t *out,
                             u16 nElements
                           )

```

Performs one of several different system resets.

_SLASetStabilizationParameters()

```
u32 SLASetStabilizationParameters ( void * context,
                                     SLASetStabilizationParameters_t * v,
                                     struct SVPOut_t * out,
                                     u16 nElements
                                   )
```

Turn on or turn off stabilization and control re-centering rate for output video.

`_SLAGetStabilizationParameters()`

```
u32 SLAGetStabilizationParameters ( void * context,
                                     SLAGetStabilizationParameters_t * v,
                                     struct SVPOut_t * out,
                                     u16 nElements
                                   )
```

Return a **SLACurrentStabilizationParameters_t** message.

`_SLAResetStabilizationParameters()`

```
u32 SLAResetStabilizationParameters ( void * context,
                                     SLAResetStabilizationParameters_t * v,
                                     struct SVPOut_t * out,
                                     u16 nElements
                                     )
```

Reset the internal motion smoothing filters that control video stabilization.

`_SLAMModifyTracking()`

[illegible]

NOTE: Newer functionality in [SLAStartTracking_t](#).

__SLASetOverlayMode()

```
u32 SLASetOverlayMode ( void *          context,
                        SLASetOverlayMode\_t * v,
                        struct SVPOut_t * out,
                        u16                nElements
                      )
```

Control display of overlay graphics.

__SLAGetOverlayMode()

```
u32 SLAGetOverlayMode ( void *          context,
                        SLAGetOverlayMode\_t * v,
                        struct SVPOut_t * out,
                        u16                nElements
                      )
```

Return a SLACurrentOverlay_t message.

__SLAStartTracking()

```
u32 SLAStartTracking ( void *          context,
                       SLAStartTracking\_t * v,
                       struct SVPOut_t * out,
                       u16                nElements
                     )
```

Command the system to start a track, stop a track, designate a track as primary, nudge a track, etc.

__SLAStopTracking()

```
u32 SLAStopTracking ( void *          context,
                     SLAStopTracking\_t * v,
```

```
    struct SVPOut_t * out,  
    u16 nElements  
)
```

Turn off all tracks.

_SLANudgeTrackingCoordinate()

```
u32 SLANudgeTrackingCoordinate ( void * context,  
    SLANudgeTrackingCoordinate_t * v,  
    struct SVPOut_t * out,  
    u16 nElements  
)
```

Adjust the primary track's coordinates by adding a nudge in pixel coordinate space to the current tracking coordinates.

_SLACoordinateReportingMode()

```
u32 SLACoordinateReportingMode ( void * context,  
    SLACoordinateReportingMode_t * v,  
    struct SVPOut_t * out,  
    u16 nElements  
)
```

Set reporting rates and telemetry types to send for telemetry packets.

_SLASetTrackingParameters()

```
u32 SLASetTrackingParameters ( void * context,  
    SLASetTrackingParameters_t * v,  
    struct SVPOut_t * out,  
    u16 nElements  
)
```

Set parameters used by tracking module.

_SLAGetTrackingParameters()

```
u32 SLAGetTrackingParameters ( void * context,  
                               SLAGetTrackingParameters_t * v,  
                               struct SVPOut_t * out,  
                               u16 nElements  
                               )
```

Return a SLACurrentTrackingParameters_t message.

_SLASetRegistrationParameters()

```
u32 SLASetRegistrationParameters ( void * context,  
                                   SLASetRegistrationParameters_t * v,  
                                   struct SVPOut_t * out,  
                                   u16 nElements  
                                   )
```

Deprecated, use SLARegistrationParameters_t

_SLAGetRegistrationParameters()

```
u32 SLAGetRegistrationParameters ( void * context,  
                                   SLAGetRegistrationParameters_t * v,  
                                   struct SVPOut_t * out,  
                                   u16 nElements  
                                   )
```

Deprecated, use SLARegistrationParameters_t

_SLASetVideoParameters()

```
u32 SLASetVideoParameters ( void * context,  
                             SLASetVideoParameters_t * v,  
                             struct SVPOut_t * out,  
                             u16 nElements  
                             )
```



```

        struct SVPOut_t *
        u16
    )

```

Sets new KLV metadata values.

_SLASetMetadataFrameValues()

```

u32 SLASetMetadataFrameValues ( void *
                                context,
                                SLASetMetadataFrameValues_t * v,
                                struct SVPOut_t *
                                u16
                                out,
                                nElements
                                )

```

Sets new KLV metadata frame data values.

_SLASetDisplayParameters()

```

u32 SLASetDisplayParameters ( void *
                                context,
                                SLASetDisplayParameters_t * v,
                                struct SVPOut_t *
                                u16
                                out,
                                nElements
                                )

```

Set new display parameter settings.

_SLAModifyTrackIndex()

```

u32 SLAModifyTrackIndex ( void *
                            context,
                            SLAModifyTrackIndex_t * v,
                            struct SVPOut_t *
                            u16
                            out,
                            nElements
                            )

```

Modify a particular track by its index (stop or designate as primary).

SLASetADCPParameters()

```
u32 SLASetADCPParameters ( void * context,
                           SLASetADCPParameters_t * v,
                           struct SVPOut_t * out,
                           u16 nElements
                           )
```

Set parameters of the video analog-to-digital converter for analog inputs and color parameters for color digital cameras.

SLAGetADCPParameters()

```
u32 SLAGetADCPParameters ( void * context,
                           SLAGetADCPParameters_t * v,
                           struct SVPOut_t * out,
                           u16 nElements
                           )
```

Return a SLACurrentADCPParameters_t message.

SLASetEthernetVideoParameters()

```
u32 SLASetEthernetVideoParameters ( void * context,
                                     SLASetEthernetVideoParameters_t * v,
                                     struct SVPOut_t * out,
                                     u16 nElements
                                     )
```

Affects the quality, size and frame rate of the individual video frames sent over Ethernet.

SLAGetEthernetVideoParameters()

```
u32 SLAGetEthernetVideoParameters ( void * context,
                                     SLAGetEthernetVideoParameters_t * v,
                                     struct SVPOut_t * out,
                                     u16 nElements
                                     )
```

Return a [SLACurrentEthernetVideoParameters_t](#) message.

_SLASetNetworkParameters()

```
u32 SLASetNetworkParameters ( void * context,  
                              SLASetNetworkParameters\_t * v,  
                              struct SVPOut_t * out,  
                              u16 nElements  
                              )
```

Configure the network settings for the device.

_SLAGetNetworkParameters()

```
u32 SLAGetNetworkParameters ( void * context,  
                              SLAGetNetworkParameters\_t * v,  
                              struct SVPOut_t * out,  
                              u16 nElements  
                              )
```

Return a [SLACurrentNetworkParameters_t](#) message.

_SLASetSDRecordingParameters()

```
u32 SLASetSDRecordingParameters ( void * context,  
                                  SLASetSDRecordingParameters\_t * v,  
                                  struct SVPOut_t * out,  
                                  u16 nElements  
                                  )
```

Modify settings for on board **video** recording.

_SLASetVideoMode()

```
u32 SLASetVideoMode ( void * context,  
                     SLASetVideoMode\_t * v,
```



```

        struct SVPOut_t * out,
        u16 nElements
    )

```

Deprecated, use [SLAVideoDisplay_t](#) and [SLAMultiDisplay_t](#).

_SLAGetVideoMode()

```

u32 SLAGetVideoMode ( void * context,
    SLAGetVideoMode\_t * v,
    struct SVPOut_t * out,
    u16 nElements
)

```

Deprecated, use [SLAVideoDisplay_t](#) and [SLAMultiDisplay_t](#).

_SLASetVideoEnhancementParameters()

```

u32 SLASetVideoEnhancementParameters ( void * context,
    SLASetVideoEnhancementParameters\_t * v,
    struct SVPOut_t * out,
    u16 nElements
)

```

Set up enhancement of the display video.

_SLAGetVideoEnhancementParameters()

```

u32 SLAGetVideoEnhancementParameters ( void * context,
    SLAGetVideoEnhancementParameters\_t * v,
    struct SVPOut_t * out,
    u16 nElements
)

```

Return a [SLACurrentVideoEnhancementParameters_t](#) message.

_SLASetH264Parameters()

```
u32 SLASetH264Parameters ( void *          context,  
                           SLASetH264Parameters_t * v,  
                           struct SVPOut_t * out,  
                           u16              nElements  
                           )
```

Modify H264, H265 and MPEG4 encoding behavior.

_SLAGetH264Parameters()

```
u32 SLAGetH264Parameters ( void *          context,  
                           SLAGetH264Parameters_t * v,  
                           struct SVPOut_t * out,  
                           u16              nElements  
                           )
```

Return a SLACurrentH264Parameters_t message.

_SLASaveParameters()

```
u32 SLASaveParameters ( void *          context,  
                        SLASaveParameters_t * v,  
                        struct SVPOut_t * out,  
                        u16              nElements  
                        )
```

Commits current parameters to flash.

_SLAGetParameters()

```
u32 SLAGetParameters ( void *          context,  
                      SLAGetParameters_t * v,  
                      struct SVPOut_t * out,  
                      u16              nElements  
                      )
```

Request return of a specified message ID.

_SLASetEthernetDisplayParameters()

```
u32 SLASetEthernetDisplayParameters ( void * context,  
                                     SLASetEthernetDisplayParameters_t * v,  
                                     struct SVPOut_t * out,  
                                     u16 nElements  
                                     )
```

Configure output format of Ethernet Video.

_SLASetDisplayAdjustments()

```
u32 SLASetDisplayAdjustments ( void * context,  
                               SLASetDisplayAdjustments_t * v,  
                               struct SVPOut_t * out,  
                               u16 nElements  
                               )
```

Set the parameters for fine-tuned adjustments of the display.

_SLASetDetectionParameters()

```
u32 SLASetDetectionParameters ( void * context,  
                                SLASetDetectionParameters_t * v,  
                                struct SVPOut_t * out,  
                                u16 nElements  
                                )
```

Set the parameters for the detection algorithms.

_SLAGetDetectionParameters()

```
u32 SLAGetDetectionParameters ( void * context,  
                                SLAGetDetectionParameters_t * v,
```

```
        struct SVPOut_t * out,  
        u16 nElements  
    )
```

Return a **SLACurrentDetectionParameters_t** message.

_SLASetBlendParameters()

```
u32 SLASetBlendParameters ( void * context,  
    SLASetBlendParameters_t * v,  
    struct SVPOut_t * out,  
    u16 nElements  
)
```

Blend an EO(visible) and an IR camera with video scaling and shifting to align frames.

_SLAGetBlendParameters()

```
u32 SLAGetBlendParameters ( void * context,  
    SLAGetBlendParameters_t * v,  
    struct SVPOut_t * out,  
    u16 nElements  
)
```

Return a **SLACurrentBlendParameters_t** message.

_SLAGetImageSize()

```
u32 SLAGetImageSize ( void * context,  
    SLAGetImageSize_t * v,  
    struct SVPOut_t * out,  
    u16 nElements  
)
```

Return a **SLACurrentImageSize_t** message.

_SLADesignateSelectedTrackPrimary()

```
u32 SLADesignateSelectedTrackPrimary ( void * context,
                                     SLADesignateSelectedTrackPrimary_t * v,
                                     struct SVPOut_t * out,
                                     u16 nElements
                                     )
```

Command the system to designate the selected track as primary.

SLA Shift Selected Track()

```
u32 SLAShiftSelectedTrack ( void * context,  
                             SLAShiftSelectedTrack_t * v,  
                             struct SVPOut_t * out,  
                             u16 nElements  
                           )
```

Command the system to shift the selected track to the next track.

SLANucParameters()

```
void SLANucParameters ( void * context,  
                        SLANucParameters_t* v,  
                        struct SVPOut_t * out,  
                        u16 nElements  
                      )
```

Set Non Uniformity Correction (NUC) and Dead Pixel Removal (DPR) parameters.

`_SLAReadWriteNuc()`

```
u32 SLAReadWriteNuc ( void * context,  
                      SLAReadWriteNuc_t* v,  
                      struct SVPOut_t * out,  
                      u16 nElements  
                    )
```

Used to save or load NUC/DPR tables as well as set a table to be applied on startup.

_SLASetAcquisitionParameters()

```
u32 SLASetAcquisitionParameters ( void * context,
                                   SLASetAcquisitionParameters_t * v,
                                   struct SVPOut_t * out,
                                   u16 nElements
                                   )
```

Configure video input for digital camera interfaces.

SLAGetAcquisitionParameters()

```
u32 SLGetAcquisitionParameters ( void * context,
                                   SLGetAcquisitionParameters_t * v,
                                   struct SVPOut_t * out,
                                   u16 nElements
                                   )
```

Return a **SLACurrentAcquisitionParameters_t** message.

SLAGetEthernetDisplayParameters()

```
u32 SLAGetEthernetDisplayParameters ( void * context,
                                     SLAGetEthernetDisplayParameters_t * v,
                                     struct SVPOut_t * out,
                                     u16 nElements
                                   )
```

Return a **SLACurrentEthernetDisplayParameters_t** message.

_SLAGetDisplayParameters()

```
u32 SLAGetDisplayParameters ( void * context,
```

```

        struct SVPOut_t *
        u16
    )
    out,
    nElements
}

```

Return a **SLACurrentDisplayParameters_t** message.

_SLADrawObject()

```

u32 SLADrawObject ( void *
                    context,
                    SLADrawObject_t * v,
                    struct SVPOut_t *
                    out,
                    u16
                    nElements
)

```

Deprecated, use **SLADrawOverlay_t**

_SLAStopSelectedTrack()

```

u32 SLAStopSelectedTrack ( void *
                           context,
                           SLAStopSelectedTrack_t * v,
                           struct SVPOut_t *
                           out,
                           u16
                           nElements
)

```

Command the system to stop the currently selected track.

_SLACommandPassThrough()

```

u32 SLACommandPassThrough ( void *
                            context,
                            SLACommandPassThrough_t * v,
                            struct SVPOut_t *
                            out,
                            u16
                            nElements
)

```

Output data payload to the port specified.

_SLASetPortConfiguration()

```
u32 SLASetPortConfiguration ( void * context,
                               SLASetPortConfiguration_t * v,
                               struct SVPOut_t * out,
                               u16 nElements
                             )
```

Configure one of the communication ports for a specific protocol or mode.

_SLAGetPortConfiguration()

```
u32 SLAGetPortConfiguration ( void * context,
                               SLAGetPortConfiguration_t * v,
                               struct SVPOut_t * out,
                               u16 nElements
                             )
```

Return a SLACurrentPortConfiguration_t message.

_SLAVersionNumber()

```
u32 SLAVersionNumber ( void * context,
                       SLAVersionNumber_t * v,
                       struct SVPOut_t * out,
                       u16 nElements
                     )
```

Reports software and hardware version numbers.

_SLACurrentStabilizationParameters()

```
u32 SLACurrentStabilizationParameters ( void * context,
                                         SLACurrentStabilizationParameters_t * v,
                                         struct SVPOut_t * out,
                                         u16 nElements
                                       )
```


Describes the current stabilization mode.

SLACurrentOverlayMode()

```
u32 SLACurrentOverlayMode ( void * context,
                             SLACurrentOverlayMode_t * v,
                             struct SVPOut_t * out,
                             u16 nElements
                           )
```

Current overlay generation mode – format identical to SLASetOverlayMode_t.

SLATrackingPosition()

```
u32 SLATrackingPosition ( void * context,
                           SLATrackingPosition_t * v,
                           struct SVPOut_t * out,
                           u16 nElements
                         )
```

Periodic report of primary track's position, measured scene translation, rotation and scale, and stabilization display offset and rotation of a camera.

SLACurrentTrackingParameters()

```
u32 SLACurrentTrackingParameters ( void * context,
                                    SLACurrentTrackingParameters_t * v,
                                    struct SVPOut_t * out,
                                    u16 nElements
                                  )
```

Current tracking parameters – format identical to SLASetTrackingParameters_t.

SLACurrentRegistrationParameters()

```
u32 SLACurrentRegistrationParameters ( void * context,
```

```

        SLACurrentRegistrationParameters_t * v,
        struct SVPOut_t * out,
        u16 nElements
    )

```

Deprecated, use **SLARegistrationParameters_t**

_SLACurrentVideoParameters()

```

u32 SLACurrentVideoParameters ( void * context,
        SLACurrentVideoParameters_t * v,
        struct SVPOut_t * out,
        u16 nElements
    )

```

Current video parameters – format identical to **SLASetVideoParameters_t**.

_SLACurrentADCPParameters()

```

u32 SLACurrentADCPParameters ( void * context,
        SLACurrentADCPParameters_t * v,
        struct SVPOut_t * out,
        u16 nElements
    )

```

Current ADC parameters – format identical to **SLASetADCPParameters_t**.

_SLACurrentEthernetVideoParameters()

```

u32 SLACurrentEthernetVideoParameters ( void * context,
        SLACurrentEthernetVideoParameters_t * v,
        struct SVPOut_t * out,
        u16 nElements
    )

```

Current Ethernet video parameters – format identical to **SLASetEthernetVideoParameters_t**.

_SLACurrentNetworkParameters()

```
u32 SLACurrentNetworkParameters ( void * context,
                                   SLACurrentNetworkParameters_t * v,
                                   struct SVPOut_t * out,
                                   u16 nElements
                                   )
```

Current network parameters – format identical to [SLASetNetworkParameters_t](#).

_SLACurrentVideoEnhancementParameters()

```
u32 SLACurrentVideoEnhancementParameters ( void * context,
                                              SLACurrentVideoEnhancementParameters_t * v,
                                              struct SVPOut_t * out,
                                              u16 nElements
                                              )
```

Current video enhancement parameters – format identical to [SLASetVideoEnhancementParameters_t](#).

_SLACurrentVideoModeParameters()

```
u32 SLACurrentVideoModeParameters ( void * context,
                                      SLACurrentVideoModeParameters_t * v,
                                      struct SVPOut_t * out,
                                      u16 nElements
                                      )
```

Deprecated, use [SLAVideoDisplay_t](#) and [SLAMultiDisplay_t](#).

_SLACurrentBlendParameters()

```
u32 SLACurrentBlendParameters ( void * context,
                                  SLACurrentBlendParameters_t * v,
                                  struct SVPOut_t * out,
                                  u16 nElements
                                  )
```

Current blend parameters – see [SLASetBlendParameters_t](#).

[_SLACurrentImageSize\(\)](#)

```
u32 SLACurrentImageSize ( void *          context,  
                          SLACurrentImageSize\_t * v,  
                          struct SVPOut_t * out,  
                          u16              nElements  
                        )
```

Current image size parameters.

[_SLACurrentAcquisitionParameters\(\)](#)

```
u32 SLACurrentAcquisitionParameters ( void *          context,  
                                       SLACurrentAcquisitionParameters\_t * v,  
                                       struct SVPOut_t * out,  
                                       u16              nElements  
                                     )
```

Sent in response to [SLAGetParameters_t\(\)](#) – see [SLASetAcquisitionParameters_t](#) for additional notes and details on the camera types.

[_SLAGetHardwareID\(\)](#)

```
u32 SLAGetHardwareID ( void *          context,  
                      SLAGetHardwareID\_t * v,  
                      struct SVPOut_t * out,  
                      u16              nElements  
                    )
```

Return a [SLAGetHardwareID_t](#) message.

[_SLATrackingPositions\(\)](#)

```
u32 SLATrackingPositions ( void *          context,
```

```

        SLATrackingPositions_t * v,
        struct SVPOut_t * out,
        u16 nElements
    )

```

Periodic report of all tracked positions.

_SLACurrentEthernetDisplayParameters()

```

u32 SLACurrentEthernetDisplayParameters ( void * context,
        SLACurrentEthernetDisplayParameters_t * v,
        struct SVPOut_t * out,
        u16 nElements
    )

```

Current Ethernet display parameters – format identical to **SLASetEthernetDisplayParameters_t**.

_SLACurrentPortConfiguration()

```

u32 SLACurrentPortConfiguration ( void * context,
        SLACurrentPortConfiguration_t * v,
        struct SVPOut_t * out,
        u16 nElements
    )

```

Current I/O port configuration parameters – format identical to **SLASetPortConfiguration_t**.

_SLACurrentDetectionParameters()

```

u32 SLACurrentDetectionParameters ( void * context,
        SLACurrentDetectionParameters_t * v,
        struct SVPOut_t * out,
        u16 nElements
    )

```

Current detection parameters – format identical to **SLASetDetectionParameters_t**.

_SLAFocusStats()

```
u32 SLAFocusStats ( void * context,
                    SLAFocusStats_t * v,
                    struct SVPOut_t * out,
                    u16 nElements
                  )
```

Periodic report of focus and brightness statistics.

_SLACurrentH264Parameters()

```
u32 SLACurrentH264Parameters ( void * context,
                               SLACurrentH264Parameters_t * v,
                               struct SVPOut_t * out,
                               u16 nElements
                             )
```

Current H264 parameters – format identical to SLASetH264Parameters_t.

_SLACurrentDisplayParameters()

```
u32 SLACurrentDisplayParameters ( void * context,
                                   SLACurrentDisplayParameters_t * v,
                                   struct SVPOut_t * out,
                                   u16 nElements
                                 )
```

Current display parameters – format identical to SLASetDisplayParameters_t.

_SLACurrentSDCardRecordingStatus()

```
u32 SLACurrentSDCardRecordingStatus ( void * context,
                                       SLACurrentSDCardRecordingStatus_t * v,
                                       struct SVPOut_t * out,
                                       u16 nElements
                                     )
```

Current SD card recording status – sent in response to [SLASetSDRecordingParameters_t](#) command.

[_SLACurrentSDCardDirectoryInfo\(\)](#)

```
u32 SLACurrentSDCardDirectoryInfo ( void *          context,
                                     SLACurrentSDCardDirectoryInfo\_t * v,
                                     struct SVPOut_t * out,
                                     u16             nElements
                                   )
```

A list of file names in the current recording directory.

[_SLASendTraceStr\(\)](#)

```
u32 SLASendTraceStr ( void *          context,
                      SLASendTraceStr\_t * v,
                      struct SVPOut_t * out,
                      u16             nElements
                    )
```

Log commands and telemetry in human-readable format.

[_SLACommandCamera\(\)](#)

```
u32 SLACommandCamera ( void *          context,
                      SLACommandCamera\_t * v,
                      struct SVPOut_t * out,
                      u16             nElements
                    )
```

Set which camera channel receives camera specific parameter settings.

[_SLADisplayAngle\(\)](#)

```
u32 SLADisplayAngle ( void *          context,
                      SLADisplayAngle\_t * v,
```

```

        struct SVPOut_t *    out,
        u16                  nElements
    )

```

Set display rotation angle settings for a single camera or for all cameras See also [SLASetDisplayParameters_t](#).

△SLACurrentSnapShot()

```

u32 SLACurrentSnapShot ( void *          context,
                        SLACurrentSnapShot\_t * v,
                        struct SVPOut_t *    out,
                        u16                  nElements
                    )

```

Current snapshot state.

△SLASetSnapShot()

```

u32 SLASetSnapShot ( void *          context,
                    SLASetSnapShot\_t * v,
                    struct SVPOut_t *    out,
                    u16                  nElements
                )

```

Set up parameters for image snapshot.

△SLAGetSnapShot()

```

u32 SLAGetSnapShot ( void *          context,
                    SLAGetSnapShot\_t * v,
                    struct SVPOut_t *    out,
                    u16                  nElements
                )

```

Return a [SLACurrentSnapShot_t](#) message.

_SLADoSnapshot()

```
u32 SLADoSnapshot ( void * context,
                    SLADoSnapshot_t * v,
                    struct SVPOut_t * out,
                    u16 nElements
                  )
```

Execute an image snapshot.

_SLASetKlvData()

```
u32 SLASetKlvData ( void * context,
                   SLASetKlvData_t * v,
                   struct SVPOut_t * out,
                   u16 nElements
                 )
```

The KLV blob data is constructed by user to be sent with MPEG2-TS stream (see [SLASetEthernetDisplayParameters_t](#)).

_SLASetMetadataRate()

```
u32 SLASetMetadataRate ( void * context,
                        SLASetMetadataRate_t * v,
                        struct SVPOut_t * out,
                        u16 nElements
                      )
```

Deprecated, use [SLATagDataRate_t](#)

_SLASetTelemetryDestination()

```
u32 SLASetTelemetryDestination ( void * context,
                                SLASetTelemetryDestination_t * v,
                                struct SVPOut_t * out,
                                u16 nElements
                              )
```

Configure the Ethernet IP address of clients for telemetry.

_SLAGetNetworkList()

```
u32 SLAGetNetworkList ( void *          context,
                        SLAGetNetworkList_t * v,
                        struct SVPOut_t *    out,
                        u16                  nElements
                      )
```

Return a **SLACurrentNetworkList_t** message.

_SLACurrentNetworkList()

```
u32 SLACurrentNetworkList ( void *          context,
                             SLACurrentNetworkList_t * v,
                             struct SVPOut_t *    out,
                             u16                  nElements
                           )
```

List of all network interfaces controller names.

_SLACurrentOverlayObjectsIds()

```
u32 SLACurrentOverlayObjectsIds ( void *          context,
                                   SLACurrentOverlayObjectsIds_t * v,
                                   struct SVPOut_t *    out,
                                   u16                  nElements
                                 )
```

A list of used object ids as a 255 long bit array.

_SLACurrentOverlayObjectParameters()

```
u32 SLACurrentOverlayObjectParameters ( void *          context,
```

```

        SLACurrentOverlayObjectParameters_t * v,
        struct SVPOut_t * out,
        u16 nElements
    )

```

Properties of a particular graphic overlay specified by its Object ID.

_SLASetLensMode()

```

u32 SLASetLensMode ( void * context,
                     SLASetLensMode_t * v,
                     struct SVPOut_t * out,
                     u16 nElements
                   )

```

Perform camera lens control.

_SLACurrentLensStatus()

```

u32 SLACurrentLensStatus ( void * context,
                           SLACurrentLensStatus_t * v,
                           struct SVPOut_t * out,
                           u16 nElements
                         )

```

Current lens focus and zoom position.

_SLASetLensParameters()

```

u32 SLASetLensParameters ( void * context,
                           SLASetLensParameters_t * v,
                           struct SVPOut_t * out,
                           u16 nElements
                         )

```

Deprecated, use **SLALensParameters_t**

SLACurrentLensParameters()

```
u32 SLACurrentLensParameters ( void * context,
                               SLACurrentLensParameters_t * v,
                               struct SVPOut_t * out,
                               u16 nElements
                               )
```

Deprecated, use SLALensParameters_t

SLASetDigitalCameraParameters()

```
u32 SLASetDigitalCameraParameters ( void * context,
                                     SLASetDigitalCameraParameters_t * v,
                                     struct SVPOut_t * out,
                                     u16 nElements
                                     )
```

Dynamically adjust a 10 to 16 bit digital camera input to an 8 bit image.

SLACurrentDigitalCameraParameters()

```
u32 SLACurrentDigitalCameraParameters ( void * context,
                                         SLACurrentDigitalCameraParameters_t * v,
                                         struct SVPOut_t * out,
                                         u16 nElements
                                         )
```

Return parameters of currently selected digital camera – similar to SLASetDigitalCameraParameters_t.

SLASetUserPalette()

```
u32 SLASetUserPalette ( void * context,
                        SLASetUserPalette_t * v,
                        struct SVPOut_t * out,
                        u16 nElements
                        )
```

Set the Y, U, V values for the user false color palette.

_SLACurrentUserPalette()

```
u32 SLACurrentUserPalette ( void * context,  
    SLACurrentUserPalette_t * v,  
    struct SVPOut_t * out,  
    u16 nElements  
)
```

Return the current user palette in YUV values – format identical to SLASetUserPalette_t.

_SLASetMultipleAlignment()

```
u32 SLASetMultipleAlignment ( void * context,  
    SLASetMultipleAlignment_t * v,  
    struct SVPOut_t * out,  
    u16 nElements  
)
```

Deprecated, use SLABlendAlign_t

_SLACurrentMultipleAlignment()

```
u32 SLACurrentMultipleAlignment ( void * context,  
    SLACurrentMultipleAlignment_t * v,  
    struct SVPOut_t * out,  
    u16 nElements  
)
```

Deprecated, use SLABlendAlign_t

_SLASetAdvancedDetectionParameters()

```
u32 SLASetAdvancedDetectionParameters ( void * context,  
    SLASetAdvancedDetectionParameters_t * v,
```

```

        struct SVPOut_t *
        u16
        )
        out,
        nElements

```

Perform advanced control of the MTI parameters.

_SLACurrentAdvancedDetectionParameters()

```

u32 SLACurrentAdvancedDetectionParameters ( void *
        context,
        SLACurrentAdvancedDetectionParameters_t * v,
        struct SVPOut_t *
        out,
        u16
        nElements
        )

```

Return the current advanced detection parameters – format identical to [SLASetAdvancedDetectionParameters_t](#).

_SLATrackingBoxPixelStats()

```

u32 SLATrackingBoxPixelStats ( void *
        context,
        SLATrackingBoxPixelStats_t * v,
        struct SVPOut_t *
        out,
        u16
        nElements
        )

```

Periodic report of pixel statistics within track boxes.

_SLADirectoryStatisticsReply()

```

u32 SLADirectoryStatisticsReply ( void *
        context,
        SLADirectoryStatisticsReply_t * v,
        struct SVPOut_t *
        out,
        u16
        nElements
        )

```

Sent in response to a [SLASetSDRecordingParameters_t](#).

SLACurrentStabilizationBias()

```
u32 SLACurrentStabilizationBias ( void * context,
                                  SLACurrentStabilizationBias_t * v,
                                  struct SVPOut_t * out,
                                  u16 nElements
                                )
```

Deprecated, use SLAStabilizationBias_t

SLAAdvancedCaptureParameters()

```
u32 SLAAdvancedCaptureParameters ( void * context,
                                     SLAAdvancedCaptureParameters_t * v,
                                     struct SVPOut_t * out,
                                     u16 nElements
                                   )
```

Set advanced analog video decoding parameters for the video decoder.

SLASetDetectionRegionOfInterestParameters()

```
u32 SLASetDetectionRegionOfInterestParameters ( void * context,
                                                  SLASetDetectionRegionOfInterestParameters_t * v,
                                                  struct SVPOut_t * out,
                                                  u16 nElements
                                                )
```

Sets the parameters for Moving Target Indication (MTI) Region of Interest(ROI).

SLACurrentDetectionRegionOfInterestParameters()

```
u32 SLACurrentDetectionRegionOfInterestParameters ( void * context,
                                                      SLACurrentDetectionRegionOfInterestParameters_t * v,
                                                      struct SVPOut_t * out,
                                                      u16 nElements
                                                    )
```

Current detection region of interest parameters – format identical to [SLASetDetectionRegionOfInterestParameters_t](#).

_SLAUserWarningLevel()

```
u32 SLAUserWarningLevel ( void * context,
                           SLAUserWarningLevel\_t * v,
                           struct SVPOut_t * out,
                           u16 nElements
                           )
```

Enable / disable User Warning Messages ([SLAUserWarningMessage_t](#)) sent from SLA hardware.

_SLASystemStatusMode()

```
u32 SLASystemStatusMode ( void * context,
                           SLASystemStatusMode\_t * v,
                           struct SVPOut_t * out,
                           u16 nElements
                           )
```

Enable / disable System Status Messages ([SLASystemStatusMessage_t](#)) and Detailed Timing Messages ([SLADetailedTimingMessage_t](#)) sent from the SLA hardware at 1 Hz.

_SLALandingAid()

```
u32 SLALandingAid ( void * context,
                    SLALandingAid\_t * v,
                    struct SVPOut_t * out,
                    u16 nElements
                    )
```

Control landing aid detection.

_SLACameraSwitch()

```
u32 SLACameraSwitch ( void * context,
```



```

        SLACameraSwitch_t * v,
        struct SVPOut_t * out,
        u16 nElements
    )

```

Deprecated, supported on 1500 and 3000 only

_SLALandingPosition()

```

u32 SLALandingPosition ( void * context,
        SLALandingPosition_t * v,
        struct SVPOut_t * out,
        u16 nElements
    )

```

Periodic report of landing aid location, orientation, distance and match score.

_SLASetVMTI()

```

u32 SLASetVMTI ( void * context,
        SLASetVMTI_t * v,
        struct SVPOut_t * out,
        u16 nElements
    )

```

Allows targets from outside SightLine to be injected into the MPEG2-TS KLV stream.

_SLAUserWarningMessage()

```

u32 SLAUserWarningMessage ( void * context,
        SLAUserWarningMessage_t * v,
        struct SVPOut_t * out,
        u16 nElements
    )

```

Message sent from the SLA hardware to notify users of any issues or system status which may not be obtained any other way.

_SLASystemStatusMessage()

```
u32 SLASystemStatusMessage ( void * context,
                             SLASystemStatusMessage_t * v,
                             struct SVPOut_t * out,
                             u16 nElements
                             )
```

Packet is sent from the SLA hardware to report diagnostic information such as CPU load and temperature, or the performance of software tools.

_SLADetailedTimingMessage()

```
u32 SLADetailedTimingMessage ( void * context,
                               SLADetailedTimingMessage_t * v,
                               struct SVPOut_t * out,
                               u16 nElements
                               )
```

Packet is sent from the SLA hardware to report DSP and ARM detailed timing measurements.

_SLAAppendedMetadata()

```
u32 SLAAppendedMetadata ( void * context,
                          SLAAppendedMetadata_t * v,
                          struct SVPOut_t * out,
                          u16 nElements
                          )
```

Append user-specified binary metadata to KLV stream embedded in network video stream.

_SLAFrameIndex()

```
u32 SLAFrameIndex ( void * context,
                   SLAFrameIndex_t * v,
                   struct SVPOut_t * out,
```

u16 nElements

)

User controlled frame index applied to specified camera index.

△SLACurrentMetadataValues()

```
u32 SLACurrentMetadataValues ( void * context,  
                               SLACurrentMetadataValues_t * v,  
                               struct SVPOut_t * out,  
                               u16 nElements  
                               )
```

Current metadata values – format nearly identical to SLASetMetadataValues_t.

△SLACurrentMetadataFrameValues()

```
u32 SLACurrentMetadataFrameValues ( void * context,  
                                     SLACurrentMetadataFrameValues_t * v,  
                                     struct SVPOut_t * out,  
                                     u16 nElements  
                                     )
```

Current metadata frame values – format nearly identical to SLASetMetadataFrameValues_t.

△SLACurrentMetadataRate()

```
u32 SLACurrentMetadataRate ( void * context,  
                             SLACurrentMetadataRate_t * v,  
                             struct SVPOut_t * out,  
                             u16 nElements  
                             )
```

Current metadata rate – format similar to SLASetMetadataRate_t.

△SLACurrentConfiguration()

```

u32 SLACurrentConfiguration ( void * context,
                               SLACurrentConfiguration_t * v,
                               struct SVPOut_t * out,
                               u16 nElements
                             )

```

Current camera and display configuration.

_SLAExternalProgram()

```

u32 SLAExternalProgram ( void * context,
                          SLAExternalProgram_t * v,
                          struct SVPOut_t * out,
                          u16 nElements
                        )

```

Configure external programs – Set names of external applications to be run on the SLA hardware.

_SLAStreamingControl()

```

u32 SLAStreamingControl ( void * context,
                           SLAStreamingControl_t * v,
                           struct SVPOut_t * out,
                           u16 nElements
                         )

```

Control to Start and Stop Network Streaming.

_SLADigitalVideoParserParameters()

```

u32 SLADigitalVideoParserParameters ( void * context,
                                       SLADigitalVideoParserParameters_t * v,
                                       struct SVPOut_t * out,
                                       u16 nElements
                                     )

```

Set parameters for the digital video SightLine protocol parser.

_SLASetSystemValue()

```
u32 SLASetSystemValue ( void *          context,
                        SLASetSystemValue_t * v,
                        struct SVPOut_t *    out,
                        u16                  nElements
                      )
```

Configure Special System Operational Values or Modes.

_SLACurrentSystemValue()

```
u32 SLACurrentSystemValue ( void *          context,
                            SLACurrentSystemValue_t * v,
                            struct SVPOut_t *    out,
                            u16                  nElements
                          )
```

Current SystemValue – format identical to SLASetSystemValue_t.

_SLAI2CCommand()

```
u32 SLAI2CCommand ( void *          context,
                    SLAI2CCommand_t * v,
                    struct SVPOut_t *    out,
                    u16                  nElements
                  )
```

Write or read I2C bus.

_SLAFourAlignPoints()

```
u32 SLAFourAlignPoints ( void *          context,
                        SLAFourAlignPoints_t * v,
                        struct SVPOut_t *    out,
                        u16                  nElements
                      )
```

)

Used to assist in the alignment of two cameras for blending purposes.

_SLATagData()

```
u32 SLATagData ( void *          context,  
                 SLATagData_t * v,  
                 struct SVPOut_t * out,  
                 u16             nElements  
                )
```

Set the value of any MISB Tag and subtag.

_SLATagDataRate()

```
u32 SLATagDataRate ( void *          context,  
                    SLATagDataRate_t * v,  
                    struct SVPOut_t *  out,  
                    u16                nElements  
                    )
```

Set the output frequency of a KLV Tag or a range of tags.

_SLATagSourceSelector()

```
u32 SLATagSourceSelector ( void *          context,  
                          SLATagSourceSelector_t * v,  
                          struct SVPOut_t *  out,  
                          u16                nElements  
                          )
```

Select the source for a particular klv tag or a range of klv tags.

_SLADecoderParameters()

```

u32 SLADecoderParameters ( void * context,
                           SLADecoderParameters_t * v,
                           struct SVPOut_t * out,
                           u16 nElements
                           )

```

Configure format of Onboard decoder.

_SLALogoParameters()

```

u32 SLALogoParameters ( void * context,
                       SLALogoParameters_t * v,
                       struct SVPOut_t * out,
                       u16 nElements
                       )

```

Used to set the opacity of the logo (watermark) and its placement relative to the bottom right corner of the output frame.

_SLADrawOverlay()

```

u32 SLADrawOverlay ( void * context,
                   SLADrawOverlay_t * v,
                   struct SVPOut_t * out,
                   u16 nElements
                   )

```

Draw user specified overlay graphics on the screen.

_SLATrackTrails()

```

u32 SLATrackTrails ( void * context,
                   SLATrackTrails_t * v,
                   struct SVPOut_t * out,
                   u16 nElements
                   )

```

Tracking Trail Information.

△SLARegistrationParameters()

```
u32 SLARegistrationParameters ( void * context,
                                SLARegistrationParameters_t * v,
                                struct SVPOut_t * out,
                                u16 nElements
                                )
```

Parameters used in frame to frame image registration matching.

△SLAStabilizationBias()

```
u32 SLAStabilizationBias ( void * context,
                            SLAStabilizationBias_t * v,
                            struct SVPOut_t * out,
                            u16 nElements
                            )
```

Adjust the stabilization solution by adding a constant bias in pixel coordinate space to the current coordinates each frame.

△SLATrackingPositionsExtended()

```
u32 SLATrackingPositionsExtended ( void * context,
                                    SLATrackingPositionsExtended_t * v,
                                    struct SVPOut_t * out,
                                    u16 nElements
                                    )
```

Periodic report of all tracked positions, with classifier information.

△SLADeadPixelStats()

```
u32 SLADeadPixelStats ( void * context,
                        SLADeadPixelStats_t * v,
                        struct SVPOut_t * out,
```



```
u16 nElements
)
```

Statistics from Dead Pixel Replacement calibration process.

△SLAVideoDisplay()

```
u32 SLAVideoDisplay ( void * context,
                      SLAVideoDisplay_t * v,
                      struct SVPOut_t * out,
                      u16 nElements
                    )
```

Configure which cameras and resolutions to display.

△SLAMultiDisplay()

```
u32 SLAMultiDisplay ( void * context,
                     SLAMultiDisplay_t * v,
                     struct SVPOut_t * out,
                     u16 nElements
                   )
```

Configure multi-camera display options such as Picture-In-Picture.

△SLAUsb3VisionFeature()

```
u32 SLAUsb3VisionFeature ( void * context,
                          SLAUsb3VisionFeature_t * v,
                          struct SVPOut_t * out,
                          u16 nElements
                        )
```

Send commands to an attached USB3Vision camera or query feature value.

△SLACustomClassifier()

```

u32 SLACustomClassifier ( void * context,
                          SLACustomClassifier_t * v,
                          struct SVPOut_t * out,
                          u16 nElements
                        )

```

Custom Classifier.

_SLADeadPixel()

```

u32 SLADeadPixel ( void * context,
                  SLADeadPixel_t * v,
                  struct SVPOut_t * out,
                  u16 nElements
                )

```

Modify Dead Pixel List Manually.

_SLAClassifierParameters()

```

u32 SLAClassifierParameters ( void * context,
                              SLAClassifierParameters_t * v,
                              struct SVPOut_t * out,
                              u16 nElements
                            )

```

Turn on/off the classifier.

_SLADoDetectSnapShot()

```

u32 SLADoDetectSnapShot ( void * context,
                          SLADoDetectSnapShot_t * v,
                          struct SVPOut_t * out,
                          u16 nElements
                        )

```

Execute a track/detection snapshot to the MicroSD card or an external FTP server.

_SLAAncillaryTextMetadata()

```
u32 SLAAncillaryTextMetadata ( void * context,
                               SLAAncillaryTextMetadata_t * v,
                               struct SVPOut_t * out,
                               u16 nElements
                             )
```

Inject text into the KLV stream.

_SLAVMTIChips()

```
u32 SLAVMTIChips ( void * context,
                   SLAVMTIChips_t * v,
                   struct SVPOut_t * out,
                   u16 nElements
                 )
```

Configure KLV VMTI Chips.

_SLAUserFont()

```
u32 SLAUserFont ( void * context,
                  SLAUserFont_t * v,
                  struct SVPOut_t * out,
                  u16 nElements
                )
```

Set user font filename.

_SLANoise3D()

```
u32 SLANoise3D ( void * context,
                 SLANoise3D_t * v,
                 struct SVPOut_t * out,
                 u16 nElements
               )
```

)

Statistics from 3D Noise calculation.

_SLACursorOnTarget()

```
u32 SLACursorOnTarget ( void *          context,
                        SLACursorOnTarget_t * v,
                        struct SVPOut_t *   out,
                        u16                  nElements
                      )
```

Cursor on Target (CoT) CoT packet contents are converted from KLV Metadata inputs.

_SLALensParameters()

```
u32 SLALensParameters ( void *          context,
                        SLALensParameters_t * v,
                        struct SVPOut_t *   out,
                        u16                  nElements
                      )
```

Lens parameters - control the lens.

_SLALensCommand()

```
u32 SLALensCommand ( void *          context,
                     SLALensCommand_t * v,
                     struct SVPOut_t *   out,
                     u16                  nElements
                   )
```

Perform a lens control command.

_SLAFocusParameters()

```

u32 SLAFocusParameters ( void * context,
                          SLAFocusParameters_t * v,
                          struct SVPOut_t * out,
                          u16 nElements
                        )

```

Focus parameters – control the focus measurement.

_SLALensRanges()

```

u32 SLALensRanges ( void * context,
                   SLALensRanges_t * v,
                   struct SVPOut_t * out,
                   u16 nElements
                 )

```

Lens focus and zoom ranges.

_SLACustomAutoFocusParameters()

```

u32 SLACustomAutoFocusParameters ( void * context,
                                   SLACustomAutoFocusParameters_t * v,
                                   struct SVPOut_t * out,
                                   u16 nElements
                                 )

```

Set Custom Autofocus Parameters.

_SLAGPIO()

```

u32 SLAGPIO ( void * context,
             SLAGPIO_t * v,
             struct SVPOut_t * out,
             u16 nElements
           )

```

Write to and read from a GPIO.

_SLAUsbWebcamFeature()

```
u32 SLAUsbWebcamFeature ( void * context,  
                          SLAUsbWebcamFeature_t * v,  
                          struct SVPOut_t * out,  
                          u16 nElements  
                          )
```

Send commands to an attached V4L2-Controlled USB WebCamera, or query feature value.

_SLACreateDeviceOverlay()

```
u32 SLACreateDeviceOverlay ( void * context,  
                             SLACreateDeviceOverlay_t * v,  
                             struct SVPOut_t * out,  
                             u16 nElements  
                             )
```

Send a command to generate and compile a devicetree overlay for a device.

_SLABlendAlign()

```
u32 SLABlendAlign ( void * context,  
                   SLABlendAlign_t * v,  
                   struct SVPOut_t * out,  
                   u16 nElements  
                   )
```

Set the alignment set for a dual camera setup.

_SLACustomAIDetect()

```
u32 SLACustomAIDetect ( void * context,  
                       SLACustomAIDetect_t * v,  
                       struct SVPOut_t * out,  
                       u16 nElements
```

Custom AI Object Detector Model.

_SLACameraCapabilities()

```
u32 SLACameraCapabilities ( void * context,  
                           SLACameraCapabilities_t * v,  
                           struct SVPOut_t * out,  
                           u16 nElements  
                           )
```

Return a message with the camera capabilities.

_SLACustomResponse()

```
u32 SLACustomResponse ( void * context,  
                       SLACustomResponse_t * v,  
                       struct SVPOut_t * out,  
                       u16 nElements  
                       )
```

A custom response that a customer can send from an overlay DLL.

_SLATrackingMultiClass()

```
u32 SLATrackingMultiClass ( void * context,  
                           SLATrackingMultiClass_t * v,  
                           struct SVPOut_t * out,  
                           u16 nElements  
                           )
```

Periodic report of all tracked multi classifier information.

_SLASendToBTS()

```
u32 SLASendToBTS ( void * context,
```

```

        SLASendToBTS_t * v,
        struct SVPOut_t * out,
        u16 nElements
    )

```

Start sending data to BTS from SLA.

_SLAVMTIFields()

```

u32 SLAVMTIFields ( void * context,
        SLAVMTIFields_t * v,
        struct SVPOut_t * out,
        u16 nElements
    )

```

Set active VMTI fields and Ontology Series Update Rate.

_SLACameraCalibration()

```

u32 SLACameraCalibration ( void * context,
        SLACameraCalibration_t * v,
        struct SVPOut_t * out,
        u16 nElements
    )

```

Set camera calibration parameters to fix distortion.

_SLAKlvClassFilters()

```

u32 SLAKlvClassFilters ( void * context,
        SLAKlvClassFilters_t * v,
        struct SVPOut_t * out,
        u16 nElements
    )

```

Set active Classification filters on KLV data.

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetStabilizationParameters_t

Struct Reference

[Stabilization](#)

Description

Turn on or turn off stabilization and control re-centering rate for output video.

```
typedef struct {
    u8 mode;
    u8 rate;
    u8 translationLimit;
    u8 angleLimit;
    u8 cameraIndex;
    u8 maxStabOff;
    u8 edgeY;
    u8 edgeU;
    u8 edgeV;
} SLASetStabilizationParameters_t;
```

Video stabilization smooths out jumpy sequences caused by camera vibration. Default value is *on* with re-centering rate = 50. A low number (approaching 0) = a slow drift to center. A high number (approaching 255) = a fast drift to center. Maximum stabilization limit is used to set an upper bound on the time-averaged stabilization solution.

See also

- [SLAStabilizationBias_t](#) which is used with constant or expected offset motion.
- [SLAResetStabilizationParameters_t](#) to reset stabilization.

Message ID 0x02

Byte Offset	Name	Description						
		Stabilization Modes & Properties						
		<table><tr><th>Bit</th><th>Description</th></tr><tr><td>0</td><td>0 = Stabilization off 1 = Stabilization on (default)</td></tr><tr><td>1</td><td>0 = Enable all processing (default) 1 = Disable all registration, stabilization, enhancement and tracking</td></tr></table>	Bit	Description	0	0 = Stabilization off 1 = Stabilization on (default)	1	0 = Enable all processing (default) 1 = Disable all registration, stabilization, enhancement and tracking
Bit	Description							
0	0 = Stabilization off 1 = Stabilization on (default)							
1	0 = Enable all processing (default) 1 = Disable all registration, stabilization, enhancement and tracking							

4	mode	2 & 4	Background Edge Mode		
			4	2	Description
			0	0	Previous images blurred and fading to Background Edge Color
			0	1	Solid Background Edge Color (default)
			1	0	Previous images, with color, without fading
			1	1	Reserved
		3	Reserved. Used in SLACurrentStabilizationParameters_t for Auto Bias.		
		5	0 = Apply display pan-tilt in zoom to track mode (default) 1 = Disable applying display pan-tilt in zoom to track mode - allows offsetting normal stabilization display, but centering on a track in Zoom to Track display. See SLASetDisplayParameters_t , bit 7 of byte 9 controls Zoom to Track and bytes 11 to 14 control pan and tilt offset.		
		6	0 = Enable registration (default). Used in stabilization, detection, tracking denoise, and gimbal scene lock. 1 = Disable registration New 3.3 . Speeds processing, but may significantly degrade other functions if the camera is moving.		
		7	Reserved		
5	rate	Screen translation re-centering rate 0..255 (default=50). (Internally divided by 1000, so 50 re-centers by 5% every frame). Higher values re-center faster, but remove less low-frequency jitter. Applies to both offset and rotation stabilization. See also SLAStabilizationBias_t which is used with constant or expected offset motion.			
6	translationLimit	Maximum displly grey edge limit in pixels (default = 0 for no limit)			
7	angleLimit	Maximum rotational stabilization limit in degrees (default none = 0).. NOTE: 0 results in much faster execution, but does not remove rotational jitter. 5 removes typical rotation jitter. Larger values smooth out large rolls.			
8	cameraIndex	Camera Index			
9	maxStabOff	Maximum stabilization offset in pixels (default = 0 for no limit).. NOTE: If the capture and display image sizes are the same, translationLimit and maxStabOff are equivalent. If the display image size is smaller, maxStabOff limits the amount of stabilization, translationLimit limits the grey border.			
10	edgeY	Background Edge Color Y (luminance) used in blur/fade and solid background edge modes			
11	edgeU	Background Edge Color U (Cb color) used in blur/fade and solid background edge modes			
12	edgeV	Background Edge Color V (Cr color) used in blur/fade and solid background edge modes			

Sightline Intelligence

Command, Control, and Script

API v3.11

Related Pages

Here is a list of all related documentation pages:

Sightline IDD & Protocol Usage	How to use the IDD in combination with the C++ API (see the command and control page) provided by Sightline Intelligence
Sightline Discover Protocol	Ethernet protocol for finding Sightline hardware on a network
Sightline File Recording	This section consolidates information related to file recording (SnapShots,Video, NUC tables, etc.)
Sightline Protocol Changes	List of parameters/function changes for old releases
Changes for 2.25	
Changes for 3.00	
Changes for 3.01	
Changes for 3.2	
Changes for 3.3	
Changes for 3.4	
Changes for 3.5	
Changes for 3.6	
Changes for 3.7	
Changes for 3.8	
Changes for 3.9	
Changes for 3.10	
Changes for 3.11	

Sightline Intelligence

Command, Control, and Script

API v3.11

Modules

Here is a list of all modules:

Blending	Two channel video blending
Capture	Camera acquisition controls
Compression	Compression, encoding and network streaming of video
Detection	Automatic object detection - moving target, color anomaly, radiometric, marine
Display	Video output display controls
Enhancement	Enhancement of moving video
Focus	Lens auto focus and brightness control
General	General system configuration - command camera, save parameters, user warnings, system profile etc
KLV Metadata	Metadata embedding into network stream
Landing Aid	Visual landing aid detection and tracking
Lua Script Functions	Lua script interface for system control
Network	Network communications and setup
Non Uniformity Correction	Non Uniformity Correction and Dead Pixel Replacement
Overlays	Graphics overlays
Recording	Video recording and snapshots to FTP server, microSD card, or other media
Serial Port	Serial port communications and setup
Stabilization	Stabilize moving video
Telemetry	Telemetry result messages and configuration
Tracking	User designated tracking of objects
Classification	AI Classifier
Message Identifiers	List of SightLine command & reply message ID (type) constants

Sightline Intelligence Command, Control, and Script API v3.11

For the most up-to-date list of supported cameras see the [camera compatibility table](#).

Sightline Intelligence

Command, Control, and Script

API v3.11

SVPEndString_t Struct Reference

Description

Text string without specified length.

```
typedef struct {
    char str[255];
} SVPEndString_t;
```

Member Data Documentation

_str
char SVPEndString_t::str[255]
Text string, length determined by end of packet and/or null-termination.

Sightline Intelligence

Command, Control, and Script

API v3.11

SVPLenString_t Struct Reference

[List of all members](#)

Description

Simple data structures used to pass data.

Text string with specified length

```
typedef struct {
    u8 len;
    char str[256];
} SVPLenString_t;
```

Member Data Documentation

_len

u8 SVPLenString_t::len

Length of valid string.

_str

char SVPLenString_t::str[256]

Text string, optionally null-terminated.

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetUserPalette_t Struct Reference

[List of all members](#)

[Display](#)

Description

Set the Y, U, V values for the user false color palette.

```
typedef struct {
    struct {
        u8 y;
        u8 u;
        u8 v;
    } palette[256];
} SLASetUserPalette_t;
```

Message ID 0x72

Byte Offset	Name	Description
varies	y	Y value
varies	u	U value
varies	v	V value

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentUserPalette_t Struct Reference

[Display](#)

Description

Return the current user palette in YUV values – format identical to [SLASetUserPalette_t](#).

```
typedef struct {
    struct {
        u8 y;
        u8 u;
        u8 v;
    } palette[256];
} SLACurrentUserPalette\_t;
```

Sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x73

Byte Offset	Name	Description
varies	y	See SLASetUserPalette_t
varies	u	
varies	v	

Sightline Intelligence

Command, Control, and Script

API v3.11

SVPLenData_t Struct Reference

Description

Data block with specified length.

```
typedef struct {  
    u8 len;  
    u8 buf[255];  
} SVPLenData_t;
```

Member Data Documentation

_len	u8 SVPLenData_t::len Length of data buffer.
_buf	u8 SVPLenData_t::buf[255] Data buffer.

Sightline Intelligence

Command, Control, and Script

API v3.11

SVPSignedData_t Struct Reference

Description

Block of signed ints of specified size.

```
typedef struct {  
    u8 len;  
    s8 buf[255];  
} SVPSignedData_t ;
```

Member Data Documentation

<u>len</u>	u8 SVPSignedData_t::len	Length of buffer.
<u>buf</u>	s8 SVPSignedData_t::buf[255]	Data buffer as signed 8bit integers.

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAVersionNumber_t Struct Reference

[General](#)

Description

Reports software and hardware version numbers.

```
typedef struct {
    u8 swMajor;
    u8 swMinor;
    u8 hwVersion;
    u8 degreesF;
    u24 hwID;
    u32 appBits;
    u8 boardType;
    u8 swRelease;
    u16 otherVersion;
    u32 srcRevision;
    u32 buildDate;
    u32 buildTime;
    u16 swBuild;
    u16 v4AppBits;
    s16 degreesC;
    u32 adapters;
} SLAVersionNumber_t;
```

The Version Number packet is automatically sent after the first three frames of acquisition on Serial Port 0.
Sent in response to [SLAGetVersionNumber_t\(\)](#) or [SLAGetParameters_t\(\)](#).
TIP: Use the Version Number(0x40) packet as a mechanism to know the system is ready to receive commands.

Application Bits

Bits specified in the license file control which features are enabled on SLA hardware.

Bit	Value	Name	Description	Requires
-	0x00000000	None	Base system - SD output to HDMI, digital SD input, network and serial communications, passthrough communications, graphics overlays	-
0	0x00000001	720 Output	720 resolution high definition (HD) output Changed 3.0	-
1	0x00000002	Stabilization	Image based video stabilization and frame to frame Registration	-

2	0x00000004	IP Encoding	Video compression and IP encoding (h.264, mpeg4, mjpeg, mpeg2-ts, rtp, rtsp)	-
3	0x00000008	Detection - Baseline	Baseline Detection - Vehicle moving target indication (MTI), Staring small target detection, Color anomaly detection, Radiometric detection, Blob detection, Drone detection	Stabilization(Registration)
4	0x00000010	Tracking	User designated tracking of objects (full featured)	Stabilization+Full Telemetry or Track Only Telemetry
5	0x00000020	Detection - Advanced	Advanced Detection - Aerial small target detection, Motion small target detection, Maritime detection, Drone detection with classification	Stabilization(Registration), Detection - Baseline
6	0x00000040	Full Telemetry	Telemetry output for gimbal control (registration, stabilization, tracking, detection)	Stabilization(Registration)
7	0x00000080	Enhancement	Image Enhancement (CLAHE, LAP, denoise, sharpen) Changed 3.0	-
8	0x00000100	Blending	Dual image blending (4000) Changed 3.0	Dual
9	0x00000200	720 Input	720 resolution high definition (HD) output Changed 3.0	-
10	0x00000400	Recording / Snapshot	Recording of video and snapshot of still images	Recording: IP Encoding, Snapshot: None
11	0x00000800	KLV	KLV metadata encoding into MPEG2-TS H.264 network video stream, VBI (KLV in analog blanking interval) metadata	IP Encoding
12	0x00001000	Focus Telemetry	Auto Focus measurement, focus telemetry output and lens control DLL	-
13	0x00002000	Precision Landing	Video Based precision Landing and telemetry output	-
14	0x00004000	Dual	Dual camera simultaneous processing New 3.0	-
15	0x00008000	Track Only Telemetry	Tracking Only telemetry output (results from 1 track, no frame registration results). Not available with IP Encoding, stabilization, recording, etc - SD input and analog output only	Tracking
16	0x00010000	H265 Encoding	H.265 IP video encoding New 3.0	IP Encoding
17	0x00020000	High Bit Depth/Temp Telemetry	High bit depth video capture and temperature telemetry output New 3.0	-
18	0x00040000	NUC/DPR	Non-uniformity correction (NUC), Dead pixel removal (DPR). New 3.0	High Bit Depth/Temp Telemetry
19	0x00080000	1080 Output	1080 resolution output New 3.0	720 Output

20	0x00100000	1080 Input	1080 resolution input) New 3.0	720 Input
21	0x00200000	4K Output	4K resolution output (4000 only) New 3.0	1080 Output, 720 Output
22	0x00400000	4K Input	4K resolution input (4000 only) New 3.0	1080 Input, 720 Input
23	0x00800000	Enhancement - Advanced	Advanced image enhancement. De-Noise Mask removes image noise and scintillation without blurring out moving objects New 3.1	Enhancement
24	0x01000000	Custom Classifier	Run a custom classifier on detection or track results New 3.2	Detection - Baseline or Tracking
25	0x02000000	Tri	Three camera simultaneous processing (ARM library and 4100 only) New 3.4	-
26	0x04000000	Tracking - Advanced	Advanced Tracking - Precision Acquisition (user initiated tracking from past frame(s)) New 3.8	Stabilization(Registration), Tracking, Full Telemetry
27	0x08000000	AI Detection	AI full frame detection - requires .slaod model file New 3.8	Stabilization(Registration), Detection - Baseline
28	0x10000000	Gas Detect	Detection for gas leaks New 3.3	Stabilization(Registration)
29	0x20000000	AiTR	Aided Target Recognition - enable AI models for first and second stage identification/classification New 3.9	Stabilization(Registration), Detection - Baseline, Classifier, AI Detection
30	0x40000000	Demo Mode	<u>Demo Mode</u> with no license. In demo mode other bits may be set.	-
31	0x80000000	Demo Mode	<u>Demo Mode</u> with a valid license file. In demo mode other bits may be set.	-

NOTE: The following changes to application bits were made between version 4 (2.25 and earlier) and version 7 (3.0 and later) of the SightLine license file:

- Split Enhance/High Bit Depth into **Enhancement** and **High Bit Depth/Temp Telemetry**
- Split NUC/DPR/Blend into **Blending** and **NUC/DPR**
- Split HD Output into **720 Output**, **1080 Output** and **Dual**
- Split HD Input into **720 Input**, **1080 Input** and **Dual**
- Removed Base Tracking, use **Track Only**

Demo Mode

Demo Mode allows users to try out most features on any SightLine system. Use [SLASetSystemValue_t](#) to enter demo mode. Restart system to leave demo mode.

The following features are limited in demo mode:

1. There will be a **SightLine Demo Mode** overlay on all video output
2. No telemetry output will be sent
3. Capture snapshots are not allowed
4. Saving parameters is not allowed
5. KLV metadata is not embedded in network video streams
6. Detect Snap Shot is not allowed

Hardware Type

Description	ID
2000-OEM	0
1500-OEM	7
3000-OEM Rev B	12
3000-OEM Rev C	14
4000-OEM	18
1750-OEM	28
4100-OEM	36 or 41
1710-OEM	27

Message ID 0x40

Byte Offset	Name	Description								
4	swMajor	Software major version number								
5	swMinor	Software minor version number								
6	hwVersion	Second FPGA version number (17XX/4000 MIPI)								
7	degreesF	Temperature in degrees F								
8-10	hwID	Hardware UID								
11-14	appBits	<u>Application bits</u> (app bits).								
15	boardType	<u>Hardware type</u>								
16	swRelease	Software revision (release number)								
17-18	otherVersion	FPGA and hardware versions								
		<table><tr><th>Bits</th><th>Description</th></tr><tr><td>0-7</td><td>FPGA Version**New 3.2**</td></tr><tr><td>8-15</td><td>Hardware Revision New 3.2</td></tr></table>	Bits	Description	0-7	FPGA Version**New 3.2**	8-15	Hardware Revision New 3.2		
		Bits	Description							
		0-7	FPGA Version**New 3.2**							
8-15	Hardware Revision New 3.2									
19-22	srcRevision	Source Code revision								
23-26	buildDate	Build date.								
		<table><tr><th>Bits</th><th>Description</th></tr><tr><td>0-7</td><td>Day of the month</td></tr><tr><td>8-15</td><td>Month of the year</td></tr><tr><td>16-31</td><td>Year the build was made</td></tr></table>	Bits	Description	0-7	Day of the month	8-15	Month of the year	16-31	Year the build was made
		Bits	Description							
		0-7	Day of the month							
		8-15	Month of the year							
16-31	Year the build was made									
27-30	buildTime	Build time.								
		<table><tr><th>Bits</th><th>Description</th></tr><tr><td>0-7</td><td>Seconds</td></tr></table>	Bits	Description	0-7	Seconds				
		Bits	Description							
0-7	Seconds									

		8-15	Minutes
		16-31	Hour value
31-32	swBuild	Build number	
33-34	v4AppBits	Version 4 Application Bits (pre 3.0 systems)	
35-36	degreesC	Temperature in degrees C	
37-40	adapters	Attached camera adapter boards	

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAGetParameters_t Struct Reference

[General](#)

Description

Request return of a specified message ID.

```
typedef struct {
    u8 id;
    u8 payload0;
    u8 payload1;
    u8 payload2;
    u8 payload3;
} SLAGetParameters_t;
```

The value of the ID field corresponds to the 'Setter' of the same type. Some commands take additional arguments. Response packets are unique and usually have their own type ID.

Id	payload0	payload1	payload2	payload3	Response type
0x00	–	–	–	–	SLAVersionNumber_t
0x01	[Camera Index] ¹	–	–	–	SLACurrentConfiguration_t
0x02 , 0x03	[Camera Index] ¹	–	–	–	SLACurrentStabilizationParameters_t
0x06 , 0x07	[Camera Index] ¹	–	–	–	SLACurrentOverlayMode_t
0x0B	[Camera Index] ¹	–	–	–	SLACoordinateReportingMode_t
0x0C , 0x0D	[Camera Index] ¹	–	–	–	SLACurrentTrackingParameters_t
0x0E , 0x0F	[Camera Index] ¹	–	–	–	SLACurrentRegistrationParameters_t
0x10 ,	[Camera	–	–	–	SLACurrentVideoParameters_t

<u>0x11</u>	Index] ¹				
<u>0x12</u>	[Camera Index] ¹	–	–	–	<u>SLACurrentStabilizationBias_t</u>
<u>0x13</u>	[Network Display ID, lsb] ²	[Network Display ID, msb] ²	–	–	<u>SLACurrentMetadataValues_t</u>
<u>0x14</u>	[Static Element ID] ³	[Network Display ID, lsb] ²	[Network Display ID, msb] ²	–	<u>SLAMetadataStaticValues_t</u>
<u>0x15</u>	[Network Display ID, lsb] ²	[Network Display ID, msb] ²	–	–	<u>SLACurrentMetadataFrameValues_t</u>
<u>0x18, 0x19</u>	[Camera Index] ¹	–	–	–	<u>SLACurrentADCPParameters_t</u>
<u>0x1A, 0x1B</u>	[Network Display ID, lsb] ²	[Network Display ID, msb] ²	–	–	<u>SLACurrentEthernetVideoParameters_t</u>
<u>0x1F, 0x20</u>	–	–	–	–	<u>SLACurrentVideoModeParameters_t</u>
<u>0x21, 0x22</u>	[Camera Index] ¹	–	–	–	<u>SLACurrentVideoEnhancementParameters_t</u>
<u>0x23, 0x24</u>	[Network Display ID, lsb] ²	[Network Display ID, msb] ²	–	–	<u>SLACurrentH264Parameters_t</u>
<u>0x29, 0x39</u>	[Network Display ID, lsb] ²	[Network Display ID, msb] ²	–	–	<u>SLACurrentEthernetDisplayParameters_t</u>
<u>0x2D, 0x2E</u>	[Camera Index] ¹	[Detection Index] ¹³	–	–	<u>SLACurrentDetectionParameters_t</u>
<u>0x2F, 0x30</u>	–	–	–	–	<u>SLACurrentBlendParameters_t</u>
<u>0x31</u>	[Camera Index] ¹	–	–	–	<u>SLACurrentImageSize_t</u>
<u>0x35</u>	[Camera Index] ¹	–	–	–	<u>SLANucParameters_t</u>
<u>0x36</u>	[NUC Read Write Mode] ¹¹	[Camera Index] ¹	–	–	<u>SLAReadWriteNuc_t</u>
<u>0x37,</u>	[Camera	–	–	–	<u>SLACurrentAcquisitionParameters_t</u>

<u>0x38</u>	Index] ¹				
<u>0x3A</u>	[Camera Index] ¹	–	–	–	<u>SLACurrentDisplayParameters_t</u>
<u>0x3B</u>	[Object ID] ⁸	–	–	–	<u>SLACurrentOverlayObjectParameters_t</u> , <u>SLACurrentOverlayObjectsIds_t</u>
<u>0x50</u>	–	–	–	–	<u>SLAGetHardwareID_t</u>
<u>0x5B</u>	[Camera Index] ¹	–	–	–	<u>SLACommandCamera_t</u>
<u>0x5C</u>	[Camera Index] ¹	–	–	–	<u>SLADisplayAngle_t</u>
<u>05E</u> , <u>0x5F</u>	[Camera Index] ¹	–	–	–	<u>SLACurrentSnapShot_t</u>
<u>0x62</u>	[Bit Field Number] ⁴	[Network Display ID, lsb] ²	[Network Display ID, msb] ²	–	<u>SLACurrentMetadataRate_t</u>
<u>0x64</u>	[Port Index] ⁵	–	–	–	<u>SLASetTelemetryDestination_t</u>
<u>0x66</u>	–	–	–	–	<u>SLACurrentNetworkList_t</u>
<u>0x6D</u>	[Camera Index] ¹	–	–	–	<u>SLACurrentLensStatus_t</u>
<u>0x6E</u> , <u>0x6F</u>	–	–	–	–	<u>SLACurrentLensParameters_t</u>
<u>0x70</u> , <u>0x71</u>	[Camera Index] ¹	–	–	–	<u>SLACurrentDigitalCameraParameters_t</u>
<u>0x72</u>	–	–	–	–	<u>SLACurrentUserPalette_t</u>
<u>0x74</u> , <u>0x75</u>	–	–	–	–	<u>SLACurrentMultipleAlignment_t</u>
<u>0x76</u>	[Camera Index] ¹	[Detection Index] ¹³	–	–	<u>SLACurrentAdvancedDetectionParameters_t</u>
<u>0x7B</u>	–	–	–	–	<u>SLAAdvancedCaptureParameters_t</u>
<u>0x7C</u>	[Camera Index] ¹	[Detection Index] ¹³	–	–	<u>SLACurrentDetectionRegionOfInterestParameters_t</u>
<u>0x7F</u>	–	–	–	–	<u>SLAUserWarningLevel_t</u>
<u>0x80</u>	–	–	–	–	<u>SLASystemStatusMode_t</u>
<u>0x81</u>	[Camera Index] ¹	–	–	–	<u>SLALandingAid_t</u>
<u>0x89</u>	[Network Display ID,	[Network Display	–	–	<u>SLAAppendedMetadata_t</u>

	lsb] ²	ID, msb] ²			
<u>0x8A</u>	[Camera Index] ¹	–	–	–	<u>SLAFrameIndex_t</u>
<u>0x8F</u>	[Program Type] ⁶	–	–	–	<u>SLAExternalProgram_t</u>
<u>0x90</u>	–	–	–	–	<u>SLAStreamingControl_t</u>
<u>0x91</u>	[Camera Index] ¹	–	–	–	<u>SLADigitalVideoParserParameters_t</u>
<u>0x92</u>	[SystemValue] ⁷	–	–	–	<u>SLACurrentSystemValue_t</u>
<u>0x96</u>	[Tag Id] ⁹	[Tag Sub Id] ¹⁰	[Network Display ID, lsb] ²	[Network Display ID, msb] ²	<u>SLATagData_t</u>
<u>0x97</u>	[Tag Id] ⁹	[Network Display ID, lsb] ²	[Network Display ID, msb] ²	–	<u>SLATagDataRate_t</u>
<u>0x98</u>	[Tag Id] ⁹	[Network Display ID, lsb] ²	[Network Display ID, msb] ²	–	<u>SLATagSourceSelector_t</u>
<u>0x99</u>	–	–	–	–	<u>SLADecoderParameters_t</u>
<u>0x9B</u>	[Camera Index] ¹	–	–	–	<u>SLALogoParameters_t</u>
<u>0x9C</u>	[Object ID] ¹	[Camera Index] ¹	–	–	<u>SLADrawOverlay_t</u>
<u>0x9D</u>	[Camera Index] ¹	–	–	–	<u>SLATrackTrails_t</u>
<u>0x9E</u>	[Camera Index] ¹	–	–	–	<u>SLARegistrationParameters_t</u>
<u>0x9F</u>	[Camera Index] ¹	–	–	–	<u>SLAStabilizationBias_t</u>
<u>0xA1</u>	[Camera Index] ¹	–	–	–	<u>SLADeadPixelStats_t</u>
<u>0xA4</u>	[Camera Index] ¹	–	–	–	<u>SLAVideoDisplay_t</u>
<u>0xA5</u>	[Camera Index] ¹	–	–	–	<u>SLAMultiDisplay_t</u>
<u>0xA7</u>	[Camera Index] ¹	–	–	–	<u>SLACustomClassifier_t</u>

<u>0xA9</u>	[Camera Index] ¹	–	–	–	<u>SLAClassifierParameters_t</u>
<u>0xAD</u>	[Network Display ID, lsb] ²	[Network Display ID, msb] ²	–	–	<u>SLAVMTIChips_t</u>
<u>0xAE</u>	[User Font Index]	–	–	–	<u>SLAUserFont_t</u>
<u>0xAF</u>	[Camera Index] ¹	–	–	–	<u>SLANoise3D_t</u>
<u>0xB0</u>	[Network Display ID, lsb] ²	[Network Display ID, msb] ²	–	–	<u>SLACursorOnTarget_t</u>
<u>0xB1</u>	[Camera Index] ¹	–	–	–	<u>SLALensParameters_t</u>
<u>0xB3</u>	[Camera Index] ¹	–	–	–	<u>SLAFocusParameters_t</u>
<u>0xB4</u>	[Camera Index] ¹	–	–	–	<u>SLALensRanges_t</u>
<u>0xB5</u>	[Camera Index] ¹	–	–	–	<u>SLACustomAutoFocusParameters_t</u>
<u>0xB6</u>	[GPIO number, lsb]	[GPIO number, msb]	–	–	<u>SLAGPIO_t</u>
<u>0xB9</u>	[Blend Align Index]	–	–	–	<u>SLABlendAlign_t</u>
<u>0xBA</u>	[Camera Index] ¹	–	–	–	<u>SLACustomAIDetect_t</u>
<u>0xBB</u>	[Camera Index] ¹	–	–	–	<u>SLACameraCapabilities_t</u>
<u>0xBE</u>	[Network Display ID, lsb] ²	[Network Display ID, msb] ²	–	–	<u>SLASendToBTS_t</u>

0xBF | [Camera Index]¹ | – | – | **SLAVMTIFields_t** **0xC0** | [Camera Index]¹ | – | – | **SLACameraCalibration_t** **0xC1** | [Camera Index]¹ | – | – | **SLAKlvClassFilters_t** **0xC2** | [Camera Index]¹ | – | – | **SLACameraParamFile_t**

Note

– No additional parameter is needed.

¹ Optional argument specifies the camera index. If the option is not specified, then the command camera is assumed.

- ² 17XX/4000/41XX: optional argument specifies Network Display ID (0x0002=Net0, 0x0080=Net1, 0x0010=Net2). If not present, then Net0 is assumed.
- ³ Static Element ID indicates identifier type, See [SLAMetadataStaticValues_t](#) type (byte4) for more information.
- ⁴ Bit field number indicates bit index of metadata mask, 17XX/4000/41XX: optional argument specifies Network Display ID (0x0002=Net0, 0x0080=Net1, 0x0010=Net2). If not present, then Net0 is assumed..
- ⁵ Port Index indicates index into list of telemetry destinations (maximum number of elements in list is 4).
- ⁶ Program Type indicates the program type to request. If not present than User Display DLL is assumed.
- ⁷ SystemValue indicates the System Value type to request. This argument may not be omitted.
- ⁸ Optional argument specifies Overlay Object ID. If not specified, returns a list of used Object IDs (see [SLACurrentOverlayObjectsIds_t](#))
- ⁹ Tag Id for the KLV Tag to use as per 0601.11 MISB STD. (See [SLATagData_t](#))
- ¹⁰ Tag Sub Id for the KLV Tag to use as per 0601.11 MISB STD. (See [SLATagData_t](#))
- ¹¹ Nuc read write mode for request (0 = None, 1 = NUC Table, 2 = Dead Table, 3 = Default NUC Table, 4 = Default Dead Table)
- ¹² CurrentConfiguration defaults to old fuctionality (displaying bitmasked information about all cameras) when no Camera Index is specified
- ¹³ Optional argument specifies the detection index. If not present, then the last detection index specified in a set command is used

Message ID 0x28

Byte Offset	Name	Description
4	id	[Required] Requested message ID
5	payload0	[Optional] First payload byte
6	payload1	[Optional] Second payload byte
7	payload2	[Optional] Third payload byte
8	payload3	[Optional] Fourth payload byte

Sightline Intelligence Command, Control, and Script API v3.11

[Main Page](#)[Related Pages](#)[Modules](#)[Classes](#) ■[Camera Configuration](#)

SLAGetVersionNumber_t Struct Reference

[General](#)

Description

Used to query for system information such as firmware version, temperature, app bits, and more.

```
typedef struct {  
} SLAGetVersionNumber t;
```

Return a [SLAVersionNumber_t](#) message.

Message ID 0x00

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetPortConfiguration_t Struct Reference

[Serial Port](#)

Description

Configure one of the communication ports for a specific protocol or mode.

```
typedef struct {
    u8 port;
    u8 baud;
    u8 dataBits;
    u8 stopBits;
    u8 parity;
    u8 maxPacket;
    u8 maxDelay;
    u8 protocol;
    u16 inputPort;
    u32be udpDestinationAddr;
    u16 udpDestinationPort;
    u16 udpAttNavPort;
} SLASetPortConfiguration\_t;
```

Current setting can be retrieved using [SLAGetPortConfiguration_t](#).

NOTE: It takes up to 10 seconds for Serial port settings to take effect. To see if the change was taken, wait for 10 seconds then use [SLAGetPortConfiguration_t](#) to retrieve the updated setting.

Port Identifier

Port Identifier	Name	4000-OEM	1750-OEM	1710-OEM	4110-OEM
0	Serial port 0	J25 - General Purpose	J1 - General Purpose	J1 - General Purpose	J1 - General Purpose
1	Serial port 1	J2 - General Purpose	J9 - Debug	J9 - Debug	J1 - General Purpose
4	Serial port 2	J6 - Camera 0	J6 - Camera 0	J7 - Camera 0 or General Purpose	J8 - Camera 0 or General Purpose
6	Serial port 3	J25 - General Purpose	J1 - General Purpose	J1 - General Purpose	MIPI2 - Dependent on adapter. See ICD-Adapter-Boards

7	Serial port 4	MIPI J2 - General Purpose	MIPI2 J2 - General Purpose	MIPI2 - General Purpose	MIPI0 - Dependent on adapter. See ICD-Adapter-Boards
8	Serial port 5	NA	NA	MIPI1 - General Purpose	NA
9	Serial port 6	J9 - Camera 1	J8 - Camera 1	J8 - Camera 1 or General Purpose	J6 - Camera 1 or General Purpose
10	Serial port 7	MIPI J2 - General Purpose	MIPI2 J2 - General Purpose	MIPI2 - General Purpose (MIPI-IN only)	MIPI0 - Dependent on adapter. See ICD-Adapter-Boards
11	Serial port USB 0	USB	USB	USB	USB
15	Ethernet Port	KLV Passthrough	KLV Passthrough	KLV Passthrough	KLV Passthrough
19	Serial port USB 1	USB	USB	USB	USB
20	Serial port USB 2	USB	USB	USB	USB
21	Serial port 8	NA	J4 LVDS	J3 LVDS	MIPI2 - Dependent on adapter. See ICD-Adapter-Boards
22	Serial port 9	NA	NA	MIPI1 - General Purpose (MIPI-IN only)	NA
23	Serial port 10	NA	NA	NA	J5 - Camera 2 or General Purpose
24	Serial port 11	NA	NA	NA	MIPI1 - Dependent on adapter. See ICD-Adapter-Boards
25	Serial port 12	NA	NA	NA	MIPI1 - Dependent on adapter. See ICD-Adapter-Boards

NOTE: All other values are reserved.

Baud Rate

Value	Baud rate
0	4800
1	9600
2	38400
3	57600 (Default)

4	115200
5	19200
6	921600
7	2400
8	230400
9	460800

Protocol Type

Value	Name	Description
0	SLA Protocol	Port will accept SightLine Application commands(0x51 0xAC) specified in this document and may also generate telemetry data.
1	Aquarius & SLA Protocol	Port will accept and parse Scan Eagle Acquarius packets and SightLine Applications commands(0x51, 0xAC).
2	SLA Protocol no telemetry	Port will accept SightLine Application commands(0x51 0xAC) specified in this document and will not generate any telemetry data.
3	<i>Reserved</i>	-
4	Port not used	Allows use by other applications running on the ARM process to access this serial port, or when Linux command line debugging is active.
5	TCP passthrough	Creates a TCP socket that waits for a connection on the <i>inputPort</i> . Any data recieved is sent to the Serial Port. Any data received on the Serial Port is sent back out the TCP socket. <i>udpDestinationAddr</i> and <i>udpDestinationPort</i> are not used as these are handled by the TCP socket.
6	Raw passthrough	A UDP socket is created to listen for any data recieved on <i>inputPort</i> , which is passed out the serial port. Any data received on the serial port is sent as a UDP packet to the IP address (<i>udpDestinationAddr</i>) at port (<i>udpDestinationPort</i>).`
7	KLV passthrough	KLV data received on serial or UDP receive port are injected into MPEG2-TS stream. KLV Passthrough receives formatted KLV metadata via serial port generated by an external processor. This KLV metadata is injected into the MPEG2-TS stream along with compressed digital video. When using this mode, internally-generated KLV metadata should be disabled via <u>SLASetMetadataRate_t</u> . See also <u>SLASetKlvData_t</u> , which allows users to set blocks of KLV data.
8	Serial-to-serial passthrough	Serial-to-Serial sends output from one serial port to another serial port specified in Destination port number (<i>udpDestinationPort</i> , must be set to one of the Port Identifier values : 4 for Serial Port 2 for example). Maximum packet length, inbound port, destination IP and AttNav are ignored. By configuring two ports to output to each other, the two ports act like they are "piped" together.
9	NMEA	Any data received on the serial port is assumed to be NMEA 0183 compatible strings. The external device which generates the strings is connected to the serial port specified. The serial port is configured for the correct baud rate and other settings to match the sender (all Ethernet

	parsing	settings are ignored and can be set to zeros). Strings are parsed and converted to appropriate Metadata values. All other data is ignored. Please see EAN-Ethernet-and-Serial-Communication.PDF for additional information.
10	Lens Protocol	Dynamically loaded lens command and control protocol
11+	<i>Reserved</i>	-

Message ID 0x3E

Byte Offset	Name	Description
4	port	Port Identifier , see SLACommandPassThrough_t.destPort
5	baud	Baud Rate - speed in bits per second
6	dataBits	Data bits - Number of data bits in each character, default 8
7	stopBits	Stop bits - Number of stop bits sent at the end of every character, default 1
8	parity	Parity - 0 = None (default), 1 = Odd, 2 = Even.
9	maxPacket	Maximum packet length in bytes for input serial or network data
10	maxDelay	Maximum timeout in milliseconds when reading from a port; when timeout is complete, any received data is sent to the destination port
11	protocol	Protocol Type
12-13	inputPort	Local inbound port number where UDP/TCP packets are expected. Ethernet port were all payload data will be received. Hardware opens a new socket to listen on this port. Must be greater than 0 (0 is reserved)
14-17	udpDestinationAddr	Destination IP address of host where Ethernet packets will be sent
18-19	udpDestinationPort	Destination port number (port number where host is listening). Hardware opens a new socket and sends data to the outbound destination IP Address at the destination port number. Must be greater than 0 (0 is reserved). For Serial-to-Serial passthrough, this is the other port ID (4 for Serial Port 2 for example).
20-21	udpAttNavPort	AttNav port number. When set to a value other than zero, a network socket is created to listen on this port (defaults 65100) for ATTNNAV packets. Any data received on this port is then sent on to the serial port defined. Only applies when Protocol Type is set to Scan Eagle Aquarius packet parsing & SLA protocol, this port is set to 65100. Requires that Destination IP address be NON - ZERO and Local inbound port number be NON - ZERO.

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetNetworkParameters_t Struct

Reference

[Network](#)

Description

Configure the network settings for the device.

```
typedef struct {
    u8 mode;
    u32be ipAddr;
    u32be subnet;
    u32be gateway;
    u16be c2replyPort;
    u16be reserved0;
    u8 reserved1;
    u8 reserved2;
    u16be listenPort;
    u16be listenPort2;
    SVPLenString\_t hostName;
} SLASetNetworkParameters\_t;
```

Note

There is only 1 network interface.

Message ID 0x1C

Byte Offset	Name	Description
4	mode	0 = Use DHCP, 1 = Use specified Static IP address
5-8	ipAddr	IP address, if mode is static; 0 = No change
9-12	subnet	Subnet mask, if mode is static, 0 = No change
13-16	gateway	Gateway, if mode is static, 0 = No change
17-18	c2replyPort	Command and control reply port. 0 = No change. Port on remote device that will be listening for replies. Default port is 14002. see Communication
19-20	reserved0	<i>Reserved</i>
21	reserved1	<i>Reserved</i>

22	reserved2	<i>Reserved</i>
23-24	listenPort	(optional) C2C command listen port, 0 indicates no change, default is 14001.
25-26	listenPort2	(optional) C2C command listen port2, 0 indicates no change, default is 14003.
27	hostName.len	string length
28-...	hostName.str	(optional) Set the Host Name returned by Discover protocol. Host name can be up to 31 bytes long, leading space characters are ignored. Setting 0 length string will reset host name to the default. If this field is omitted, then no change will be made to hostname.

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAUserWarningMessage_t Struct Reference

[General](#)

Description

Message sent from the SLA hardware to notify users of any issues or system status which may not be obtained any other way.

```
typedef struct {
    u16 warnLevel;
    SVPEndString\_t str;
} SLAUserWarningMessage\_t;
```

Examples could be : an un-licensed feature was requested, or an out-of-index-range camera was specified.
NOTE: This packet is sent only to Ethernet Command and Control Reply Port, and never sent to a Serial port.

Message ID 0x86

Byte Offset	Name	Description
4-5	warnLevel	Warning level of this message (see SLAUserWarningLevel_t.warnLevel)
6	str	Text characters for the message itself, comprised of one or more lines delimited by '\n' character (terminating null character may or may not be present)

Sightline Intelligence

Command, Control, and Script

API v3.11

Telemetry

[Classes](#)

Classes

struct	SLACoordinateReportingMode_t
struct	SLATrackingPosition_t
struct	SLATrackingPositions_t
struct	SLAFocusStats_t
struct	SLASetTelemetryDestination_t
struct	SLATrackingBoxPixelStats_t
struct	SLALandingPosition_t
struct	SLATrackingPositionsExtended_t
struct	SLATrackingMultiClass_t

Detailed Description

Telemetry result messages and configuration.

SightLine hardware can output per frame telemetry results measuring:

- Frame to frame scene motion (registration) in offset, angle and zoom for use in gimbal scene lock
- Stabilized display offset and angle (stabilization)
- Multiple user designated track locations
- Multiple automatic object detection locations and velocities
- Pixel statistics for a stationary or moving box for measuring IR temperature
- Lens focus and brightness measures for use in auto focus and aperture/exposure control
- Landing aid position and angle for use in auto landing systems

NOTE: Availability depends on the application bits enabled in each system

NOTE: Telemetry is not available in [Demo Mode](#).

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetTelemetryDestination_t Struct Reference

[Telemetry](#) | [Network](#)

Description

Configure the Ethernet IP address of clients for telemetry.

```
typedef struct {
    u8 mode;
    u8 cameraID;
    u32be ipAddr;
    u16 udpDestinationPort;
    u8 flags;
    u8 reserved;
} SLASetTelemetryDestination_t;
```

Up to thirty-two (32) IP addresses + port + camIdx combinations may be added to the list. All telemetry enabled by [SLACoordinateReportingMode_t](#) will be sent to each receiver.

Also sent in response to [SLAGetParameters_t\(\)](#).

NOTE: Telemetry output is not available in [Demo Mode](#).

See also

[SLACoordinateReportingMode_t](#) for the types of [Telemetry](#) output that are available.

Message ID 0x64

Byte Offset	Name	Description												
4	mode	Function												
		<table><tr><th>Value</th><th>Description</th></tr><tr><td>0</td><td>Reserved</td></tr><tr><td>1</td><td>Add an IP address as telemetry output destination</td></tr><tr><td>2</td><td>Remove an IP address from receiving telemetry output</td></tr><tr><td>3</td><td>Remove all IP addresses from telemetry output list</td></tr><tr><td>4-15</td><td>Reserved</td></tr></table>	Value	Description	0	Reserved	1	Add an IP address as telemetry output destination	2	Remove an IP address from receiving telemetry output	3	Remove all IP addresses from telemetry output list	4-15	Reserved
		Value	Description											
		0	Reserved											
		1	Add an IP address as telemetry output destination											
		2	Remove an IP address from receiving telemetry output											
		3	Remove all IP addresses from telemetry output list											
		4-15	Reserved											

5	cameraID	Camera for settings (0 to 2).								
6-9	ipAddr	IP address of receiver to be added/removed; 3000 only: 0 = send telemetry to serial port								
10-11	udpDestinationPort	Destination port – data is sent to this port on the receiver, 0 = <i>Reserved</i>								
12	flags	<table><tr><th>Bit</th><th>Description</th></tr><tr><td>0</td><td>0 = Do not save this packet destination 1 = save packet destination when parameter file is saved and restore when loaded</td></tr><tr><td>1</td><td>0 = Do not apply camera ID (instead send actual camera index in telemetry packets) 1 = send cameraID in telemetry packets (1500 only)</td></tr><tr><td>2-7</td><td><i>Reserved</i></td></tr></table>	Bit	Description	0	0 = Do not save this packet destination 1 = save packet destination when parameter file is saved and restore when loaded	1	0 = Do not apply camera ID (instead send actual camera index in telemetry packets) 1 = send cameraID in telemetry packets (1500 only)	2-7	<i>Reserved</i>
		Bit	Description							
		0	0 = Do not save this packet destination 1 = save packet destination when parameter file is saved and restore when loaded							
		1	0 = Do not apply camera ID (instead send actual camera index in telemetry packets) 1 = send cameraID in telemetry packets (1500 only)							
2-7	<i>Reserved</i>									
13	reserved	<i>Reserved</i>								

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

Sightline Discover Protocol

Ethernet protocol for finding Sightline hardware on a network.

Overview

- The client (PC) sends a [Request Packet](#) to broadcast address of 255.255.255.255
- The Sightline listens for Request Packets on port **51000**.
- A [Reply Packet](#) is then sent back to the orginator's IP address and port 51000.

Transport Layer	UDP
IP Address	255.255.255.255
Port	51000

Request Packet

- A request is generated by broadcasting a UDP packet with the string "SLDISCOVER" or "EXTSLDISCOVER" to port 51000.
- The extended discover request, "EXTSLDISCOVER", will add Reply Port and Video Port Net 1 and 2 to the response.
- A request is also sent as a directed broadcast based on the subnet mask of the client. e.g. 192.168.3.255
- All Sightline hardware units on the network are listening for that message.
- All Sightline hardware units will send the [Reply Packet](#) as a response directly to the originator of the request packet.

Note

Because a request packet is sent to both the general broadcast address and the directed broadcast address, the target hardware will send two Reply Packets.

Reply Packet

Byte	Length	Name	Description
0	4	ID	Magic identifier number
4	4	Length	Discover message length
8	2	Minor Version	Discover protocol minor version. (0x04)
10	2	Major Version	Discover protocol major version. (0x00)
12	2	Type	Services provided (internal use only)

14	2	Hardware Type	Hardware type index
16	20	MAC	MAC address of sender
36	16	IP Address	IPv4 address string of sender
52	16	netmask	IP Subnet mask associated with this IP address
68	32	Host Name	Human readable name of hardware
100	2	Video Port 0	Port number where images are sent on network 0
102	2	Command and Control Port	Port number open to receive commands (default 14001)
104	2	Reply Port	Port number open to send responses (default 14002). Only sent in response to the extended discover request.
106	2	Video Port 1	Port number where images are sent on network 1. Only sent in response to the extended discover request.
108	2	Video Port 2	Port number where images are sent on network 2. Only sent in response to the extended discover request.

Magic Number

A 4-byte code (0x51acd00d) indicating this is a reply packet.

Length

Number of bytes to follow.

Minor and Major Version Number

Current Reply Packet is version 0.4

Type

A list of features that this software provides. Internal Use Only.

Hardware Type

Identifies the class of hardware. See [Sightline Hardware Types](#)

MAC address

The MAC address string is a unique 6 octect value assigned to the hardware network interface control. Example:
"78:D8:00:91:2f:d2"

IPv4 address

See [IP Address Assignment](#) Example: "192.168.1.176"

subnet mask

Example: "255.255.255.0"

Hostname

See [host name](#) for more information.

Video Port 0

Port number where images are sent on network 0.

Command and Control Port

Network port on which this device will be listening to command and control packets. see [SLASetNetworkParameters_t](#)

Reply Port

Network port on which this device will be sending out responses. Only sent in response to the extended discover request.

Video Port 1

Port number where images are sent on network 1. Only sent in response to the extended discover request.

Video Port 2

Port number where images are sent on network 2. Only sent in response to the extended discover request.

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACommandCamera_t Struct

Reference

[General](#)

Description

Set which camera channel receives camera specific parameter settings.

```
typedef struct {
    u8 cameraIndex;
} SLACommandCamera t;
```

Also sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x5B

Byte Offset	Name	Description
4	cameraIndex	Camera Index Required

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentConfiguration_t Struct Reference

[General](#)

Description

Current camera and display configuration.

```
typedef struct {
    u8 maxCameras;
    u8 maxVirtCameras;
    u8 maxStreams;
    u8 maxProcessed;
    u16 cameraConfiguredBits;
    u16 cameraConnectedBits;
    u32 displayPresentBits;
    u32 captureStateBits;
} SLACurrentConfiguration_t;
```

- NOTE:** Analog cameras will always have display an 'on' signal.
- NOTE:** After startup, the values of **cameraConnectedBits** and **captureStateBits** will only be updated for cameras which are streaming video to any output. Sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x8E

Byte Offset	Name	Description
4	maxCameras	Maximum number of camera inputs
5	maxVirtCameras	Maximum number of virtual camera inputs (the blend 'camera') – 4000: 0, 17XX: 0
6	maxStreams	Maximum number of streams the system can display simultaneously
7	maxProcessed	Maximum number of cameras the system can process simultaneously
8-9	cameraConfiguredBits	1 if the Camera Index camera is configured, 0 otherwise. Without Camera Index Parameter: Camera configured bit pattern, bit0=Cam0, bit1=Cam1, bit2=Cam2 (e.g. 0x00000005 indicates Cam0 and Cam2 are configured)
10-11	cameraConnectedBits	1 if the Camera Index camera is connected and acquiring, 0 otherwise. Without Camera Index Parameter: Camera connected bit pattern, bit0=Cam0, bit1=Cam1, bit2=Cam2 (e.g. 0x00000005 indicates Cam0 and Cam2 are connected and acquiring)

		NOTE: Analog cameras will indicate true even if the camera is not connected.												
12-15	displayPresentBits	The currently available display types. Note: See SLAVideoDisplay_t destination for display type values (e.g. 0x00000083 indicates Analog, Net0 and Net1 are available).												
16-19	captureStateBits	The CameraIndex camera's capture state. Without Camera Index Parameter: Camera state bit pattern. (e.g. 0x00000001100000000 indicates Cam1 is actively capturing and cam0 is in an Unknown state) <table><tr><th>Value</th><th>Description</th></tr><tr><td>0</td><td>Unknown</td></tr><tr><td>1</td><td>Initializing</td></tr><tr><td>2</td><td>Re-initializing</td></tr><tr><td>3</td><td>Active</td></tr><tr><td>4</td><td>Timeout</td></tr></table>	Value	Description	0	Unknown	1	Initializing	2	Re-initializing	3	Active	4	Timeout
Value	Description													
0	Unknown													
1	Initializing													
2	Re-initializing													
3	Active													
4	Timeout													

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetEthernetDisplayParameters_t

Struct Reference

[Compression](#)

Description

Configure output format of Ethernet Video.

```
typedef struct {
    u8 protocol;
    u32be ipAddr;
    u16 port;
    u16 displayId;
    u16 maxPacket;
    u16 maxRawPacket;
} SLASetEthernetDisplayParameters_t;
```

Sets the destination IP address and port number where the video will be sent.

See also [SLAVideoDisplay_t](#) to set the Display Destination mode.

Video quality settings can be set using [SLASetEthernetVideoParameters_t](#) and/or [SLASetH264Parameters_t](#) depending on which Video Mode is selected.

Message ID 0x29

Byte Offset	Name	Description	
		Video encoding and transport	
		Bits	Description
		0-3	0 = <i>Reserved</i>
			1 = MPEG2-TS H.264 Baseline/High/Main Profile SD/HD video encoding
			2 = <i>Reserved</i>
			3 = <i>Reserved</i>
			4 = <i>Reserved</i>
			5 = RTP H.264 for use with RTSP
			6 = RTP MPEG2-TS H.264 for use with RTSP
			7 = <i>Reserved</i>
			8 = MPEG2-TS H.265

4	protocol		9 = RTP H.265 for use with RTSP 10 = RTP MPEG-TS H.265 for use with RTSP 11 = MPEG2-TS KLV Only. Use this mode to stream an MPEG2-TS with only KLV data. 12-14 = <i>Reserved</i> 15 = None	
		4-5	Aggregate Packet Mode - RTP H.264 and RTP H.265 only, set to 0 otherwise.	
			0	Default - Aggregate packets enabled for RTP H.265 and disabled for RTP H.264
			1	Enable Aggregate Packets
		2	Disable Aggregate Packets	
6	0 = Standard encapsulation (default), 1= for MPEG2-TS types output only raw encoded video w/o encapsulation			
7	0 = disable broadcast video (default), 1= enable broadcast video. NOTE: Enabling broadcast video may interact strongly with other network traffic. Use with caution.			
5-8	ipAddr	Any valid IPv4 address. If address is within the valid Multicast range the video will be sent as a multicast packet. For example: the address 224.10.10.10 is in the Multicast range, and setting ipAddr to this address will cause multicast video to be sent. The Time To Live(TTL) for all Multicast packets is set to 64, so this will limit how many intermediate hops between the source(SLA Hardware) and the destination(the PC). If Broadcast Ethernet Video bit is set to 1, then you must set the IP address to to accommodate either limited or directed broadcast.		
9-10	port	Any value > 0, port 5004 recommended for RTP MJPEG; port 15004 recommended for other protocols. NOTE: Per RFC 3550 port number should be even for RTP protocols. Odd port will be rejected and system will generate a SLAUserWarningMessage_t .		
11-12	displayId	Network Display ID 4000/17XX: (0x0002 = Net0, 0x0080 = Net1, 0x0082 = both) 41XX: (0x0002 = Net0, 0x0080 = Net1, 0x0010 = Net2, 0x0092 = all)		
13-14	maxPacket	Maximum number of bytes per UDP packet in MPEG2-TS protocols. (188..1316). Only valid when <i>protocol</i> is 1, 3, 4, 6, 8, 10, or 11. A value of 0 defaults to the maximum value. This still affects the same protocols when <i>Standard Encapsulation</i> is enabled.		
15-16	maxRawPacket	Maximum number of bytes per UDP packet Non-MPEG2-TS protocols. (300..1472). Only valid when <i>protocol</i> is 5 or 9. A value below the minimum defaults to the maximum value. New 3.5.8		

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetSDRecordingParameters_t

Struct Reference

[Recording](#)

Description

Modify settings for on board **video** recording.

```
typedef struct {
    u8 recordMode;
    u8 clearFlash;
    u8 getStatus;
    u8 getDirectory;
    u16 videoLength;
    u8 recordType;
    u8 displayId;
    SVPLenString\_t lbl;
} SLASetSDRecordingParameters\_t;
```

Video can be recorded to FTP, microSD cards (all platforms) or USB drive (17XX/4000 only). For SnapShot recording see [SLASetSnapShot_t](#).

To check if an SD card is present see ***getStatus***.

See also
[File Recording](#) for additional details.

Message ID 0x1E

Byte Offset	Name	Description														
4	recordMode	Modify the recording state														
		<table><tr><th>Value</th><th>Description</th></tr><tr><td>0</td><td>No change</td></tr><tr><td>1</td><td>Start recording</td></tr><tr><td>2</td><td>Stop recording</td></tr><tr><td>3</td><td>Enable network debug trace of commands and responses</td></tr><tr><td>4</td><td>Disable network debug trace of commands and responses</td></tr><tr><td></td><td></td></tr></table>	Value	Description	0	No change	1	Start recording	2	Stop recording	3	Enable network debug trace of commands and responses	4	Disable network debug trace of commands and responses		
		Value	Description													
		0	No change													
		1	Start recording													
		2	Stop recording													
		3	Enable network debug trace of commands and responses													
		4	Disable network debug trace of commands and responses													

		<table><tr><td>5</td><td>Enable network debug trace of telemetry</td></tr><tr><td>6</td><td>Disable network debug trace of telemetry</td></tr></table>	5	Enable network debug trace of telemetry	6	Disable network debug trace of telemetry										
5	Enable network debug trace of telemetry															
6	Disable network debug trace of telemetry															
5	clearFlash	Clear file(s) from flash <table><tr><th>Value</th><th>Description</th></tr><tr><td>0</td><td>Do not clear flash</td></tr><tr><td>1</td><td>Clear flash. If a recording is in progress, it will be stopped.</td></tr><tr><td>2</td><td>If a file name is specified, only that file will be deleted.</td></tr></table>	Value	Description	0	Do not clear flash	1	Clear flash. If a recording is in progress, it will be stopped.	2	If a file name is specified, only that file will be deleted.						
Value	Description															
0	Do not clear flash															
1	Clear flash. If a recording is in progress, it will be stopped.															
2	If a file name is specified, only that file will be deleted.															
6	getStatus	<table><tr><th>Value</th><th>Description</th></tr><tr><td>0</td><td>Do not get status</td></tr><tr><td>1</td><td>Get the recording status, system will return SLACurrentSDCardRecordingStatus_t. NOTE: This can be used to check if an SD card is present.</td></tr></table>	Value	Description	0	Do not get status	1	Get the recording status, system will return SLACurrentSDCardRecordingStatus_t . NOTE: This can be used to check if an SD card is present.								
Value	Description															
0	Do not get status															
1	Get the recording status, system will return SLACurrentSDCardRecordingStatus_t . NOTE: This can be used to check if an SD card is present.															
7	getDirectory	Get file directory information. For non zero values if the lbl is a path starting with "/", reply will be for that directory. Otherwise, the directory implied by the last SetSnapShot will be used. (/media/mmcblk0p1 for SD card, /media/sdg1 for USB) <table><tr><th>Value</th><th>Description</th></tr><tr><td>0</td><td>Do not get directory information</td></tr><tr><td>1</td><td>Get directory information. See SLACurrentSDCardDirectoryInfo_t.</td></tr><tr><td>2</td><td>Get directory statistics. See SLADirectoryStatisticsReply_t.</td></tr></table>	Value	Description	0	Do not get directory information	1	Get directory information. See SLACurrentSDCardDirectoryInfo_t .	2	Get directory statistics. See SLADirectoryStatisticsReply_t .						
Value	Description															
0	Do not get directory information															
1	Get directory information. See SLACurrentSDCardDirectoryInfo_t .															
2	Get directory statistics. See SLADirectoryStatisticsReply_t .															
8-9	videoLength	Length of video clip at which to start a new video file specified in either kb or number of frames (see recordType); 0 = use default 1,048,576 kb/150 frames														
10	recordType	Recording type. NOTE: Specified as bits <table><tr><th>Bit</th><th>Description</th></tr><tr><td>0</td><td>H.264</td></tr><tr><td>1 - 3</td><td>Reserved</td></tr><tr><td>4</td><td>0 = use bit 6 for video length units, 1 = video length in MegaBytes.</td></tr><tr><td>5</td><td>0 = record multiple files of specified length, 1 = stop recording after 1 file of the specified length in KiloBytes or frames</td></tr><tr><td>6</td><td>0 = video length in KiloBytes, 1 = video length in frames</td></tr><tr><td>7</td><td>When using file auto-numbering, begin file numbering after highest-numbered existing filename</td></tr></table>	Bit	Description	0	H.264	1 - 3	Reserved	4	0 = use bit 6 for video length units, 1 = video length in MegaBytes.	5	0 = record multiple files of specified length, 1 = stop recording after 1 file of the specified length in KiloBytes or frames	6	0 = video length in KiloBytes, 1 = video length in frames	7	When using file auto-numbering, begin file numbering after highest-numbered existing filename
Bit	Description															
0	H.264															
1 - 3	Reserved															
4	0 = use bit 6 for video length units, 1 = video length in MegaBytes.															
5	0 = record multiple files of specified length, 1 = stop recording after 1 file of the specified length in KiloBytes or frames															
6	0 = video length in KiloBytes, 1 = video length in frames															
7	When using file auto-numbering, begin file numbering after highest-numbered existing filename															
11	displayId	<u>Network Display ID</u> (Does not follow normal displayID rules) 4000/17XX: (0 = Net0, 1 = Net1) 41XX: (0 = Net0, 1 = Net1, 2 = Net2)														
12	lbl.len	string length														

13-...	lbl.str	Filename or path name – when recording a video, file extension '.ts' is appended to the filename.
--------	---------	---

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentSDCardRecordingStatus_t Struct Reference

[Recording](#)

Description

Current SD card recording status – sent in response to [SLASetSDRecordingParameters_t](#) command.

```
typedef struct {
    u8 state;
    u32 length;
    u32 space;
    u8 displayId;
} SLACurrentSDCardRecordingStatus_t;
```

See also [SLADirectoryStatisticsReply_t](#).

Message ID 0x58

Byte Offset	Name	Description										
4	state	Recording state. NOTE: If bit 0 and bit 1 are both 0 it indicates that no SD card is present										
		<table><tr><th>Bit</th><th>Description</th></tr><tr><td>0</td><td>1 = Recording - see note above</td></tr><tr><td>1</td><td>1 = Stopped - see note above</td></tr><tr><td>2</td><td>1 = Network trace commands enabled</td></tr><tr><td>3</td><td>1 = Network trace telemetry enabled</td></tr></table>	Bit	Description	0	1 = Recording - see note above	1	1 = Stopped - see note above	2	1 = Network trace commands enabled	3	1 = Network trace telemetry enabled
		Bit	Description									
		0	1 = Recording - see note above									
		1	1 = Stopped - see note above									
		2	1 = Network trace commands enabled									
3	1 = Network trace telemetry enabled											
5-8	length	<i>Deprecated</i> - See SLADirectoryStatisticsReply_t .										
9-12	space	<i>Deprecated</i> - See SLADirectoryStatisticsReply_t .										
13	displayId	Network Display ID (<i>Does not follow normal displayID rules</i>) Only applies to recording state 4000/17XX: (0 = Net0, 1 = Net1) 41XX: (0 = Net0, 1 = Net1, 2 = Net2)										

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAResetAllParameters_t Struct

Reference

[List of all members](#)

[General](#)

Description

Performs one of several different system resets.

```
typedef struct {  
    u8 resetType;  
} SLAResetAllParameters_t;
```

See [SLASaveParameters_t](#) for information on persisting parameters.

Message ID 0x01

Byte Offset	Name	Description	
4	resetType	Value	Description
		0	Reset parameters to factory defaults
		1	Reset onboard video decoder
		2	Reboot board with saved parameters
		3	Deprecated
		4	Delete saved parameters and reboot board
		5	Send reset command to known attached camera (Tau, Sony, etc.)
		6	Reset parameters to factory defaults – does not change network serial port or camera type settings
		7	Sync the file system
		8-255	Reserved

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAAdvancedCaptureParameters_t

Struct Reference

[Capture](#)

Description

Set advanced analog video decoding parameters for the video decoder.

```
typedef struct {
    u16 horzControl;
    s16 xstart;
    s16 ystart;
} SLAAdvancedCaptureParameters_t;
```

For NTSC/PAL cameras that have significant amounts of black data on the left or right side or incorrect aspect ratio, setting these values will stretch the image horizontally. If the image is stretched, the pixels may no longer be square, and objects in the scene can appear wider or narrower.

This parameter is not saved in the parameters and must be set every time the system is started.
This parameter is per-camera and will be set for the command camera only.

See also [SLASetVideoParameters_t](#) as changes here can affect the need to remove black edge pixels with Region of Interest in [SLASetAcquisitionParameters_t](#).

This message is also sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x7B

Byte Offset	Name	Description
4-5	horzControl	Horizontal Scaling Control, scaled up by 10 bits. Used for horizontal rescale of NTSC/PAL images Value = 720 * 1024 / Desired_Pixels. Default = 0 for standard square pixel operation. Standard for NTSC square pixels : 720 * 1024 / 640 = 1152 Make the value slightly less to get rid of black edge on the right, but not quite square Standard for PAL square pixels : 720 * 1024 / 768 = 960
		Horizontal Pixel Start - adjust this value to move the starting left edge of the image. Multiple

6-7	xstart	of 2. Standard for square pixels : 0 Use a small value like 2 to 8 to remove black pixels on the left <i>Ignored if horzControl is 0</i>
8-9	ystart	<i>Reserved</i>

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAAncillaryTextMetadata_t Struct Reference

[KLV Metadata](#)

Description

Inject text into the KLV stream.

```
typedef struct {
    u64 creationTime;
    SVPLenString\_t source;
    SVPLenString\_t originator;
    SVPLenString\_t messageBody;
    u16 displayId;
} SLAAncillaryTextMetadata\_t;
```

Allow for the insertion of customer defined "text" (block of bytes) that then get sent into the appropriately formatted KLV block defined by MISB Standard (ST 0808) Ancillary Text Metadata. Message is sent with the next frame.

- messageBody(text) - a string of UTF-8 characters
 - 0x09 - 0x0D** Tab, Line Feed, Vertical Tab, Form Feed, Carriage Return
 - 0x20 - 0x7E** Space through tilde(all printable characters)
 - customer is expected to abide by above. content will not be validated for compliance.
- precisionTimeStamp is set automatically to ST 0601 TAG 2 (See [SLATagData_t](#) or [SLASetMetadataValues_t](#))

No [SLATagDataRate_t](#) available (yet). [SLATagSourceSelector_t](#) = VT.

Message ID 0xAC

Byte Offset	Name	Description
4-11	creationTime	UTC time time at which the ancillary text message was created
12	source.len	string length
13-...	source.str	type of the Originator (e.g. "human", "file", "email", etc.). max = 8 characters. If len is 0, packet is not sent.
varies	originator.len	string length
varies	originator.str	Text identifier for the creator of the message. up to 16 characters. [Optional] Set len to

		0.
varies	messageBody.len	string length
varies	messageBody.str	Free text UTF 8-Bit coded character set up to <i>TBD</i> characters
varies	displayId	<u>Network Display ID</u> 4000/17XX: (0x0002 = Net0, 0x0080 = Net1, 0x0082 = both) 41XX: (0x0002 = Net0, 0x0080 = Net1, 0x0010 = Net2, 0x0092 = all)

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

[List of all members](#)

SLAAppendedMetadata_t Struct

Reference

[KLV Metadata](#)

Description

Append user-specified binary metadata to KLV stream embedded in network video stream.

```
typedef struct {
    SVPLenString\_t data;
    u16 displayId;
} SLAAppendedMetadata\_t;
```

Also sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x89

Byte Offset	Name	Description
4	data.len	string length
5-...	data.str	Binary data to append
varies	displayId	Network Display ID 4000/17XX: (0x0002 = Net0, 0x0080 = Net1, 0x0082 = both) 41XX: (0x0002 = Net0, 0x0080 = Net1, 0x0010 = Net2, 0x0092 = all)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLABlendAlign_t Struct Reference

[Blending](#)

Description

Set the alignment set for a dual camera setup.

```
typedef struct {
    u8 index;
    s16 vertical;
    s16 horizontal;
    u16 rotate;
    u16 zoom;
    u16 hzoom;
} SLABlendAlign_t;
```

Message ID 0xB9

Byte Offset	Name	Description
4	index	Index of which alignment to change
5-6	vertical	Vertical offset in pixels.
7-8	horizontal	Horizontal offset
9-10	rotate	Rotate angle in degrees (0 to 360) * 128
11-12	zoom	Vertical and horizontal zoom. (0.01x to 15.99x) * 4096
13-14	hzoom	Just horizontal zoom. (0.01x to 15.99x) * 4096

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACameraCalibration_t Struct Reference

[List of all members](#)

[Enhancement](#)

Description

Set camera calibration parameters to fix distortion.

```
typedef struct {
    u8 cameraIndex;
    u16 distortionCenterRow;
    u16 distortionCenterCol;
    s16 distortionCorrection;
    s16 reserved;
} SLACameraCalibration_t;
```

Message ID 0xC0

Byte Offset	Name	Description
4	cameraIndex	Camera Index
5-6	distortionCenterRow	Pixel location of optical principal point. Default zero = center.
7-8	distortionCenterCol	Pixel location of optical principal point. Default zero = center.
9-10	distortionCorrection	Amount of barrel distortion to correct. Default zero = disabled.
11-12	reserved	Reserved for future use

Sightline Intelligence

Command, Control, and Script

API v3.11

[Main Page](#)
[Related Pages](#)
[Modules](#)
[Classes](#)
[Camera Configuration](#)

List of all members

SLACameraCapabilities_t Struct Reference

Capture

Description

Return a message with the camera capabilities.

```
typedef struct {
    u8 cameraIndex;
    u32 capabilities;
    u32 autoGainCapabilities;
} SLACameraCapabilities_t;
```

Only the camera index is necessary to get the capabilities. The capabilities values are ignored by Video Track in the request.

Message ID 0xBB

Byte Offset	Name	Description																						
4	cameraIndex	<u>Camera Index</u>																						
5-8	capabilities	Camera capabilities																						
		<table><tr><th>Bit</th><th>Option</th></tr><tr><td>4</td><td>Brightness</td></tr><tr><td>5</td><td>Contrast</td></tr><tr><td>6</td><td>Saturation</td></tr><tr><td>7</td><td>Hue</td></tr><tr><td>8</td><td>Analog Gain or Luma Target</td></tr><tr><td>9</td><td>Red</td></tr><tr><td>10</td><td>Green</td></tr><tr><td>11</td><td>Blue</td></tr><tr><td>12</td><td>Digital Gain</td></tr><tr><td>15</td><td>Exposure</td></tr></table>	Bit	Option	4	Brightness	5	Contrast	6	Saturation	7	Hue	8	Analog Gain or Luma Target	9	Red	10	Green	11	Blue	12	Digital Gain	15	Exposure
		Bit	Option																					
		4	Brightness																					
		5	Contrast																					
		6	Saturation																					
		7	Hue																					
		8	Analog Gain or Luma Target																					
		9	Red																					
		10	Green																					
		11	Blue																					
		12	Digital Gain																					
15	Exposure																							
		Camera Gain capabilities																						

9-12	autoGainCapabilities	Bit	Option
		0	SLA High Bit Depth Auto Gain
		1	Manual Gain
		2	SLA Auto Gain
		3	On-Camera Auto Gain

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) ■ [Camera Configuration](#)

[List of all members](#)

SLACameraSwitch_t Struct

Reference

[Capture](#)

Description

Deprecated, supported on 1500 and 3000 only

```
typedef struct {  
    u16 mode;  
    u16 cameraIndex;  
    u16 cameraType;  
} SLACameraSwitch_t;
```

Deprecated, supported on 1500 and 3000 only

Switch a digital camera on the fly - used for customer boards where they are muxing two or more digital cameras into a single digital video port using an FPGA.

When camera is switched using **mode** == 1, it uses predefined camera settings values.

If you need settings different from the predefined ones, send the [SLASetAcquisitionParameters_t](#) (0x37) command first, then send the Camera Switch command with **mode** == 2.

GPIO

When a camera is switched, the 3000 & 1500 also change GPIOs to indicate the **Camera Type** selected (see below).

Three GPIO pins are used to signal to the FPGA that the camera has changed. This is can be used by an external FPGA to select which a **Camera Type** to provide video from.

3000 - J2 (VOUT) connector GPIO are used to signal that the **Camera Type** has changed.

Vin0 (Cam0): J2P51 (GPIO 22), J2P50 (BPIO 17), J2P49 (GPIO 16)

Vin1 (Cam2): J2P54 (GPIO 28), J2P53 (GPIO 24), J2P52 (GPIO 23)

1500 J4 (Digital Connector) GPIO are used to signal that the **Camera Type** has changed.

Cam2: J4P13 (GPIO 174), J4P14 (GPIO 173), J4P40 (GPIO 172)

Laser Spot Detection

Up to 10 detected LaserSpots are highlighted in the captured image.

See also

[ICD-3000-OEM.PDF](#)

Message ID 0x82

Byte Offset	Name	Description																											
4-5	mode	Operations to perform:																											
		<table><tr><th>Value</th><th>Description</th></tr><tr><td>0</td><td>Restart the capture without switching the camera or changing camera settings.</td></tr><tr><td>1</td><td>Apply Camera Settings - Switch to the camera specified by Camera Type below.</td></tr><tr><td>2</td><td>Build Camera Table - Updates the camera settings using the values set by <u>SLASetAcquisitionParameters_t</u></td></tr><tr><td>3 - 8</td><td><i>Reserved</i></td></tr><tr><td>9</td><td>Enables LaserSpotDetect and switch to the camera specified by Camera Type (same as Mode 1)</td></tr><tr><td>10 - 15</td><td><i>Reserved</i></td></tr></table>	Value	Description	0	Restart the capture without switching the camera or changing camera settings.	1	Apply Camera Settings - Switch to the camera specified by Camera Type below.	2	Build Camera Table - Updates the camera settings using the values set by <u>SLASetAcquisitionParameters_t</u>	3 - 8	<i>Reserved</i>	9	Enables LaserSpotDetect and switch to the camera specified by Camera Type (same as Mode 1)	10 - 15	<i>Reserved</i>													
		Value	Description																										
		0	Restart the capture without switching the camera or changing camera settings.																										
		1	Apply Camera Settings - Switch to the camera specified by Camera Type below.																										
		2	Build Camera Table - Updates the camera settings using the values set by <u>SLASetAcquisitionParameters_t</u>																										
		3 - 8	<i>Reserved</i>																										
		9	Enables LaserSpotDetect and switch to the camera specified by Camera Type (same as Mode 1)																										
10 - 15	<i>Reserved</i>																												
6-7	cameraIndex	<u>Camera Index</u> 1500: applies to camera index 2. 3000 applies to camera index 0 or 2.																											
8-9	cameraType	<table><tr><th>Value</th><th>Bits</th><th>Camera Type</th></tr><tr><td>0</td><td>000</td><td>None</td></tr><tr><td>1</td><td>001</td><td>Custom 1 (default Hitachi)</td></tr><tr><td>2</td><td>010</td><td>Custom 2 (default Fury)</td></tr><tr><td>3</td><td>011</td><td>Custom 3 (default NightwarriorZoom)</td></tr><tr><td>4</td><td>100</td><td>Custom 4 (default NightwarriorTelescope)</td></tr><tr><td>5</td><td>101</td><td>Custom 5 (default Quantum)</td></tr><tr><td>6</td><td>110</td><td>Custom 6 (default Quantum Laser)</td></tr><tr><td>7</td><td>111</td><td><i>Reserved</i> / Big Snap</td></tr></table>	Value	Bits	Camera Type	0	000	None	1	001	Custom 1 (default Hitachi)	2	010	Custom 2 (default Fury)	3	011	Custom 3 (default NightwarriorZoom)	4	100	Custom 4 (default NightwarriorTelescope)	5	101	Custom 5 (default Quantum)	6	110	Custom 6 (default Quantum Laser)	7	111	<i>Reserved</i> / Big Snap
		Value	Bits	Camera Type																									
		0	000	None																									
		1	001	Custom 1 (default Hitachi)																									
		2	010	Custom 2 (default Fury)																									
		3	011	Custom 3 (default NightwarriorZoom)																									
		4	100	Custom 4 (default NightwarriorTelescope)																									
		5	101	Custom 5 (default Quantum)																									
6	110	Custom 6 (default Quantum Laser)																											
7	111	<i>Reserved</i> / Big Snap																											

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAClassifierParameters_t Struct

Reference

[Classification](#)

Description

Turn on/off the classifier.

```
typedef struct {
    u8 cameraIndex;
    u8 trackFlags;
    u8 detectionFlags;
    u8 maxClassPerFrame;
    u64 slaFilters;
    u64 customFilters;
    u16 minClassDim;
    u8 oodClassThresh;
    u64 subClassThreshA;
    u64 subClassThreshB;
    u64 subClassThreshC;
} SLAClassifierParameters_t;
```

Used to control under what circumstances the classifier runs. Filter controls for classifications and sub classifications

NOTE: 17XX and 4X00 Only

Message ID 0xA9

Byte Offset	Name	Description				
4	cameraIndex	<u>Camera Index</u>				
5	trackFlags	Tracking flags - Enables the classifier for object tracks based on the bits set. Results can be read using <u>SLATrackingPositionsExtended_t</u> . NOTE: If drone detection is enabled with classification through the <u>SLASetDetectionParameters_t</u> message that takes precedence over settings here.				
		<table><tr><th>Bits</th><th>Description</th></tr><tr><td></td><td>Classification Mode 0 = Disable classifier for all tracks </td></tr></table>	Bits	Description		Classification Mode 0 = Disable classifier for all tracks
		Bits	Description			
	Classification Mode 0 = Disable classifier for all tracks					

		<table><tr><td>0..3</td><td>1 = Run Classifier on Primary Track Only 2 = Run Classifier on All User Tracks 3 = Run Classifier on All Tracks - this includes ones started from Vehicle or Drone detection </td></tr><tr><td>4..7</td><td>*Reserved*</td></tr></table>	0..3	1 = Run Classifier on Primary Track Only 2 = Run Classifier on All User Tracks 3 = Run Classifier on All Tracks - this includes ones started from Vehicle or Drone detection	4..7	*Reserved*		
0..3	1 = Run Classifier on Primary Track Only 2 = Run Classifier on All User Tracks 3 = Run Classifier on All Tracks - this includes ones started from Vehicle or Drone detection							
4..7	*Reserved*							
6	detectionFlags	Detection flags - Enables the classifier for detection modes other than Drone, Vehicle, and Person. Results can be read using SLATrackingPositionsExtended_t. <table><tr><th>Bits</th><th>Description</th></tr><tr><td>0..3</td><td>Classification Mode 0 = Disable classifier for all tracks 1 = Run Classifier on all detections. 2 = Run Classifier on Selectable detections only. These are generally the 10 best detections, larger in size, and persisting longer. </td></tr><tr><td>4..7</td><td>*Reserved*</td></tr></table>	Bits	Description	0..3	Classification Mode 0 = Disable classifier for all tracks 1 = Run Classifier on all detections. 2 = Run Classifier on Selectable detections only. These are generally the 10 best detections, larger in size, and persisting longer.	4..7	*Reserved*
Bits	Description							
0..3	Classification Mode 0 = Disable classifier for all tracks 1 = Run Classifier on all detections. 2 = Run Classifier on Selectable detections only. These are generally the 10 best detections, larger in size, and persisting longer.							
4..7	*Reserved*							
7	maxClassPerFrame	Maximum number of objects to classify per frame when detection classification mode is not disabled. Valid range is 1-10. Deafult is 1.						
8-15	slaFilters	SightLine Classification Filters - Allows the classifier to filter out detections based on user preference. Setting the bit: The value of the label corresponds to the bit you need to set to filter out that label. You can find the full list of labels here. For example - to filter out background/other, you would set bit 0 (the least significant bit). Default: 0 - No Filtering. NOTE: Not used when Drone Classifier Filtering is enabled with SLASetDetectionParameters_t. NOTE: There are 100 options but only the first 64 are supported here.						
16-23	customFilters	Custom Classification Filters - Allows the classifier to filter out detections based on user preference. Setting the bit: The custom filters in the list of labels are referred to as "User Types" and are values 100-199. You can find the full list of labels here. (The value of the label - 100) corresponds to the bit you need to set to filter out that label. For example - to filter out the first user types label, you would set bit 0 (the least significant bit) because 100 - 100 = 0. Default: 0 - No Filtering. NOTE: There are 100 options but only the first 64 are supported here.						
24-25	minClassDim	Minimum classification chip dimensions before classifier will attempt to classify. 0 = no limit						
26	oodClassThresh	Out of distribution subclass model primary class threshold. If out of distribution filters are on, the primary class confidence from the classifier must exceed this threshold. Otherwise, it will be marked as Background.						

27-34	subClassThreshA	Sub classification threshold filtering. Up to 4 SightLine Classification IDs and classification confidence pairs (0 = no filter/disable). All non-zero conditions in the table below must be met to show the detection.		
		Bytes	Description	Valid Data
		0	SightLine Classification Filter ID 0. Full list of SightLine Classification IDs here	0-255
		1	Subclass Minimum Threshold Confidence for ID 0 (Percent)	0-100
		2	SightLine Classification Filter ID 1	0-255
		3	Subclass Minimum Threshold Confidence for ID 1	0-100
		4	SightLine Classification Filter ID 2	0-255
		5	Subclass Minimum Threshold Confidence for ID 2	0-100
		6	SightLine Classification Filter ID 3	0-255
		7	Subclass Minimum Threshold Confidence for ID 3	0-100
35-42	subClassThreshB	Sub classification threshold filtering, option B. Same as subClassThreshA, but another optional match for filtering objects. This option is OR'd with the other threshold options (A OR B OR C must be true to show detection).		
43-50	subClassThreshC	Sub classification threshold filtering, option C. Same as subClassThreshA/B, another optional match for filtering objects.		

Sightline Intelligence

Command, Control, and Script

API v3.11

[Main Page](#)
[Related Pages](#)
[Modules](#)
[Classes](#)

[Camera Configuration](#)

List of all members

SLACCommandPassThrough_t Struct Reference

Serial Port

Description

Output data payload to the port specified.

```
typedef struct {
    u8 destPort;
    SVPEndData_t data;
} SLACommandPassThrough_t;
```

Use **SLASetPortConfiguration_t** to setup the outbound physical ports.

NOTE: actual payload length can be anywhere between 1 and 80 bytes.

NOTE: set the length in the header = payload_length + 3

NOTE: no assumptions are made on terminating characters such as carriage return (0x0D), line feed (0x0A), or null(0x00).

NOTE: destPort's protocol needs to be configured something other than "Port not used (value 4)". If the port is configured as "Port not used", then the command is ignored.

Message ID 0x3D

Byte Offset	Name	Description	
4	destPort	Destination Port Identifier	
		Port Identifier	Description
		0	Serial port 0
		1	Serial port 1 (4000 only)
		2	Ethernet port (c2replyPort - see SLASetNetworkParameters_t)
		4	Serial port 2
		6	Serial port 3 (17XX/4000 only)
		7	Serial port 4 (17XX/4000 only)
		8	Serial port 5 (17XX only)
		9	Serial port 6 (17XX/4000 only)

		10	Serial port 7 (17XX/4000 only)
		12	Network port 14003 for external programs to do SVP with VideoTrack
		21	Serial port 8 (17XX only)
		22	Serial port 9 (1710 only)
		others	<i>Reserved</i>
5	data	Data buffer, length = header_length - 3	

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACoordinateReportingMode_t

Struct Reference

[Telemetry](#)

Description

Set reporting rates and telemetry types to send for telemetry packets.

```
typedef struct {
    u8 framePeriod;
    u16 flags;
    u8 cameraIndex;
} SLACoordinateReportingMode_t;
```

Also sent in response to [SLAGetParameters_t\(\)](#).
NOTE: This setting applies to the Command Camera and can be different for each camera.
NOTE: Telemetry output is not available in [Demo Mode](#).

Message ID 0x0B

Byte Offset	Name	Description														
4	framePeriod	0 = no telemetry reporting; 1 = report telemetry for every input frame; 2 = report telemetry every 2nd input frame; 3 = report every 3rd frame, ...														
		Telemetry types (DEFAULT 0 is identical to setting bits 0 and 1) <table><tr><th>Bit</th><th>Description</th></tr><tr><td>0</td><td>Enable <u>SLATrackingPosition_t</u></td></tr><tr><td>1</td><td>Enable <u>SLATrackingPositions_t</u> for primary track</td></tr><tr><td>2</td><td>Enable <u>SLATrackingPositions_t</u> for non-primary tracks</td></tr><tr><td>3</td><td>*Reserved* = 0</td></tr><tr><td>4</td><td>Enable <u>SLALandingPosition_t</u></td></tr><tr><td></td><td>Prioritize telemetry over video output. If this bit is on and telemetry is reported every</td></tr></table>	Bit	Description	0	Enable <u>SLATrackingPosition_t</u>	1	Enable <u>SLATrackingPositions_t</u> for primary track	2	Enable <u>SLATrackingPositions_t</u> for non-primary tracks	3	*Reserved* = 0	4	Enable <u>SLALandingPosition_t</u>		Prioritize telemetry over video output. If this bit is on and telemetry is reported every
Bit	Description															
0	Enable <u>SLATrackingPosition_t</u>															
1	Enable <u>SLATrackingPositions_t</u> for primary track															
2	Enable <u>SLATrackingPositions_t</u> for non-primary tracks															
3	*Reserved* = 0															
4	Enable <u>SLALandingPosition_t</u>															
	Prioritize telemetry over video output. If this bit is on and telemetry is reported every															

5-6	flags	5	frame, the system may skip displaying some frames so that it can maintain 30Hz telemetry output rate. NOTE: Telemetry frames can still occasionally skip even with this feature enabled with high processor load. NOTE: To get full rate telemetry with high processor load, the best practice is to increase display frame step in SLASetEthernetVideoParameters_t until telemetry is full rate. This additionally provides evenly spaced output frames and consistent system timing. NOTE: Not recommended for 17XX or 4000.
		6	Enable SLATrackingBoxPixelStats_t . Statistics area adjustable via SLASetTrackingParamters_t. Available for manual tracks as well as radiometric and blob (NEW 3.7) detections. Limited to 10 tracks.
		7	Enable SLAFocusStats_t
		8	Enable SLATrackingPositionsExtended_t NOTE: SLATrackingPositions_t must also be enabled
		9	Enable SLATrackingMultiClass_t NOTE: SLATrackingPositions_t must also be enabled
		10..14	<i>Reserved</i>
		15	Include auxiliary data, additional 12 bytes in telemetry packets comprised of: 1. User managed frame index (does not automatically increment). See SLAFrameIndex_t. 2. Frame time-stamp. Unsigned 64 bit value. System time at which this frame was captured, in micro-second resolution.
7	cameraIndex	<u>Camera Index</u> 255 = All	

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACreateDeviceOverlay_t Struct Reference

[Capture](#)

Description

Send a command to generate and compile a devicetree overlay for a device.

```
typedef struct {
    u8 index;
    u8 settle;
    u8 clk;
    u8 lanes;
    SVPLenString\_t driverName;
} SLACreateDeviceOverlay\_t;
```

Note

Only supported on 1750.

Message ID 0xB8

Byte Offset	Name	Description
4	index	Camera Index Value representing the camera index. Can be 0 or 1.
5	settle	A number representing the settle count (hs-settle)
6	clk	A number representing the clk_settle value
7	lanes	Value representing the number of data-lanes
8	driverName.len	string length
9-...	driverName.str	Optional: Drivename field. Used when associating the overlay with a driver. (Defaults to generic driver)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentAcquisitionParameters_t

Struct Reference

[Capture](#)

Description

Sent in response to [SLAGetParameters_t\(\)](#) – see [SLASetAcquisitionParameters_t](#) for additional notes and details on the camera types.

```
typedef struct {
    u8 numCamera;
    u8 cameraIndex;
    u8 cameraType;
    u16 high;
    u16 wide;
    u8 bitDepth;
    u16 vertFrontPorch;
    u16 horzFrontPorch;
    u16 flags;
    u8 frameStep;
    u8 reserved;
    u16 validRow;
    u16 validCol;
    u16 validHigh;
    u16 validWide;
    SVPLenString\_t optArgs;
    u16 bigHigh;
    u16 bigWide;
    u16 bigVertFrontPorch;
    u16 bigHorzFrontPorch;
    u8 camSwitchIndex;
    SVPLenString\_t cameraName;
} SLACurrentAcquisitionParameters\_t;
```

Message ID 0x4F

Byte Offset	Name	Description
4	numCamera	Number of configured cameras
5	cameraIndex	Camera Index
6	cameraType	Camera type see SLASetAcquisitionParameters_t cameraType
7-8	high	Image height (0 for default)
9-10	wide	Image width (0 for default)

11	bitDepth	Bit depth (0 for default)
12-13	vertFrontPorch	Vertical front porch in lines (0 for default)
14-15	horzFrontPorch	Horizontal front porch in pixels (0 for default)
16-17	flags	Digital Video Flags – see SLASetAcquisitionParameters_t.flags
18	frameStep	Step between frames – 0 = ignore, 1 = every frame, 2 = every other frame
19	reserved	<i>Reserved</i>
20-21	validRow	Valid ROI row
22-23	validCol	Valid ROI column
24-25	validHigh	Valid ROI height
26-27	validWide	Valid ROI width
28	optArgs.len	string length
29-...	optArgs.str	Optional arguments for camera script. The first word should be the name of the camera.
varies	bigHigh	JPEG Big Snapshot High
varies	bigWide	JPEG Big snapshot Wide
varies	bigVertFrontPorch	JPEG Big Vertical Front Porch in lines
varies	bigHorzFrontPorch	JPEG Big Horizontal Front Porch in pixels
varies	camSwitchIndex	For Duplicate camera type, used to indicate which camera is being duplicated.
varies	cameraName.len	string length
varies	cameraName.str	Optional: The name of the camera used. Used to keep track of the camera these settings are for, since most cameras have a 'generic digital' type now.

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) [Camera Configuration](#)

[List of all members](#)

SLASetAcquisitionParameters_t

Struct Reference

[Capture](#)

Description

Configure video input for digital camera interfaces.

```
typedef struct {
    u8 cameraIndex;
    u8 cameraType;
    u16 high;
    u16 wide;
    u8 bitDepth;
    u16 vertFrontPorch;
    u16 horzFrontPorch;
    u16 flags;
    u8 frameStep;
    u8 reserved;
    u16 validRow;
    u16 validCol;
    u16 validHigh;
    u16 validWide;
    SVPLenString\_t optArgs;
    u16 bigHigh;
    u16 bigWide;
    u16 bigVertFrontPorch;
    u16 bigHorzFrontPorch;
    u8 camSwitchIndex;
    SVPLenString\_t cameraName;
    u8 portID;
    u8 deviceID;
} SLASetAcquisitionParameters\_t;
```

See [EAN-Digital-Video-Configuration.pdf](#) for more information.

See also [camera configuration table](#) for detailed parameter settings.

See also [SLAAdvancedCaptureParameters_t](#).

NOTE: For changes to the acquisition settings to take effect, a board reboot may be required by issuing [SLASaveParameters_t](#) (0x25) and [SLAResetAllParameters_t](#) (0x01) (Reset Type = 2, reboot) messages.

Camera initialization routine

Assuming the appropriate adapter board or communication signals are connected, the VideoTrack software will try to configure the camera with the specified values or other known settings. See also [SLACommandPassThrough_t](#).

Camera Type

Type ID	Description	Hardware Support
0	None	No camera
1	<i>Reserved</i>	
2	Generic Digital	Generic Digital Interface. Uses high , wide , bitDepth , vertFrontPorch , horzFrontPorch , flags , and optArgs Automatic fill parameters are available in Panel+ and camera configuration table .
3 - 34	<i>Reserved</i>	
35	IP Camera	Receives IP video camera input. See EAN-Decoder.pdf and SLADecoderParameters_t for more information.
36	USB3 Vision	Cameras compatible with the USB3 Vision Standard Specification .
37	USB Webcam	USB Video Device Class (UVC) compatible webcams. Use optArgs to set internal format and frame rate settings.
39	Duplicate	Create a duplicate processing stream for a camera. Usually used to create two streams of the same camera, one with and one without enhancement/graphics or at two different resolutions Use camSwitchIndex to set which camera is being duplicated. Available with Dual Application Bit NOTE: The duplicated camera must be processed for the duplicate camera to be processed or displayed
132	Test Pattern Logo	SightLine Logo. NOTE: Test patterns may take significant time to render, affecting system performance.
136	Test Pattern Car	Single moving car with jittering and moving background
137	Test Pattern Still	Stationary background
141	Test Pattern Multi Car	Multiple moving cars with a slowly moving background
143	Test Pattern Landing Aid	Landing aid pattern on a stationary background
145	Test Pattern Color Bar	Color Bar
146	Test Pattern Low Contrast Boxes	Varying intensity boxes

Digital Video Flags

Bits	Description
	Input data mode

0 - 2	0	Grayscale
	1	YUV color
	2	8 bit grayscale, but input as 16 bit
	3	Bayer color, normally in RGGB order. BGGR and GBRG are also supported for USB3Vision
	4	15 bit grayscale plus 1 bit "laser" indicator in the msb
	5 - 7	*Reserved*
3	0 = progressive(default), 1 = interlaced	
4	Vertical Sync Polarity(0 = negative, 1 = positive for Sony 7500, etc.) (default 0)	
5	Horizontal Sync Polarity(0 = negative, 1 = positive for Sony 7500, etc.) (default 0)	
6 - 9	Camera Initialization Routine. Called with parameters specified above in this message.	
	0	No Initialization Routine (default)
	1	InitVISCA / Sony. See EAN-Sony-Block-Cameras.pdf
	2	InitDRS. See EAN-DRS-Cameras.pdf .
	3-4	<i>Reserved</i>
	5	InitCameraLinkLowSpeed
	6	InitTau. See EAN-FLIR-Cameras.pdf
	7-8	<i>Reserved</i>
	9	InitCameraLink
	10	InitBoson. See EAN-FLIR-Cameras.pdf
	11	<i>Reserved</i>
	12	InitFPC. Initialize the FPC serial port for user control of Boson, Tau, DRS FPC cameras.
10 - 11	Sync / Crop Option(17XX/4000)	
	0	None (default)
	1	use Data Valid signal
	2	use Embedded Sync
	3	use Hardware Crop
12	<i>Reserved</i>	
13	2x Hardware Binning	
14	Byte Swap	
15	Always acquire digital camera. Useful for faster network switching and display.	

NTSC Image Size: The analog video decoder hardware captures non-"square pixel" 720 wide by 480 high digital images. **Capture** snapshots are this 720x480 size and objects will appear slightly stretched horizontally.

The captured images are resampled horizontally into 640x480 images before Analyze, Render and Display. This creates a "square pixel" image. Objects will not appear stretched.

Network video output and **Display** snapshots are 640x480 and all other processing (such as tracking) is based on the 640x480 image size.

It is possible (though not usually recommended) to modify the horizontal resampling with

SLAAdvancedCaptureParameters_t.

PAL Image Size: Similar, but the captured images are 720 wide by 576 pixels high which get resampled to 768 pixels wide by 576 pixels high "square pixel" images.

Message ID 0x37

Byte Offset	Name	Description										
4	cameraIndex	<u>Camera Index</u>										
5	cameraType	<u>Camera Type</u>										
6-7	high	Image Height (0 for default). NOTE: Must be a multiple of 8										
8-9	wide	Image Width (0 for default). NOTE: Must be a multiple of 16, except in some 10-bit input cases where it must be a multiple of 8.										
10	bitDepth	Bit Depth (0 for default, 8 to 16) If the camera has more than 8, but less than 16 active bits, it is ok to set it to 15 or less.										
11-12	vertFrontPorch	Vertical Front Porch in lines (0 for default)										
13-14	horzFrontPorch	Horizontal Front Porch in pixels (0 for default)										
15-16	flags	<u>Digital Video Flags</u>										
17	frameStep	Frame Step – step between captured frames; 0 = ignore, 2 = every other (default 1)										
18	reserved	<i>Reserved</i>										
19-20	validRow	Acquisition ROI row The Acquisition Region of Interest (ROI) specifies a dynamically changeable subregion of the complete acquired image frame. Use it to process a smaller sub-region to speed processing or to remove blank or bad rows or columns from the captured image. NOTE: ROI parameters take effect immediately, and do not require a system restart. NOTE For some image types, validRow must be a multiple of 2 (eg. Bayer).										
21-22	validCol	Acquisition ROI column NOTE For some image types, validCol must be a multiple of 2 or 4 (eg. Bayer).										
23-24	validHigh	Acquisition ROI height - Minimum 120, Maximum 1080 (2160 on 4000) (0 - use image size). NOTE: Must be a multiple of 8										
25-26	validWide	Acquisition ROI width - Minimum 160, Maximum 1920 (3840 on 4000) (0 - use image size). NOTE: Must be a multiple of 8										
27	optArgs.len	string length										
		Comma separated optional arguments for camera setup. Specify with name1=value1[, name2=value2]. Example: fmt=UYVY, fps=60										
		<table><tr><th>Platform</th><th>Camera</th><th>Name</th><th>Description</th><th>Example</th></tr><tr><td></td><td>USB</td><td></td><td>Frame rate in frames per</td><td></td></tr></table>	Platform	Camera	Name	Description	Example		USB		Frame rate in frames per	
Platform	Camera	Name	Description	Example								
	USB		Frame rate in frames per									

17XX/4000	Webcam (UVC)	fps	second. System will choose the closest supported frame rate.	fps=30
17XX/4000	USB Webcam (UVC)	fmt	Image format: MPEG2, MPEG4, H264, H265, MJPEG, NV12, YUYV, UYVY, YUV420 NOTE: The system will look for a format in the above order and will not capture if given an invalid format List formats with: v4l2-ctl -l list-formats-ext -d /dev/videoN (N is 0 to 10)	fmt=MJPEG
17XX/4000	USB3 Vision Camera	fps	Frame rate in frames per second. If the specified fps is invalid, the camera will default to 30. NOTE: Specifying a fps higher than 30 may result in no video for some cameras (such as the Basler Dart), especially when combined with low exposure time.	fps=60
17XX/4000	Mipi Board Camera	mipi	Cam1 only mipi driver name. Options: ar0234 - E-con eCAM24 imx219 - Raspberry Pi Camera Module V2.0 imx412 - VC MIPI IMX412 imx412_4lane_v2 - MIPI IMX412 in FLIR Hadron 320 ov5647 - Raspberry Pi Camera Module V1.0	mipi=imx412
17XX/4000	Sony FCB-ER 8000 series	mipi	Cam1 only mipi driver name. Options: lt6911 - Sony FCB-ER 8000 series	mipi=lt6911
17XX/4000	Camera AB	cam	MIPI-AB/ADV7280 Chip only. Selecting Camera a or b	cam=a
	Sony FCB-ER		Frame rate in frames per second. System will choose the closest supported frame rate. Valid options:	

28-...	optArgs.str	17XX/4000	8000 series	fps	720: 50, 60 (default 60) 1080: 24, 25, 30, 50, 60 (default 30) 2160p: 24, 25, 30 (default 24)	fps=30
		17XX/4000	HDMI	fps	Frame rate in frames per second. For 1080p, can be set to 60 (default 30)	fps=60
		17XX/4000	Visca-Compliant cameras	vendor	Override the vendor type obtained from the camera. A supported vendor type that is similar in setup needs to the current camera. Use when current camera is not from a supported vendor. Valid options: "sony", "tamron", "savgood", "KTC".	vendor=sony
		17XX/4000	Visca-Compliant cameras	model	Used with option vendor=sony. Override the model type obtained from the camera. The model should be a supported model type that is similar in setup needs to the current camera. Use when current camera is not from a supported vendor or sony model. Valid options: "ER85XX", "EW9500H", "EV7520A".	model=ER85XX
		17XX/4000	Visca-Compliant cameras	mm	Override the monitoring mode to use to set the resolution of the camera. Defaults are normally calculated based on vendor, model, fps, or provided resolution.	mm=0x0006
		17XX/4000	Visca-Compliant cameras	lvds	Override the default behavior of cameras enabling single-channel LVDS. Specifying default prevents VideoTrack from explicitly sending a command to enable LVDS single channel. Valid option: "single", "default".	lvds=single
		17XX/4000	Advanced	mipi_txl	Mipi Transmit Lanes 1,2,4	mipi_txl=4

					Camera pixel clock rate Mhz This affects both the MIPI CSI frequency and the TSettle count <i>pixel_clk</i> is now the preferred name. For backward compatibility, both are supported during this transition period. See EAN-4000-OEM-MIPI-Cameras	pixel_clk=200
			17XX/4000	Advanced	mipi_clk or pixel_clk (new) <i>pixel_clk is preferred; both are supported</i>	
			17XX/4000	Advanced	fpga	prefix of fpga files (.iea,.ied) to load fpga=GEN_HD
			41xx	Mipi Board Camera	csi	Override the default CSI (Camera Serial Interface) assignment for the camera. Use when the camera is connected to a non-default CSI port or when remapping is required for custom hardware configurations. Allows the user to specify which MIPI CSI camera interface will correspond with a customer Cam#. Used when upgrading from older platforms or for custom OEM boards to keep user camera numbering consistent. <i>Currently limited to MIPI Cams using the Default MIPI Driver.</i> Default Mapping: Cam0 - CSI2 Cam1 - CSI0 Cam2 - CSI1 (4110 only) csi=0
						Specifies which USB camera index to use. Identifies the USB camera position when one or more USB cameras are connected. Use when assigning a USB camera to a specific Cam# during PanelPlus configuration. <i><First detected USB camera is assigned index 0</i>

		41xx	USB Camera	usb	Use to remap which USB interface will be mapped to a user Cam# - for multiple USB cameras, the Sightline processor will determine which USB camera will be index=0 or 1 - this will remain consistent across power cycles. <i>Note:</i> if a USB camera is assigned to say Cam1, then CSI0 will be unavailable for mapping to other Cam#	usb=0									
		Specific Camera Examples:													
		<table><tr><th>Camera</th><th>Example</th><th>Description</th></tr><tr><td>e-con See3Cam_CU30</td><td>fmt=UYVY, fps=30</td><td>Send raw (at 30 fps) instead of default MJPEG (at ~21 fps)</td></tr><tr><td>Raspberry-Pi V2</td><td>mipi=imx219</td><td>Raspberry-Pi V2 uses the imx219 driver</td></tr></table>					Camera	Example	Description	e-con See3Cam_CU30	fmt=UYVY, fps=30	Send raw (at 30 fps) instead of default MJPEG (at ~21 fps)	Raspberry-Pi V2	mipi=imx219	Raspberry-Pi V2 uses the imx219 driver
Camera	Example	Description													
e-con See3Cam_CU30	fmt=UYVY, fps=30	Send raw (at 30 fps) instead of default MJPEG (at ~21 fps)													
Raspberry-Pi V2	mipi=imx219	Raspberry-Pi V2 uses the imx219 driver													
		NOTE: Users may use the keyword user for their own purposes of saving a string per camera. For example: user=myCam18													
varies	bigHigh	JPEG Big Snapshot High													
varies	bigWide	JPEG Big snapshot Wide													
varies	bigVertFrontPorch	JPEG Big Vertical Front Porch in lines													
varies	bigHorzFrontPorch	JPEG Big Horizontal Front Porch in pixels													
varies	camSwitchIndex	For Duplicate camera type, used to indicate which camera is being duplicated.													
varies	cameraName.len	string length													
varies	cameraName.str	Optional: The name of the camera used. Used to keep track of the camera these settings are for, since most cameras have a 'generic digital' type now.													
varies	portID	Reserved for future use.													
varies	deviceID	Reserved for future use.													

Sightline Intelligence

Command, Control, and Script

API v3.11

[List of all members](#)

SLACurrentADCParameters_t Struct Reference

[Capture](#)

Description

Current ADC parameters – format identical to [SLASetADCParameters_t](#).

```
typedef struct {
    u8 brightness;
    u8 contrast;
    u8 saturation;
    u8 hue;
    u8 luma1;
    u8 luma2;
    u8 luma3;
    u8 chroma1;
    u8 chroma2;
    u8 mode;
    u8 cameraIndex;
    u8 exposure;
    u8 exposureFrac;
    u16 setMask;
    u8 whiteBalanceCount;
    u8 whiteBalanceExclusion;
} SLACurrentADCParameters\_t;
```

Sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x47

Byte Offset	Name	Description
4	brightness	See SLASetADCParameters_t
5	contrast	
6	saturation	
7	hue	
8	luma1	
9	luma2	
10	luma3	
11	chroma1	

12	chroma2	
13	mode	
14	cameraIndex	
15	exposure	
16	exposureFrac	
17-18	setMask	
19	whiteBalanceCount	
20	whiteBalanceExclusion	

Firmware Version: 3.11

SLASetADCParameters_t Struct Reference

Capture

Description

Set parameters of the video analog-to-digital converter for analog inputs and color parameters for color digital cameras.

```
typedef struct {
    u8 brightness;
    u8 contrast;
    u8 saturation;
    u8 hue;
    u8 luma1;
    u8 luma2;
    u8 luma3;
    u8 chroma1;
    u8 chroma2;
    u8 mode;
    u8 cameraIndex;
    u8 exposure;
    u8 exposureFrac;
    u16 setMask;
    u8 whiteBalanceCount;
    u8 whiteBalanceExclusion;
} SLASetADCParameters_t;
```

A lower noise image is achieved by increasing exposure time first then increasing analog gain before increasing digital gain to make a bright enough image. Long exposures may result in frame blur for high speed camera or scene motion, so you may need to reduce the exposure in some cases.

See also

- [SLASetDigitalCameraParameters_t](#) for auto gain control which uses a few of these parameters.
- [SLASetVideoEnhancementParameters_t](#) for enhancement modes that use a few of these parameters in some cases.

Sensor Controls and Gain Modes

ADC parameter mapping to on chip and auto-gain controls.

Type	Camera/Sensor	brightness	contrast	saturation	hue	luma1	luma2	luma3	chroma1	chroma2	exposure	Auto Gain Mode			
												High Depth	Manual	SightLine	Camera
Analog ₁	NTSC/PAL	Brightness ₁	Contrast ₁	Saturation ₁	Hue ₁	N	N	N	N	N	N	N	Manual	N	N
HDMI ₁	Many	Brightness ₁	Contrast ₁	Saturation ₁	Hue _{1,7}	N	N	N	N	N	N	N	Manual	N	N
	VC-IMX412	Brightness _{3,5}	Contrast ₃	N	N	Analog Gain _{3,4}	Red _{3,4,5}	Green _{3,4,5}	Blue _{3,4,5}	Digital Gain _{3,4}	Exposure _{3,4}	10 Bit AGC	Manual	SLA AGC	N
	RPiV1 OV5647	Brightness ₅	N	N	N	Analog Gain ₄	N	N	N	N	Exposure ₄	N	Manual	SLA AGC	N
	RPiV2 IMX219	Brightness _{3,5}	Contrast ₃	N	N	Analog Gain	Red ₃	Green ₃	Blue ₃	Digital Gain	Exposure _{3,4}	10 Bit	Manual	SLA AGC	N

MIPI						3,4				3,4		AGC			
	e-cam24 AR0234	Brightness _{4,5,6}	Contrast _{4,5,6}	Saturation _{4,5,6}	N	Sharpness _{4,6}	N	N	N	Gain ₄	Exposure ₄	N	Manual	SLA AGC	Cam AGC
	VeyeEye IMX385	Brightness	Contrast	Saturation	N	N	Red	N	Blue	N	N	N	Manual	N	N
	IMX900	Brightness _{3,5}	Contrast ₃	N	N	Analog Gain _{3,4}	Red _{3,4,5}	Green _{3,4,5}	Blue _{3,4,5}	Digital Gain _{3,4}	Exposure _{3,4}	10 Bit AGC	Manual	SLA AGC	N
	Hadron 640R	Brightness _{3,5}	Contrast ₃	N	N	Analog Gain _{3,4}	Red _{3,4,5}	Green _{3,4,5}	Blue _{3,4,5}	Digital Gain _{3,4}	N	10 Bit AGC	Manual	SLA AGC	N
USB	USB Webcam(UVC) ₁	Brightness ₁	Contrast ₁	Saturation ₁	N	Sharpness ₁	N	N	N	N	N	N	Manual	N	N
	USB3 Vision	Brightness ₅	N	N	N	N	N	N	N	Gain ₄	Exposure ₄	N	Manual	SLA AGC	N
High Bit Depth CL, HD-SDI, Boson, etc. ₂		Brightness _{3,4}	Contrast _{3,4}	N	N	N	N	N	N	N	N	High Bit AGC	Manual	N	N
8 bit CL, HD-SDI, Sony, etc. ₂		N	N	N	N	N	N	N	N	N	N	N	Manual	N	N

1. HDMI, Analog, and USB Webcam(UVC) ranges and default are returned from the driver for that sensor and mapped to input values between 0 (minimum) to 255 (maximum).
2. Other cameras (including Camera Link, HD-SDI, Hitachi, Sony, Tau, Boson, DRS) do not set these parameters on the sensor.
3. Available in High Bit Depth Auto Gain. See [SLASetDigitalCameraParameters_t](#).
4. Available in manual gain mode. See [SLASetDigitalCameraParameters_t](#).
5. Available in SightLine Auto Gain mode. See [SLASetDigitalCameraParameters_t](#).
6. Available in Camera Auto Gain mode.
7. HDMI Hue range is 0 to 127.

Message ID 0x18

Byte Offset	Name	Description
4	brightness	0 = dark, 128 = default, 255 = bright - see Sensor Controls and SLASetDigitalCameraParameters_t In High Bit Depth Auto Gain and SightLine Auto Gain, this acts as a target for image brightness used by the auto gain algorithm. (see Sensor Controls for camera support), controls the SLA auto gain In manual gain mode, this value is set on the sensor if supported, see Sensor Controls
5	contrast	0 = minimum contrast, 128 = default, 255 = maximum - see Sensor Controls and SLASetDigitalCameraParameters_t < In High Bit Depth Auto Gain mode, controls the auto gain algorithm In manual gain mode, this value is set on the sensor if supported, see Sensor Controls
6	saturation	Color Saturation. 0 = no color, 128 = default, 255= maximum - see Sensor Controls For High Bit Depth Cameras: When the first bit of highBitDepthFlags in SetDigitalCameraParameters is set, this value changes the max percent of saturated/ undersaturated pixels. The total % is (saturation / (256 *10)). Default = 128 (max 5% of pixels will be mapped to pure black or white). This value is set on the sensor if supported, see Sensor Controls Also used in enhancement modes, see SLASetVideoEnhancementParameters_t
7	hue	Color Hue control - see Sensor Controls This value is set on the sensor if supported, see Sensor Controls
		Analog Gain or Sharpness - see Sensor Controls

8	luma1	In SightLine Auto Gain mode (see Sensor Controls for camera support), this value is controlled by SLA auto gain In Manual Gain mode, this value is set on the sensor if supported, see Sensor Controls																																							
9	luma2	Red Gain - see Sensor Controls																																							
10	luma3	Green Gain - see Sensor Controls																																							
11	chroma1	Blue Gain - see Sensor Controls																																							
12	chroma2	Digital Gain - see Sensor Controls In SightLine Auto Gain mode (see Sensor Controls for camera support), this value is controlled by SLA auto gain In Manual Gain mode, this value is set on the sensor if supported, see Sensor Controls																																							
13	mode	Reset Mode 0: Normal setting of ADC parameters 1: Reset ADC parameters to camera defaults																																							
14	cameraIndex	Camera Index																																							
15	exposure	Exposure or integration time. 0 = minimum to 255 = maximum, default 128. New 3.5 (previously used saturation) - see Sensor Controls IMX412: Exposure in image lines = 8 + 16*exposure. New 3.5 (previously 8 + 8*exposure) IMX219: Exposure in image lines = 8 + 6.5*exposure OV5647: Exposure time = 8 + 48*exposure USB3Vision: Exposure time = map between minimum and maximum as defined by the sensor In SightLine Auto Gain mode (see Sensor Controls for camera support), this value is controlled by SLA auto gain In Manual Gain mode, this value is set on the sensor if supported, see Sensor Controls																																							
16	exposureFrac	Fractional Exposure times 256. totalExposure = exposure + (exposureFrac/256.0) New 3.5.6																																							
17-18	setMask	Mask indicating which parameters in this message to set New 3.5.6 To set a limited number of values, pass only the bits for the values to be set (OR these together), 0 or 0xFFFF to set all Purpose is to prevent setting exposure, etc when changing some other value like brightness while in auto gain mode <table border="1"> <thead> <tr> <th>Bit</th><th>Value</th><th>Name see</th></tr> </thead> <tbody> <tr> <td>all</td><td>0x0000</td><td>set all parameters (legacy behavior)</td></tr> <tr> <td>4</td><td>0x0010</td><td>brightness</td></tr> <tr> <td>5</td><td>0x0020</td><td>contrast</td></tr> <tr> <td>6</td><td>0x0040</td><td>saturation</td></tr> <tr> <td>7</td><td>0x0080</td><td>hue</td></tr> <tr> <td>8</td><td>0x0100</td><td>luma1 / analog gain / luma target / sharpness</td></tr> <tr> <td>9</td><td>0x0200</td><td>luma2 / red gain</td></tr> <tr> <td>10</td><td>0x0400</td><td>luma3 / green gain</td></tr> <tr> <td>11</td><td>0x0800</td><td>chroma1 / blue gain</td></tr> <tr> <td>12</td><td>0x1000</td><td>chroma2 / digital gain</td></tr> <tr> <td>15</td><td>0x8000</td><td>exposure</td></tr> <tr> <td>all</td><td>0xFFFF</td><td>set all parameters</td></tr> </tbody> </table>	Bit	Value	Name see	all	0x0000	set all parameters (legacy behavior)	4	0x0010	brightness	5	0x0020	contrast	6	0x0040	saturation	7	0x0080	hue	8	0x0100	luma1 / analog gain / luma target / sharpness	9	0x0200	luma2 / red gain	10	0x0400	luma3 / green gain	11	0x0800	chroma1 / blue gain	12	0x1000	chroma2 / digital gain	15	0x8000	exposure	all	0xFFFF	set all parameters
Bit	Value	Name see																																							
all	0x0000	set all parameters (legacy behavior)																																							
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6	0x0040	saturation																																							
7	0x0080	hue																																							
8	0x0100	luma1 / analog gain / luma target / sharpness																																							
9	0x0200	luma2 / red gain																																							
10	0x0400	luma3 / green gain																																							
11	0x0800	chroma1 / blue gain																																							
12	0x1000	chroma2 / digital gain																																							
15	0x8000	exposure																																							
all	0xFFFF	set all parameters																																							
19	whiteBalanceCount	0 = Disabled (Default), 255 = Continuous White Balance, Otherwise do this number of white balance updates New 3.8 If enabled and in High Bit Depth Auto Gain mode or (for supported cameras) SightLine Auto Gain or Manual Gain mode, colors will be automatically balanced using the Gray World algorithm. Note: Midpoint will not function while white balance is enabled. (See SLASetDigitalCameraParameters_t for Midpoint).																																							

		Note: If the image has a large portion filled by a single color (like a blue sky or ocean), the white balance algorithm may not perform well.
20	whiteBalanceExclusion	0 = 40% of the darkest and brightest pixels are excluded from the calculation, 128 = 20% (Default), 255 = 0% New 3.8 If Whitebalance is enabled and in High Bit Depth Auto Gain, excludes a proportion of the darkest and brightest pixels from the white balance calculation. This helps if the image contains a large quantity of under or over exposed pixels.

Sightline Intelligence

Command, Control, and Script

API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) [Camera Configuration](#)

[List of all members](#)

SLACurrentAdvancedDetectionParameters_t Struct Reference

[Detection](#)

Description

Return the current advanced detection parameters – format identical to [SLASetAdvancedDetectionParameters_t](#).

```
typedef struct {
    s16 minVel8;
    s16 maxVel8;
    s16 reserved1;
    s16 minWide;
    s16 maxWide;
    s16 minHigh;
    s16 maxHigh;
    u8 hideOverlapTrks;
    u8 nFramesBack;
    u8 mergeRadius;
    u8 mergeDirTol;
    u8 useRegistration;
    u8 detUpdateRate;
    u8 surroundSize;
    u8 detIdx;
    u16 bgTimeConst;
    u8 bgEdgePenalty6;
    u8 bgResetConf;
    u8 bgResetOff;
    u8 bgResetAng;
    u8 bgResetFrames;
    u8 bgWarpConf;
    u8 bgWarpOff;
    u8 bgWarpAng;
    u8 bgWarpFrames;
    u8 maxTrackFrames;
    u8 debugFiltering;
    u8 downsample;
    u8 maxTelemTrks;
    u8 maxKlvTrks;
    u8 cameraIndex;
    u8 blocksHigh;
    u8 blocksWide;
    u8 maxDisplayTrks;
    u8 ellipsePadding;
    u8 closeFilter;
    u8 detDirection;
    u8 detBitDepth;
    u8 aiDetectIOU;
    u8 detNearTrk;
    u8 frameDepth;
} SLACurrentAdvancedDetectionParameters\_t;
```

Sent in response to [SLAGetParameters_t\(\)](#).

Byte Offset	Name	Description
4-5	minVel8	See SLASetAdvancedDetectionParameters_t
6-7	maxVel8	
8-9	reserved1	
10-11	minWide	
12-13	maxWide	
14-15	minHigh	
16-17	maxHigh	
18	hideOverlapTrks	
19	nFramesBack	
20	mergeRadius	
21	mergeDirTol	
22	useRegistration	
23	detUpdateRate	
24	surroundSize	
25	detIdx	
26-27	bgTimeConst	
28	bgEdgePenalty6	
29	bgResetConf	
30	bgResetOff	
31	bgResetAng	
32	bgResetFrames	
33	bgWarpConf	
34	bgWarpOff	
35	bgWarpAng	
36	bgWarpFrames	
37	maxTrackFrames	
38	debugFiltering	
39	downsample	
40	maxTelemTrks	
41	maxKlvTrks	
42	cameraIndex	
43	blocksHigh	
44	blocksWide	

45	maxDisplayTrks	
46	ellipsePadding	
47	closeFilter	
48	detDirection	
49	detBitDepth	
50	aiDetectIOU	
51	detNearTrk	
52	frameDepth	

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) [Camera Configuration](#)

[List of all members](#)

SLASetAdvancedDetectionParameters_t Struct Reference

[Detection](#)

Description

Perform advanced control of the MTI parameters.

```
typedef struct {
    s16 minVel8;
    s16 maxVel8;
    s16 reserved1;
    s16 minWide;
    s16 maxWide;
    s16 minHigh;
    s16 maxHigh;
    u8 hideOverlapTrks;
    u8 nFramesBack;
    u8 mergeRadius;
    u8 mergeDirTol;
    u8 useRegistration;
    u8 detUpdateRate;
    u8 surroundSize;
    u8 detIdx;
    u16 bgTimeConst;
    u8 bgEdgePenalty6;
    u8 bgResetConf;
    u8 bgResetOff;
    u8 bgResetAng;
    u8 bgResetFrames;
    u8 bgWarpConf;
    u8 bgWarpOff;
    u8 bgWarpAng;
    u8 bgWarpFrames;
    u8 maxTrackFrames;
    u8 debugFiltering;
    u8 downsample;
    u8 maxTelemTrks;
    u8 maxKlvTrks;
    u8 cameraIndex;
    u8 blocksHigh;
    u8 blocksWide;
    u8 maxDisplayTrks;
    u8 ellipsePadding;
    u8 closeFilter;
    u8 detDirection;
    u8 detBitDepth;
    u8 aiDetectIOU;
    u8 detNearTrk;
    u8 frameDepth;
} SLASetAdvancedDetectionParameters\_t;
```

These parameters apply to all modes except where explicitly stated otherwise.

See also [SLASetDetectionParameters_t](#).

NOTE: Applied to the Command Camera.

Message ID 0x76

Byte Offset	Name	Description
4-5	minVel8	Minimum target velocity (256*pixels/frame) (-1 = not set) (not used in <i>vehicle</i> , <i>person</i> , or <i>drone</i> modes)
6-7	maxVel8	Maximum target velocity (256*pixels/frame) (-1 = not set) (not used in <i>vehicle</i> , <i>person</i> , or <i>drone</i> modes)
8-9	reserved1	<i>Reserved</i> previously used for maximum acceleration, which was removed in version 3.5
10-11	minWide	Minimum target width (pixels) (-1 = not set)
12-13	maxWide	Maximum target width (pixels) (-1 = not set)
14-15	minHigh	Minimum target height (pixels) (-1 = not set)
16-17	maxHigh	Maximum target height (pixels) (-1 = not set)
18	hideOverlapTrks	1 = Hide targets that overlap with object tracks – these will not be displayed or reported in telemetry; 0 = Don't hide (not used in <i>vehicle</i> , <i>person</i> , or <i>drone</i> modes)
19	nFramesBack	Number of frames to go back in time to compare frames in MTI difference mode, default 15. Parameter only applies to <i>aerial</i> mode. <i>Aerial</i> : (5 to 15) Use 15 for small slow moving object detection and smaller numbers for faster larger objects.
20	mergeRadius	Maximum distance between targets at which they can be merged into a single target (0 to 255 pixels), 0 = automatic, default = 0, typical is 10 to 25 (not used in <i>vehicle</i> , <i>person</i> , or <i>drone</i> modes)
21	mergeDirTol	<i>Reserved</i>
22	useRegistration	Use registration information to track detections from frame to frame (used in <i>anomaly</i> , <i>maritime</i> , <i>radiometric</i> , and <i>blob</i> modes)
23	detUpdateRate	Rate we update the background model for detection algorithms, 0-255. Larger values put more emphasis to the current frame. (used in <i>anomaly</i> and <i>maritime</i> modes)
24	surroundSize	Blob Detection: size of surround for high pass processing default = 25. Range = 7-79 (odd numbers) (used in <i>blob</i> mode) NOTE: Default changed from 7 to 25 in version 3.5
25	detIdx	Detection index Pass 0 for the primary detection, 1 for control of a second detection.
26-27	bgTimeConst	Time constant (in frames) at which frames are averaged into the background model (at the default value of 600, the background will mostly be replaced after about 20 seconds) (applies only to <i>staring</i> and <i>gas</i> modes)
		Penalty to apply to edges in background model mode – typical values are 0 (no edge

28	bgEdgePenalty6	penalty) to 64 (full edge penalty), default 64 (applies only to <i>staring</i> and <i>gas</i> modes)												
29	bgResetConf	Reset background model if registration confidence falls below this value; 0 to 100, default 40 (applies only to <i>staring</i> and <i>gas</i> modes)												
30	bgResetOff	Reset background model if the registration column or row offset exceeds this value; 0 to 255, default 100 (applies only to <i>staring</i> and <i>gas</i> modes)												
31	bgResetAng	Reset background model if the registration angle in degrees exceeds this value; 0 to 180, default 1 (applies only to <i>staring</i> and <i>gas</i> modes)												
32	bgResetFrames	<i>Reserved</i>												
33	bgWarpConf	Warp the background model if registration confidence falls below this value; 0 to 100, default 85 (applies only to <i>staring</i> and <i>gas</i> mode)												
34	bgWarpOff	Warp the background model if the registration column or row offset exceeds this value; 0 to 255, default 32 (applies only to <i>staring</i> and <i>gas</i> modes)												
35	bgWarpAng	Warp the background model if the registration angle in degrees exceeds this value; 0 to 180, default 1 (applies only to <i>staring</i> and <i>gas</i> modes)												
36	bgWarpFrames	<i>Reserved</i>												
37	maxTrackFrames	Number of frames to keep tracking once target has disappeared, 1 to 253; 0 = automatic, 254 = off, 255 = reserved												
38	debugFiltering	Debug flags to help user understand filtering based on size, velocity, and acceleration. (not used in <i>vehicle</i>, <i>person</i>, or <i>drone</i> modes) <table><tr><th>Bit</th><th>Description</th></tr><tr><td>0-3</td><td>0 = No debug drawing, 1 = Show Size (high,wide), 2 = Show Velocity, 3 = Show Acceleartion</td></tr><tr><td>4</td><td><i>Reserved</i></td></tr><tr><td>5</td><td><i>Reserved</i> SightLine internal use only.</td></tr><tr><td>6</td><td>1 = Show tracks that don't make it in telemetry or klv in orange. 0 = Show tracks in default color set by user. See <i>maxTelemTrks</i> and <i>maxKlvTrks</i>.</td></tr><tr><td>7</td><td>1 = Show filtered tracks in red. 0 = Don't show filtered tracks: NOTE: When enabling this, filtered tracks also appear in telemetry and KLV streams but will have confidence of 0.</td></tr></table>	Bit	Description	0-3	0 = No debug drawing, 1 = Show Size (high,wide), 2 = Show Velocity, 3 = Show Acceleartion	4	<i>Reserved</i>	5	<i>Reserved</i> SightLine internal use only.	6	1 = Show tracks that don't make it in telemetry or klv in orange. 0 = Show tracks in default color set by user. See <i>maxTelemTrks</i> and <i>maxKlvTrks</i> .	7	1 = Show filtered tracks in red. 0 = Don't show filtered tracks: NOTE: When enabling this, filtered tracks also appear in telemetry and KLV streams but will have confidence of 0.
Bit	Description													
0-3	0 = No debug drawing, 1 = Show Size (high,wide), 2 = Show Velocity, 3 = Show Acceleartion													
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7	1 = Show filtered tracks in red. 0 = Don't show filtered tracks: NOTE: When enabling this, filtered tracks also appear in telemetry and KLV streams but will have confidence of 0.													
		Select the amount of downsampling that will be applied to the capture image (not used in <i>anomaly</i> mode) NOTE: Hardware limitations exist based on the platform, algorithm, and frame size. <table><tr><th>Bits</th><th>Description</th></tr><tr><td></td><td></td></tr></table>	Bits	Description										
Bits	Description													

39	downsample	0-6	Value	Description
			0	none
			1	2x2
			2	4x4
			3	8x8
			4-127	<i>Reserved</i>
		7	automatically determine based on frame size.	
40	maxTelemTrks	Maximum number of MTI tracks reported in <u>SLATrackingPositions_t</u> , default 10. (not used in <i>vehicle</i> , <i>person</i> , or <i>drone</i> modes) If the total number of tracks is greater than maxTelemTrks, then the reported tracks are chosen based on which tracks have the highest confidence.		
41	maxKlvTrks	Maximum number of MTI tracks reported in the KLV data, default 10. (not used in <i>vehicle</i> , <i>person</i> , or <i>drone</i> modes) If the total number of tracks is greater than maxKlvTrks, then the reported tracks are chosen based on which tracks have the highest confidence.		
42	cameraIndex	<u>Camera Index</u>		
43	blocksHigh	Number of vertical blocks to use for region based processing, range 1 to 16, default 1. (used in <i>blob</i> mode)		
44	blocksWide	Number of horizontal blocks to use for region based processing, range 1 to 16, default 1. (used in <i>blob</i> mode)		
45	maxDisplayTrks	Maximum number of MTI tracks displayed, default 100. (not used in <i>vehicle</i> , <i>person</i> , or <i>drone</i> modes) If the total number of tracks is greater than maxDisplayTrks, then the displayed tracks are chosen based on which tracks have the highest confidence.		
46	ellipsePadding	Padding added to the estimated target size when drawing overlays. Range 0 to 100, default is 5. (not used in <i>vehicle</i> , <i>person</i> , or <i>drone</i> modes)		
47	closeFilter	Additional filtering step done after main detection processing. (not used in <i>gas</i> mode) 0 = default based on algorithm - see below. 1 = None (Default for Maritime, Gas, and Anomaly) 2 = 3x3 (Default for Aerial, Staring, Motion, Vehicle, Drone, and Person) 3 = 5x5 (Default for Blob, and Radiometric) 4 = Open filter 3x3 5 = Open filter 5x5 6 = Erode filter 3x3 7 = Erode filter 5x5 8 = Dilate filter 3x3 9 = Dilate filter 5x5		
48	detDirection	Limit detection to bright or dark objects or find all. (used in <i>blob</i> mode) 1 = Find bright objects only 2 = Find dark objects only		

		3 = Find all objects (default)
49	detBitDepth	Policy for bit-depth on images used for detection. (used in <i>anomaly</i>, <i>blob</i>, <i>radiometric</i>, <i>staring</i>, and <i>gas</i> modes) NOTE: Only valid when the source image for detection is greater than 8 bits. 0 = Default. Use 16-bit images for detection. 1 = Use 8-bit images for detection.
50	aiDetectIOU	AI Detection intersection over union. Default 1. Valid values 0 to 10. Smaller values merge AI boxes with smaller overlaps. Larger values merge AI boxes with larger overlaps
51	detNearTrk	This enables AI data transfer between detection and tracking
52	frameDepth	Frame Depth For Motion Detection Modes. 2-7. Smaller or faster objects will benefit from lower frame depth.

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentBlendParameters_t Struct Reference

[List of all members](#)

[Blending](#)

Description

Current blend parameters – see [SLASetBlendParameters_t](#).

```
typedef struct {
    u8 absOffZoom;
    u8 up;
    u8 right;
    u8 down;
    u8 left;
    u8 rotation;
    u8 zoom;
    u8 mode;
    u8 amt;
    u8 hue;
    u8 flags;
    u8 reserved;
    u8 warpIndex;
    u8 fixedIndex;
    u8 usePresetAlign;
    u8 presetAlignIndex;
    u8 hzoom;
    u8 hotStart;
    u8 coldEnd;
} SLACurrentBlendParameters_t;
```

Sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x4D

Byte Offset	Name	Description
4	absOffZoom	Bit 1: Zoom mode; 0 = (1..255) maps to (0.9..1.1), 1 = (1..255) maps to (0.004..0.996) Bits 0 and 2-7 reserved
5	up	Number of pixels warp image is shifted up
6	right	Number of pixels warp image is shifted right
7	down	Number of pixels warp image is shifted down
8	left	Number of pixels warp image is shifted left

9	rotation	Rotation of warp image (1..255) maps to (-5..5) degrees
10	zoom	Scale factor applied to warp image (1..255) maps to (0.9..1.1) or to (0.004..0.996) depending on value of absOffZoom
11	mode	Blend mode (see SLASetBlendParameters_t)
12	amt	Amount of luminance information from the EO (visible) camera to include in the blended result. Applies to: Frame Blend, Night Blend, Color Blend, Thermal Blend and Color IR Blend modes.
13	hue	Amount of yellow hue to apply from bright areas in the EO camera. Applies to: FrameBlend and NightBlend modes
14	flags	See SLASetBlendParameters_t
15	reserved	<i>Reserved</i>
16	warpIndex	Image from camera with this index into the space of the other camera through the calibration settings before blending
17	fixedIndex	Image from camera with this index is not warped before blending with warp image
18	usePresetAlign	Image alignment parameter index. 0 Use the alignment parameters in this message. 1 Use a preset alignment (defined by SLABlendAlign_t or SLAFourAlignPoints_t) and set index of preset alignment parameters to value of presetAlignIndex
19	presetAlignIndex	Indicate index of preset alignment parameters – ignored if usePresetAlign is 0. Indexes 0..4 are used for alignment sets defined by SLABlendAlign_t . Indexes 10..14 are used for alignment sets defined by SLAFourAlignPoints_t .
20	hzoom	Horizontal zoom scale applied to EO camera (1..255, 0=no change) on top of zoom . maps to (0.9..1.1) Verical zoom = zoom Horizontal zoom = hzoom * zoom
21	hotStart	Applies to: ThermalBlend, beginning value for hot palette.
22	coldEnd	Applies to: ThermalBlend, ending value for cold palette.

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetBlendParameters_t Struct

Reference

[Blending](#)

Description

Blend an EO(visible) and an IR camera with video scaling and shifting to align frames.

```
typedef struct {
    u8 absOffZoom;
    s8 vertical;
    s8 horizontal;
    u8 rotation;
    u8 zoom;
    u8 mode;
    u8 amt;
    u8 hue;
    u8 flags;
    u8 reset;
    u8 reserved;
    u8 warpIndex;
    u8 fixedIndex;
    u8 usePresetAlign;
    u8 presetAlignIndex;
    u8 hzoom;
    u8 hotStart;
    u8 coldEnd;
} SLASetBlendParameters_t;
```

When both cameras are standard definition with similar fields of view, set Absolute Zoom Mode off to scale the EO camera to match the IR. When the warp camera is HD, Absolute Zoom Mode can be turned on for larger zoom factors. For example, if a 1280x720 HD camera is blending with a 640x480 camera where the horizontal fields of view match, pass zoom = 128 and zoom absolute = 1 to get a zoom factor of 0.5 which will scale the HD image to 640x360 before blending.

Requires [SLAVideoDisplay_t](#) be used to set Display Modes to Blend. Also requires [SLABlendAlign_t](#) to set alignment. See also [SLAFourAlignPoints_t](#) and [SLACurrentBlendParameters_t](#).

NOTE: 4000 Only

Message ID 0x2F

Byte Offset	Name	Description
		Offset and zoom modes
	Bit	Description

4	absOffZoom	0	0. interpret vertical and horizontal as incremental offsets 1. interpret as absolute offsets	
		1-3	DEPRECATED - Zoom multiplier 0 - zoom (1..255) maps to (0.9..1.1) – useful when both cameras are SD or have matching pixel size on target 1-7 - zoom maps to N*(0.004..0.996) – useful when one camera is HD and the other is SD	
		4-7	Reserved	
5	vertical	DEPRECATED - USE SLABlendAlign. Shift warp camera index video position vertically.		
6	horizontal	DEPRECATED - USE SLABlendAlign. Shift warp camera index video position horizontally		
7	rotation	DEPRECATED - USE SLABlendAlign. Rotate warp camera index video; (1..255) maps to (-5..5) degrees; 0 = no change		
8	zoom	DEPRECATED - USE SLABlendAlign. Scale warp camera index video. (1..255) maps to (0.9..1.1) or (0.004..0.996) depending on absOffZoom		
9	mode	Value	Blend Mode	Description
		0	No change	-
		1	Frame Blend Warped EO	Basic percentage (controlled by amt) blending of the warped EO and fixed IR images. See flags below for optional histogram equalization
		2	Thermal Blend Warped EO	Blends hot pixels from the fixed IR camera (shown in red) with the warped EO image. Typically used to highlight hot areas in a daytime EO image.
		3	Night Blend Warped EO	Blends bright areas from the warped EO camera (modify color with hue) with the fixed IR image. Typically used to highlight bright lights in a night time IR image.
		4	Color Blend Warped EO	Percentage (controlled by amt) blending of the warped EO and fixed IR images. Like Frame Blend, but the color from the warped EO image is also passed to the output (controlled by amt or hue , see flags).
		5	Reserved	-
		6	Frame Blend Fixed EO	Basic percentage (controlled by amt) blending of the fixed EO and warped IR images.

		7	Thermal Blend Fixed EO	Blends hot pixels from the fixed EO camera (shown in red) with the warped IR image. Typically used to highlight hot areas in a daytime EO image. See flags below for optional histogram equalization
		8	Night Blend Fixed EO	Blends bright areas from the fixed EO camera (modify color with hue) with the warped IR image. Typically used to highlight bright lights in a night time IR image.
		9	Color Blend Fixed EO	Percentage (controlled by amt) blending of the fixed EO and warped IR images. Like Frame Blend, but the color from the fixed EO image is also passed to the output (controlled by amt or hue , see flags).
		10	Color IR Blend Fixed EO	Percentage (controlled by amt) blending of the fixed EO and the user palette colored warped IR images. Like Frame Blend, with the color from the fixed EO and user palette colored IR also blended(controlled by amt or hue , see flags). See SLASetUserPalette_t
		11	Color IR Blend Warped EO	Percentage (controlled by amt) blending of the warped EO and the user palette colored fixed IR images. Like Frame Blend, with the color from the fixed EO and user palette colored IR also blended (controlled by amt or hue , see flags). See SLASetUserPalette_t
		12	Color EO Blend EO	Percentage blending of the first EO and second EO. (controlled by amt or hue , see flags)
		others	<i>Reserved</i>	-
10	amt	Amount of luminance information from the EO (visible) camera to include in the blended result. 0 = no change, (1..255) maps to (0..100%). Applies to Frame Blend, Night Blend, Color Blend, Thermal Blend and Color IR Blend modes		
11	hue	Night Blend mode - Amount of yellow hue to apply from bright areas of the EO camera. 0 = no change, 1 = no hue, 255 = full hue Color Blend and Color IR Blend modes - when flags bit 1 is 1 0 = no change, (1..255) maps to (0..100%) for the amount of color to show.		
12	flags	Bits	Value	Description
		0	0	Don't apply histogram equalization to infrared image in Thermal Blend modes New 3.2
			1	Apply histogram equalization to infrared image in Thermal Blend modes. No Longer Supported. (legacy/default behavior)
		1	0	Use amt to control color in Color Blend and Color IR Blend modes (legacy/default behavior)
1	Use hue to control color in Color Blend and Color IR Blend modes New 3.5			

		2-7	-	Reserved
13	reset	Calibration reset (0 or 1). Resets the image warp calibration (zoom, rotate, shift up / down / left / right) back to default		
14	reserved	Reserved		
15	warpIndex	Warp camera index – video from this camera is warped into the space of the other camera through the calibrations settings before blending		
16	fixedIndex	Fixed camera index – video from this camera is not warped before blending		
17	usePresetAlign	DEPRECATED - Should always be 1. Image alignment parameter index. 0 Use the alignment parameters in this message. 1 Use a preset alignment (defined by SLABlendAlign_t or SLAFourAlignPoints_t) and set index of preset alignment parameters to value of presetAlignIndex		
18	presetAlignIndex	Indicate index of preset alignment parameters – ignored if usePresetAlign is 0. Indexes 0..4 are used for alignment sets defined by SLABlendAlign_t . Indexes 10..14 are used for alignment sets defined by SLAFourAlignPoints_t .		
19	hzoom	DEPRECATED - USE SLABlendAlign . Horizontal zoom scale applied to EO camera (1..255, 0=no change) on top of zoom . maps to (0.9..1.1) Verical zoom = zoom Horizontal zoom = hzoom * zoom		
20	hotStart	Pixel value to start hot pallet of Thermal Blend. Only applies to Thermal blend mode (ignored otherwise). Should be more than coldEnd.		
21	coldEnd	Pixel value to end cold pallet of Thermal Blend. Only applies to Thermal blend mode (ignored otherwise). Should be less than hotStart.		

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentDetectionParameters_t

Struct Reference

[Detection](#)

Description

Current detection parameters – format identical to [SLASetDetectionParameters_t](#).

```
typedef struct {
    u8 mode;
    u8 debugCode;
    u8 threshold;
    u8 manualThreshold;
    u8 manualWatchFrames;
    u8 suspScore;
    u8 detIdx;
    u8 modeUpperByte;
    u16 minTemperature;
    u16 maxTemperature;
    u8 cameraIndex;
    u8 classSensitivity;
    u8 enableAiMtd;
} SLACurrentDetectionParameters\_t;
```

Sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x54

Byte Offset	Name	Description
4	mode	See SLASetDetectionParameters_t
5	debugCode	
6	threshold	
7	manualThreshold	
8	manualWatchFrames	
9	suspScore	
10	detIdx	
11	modeUpperByte	
12-13	minTemperature	
14-15	maxTemperature	

16	cameraIndex	
17	classSensitivity	
18	enableAiMtd	

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) [Camera Configuration](#)

[List of all members](#)

SLASetDetectionParameters_t Struct Reference

[Detection](#)

Description

Set the parameters for the detection algorithms.

```
typedef struct {
    u8 mode;
    u8 debugCode;
    u8 threshold;
    u8 manualThreshold;
    u8 manualWatchFrames;
    u8 suspScore;
    u8 detIdx;
    u8 modeUpperByte;
    u16 minTemperature;
    u16 maxTemperature;
    u8 cameraIndex;
    u8 classSensitivity;
    u8 enableAiMtd;
} SLASetDetectionParameters_t;
```

See [SLATrackingPositions_t](#) for results. See also [SLASetAdvancedDetectionParameters_t](#).

NOTE: Applied to the Command Camera.

Detection Modes

Multiple modes of automatic moving target indication are available, depending on the security bits enabled.

Vehicle Medium Moving Target Indication

This mode works best when finding moving vehicles in a moving scene. It works well from a moving aerial platform for finding vehicles that are typically 10 to 100 pixels long in the scene. It does not work as well at finding very small, very large, slow or erratically moving objects. Available with [SLA Baseline Detection Application Bits](#).

Motion Small Moving Target Indication

This mode works from a moving platform and can find small and slow, fast or erratically moving objects. This uses the same detection algorithm as Vehicle, Drone and Person Trackers ([SLASetTrackingParameters_t](#)) but with any tracker mode. This is useful for finding small linear moving objects or erratic objects with optional MTD [EnableAIMTD](#). [SLA Enhanced](#)

Detection Application Bits.

Staring Small Moving Target Indication

This mode works well for a ground camera that is either stationary or moves then stares for an extended time at a scene. It can find objects that are small and move at slow, fast or varying rates or that temporarily stop. Available with [SLA Baseline Detection Application Bits](#).

Aerial Small Moving Target Indication

This mode works from a moving platform and can find small and slow, fast or erratically moving objects. It is capable of handling scene motion with frame to frame perspective change. It will lose objects that stop moving. This mode takes the most compute resource and is the most likely to run at less than full frame rate if other processing is also enabled(such as network video output). Available with [SLA Enhanced Detection Application Bits](#).

Anomaly Detection

This mode works from a moving or stationary platform and finds objects that are unique anomalies or outliers in the scene based on color and / or intensity. This mode doesn't require objects to be moving to be detected. Available with [SLA Baseline Detection Application Bits](#).

Maritime Detection

This mode works from a moving or stationary platform and finds objects in a maritime environment such as ships or life jackets. Available with [SLA Enhanced Detection Application Bits](#).

Radiometric Detection

This mode works from a moving or stationary platform and finds objects within a specified temperature range. This mode doesn't require objects to be moving to be detected. Available with [SLA Baseline Detection Application Bits](#).

Blob Detection

This mode works from a moving or stationary platform and finds objects that are light or dark compared to their surroundings. This mode doesn't require objects to be moving to be detected. Available with [SLA Baseline Detection Application Bits](#).

Drone Medium Moving Target Indication

This mode works best when finding drones in a moving or stationary scene. It works well from a moving or stationary platform for finding drones that are typically 10 to 100 pixels long in the scene. Available with [SLA Enhanced Detection Application Bits](#).

Gas Leak Detection

This mode enhances and detects moving gasses such as those visible with specialized cameras that are sensitive in the methane absorption band looking for Gas Leaks. Available with [SLA Gas Detection Application Bits](#). Set the gas color with [SLASetSystemValue_t](#).

Person Medium Moving Target Indication

This mode is the same as vehicle detection mode but will start a person track instead of a vehicle track. This can be useful if the objects to track change direction and appearance more quickly. Available with [SLA Baseline Detection Application Bits](#).

AI Detection

This mode works based on full frame AI detection. Requires a model file (.slaod). This file can contain a trained model to enable detections. This mode doesn't require objects to be moving to be detected. Available with [SLA AI Detection Application Bits](#).

Parameter Details

Detection Mode

Bit	Value	Option
0	0x0001	1 = Disable all (other bits ignored)
1	0x0002	1 = Reset Detect – will clear all tracks, except in <i>Vehicle</i> mode the primary track will persist
2	0x0004	<i>Reserved</i> SightLine internal use only.
3	0x0008	<i>Motion</i> Basic Motion Detection using the underlying Vehicle, Drone and Person detection algorithm. Can be used with any track mode (see SLASetTrackingParameters_t).
4	0x0010	<i>Vehicle</i> Medium Moving Target Indication – for use on a mobile platform NOTE : if enabled, <i>tracking mode</i> will be set to <i>vehicle</i> (see SLASetTrackingParameters_t).
5	0x0020	<i>Staring</i> Camera Small Moving Target Indication – for use with a fixed camera or a ground mounted camera that moves then `stares'
6	0x0040	<i>Aerial</i> Platform Small Moving Target Indication – for use with an aerial moving camera
7	0x0080	<i>Anomaly Detection</i> in color and/or intensity histogram – for use on a mobile or fixed platform
8	0x0100	<i>Maritime Detect</i> for use on a mobile or fixed platform to detect objects at sea
9	0x0200	<i>Radiometric Detect</i> – for use on a mobile or fixed platform to detect objects within a specified temperature range
10	0x0400	<i>Blob Detect</i> – for use on a mobile or fixed platform to detect objects that are either light or dark compared to their surroundings
11	0x0800	<i>Drone</i> – Medium Moving Target Indication For Drones - for use on a mobile or stationary platform NOTE : if enabled, tracking mode will be set to <i>drone</i> (see SLASetTrackingParameters_t).
12	0x1000	<i>Gas</i> – Gas Leak Detection - for use on a mobile or stationary platform with a specialized gas leak detection camera.
13	0x2000	<i>AI Detect</i> – AI full frame detection mode. For detecting full frame based on the .slaod model file. (Currently Available on 4100/17XX Only) Beta
14	0x4000	<i>Person</i> Medium Moving Target Indication – for use on a mobile or stationary platform NOTE : if enabled, <i>tracking mode</i> will be set to <i>person</i> (see SLASetTrackingParameters_t).
15	0x8000	<i>With Classification</i> – Only Applicable for Drone Mode - filter out false positives with neural network classifier. Classification results can be read using SLATrackingPositionsExtended_t . This mode is unavailable when filters are set in SLAClassifierParameters_t .

Byte Offset	Name	Description																				
4	mode	Detection Mode LSB – See byte 11 for MSB For each bit: 0 disables feature, 1 enables feature. 0 = Don't change detection modes Multiple detection mode combinations allowed include one of Vehicle/Drone/Person plus one of Staring/Aerial plus one of Anomaly/Maritime/Radiometric/Blob																				
5	debugCode	Debug Flags <table><tr><th>Bit</th><th>Option</th></tr><tr><td>0</td><td>Debug display – required to visualize any of the following options</td></tr><tr><td>1</td><td>Threshold – display points above threshold as white (not used in <i>vehicle</i>, <i>drone</i>, or "person" modes)</td></tr><tr><td>2</td><td>Peaks – display peaks in cyan (not used in <i>vehicle</i>, <i>drone</i> or <i>person</i> modes)</td></tr><tr><td>3</td><td>Blobs – display blobs in yellow (not used in <i>vehicle</i> or <i>drone</i> or <i>person</i> modes)</td></tr><tr><td>4</td><td><i>Reserved</i></td></tr><tr><td>5</td><td>Indices – display index number next to each blob or peak (not used in <i>vehicle</i>, <i>drone</i>, or "person" modes)</td></tr><tr><td>6</td><td><i>Reserved</i></td></tr><tr><td>7</td><td><i>Reserved</i></td></tr></table>	Bit	Option	0	Debug display – required to visualize any of the following options	1	Threshold – display points above threshold as white (not used in <i>vehicle</i> , <i>drone</i> , or "person" modes)	2	Peaks – display peaks in cyan (not used in <i>vehicle</i> , <i>drone</i> or <i>person</i> modes)	3	Blobs – display blobs in yellow (not used in <i>vehicle</i> or <i>drone</i> or <i>person</i> modes)	4	<i>Reserved</i>	5	Indices – display index number next to each blob or peak (not used in <i>vehicle</i> , <i>drone</i> , or "person" modes)	6	<i>Reserved</i>	7	<i>Reserved</i>		
Bit	Option																					
0	Debug display – required to visualize any of the following options																					
1	Threshold – display points above threshold as white (not used in <i>vehicle</i> , <i>drone</i> , or "person" modes)																					
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6	<i>Reserved</i>																					
7	<i>Reserved</i>																					
6	threshold	Sensitivity - The moving target indication sensitivity refers to the algorithm's threshold for deciding whether a particular candidate target is a moving target or a false positive. <table><tr><th>Value</th><th>Description</th></tr><tr><td>0</td><td>Manual mode (use <i>manualThreshold</i>, <i>manualWatchFrames</i>, and <i>suspScore</i> directly)</td></tr><tr><td>1</td><td>Highest sensitivity - more false detects, faster detection, less missed detections.</td></tr><tr><td>10</td><td>Lowest sensitivity - less false detects, slower detection, more missed detections.</td></tr></table> When AI Detection Mode is selected, the values of threshold map to the following: <table><tr><th>Value</th><th>AI Confidence Percentage</th></tr><tr><td>1</td><td>1% - Show AI detection >= 1% confident</td></tr><tr><td>2</td><td>5%</td></tr><tr><td>3</td><td>10%</td></tr><tr><td>4</td><td>15%</td></tr><tr><td></td><td></td></tr></table>	Value	Description	0	Manual mode (use <i>manualThreshold</i> , <i>manualWatchFrames</i> , and <i>suspScore</i> directly)	1	Highest sensitivity - more false detects, faster detection, less missed detections.	10	Lowest sensitivity - less false detects, slower detection, more missed detections.	Value	AI Confidence Percentage	1	1% - Show AI detection >= 1% confident	2	5%	3	10%	4	15%		
Value	Description																					
0	Manual mode (use <i>manualThreshold</i> , <i>manualWatchFrames</i> , and <i>suspScore</i> directly)																					
1	Highest sensitivity - more false detects, faster detection, less missed detections.																					
10	Lowest sensitivity - less false detects, slower detection, more missed detections.																					
Value	AI Confidence Percentage																					
1	1% - Show AI detection >= 1% confident																					
2	5%																					
3	10%																					
4	15%																					

		<table><tr><td>5</td><td>20% - Default</td></tr><tr><td>6</td><td>30%</td></tr><tr><td>7</td><td>40%</td></tr><tr><td>8</td><td>55%</td></tr><tr><td>9</td><td>70%</td></tr><tr><td>10</td><td>90% - Show AI detection >= 90% confident</td></tr></table>	5	20% - Default	6	30%	7	40%	8	55%	9	70%	10	90% - Show AI detection >= 90% confident
5	20% - Default													
6	30%													
7	40%													
8	55%													
9	70%													
10	90% - Show AI detection >= 90% confident													
7	manualThreshold	Threshold (not used <i>vehicle</i> , <i>drone</i> , or <i>radiometric</i> modes). This is an advanced parameter. It controls the threshold at which potential moving targets are generated. This is ignored unless the sensitivity parameter is set to 0. In the <u>SLACurrentDetectionParameters_t</u> message this will be filled with the value calculated based on the sensitivity.												
8	manualWatchFrames	Watch frames (not used <i>vehicle</i> , <i>drone</i> , or <i>person</i> modes). This is an advanced parameter. After a potential moving target is generated by the moving target indication system, it is watched by the system for a number of frames. This controls the number of frames the system watches a moving target before displaying it. This is ignored unless the sensitivity parameter is set to 0. In the <u>SLACurrentDetectionParameters_t</u> message this will be filled with the value calculated based on the sensitivity.												
9	suspScore	Suspicious Score - (only used in <i>motion</i>, <i>aerial</i> and <i>staring</i> modes). This is an advanced parameter. It controls the level at which we consider a track "suspicious". Valid range is 1-10. Setting this value higher will result in more false positives. This is ignored unless the <u>sensitivity</u> parameter is set to 0. In the <u>SLACurrentDetectionParameters_t</u> message this will be filled with the value calculated based on the sensitivity. <table><tr><th>Value</th><th>Description</th></tr><tr><td>1</td><td>Good for cars - Track moves at a constant velocity</td></tr><tr><td>3-5</td><td>Good for people</td></tr><tr><td>10</td><td>Track can move very erratically</td></tr></table>	Value	Description	1	Good for cars - Track moves at a constant velocity	3-5	Good for people	10	Track can move very erratically				
Value	Description													
1	Good for cars - Track moves at a constant velocity													
3-5	Good for people													
10	Track can move very erratically													
10	detIdx	Detection index - Pass 0 for the primary detection, 1 for control of a second detection. Vehicle, Drone, Person and Gas detection modes are only available for detection index 0. Motion Mask (from <u>SLASetVideoEnhancementParameters_t</u>) uses detection index 1. NOTE: previously detection frame step removed in version 3.3. Use the capture frame step found in <u>SLASetAcquisitionParameters_t</u>.												
11	modeUpperByte	<u>Detection Mode</u> MSB – combines with Detection Mode LSB above												
12-13	minTemperature	Minimum temperature value for a detection when running in <i>radiometric detect</i> mode.												
14-15	maxTemperature	Maximum temperature value for a detection when running in <i>radiometric detect</i> mode.												
16	cameraIndex	<u>Camera Index</u>												
		Classifier sensitivity - (only used in <i>Drone Mode with Classification</i>).												

17	classSensitivity	If not present or 0 the classification sensitivity is derived from the threshold . A valid range is 1-10.	
		Value	Description
		0	Manual mode (use threshold to derive classSensitivity)
		1	Highest sensitivity - more likely to classify object as drone, as a result more false detections
18	enableAiMtd	10	Lowest sensitivity - less likely to classify object as drone, as a result less false detection
		Bool for toggling AI tracking OR MTD Tracking, allowing detections to start tracks. default = 0 (disabled).	
		Bit	Option
		0	AI MTI Mode Tracking
		1	Staring, Aerial, Motion Tracking
		2	<i>Reserved</i>
		3	<i>Reserved</i>
		4	<i>Reserved</i>
		5	<i>Reserved</i>
		6	<i>Reserved</i>
		7	<i>Reserved</i>

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentDetectionRegionOfInterestParameters_t Struct Reference

[Detection](#)

Description

Current detection region of interest parameters – format identical to [SLASetDetectionRegionOfInterestParameters_t](#).

```
typedef struct {
    u8 flags;
    u16 searchRowUl;
    u16 searchColUl;
    u16 searchHeight;
    u16 searchWidth;
    u16 detectRowUl;
    u16 detectColUl;
    u16 detectHeight;
    u16 detectWidth;
    u8 cameraIndex;
    u8 maskBlockHigh;
    u8 maskBlockWide;
    u64 mask0;
    u64 mask1;
    u64 mask2;
    u64 mask3;
} SLACurrentDetectionRegionOfInterestParameters_t;
```

Sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x7D

Byte Offset	Name	Description
4	flags	Flags – see SLASetDetectionRegionOfInterestParameters_t flags
5-6	searchRowUl	Upper left row of the search region. Default = 0
7-8	searchColUl	Upper left column of the search region. Default = 0
9-10	searchHeight	Height of the search region. Default = 0 (use entire height)
11-12	searchWidth	Width of the search region. Default = 0 (use entire width)
13-14	detectRowUl	Upper left row of the detect region. Default = 0
15-16	detectColUl	Upper left column of the detect region. Default = 0
17-18	detectHeight	Height of the detect region. Default = 0 (use entire height)
19-20	detectWidth	Width of the detect region. Default = 0 (use entire width).

21	cameraIndex	<u>Camera Index</u>
22	maskBlockHigh	Mask blocks high. Default = 0, Max value of maskBlocksHigh*maskBlocksWide is 256.
23	maskBlockWide	Mask blocks high. Default = 0, Max value of maskBlocksHigh*maskBlocksWide is 256.
24-31	mask0	Mask bits 0-63, 1 = mask this region out, 0 = don't mask this region out.
32-39	mask1	Mask bits 64-127, 1 = mask this region out, 0 = don't mask this region out.
40-47	mask2	Mask bits 128-191, 1 = mask this region out, 0 = don't mask this region out.
48-55	mask3	Mask bits 192-255, 1 = mask this region out, 0 = don't mask this region out.

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetDetectionRegionOfInterestParameters_t Struct Reference

[Detection](#)

Description

Sets the parameters for Moving Target Indication (MTI) Region of Interest(ROI).

```
typedef struct {
    u8 flags;
    u16 searchRowUl;
    u16 searchColUl;
    u16 searchHeight;
    u16 searchWidth;
    u16 detectRowUl;
    u16 detectColUl;
    u16 detectHeight;
    u16 detectWidth;
    u8 cameraIndex;
    u8 maskBlockHigh;
    u8 maskBlockWide;
    u64 mask0;
    u64 mask1;
    u64 mask2;
    u64 mask3;
} SLASetDetectionRegionOfInterestParameters_t;
```

The search ROI is the area that the MTI algorithms will look for motion in the video stream. The detection ROI is the area of the video in which the MTI algorithms will report detected motion. Having a search area larger then the detection area allows the MTI to report motion as soon as it enters the detection area, whereas having the two areas equal in size reduces the overall processing load. The detect region must be fully contained inside of the search region. If it is not the settings will be ignored.

NOTE: Applied to the Command Camera.

NOTE: Values are in **capture** image coordinates. If you are setting this up with the mouse from Panel+, first set the display resolution to match the capture resolution and zero video rotation, zoom and pan/tilt, and zero out any chop that has been applied.

NOTE: Masked regions don't apply to Vehicle/Drone/Person detection modes.

Message ID 0x7C

Byte Offset	Name	Description	
		Bit	Description

4	flags	0	0 = Disable debug display, 1 = enable debug display
		1	0 = Detection region rectangle (default), 1 = detection region above/below line
		2	0 = Ignore detections above line (default), 1 = Ignore detections below line. This option only make sense if bit 1 is also set to 1.
		3	0 = Disable masked regions, 1 = enabled masked regions.
		4-5	Detection Index: 0 = detection 0, 1 = detection 1
		6-7	<i>Reserved</i>
5-6	searchRowUI	Upper left row of the search region. Default = 0	
7-8	searchColUI	Upper left column of the search region. Default = 0	
9-10	searchHeight	Height of the search region. Default = 0 (use entire height)	
11-12	searchWidth	Width of the search region. Default = 0 (use entire width)	
13-14	detectRowUI	Upper left row of the detect region. Default = 0 If flags Bit 1 is set this becomes Left Y point.	
15-16	detectColUI	Upper left column of the detect region. Default = 0 If flags Bit 1 is set this becomes Left X point.	
17-18	detectHeight	Height of the detect region. Default = 0 (use entire height) If flags Bit 1 is set this becomes Right Y point.	
19-20	detectWidth	Width of the detect region. Default = 0 (use entire width). If flags Bit 1 is set this becomes Right X point.	
21	cameraIndex	<u>Camera Index</u>	
22	maskBlockHigh	Mask blocks high. Default = 0, Max value of maskBlocksHigh*maskBlocksWide is 256.	
23	maskBlockWide	Mask blocks high. Default = 0, Max value of maskBlocksHigh*maskBlocksWide is 256.	
24-31	mask0	Mask bits 0-63, 1 = mask this region out, 0 = don't mask this region out.	
32-39	mask1	Mask bits 64-127, 1 = mask this region out, 0 = don't mask this region out.	
40-47	mask2	Mask bits 128-191, 1 = mask this region out, 0 = don't mask this region out.	
48-55	mask3	Mask bits 192-255, 1 = mask this region out, 0 = don't mask this region out.	

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentDigitalCameraParameters_t Struct Reference

[Capture](#)

Description

Return parameters of currently selected digital camera – similar to [SLASetDigitalCameraParameters_t](#).

```
typedef struct {
    u8 mode;
    u16 agHoldmax;
    u16 agHoldmin;
    u8 rowROIpct;
    u8 colROIpct;
    u8 highROIpct;
    u8 wideROIpct;
    u8 cameraIndex;
    u16 minAGRange;
    u8 agRate8;
    u8 minExp;
    u8 maxExp;
    u8 rejectDarkTail;
    u8 rejectBrightTail;
    u8 midpoint;
    u32 highBitDepthFlags;
} SLACurrentDigitalCameraParameters_t;
```

Sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x71

Byte Offset	Name	Description								
4	mode	Mode for automatic and manual gain/exposure control. See Sensor Controls and Gain Modes								
		<table><tr><th>Value</th><th>Description</th></tr><tr><td>0</td><td>High bit depth automatic gain control (for 10 to 16 bit image conversion to 8 bits)</td></tr><tr><td>1</td><td>Manual gain and exposure control - set agHoldmax/agHoldmin for high bit depth imagers or set sensor Exposure and Gains if available in SLASetADCPParameters_t</td></tr><tr><td></td><td>SightLine automatic exposure and gain control (SLA auto gain algorithm adjusts</td></tr></table>	Value	Description	0	High bit depth automatic gain control (for 10 to 16 bit image conversion to 8 bits)	1	Manual gain and exposure control - set agHoldmax/agHoldmin for high bit depth imagers or set sensor Exposure and Gains if available in SLASetADCPParameters_t		SightLine automatic exposure and gain control (SLA auto gain algorithm adjusts
		Value	Description							
		0	High bit depth automatic gain control (for 10 to 16 bit image conversion to 8 bits)							
		1	Manual gain and exposure control - set agHoldmax/agHoldmin for high bit depth imagers or set sensor Exposure and Gains if available in SLASetADCPParameters_t							
	SightLine automatic exposure and gain control (SLA auto gain algorithm adjusts									

		2	parameter registers on the sensor, see SLASetADCParameters_t) 4000/17XX only, New 3.1
		3	Camera automatic exposure and gain control (camera controls its gain and exposure, see SLASetADCParameters_t) 4000/17XX only, New 3.6
5-6	agHoldmax	Autogain Max value used with Manual Gain mode control for high bit depth images (max value maps to 255 in 8 bit output)	
7-8	agHoldmin	Autogain Min Value with Manual Gain mode control for high bit depth images (min value maps to 0 in 8 bit output)	
9	rowROIpct	Row ROI for Autogain stats. Row Offset in % of image height (255=100%); Full image(default) = 0	
10	colROIpct	Col ROI for Autogain stats. Col Offset in % of image width (255=100%); Full image(default) = 0	
11	highROIpct	High ROI for Autogain stats. Height in % of image height (255=100%); Full image(default) = 255	
12	wideROIpct	Wide ROI for Autogain stats. Width in % of image width (255=100%); Full image(default) = 255	
13	cameraIndex	Camera Index	
14-15	minAGRange	High Bit Depth Autogain mode: Minimum spread of auto gain. 0 = does nothing, 200 = max - min scale range must be at least 200. Prevents over-gaining of flat scenes. New 3.0 SightLine Auto Gain Mode mode: Difference limit for updates. 0 = does nothing, 10 = if image average is within 10 of target Brightness, don't update exposure and gain. Prevents image flashing and excessive setting of sensor registers. New 3.1	
16	agRate8	Auto gain update rate. 1 = minimum rate, 255 = maximum rate, 150 = default. New 3.0 Use Manual mode to stop AGC updates	
17	minExp	Minimum exposure or integration time 0 to 255. Default(0) = ignored. For cameras that support exposure control, see SLASetADCParameters_t . New 3.2	
18	maxExp	Maximum exposure or integration time 0 to 255. Default(0) = ignored. For cameras that support exposure control, see SLASetADCParameters_t . New 3.2	
19	rejectDarkTail	Reject or ignore this percent/10 of the left/dark side of the histogram when calculating Autogain. 0 = reject none, 10 = reject 1.0%, 255 = reject 25.5%. Useful when there are a small number of very dark pixels in a scene. New 3.4	
20	rejectBrightTail	Reject or ignore this percent/10 of the right/bright side of the histogram when calculating Autogain. 0 = reject none, 10 = reject 1.0%, 255 = reject 25.5%. Useful when there are a small number of very bright pixels in a scene. New 3.4	
21	midpoint	Autogain Central value used for high bit depth images. Adjusts which input value corresponds to central output value. New 3.7	
22-25	highBitDepthFlags	Bit	Description
		0	Enable contrast stretching: calculate the agHoldmin and agHoldmax directly from the max % of saturated pixels. Use with Saturation in SLASetADCParameters_t .

Sightline Intelligence

Command, Control, and Script

API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) [Camera Configuration](#)

[List of all members](#)

SLASetDigitalCameraParameters_t

Struct Reference

[Capture](#)

Description

Dynamically adjust a 10 to 16 bit digital camera input to an 8 bit image.

```
typedef struct {
    u8 cameraIndex;
    u8 mode;
    u16 agHoldmax;
    u16 agHoldmin;
    u8 rowROIpct;
    u8 colROIpct;
    u8 highROIpct;
    u8 wideROIpct;
    u16 minAGRange;
    u8 agRate8;
    u8 minExp;
    u8 maxExp;
    u8 rejectDarkTail;
    u8 rejectBrightTail;
    u8 midpoint;
    u32 highBitDepthFlags;
} SLASetDigitalCameraParameters_t;
```

- AGC (Automatic Gain Control) is applied to the capture image before all other processing. agRate8 determines the smoothing filter time constant and minAGRange prevents excessive changes.
- ROI (Region Of Interest) is used to indicate a part of image over which to perform autogain statistics . This ROI is defined as an offset from the relevant edge as a percentage of the full width / height of image.

To get the current values, use [SLAGetParameters_t](#). This will return a [SLACurrentDigitalCameraParameters_t](#).

Note

Setting both 'Autogain Max value' and 'Autogain Min Value' to 0 will result in these parameters being ignored.

See also

[SLASetADCParameters_t](#) to set the brightness (offset) and contrast (gain) used for AGC.

[SLASetOverlayMode_t](#) to enable histogram overlays for Capture & Display images.

[EAN-Infrared-Temperature](#) for more information.

Byte Offset	Name	Description												
4	cameraIndex	<u>Camera Index</u>												
5	mode	Mode for automatic and manual gain/exposure control. See <u>Sensor Controls and Gain Modes</u> <table><tr><th>Value</th><th>Description</th></tr><tr><td>0</td><td>High bit depth automatic gain control (for 10 to 16 bit image conversion to 8 bits)</td></tr><tr><td>1</td><td>Manual gain and exposure control - set agHoldmax/agHoldmin for high bit depth imagers or set sensor Exposure and Gains if available in <u>SLASetADCParameters_t</u></td></tr><tr><td>2</td><td>SightLine automatic exposure and gain control (SLA auto gain algorithm adjusts parameter registers on the sensor, see <u>SLASetADCParameters_t</u>) 4000/17XX only, New 3.1</td></tr><tr><td>3</td><td>Camera automatic exposure and gain control (camera controls its gain and exposure, see <u>SLASetADCParameters_t</u>) 4000/17XX only, New 3.6</td></tr><tr><td>255</td><td>Unknown - state after reset or after setting up a new camera model in <u>SLASetAcquisitionParameters_t</u> 4000/17XX only, New 3.6</td></tr></table>	Value	Description	0	High bit depth automatic gain control (for 10 to 16 bit image conversion to 8 bits)	1	Manual gain and exposure control - set agHoldmax/agHoldmin for high bit depth imagers or set sensor Exposure and Gains if available in <u>SLASetADCParameters_t</u>	2	SightLine automatic exposure and gain control (SLA auto gain algorithm adjusts parameter registers on the sensor, see <u>SLASetADCParameters_t</u>) 4000/17XX only, New 3.1	3	Camera automatic exposure and gain control (camera controls its gain and exposure, see <u>SLASetADCParameters_t</u>) 4000/17XX only, New 3.6	255	Unknown - state after reset or after setting up a new camera model in <u>SLASetAcquisitionParameters_t</u> 4000/17XX only, New 3.6
Value	Description													
0	High bit depth automatic gain control (for 10 to 16 bit image conversion to 8 bits)													
1	Manual gain and exposure control - set agHoldmax/agHoldmin for high bit depth imagers or set sensor Exposure and Gains if available in <u>SLASetADCParameters_t</u>													
2	SightLine automatic exposure and gain control (SLA auto gain algorithm adjusts parameter registers on the sensor, see <u>SLASetADCParameters_t</u>) 4000/17XX only, New 3.1													
3	Camera automatic exposure and gain control (camera controls its gain and exposure, see <u>SLASetADCParameters_t</u>) 4000/17XX only, New 3.6													
255	Unknown - state after reset or after setting up a new camera model in <u>SLASetAcquisitionParameters_t</u> 4000/17XX only, New 3.6													
6-7	agHoldmax	Autogain Max value used with Manual Gain mode control for high bit depth images (max value maps to 255 in 8 bit output)												
8-9	agHoldmin	Autogain Min Value with Manual Gain mode control for high bit depth images (min value maps to 0 in 8 bit output)												
10	rowROIPct	Row ROI for Autogain stats. Row Offset in % of image height (255=100%); Full image(default) = 0												
11	colROIPct	Col ROI for Autogain stats. Col Offset in % of image width (255=100%); Full image(default) = 0												
12	highROIPct	High ROI for Autogain stats. Height in % of image height (255=100%); Full image(default) = 255												
13	wideROIPct	Wide ROI for Autogain stats. Width in % of image width (255=100%); Full image(default) = 255												
14-15	minAGRange	High Bit Depth Autogain mode: Minimum spread of auto gain. 0 = does nothing, 200 = max - min scale range must be at least 200. Prevents over-gaining of flat scenes. New 3.0 SightLine Auto Gain Mode mode: Difference limit for updates. 0 = does nothing, 10 = if image average is within 10 of target Brightness, don't update exposure and gain. Prevents image flashing and excessive setting of sensor registers. New 3.1												
16	agRate8	Auto gain update rate. 1 = minimum rate, 255 = maximum rate, 150 = default. New 3.0 Use Manual mode to stop AGC updates												
17	minExp	Minimum exposure or integration time 0 to 255. Default(0) = ignored. For cameras that support exposure control, see <u>SLASetADCParameters_t</u> . New 3.2												

18	maxExp	Maximum exposure or integration time 0 to 255. Default(0) = ignored. For cameras that support exposure control, see SLASetADCParameters_t . New 3.2				
19	rejectDarkTail	Reject or ignore this percent/10 of the left/dark side of the histogram when calculating Autogain. 0 = reject none, 10 = reject 1.0%, 255 = reject 25.5%. Useful when there are a small number of very dark pixels in a scene. New 3.4				
20	rejectBrightTail	Reject or ignore this percent/10 of the right/bright side of the histogram when calculating Autogain. 0 = reject none, 10 = reject 1.0%, 255 = reject 25.5%. Useful when there are a small number of very bright pixels in a scene. New 3.4				
21	midpoint	Autogain Central value used for high bit depth images. Adjusts which input value corresponds to central output value. New 3.7 0 = Black pixels convert to gray, 128 = pixels (legacy behavior), 255 = White pixels convert to gray.				
22-25	highBitDepthFlags	<table><tr><th>Bit</th><th>Description</th></tr><tr><td>0</td><td>Enable contrast stretching: calculate the agHoldmin and agHoldmax directly from the max % of saturated pixels. Use with Saturation in SLASetADCParameters_t.</td></tr></table>	Bit	Description	0	Enable contrast stretching: calculate the agHoldmin and agHoldmax directly from the max % of saturated pixels. Use with Saturation in SLASetADCParameters_t .
Bit	Description					
0	Enable contrast stretching: calculate the agHoldmin and agHoldmax directly from the max % of saturated pixels. Use with Saturation in SLASetADCParameters_t .					

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentDisplayParameters_t

Struct Reference

[Display](#)

Description

Current display parameters – format identical to [SLASetDisplayParameters_t](#).

```
typedef struct {
    u16 rotationDegrees;
    u16 rotationLimit;
    u8  decayRate;
    u8  falseColorZTT;
    u8  zoom;
    s16 panCol;
    s16 tiltRow;
    u8  index;
    u16 extendedZoom10;
    u16 zoomRate;
    u8  flipMode;
} SLACurrentDisplayParameters_t;
```

Sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x57

Byte Offset	Name	Description
4-5	rotationDegrees	See SLASetDisplayParameters_t
6-7	rotationLimit	
8	decayRate	
9	falseColorZTT	
10	zoom	
11-12	panCol	
13-14	tiltRow	
15	index	
16-17	extendedZoom10	
18-19	zoomRate	

20	flipMode	
----	----------	--

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetDisplayParameters_t Struct

Reference

[Display](#)

Description

Set new display parameter settings.

```
typedef struct {
    u16 rotationDegrees;
    u16 rotationLimit;
    u8 decayRate;
    u8 falseColorZTT;
    u8 zoom;
    s16 panCol;
    s16 tiltRow;
    u8 cameraIndex;
    u16 extendedZoom10;
    u16 zoomRate;
    u8 flipMode;
} SLASetDisplayParameters_t;
```

Current rotation is smoothly changed to specified value, digital zoom, and false color are directly controlled.
See also See [SLADisplayAngle_t](#).

False Color Modes

Value	Description	Value	Description
0, 1	no false color	22, 23	Color3, Color3 inverted
2, 3	white hot, black hot	24, 25	hot iron, hot iron inverted
4, 5	rainbow, rainbow inverted	26, 27	ice fire, ice fire inverted 0=blue(red), 255=red(blue), else=grey level
6, 7	iron, iron inverted	28, 29	IDDEF, IDDEF inverted
8, 9	hot / cold, hot / cold inverted	30, 31	Iron256, Iron256 inverted
10, 11	jet, jet inverted	32, 33	Rain256, Rain256 inverted
12, 13	hot, hot inverted	34, 35	XVolcano, XVolcano inverted
14, 15	HSV, HSV inverted	36, 37	Red, Red inverted
16, 17	470CLR_S, 470CLR_S inverted	38, 39	Green, Green inverted

18, 19	Color1, Color1 inverted	40, 41	Blue, Blue inverted
20, 21	Color2, Color2 inverted	127	User Palette. See SLASetUserPalette_t

Message ID 0x16

Byte Offset	Name	Description														
4-5	rotationDegrees	Rotation angle in degrees (0..360) * 128 Display will smoothly/gradually rotate to this angle based on rotationLimit and decayRate controls														
6-7	rotationLimit	Rotation rate limit in degrees (0..360) * 128 per frame, (default 256) Limits the maximum display angle step taken per frame to get from the current rotation to rotationDegrees														
8	decayRate	Rotation decay rate per frame, 0(0%) to 255(100%) (default 50) Controls the rate at which the system moves from the current rotation to rotationDegrees														
9	falseColorZTT	Bits 0 - 6: False Color Modes Bit 7: Zoom mode: 0 = zoom to center of display, 1 = zoom to tracking box														
10	zoom	Zoom – digital image magnification factor times 64. NOTE: Zoom may be further limited for some camera and display combinations. NOTE: If extendedZoom10 and zoomRate are set, this field is ignored <table><tr><th>Value</th><th>Result</th></tr><tr><td>0 to 15</td><td>No zoom(1X) (subject to change)</td></tr><tr><td>16 to 63</td><td>Zoom out by 0.25 to 0.98</td></tr><tr><td>64</td><td>No zoom(1X)</td></tr><tr><td>65 - 255</td><td>Zoom in by 1.01 to 3.98</td></tr></table>	Value	Result	0 to 15	No zoom(1X) (subject to change)	16 to 63	Zoom out by 0.25 to 0.98	64	No zoom(1X)	65 - 255	Zoom in by 1.01 to 3.98				
Value	Result															
0 to 15	No zoom(1X) (subject to change)															
16 to 63	Zoom out by 0.25 to 0.98															
64	No zoom(1X)															
65 - 255	Zoom in by 1.01 to 3.98															
11-12	panCol	Display pan column offset in pixels														
13-14	tiltRow	Display tilt row offset in pixels														
15	cameraIndex	<u>Camera Index</u> 255 = All cameras.														
16-17	extendedZoom10	<table><tr><th>Value</th><th>Result</th></tr><tr><td>0 to 255</td><td>No zoom(1X) (subject to change)</td></tr><tr><td>256 to 511</td><td>Zoom out by 0.25 to 0.499</td></tr><tr><td>512 to 1023</td><td>Zoom out by 0.5 to 0.999</td></tr><tr><td>1024</td><td>No zoom(1X)</td></tr><tr><td>1025 to 4095</td><td>Zoom in by 1.001 to 3.999</td></tr><tr><td>4096 to 16383</td><td>Zoom in by 4.0 to 15.999</td></tr></table>	Value	Result	0 to 255	No zoom(1X) (subject to change)	256 to 511	Zoom out by 0.25 to 0.499	512 to 1023	Zoom out by 0.5 to 0.999	1024	No zoom(1X)	1025 to 4095	Zoom in by 1.001 to 3.999	4096 to 16383	Zoom in by 4.0 to 15.999
Value	Result															
0 to 255	No zoom(1X) (subject to change)															
256 to 511	Zoom out by 0.25 to 0.499															
512 to 1023	Zoom out by 0.5 to 0.999															
1024	No zoom(1X)															
1025 to 4095	Zoom in by 1.001 to 3.999															
4096 to 16383	Zoom in by 4.0 to 15.999															
18-19	zoomRate	0 to 1023 Measured in frames per X16														

		For example, a display with a zoom rate of 45 and a framerate of 30fps will move from X1 to X16 zoom in 1.5 seconds.
20	flipMode	Mirror and Flip mode: 0 = normal 1 = mirror (flip left to right) 2 = flip (flip top to bottom) 3 = mirror and flip (flip left to right and top to bottom)

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentEthernetDisplayParameters_t Struct Reference

[Compression](#)

Description

Current Ethernet display parameters – format identical to [SLASetEthernetDisplayParameters_t](#).

```
typedef struct {
    u8 protocol;
    u32be ipAddr;
    u16 port;
    u16 displayId;
    u16 maxPacket;
    u16 maxRawPacket;
} SLACurrentEthernetDisplayParameters_t;
```

Sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x52

Byte Offset	Name	Description
4	protocol	Video encoding and transport see SLASetEthernetDisplayParameters_t .
5-8	ipAddr	Current IPv4 address.
9-10	port	Network port number.
11-12	displayId	Network Display ID 4000/17XX: (0x0002 = Net0, 0x0080 = Net1) 41XX: (0x0002 = Net0, 0x0080 = Net1, 0x0010 = Net2)
13-14	maxPacket	Maximum number of bytes transferred in each MPEG2-TS packet.
15-16	maxRawPacket	Maximum number of bytes transferred in each Non-MPEG2-TS packet. New 3.5.8

Sightline Intelligence

Command, Control, and Script

API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) [Camera Configuration](#)

[List of all members](#)

SLACurrentEthernetVideoParameters_t Struct Reference

[Compression](#)

Description

Current Ethernet video parameters – format identical to [SLASetEthernetVideoParameters_t](#).

```
typedef struct {  
    u8 quality;  
    u8 foveal;  
    u8 frameStep;  
    u8 frameSize;  
    u16 displayId;  
    u16 customWide;  
    u16 customHigh;  
} SLACurrentEthernetVideoParameters_t;
```

Sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x48

Byte Offset	Name	Description
4	quality	0: lowest image quality to 100: highest image quality (default 80); applies only to MJPEG video
5	foveal	Reduces image quality for pixels away from image center – 0: no quality reduction, 100 : maximum quality reduction; applies only to MJPEG video
6	frameStep	1 shows every frame, 2 shows every other frame, etc.
7	frameSize	Specifies down sample and frame size. see SLASetEthernetVideoParameters_t .
8-9	displayId	Network Display ID 4000/17XX: (0x0002 = Net0, 0x0080 = Net1) 41XX: (0x0002 = Net0, 0x0080 = Net1, 0x0010 = Net2)
10-11	customWide	Width of video in pixels when Output Frame Size is set to Custom . Range: 224 to 1920/3840 (17XX/4000), must be a multiple of 32.
12-13	customHigh	Height of video in pixels when Output Frame Size is set to Custom . Range: 224 to 1080/2160 (17XX/4000), must be a multiple of 8.

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetEthernetVideoParameters_t

Struct Reference

[Compression](#)

[List of all members](#)

Description

Affects the quality, size and frame rate of the individual video frames sent over Ethernet.

```
typedef struct {
    u8 quality;
    u8 foveal;
    u8 frameStep;
    u8 frameSize;
    u16 displayId;
    u16 customWide;
    u16 customHigh;
} SLASetEthernetVideoParameters_t;
```

See [SLASetH264Parameters_t](#) for additional H.264 specific parameters.

Output Frame Size

Size of destination video. Works only when Video Format is set to a compatible format (see [SLASetEthernetDisplayParameters_t](#))

Value	Description
0	Output size = input size
1	640x480 (NTSC)
2	768x576 (PAL)
3	720p (1280x720)
4	720p cropped (960x720)
5	1080p (1920x1080)
6	1080p cropped (1440x1080)
7	1280x960
8	240 (320x240)
9	4000 only: 2560x1440 (H264 and H265 only)

10	4000 only: 3840x2160 (4K) (H264 and H265 only)
11	Custom size specified by customWide and customHigh parameters

Message ID 0x1A

Byte Offset	Name	Description
4	quality	0: lowest image quality to 100: highest image quality 0 to 100 (default 80); applies only to MJPEG video
5	foveal	Reduces image quality for pixels away from image center – 0: no quality reduction, 100 : maximum quality reduction; applies only to MJPEG video
6	frameStep	1 shows every frame, 2 shows every other frame, etc. Applies to Analog, HDMI, HD-SDI and Ethernet video display. 1 to 120 (default 1). NOTE: This feature is useful for freeing processor load from display so that telemetry output can maintain full rate output for camera gimbal control.It also provides consistent display timing that can look better than alternating between showing and skipping frames.
7	frameSize	Specifies frame size. Bits 0..3: set to 0, Bits 4..7: Output Frame Size
8-9	displayId	Network Display ID 4000/17XX: (0x0002 = Net0, 0x0080 = Net1, 0x0082 = both) 41XX: (0x0002 = Net0, 0x0080 = Net1, 0x0010 = Net2, 0x0092 = all)
10-11	customWide	Width of video in pixels when Output Frame Size is set to Custom . Range: 224 to 1920(17xx)/3840(4000), must be a multiple of 32.
12-13	customHigh	Height of video in pixels when Output Frame Size is set to Custom . Range: 224 to 1080(17xx)/2160(4000), must be a multiple of 8.

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentH264Parameters_t Struct

Reference

[List of all members](#)

[Compression](#)

Description

Current H264 parameters – format identical to [SLASetH264Parameters_t](#).

```
typedef struct {
    u32 targetBitRate;
    u8 intraFrameInterval;
    u8 lfDisableIdc;
    u8 airMbPeriod;
    u8 sliceRefreshRowNumber;
    u8 flags;
    u16 displayId;
    u8 minQp;
    u8 maxQp;
} SLACurrentH264Parameters_t;
```

Sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x56

Byte Offset	Name	Description
4-7	targetBitRate	See SLASetH264Parameters_t
8	intraFrameInterval	
9	lfDisableIdc	
10	airMbPeriod	
11	sliceRefreshRowNumber	
12	flags	
13-14	displayId	
15	minQp	
16	maxQp	

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetH264Parameters_t Struct Reference

[Compression](#)

Description

Modify H264, H265 and MPEG4 encoding behavior.

```
typedef struct {
    u32 targetBitRate;
    u8 intraFrameInterval;
    u8 lfDisableIdc;
    u8 airMbPeriod;
    u8 sliceRefreshRowNumber;
    u8 flags;
    u16 displayId;
    u8 minQp;
    u8 maxQp;
} SLASetH264Parameters_t;
```

SightLine's H264 (AVC), H265 (HEVC) and MPEG4 compression encodes the video stream into I-frames and P-frames. I-Frames are a compression of the entire video frame, while P-Frames only include the delta between the video frame that was compressed in the last I-frame and the current video frame.

H264 and H265 compression can also use **Intra Refresh** mode to put blocks of I data in every frame.

See also

- [SLASetEthernetDisplayParameters_t](#) to set the receiver IP address.
- [SLASetKlvData_t](#) to set KLV metadata.
- [SLASetEthernetVideoParameters_t](#) to change Down Sample and Frame Step.
- [SLAStreamingControl_t](#) for starting and stopping video.
- [EAN-Encoding.pdf](#) for additional details on SightLine Ethernet encoding.

Message ID 0x23

Byte Offset	Name	Description
4-7	targetBitRate	Target bit rate, bits per second (default 3000000). Actual bit rate will vary (see flags below). NOTE: Maximum bitrates are:

		17XX/4000 - 30 Mbps																								
8	intraFrameInterval	1-255 = Intra Frame (I-frame) interval (default 30). The rate at which I-frames are produced. 0 = Intra Refresh mode where there are blocks of I data in every frame resulting in a smoother overall bitrate. Slower to recover if data packets are dropped. NOTE: Not all decoders can connect to an intra refresh stream.																								
9	IfDisableIdc	H264 in-loop filter control <table><tr><th>Bits</th><th>Description</th></tr><tr><td>0-3</td><td>0 = filter all edges (default), 1 = disable all filtering, 2 = disable slice edge filter, other values = <i>Reserved</i></td></tr><tr><td>4-7</td><td><i>Reserved</i> set to 0</td></tr></table>	Bits	Description	0-3	0 = filter all edges (default), 1 = disable all filtering, 2 = disable slice edge filter, other values = <i>Reserved</i>	4-7	<i>Reserved</i> set to 0																		
Bits	Description																									
0-3	0 = filter all edges (default), 1 = disable all filtering, 2 = disable slice edge filter, other values = <i>Reserved</i>																									
4-7	<i>Reserved</i> set to 0																									
10	airMbPeriod	Only used if intraFrameInterval is 0 (Intra Refresh mode) 17XX/4000 - Cyclic Intra Refresh (CIR) mega-block period, if 0 a default value of 20 will be used																								
11	sliceRefreshRowNumber	Only used if intraFrameInterval is 0 (Intra Refresh mode) 17XX/4000 - Not available																								
12	flags	<table><tr><th>Bits</th><th>Description</th></tr><tr><td rowspan="4">0-1</td><td>Profile (H.264)<table><tr><td>0</td><td>Baseline profile</td></tr><tr><td>1</td><td>Main profile</td></tr><tr><td>2</td><td>High profile (default)</td></tr></table></td></tr><tr><td>2-3</td><td><i>Reserved</i> set to 0</td></tr><tr><td rowspan="4">4-5</td><td>Bit Rate Mode (H.264/H.265 only, set to 0 otherwise)<table><tr><td>0</td><td>Legacy Approximates Variable Bit Rate (VBR) with loose constraints but a fairly short time constant. NOTE On 17XX same as VBR</td></tr><tr><td>1</td><td>Variable Bit Rate (VBR) (default). Bit rate dynamically changes based on the video content and motion in order to acheive high quality video.</td></tr><tr><td>2</td><td>Constrained Bit Rate Mode - Not available on 17XX/4000, try Intra Refresh mode (intraFrameInterval = 0).</td></tr><tr><td>3</td><td>Balanced Bit Rate Mode - Only available on 17XX. Goal is to better distribute data between I and P Frames than Variable/Legacy</td></tr></table></td></tr><tr><td>6-7</td><td><i>Reserved</i> set to 0</td></tr></table>	Bits	Description	0-1	Profile (H.264) <table><tr><td>0</td><td>Baseline profile</td></tr><tr><td>1</td><td>Main profile</td></tr><tr><td>2</td><td>High profile (default)</td></tr></table>	0	Baseline profile	1	Main profile	2	High profile (default)	2-3	<i>Reserved</i> set to 0	4-5	Bit Rate Mode (H.264/H.265 only, set to 0 otherwise) <table><tr><td>0</td><td>Legacy Approximates Variable Bit Rate (VBR) with loose constraints but a fairly short time constant. NOTE On 17XX same as VBR</td></tr><tr><td>1</td><td>Variable Bit Rate (VBR) (default). Bit rate dynamically changes based on the video content and motion in order to acheive high quality video.</td></tr><tr><td>2</td><td>Constrained Bit Rate Mode - Not available on 17XX/4000, try Intra Refresh mode (intraFrameInterval = 0).</td></tr><tr><td>3</td><td>Balanced Bit Rate Mode - Only available on 17XX. Goal is to better distribute data between I and P Frames than Variable/Legacy</td></tr></table>	0	Legacy Approximates Variable Bit Rate (VBR) with loose constraints but a fairly short time constant. NOTE On 17XX same as VBR	1	Variable Bit Rate (VBR) (default). Bit rate dynamically changes based on the video content and motion in order to acheive high quality video.	2	Constrained Bit Rate Mode - Not available on 17XX/4000, try Intra Refresh mode (intraFrameInterval = 0).	3	Balanced Bit Rate Mode - Only available on 17XX. Goal is to better distribute data between I and P Frames than Variable/Legacy	6-7	<i>Reserved</i> set to 0
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13-14	displayId	Network Display ID 4000/17XX: (0x0002 = Net0, 0x0080 = Net1, 0x0082 = both) 41XX: (0x0002 = Net0, 0x0080 = Net1, 0x0010 = Net2, 0x0092 = all)																								
		Minumum Quantization Parameter for I and P frames on H264. Should be between 1																								

15	minQp	and 30. 0 = default, minimum will be set based on platform and other encoding parameters. (H.264 only) New 3.7: Now supported for 1750 and 4100.
16	maxQp	Maximum Quantization Parameter for I and P frames on H264. The maximum allowed value is 51 and it must be larger than minimum QP. 0 = default, maximum will be set based on platform and other encoding parameters. (H.264 only) New 3.7: Now supported for 1750 and 4100.

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentImageSize_t Struct

Reference

[General](#)

Description

Current image size parameters.

```
typedef struct {
    s16 capWide;
    s16 capHigh;
    s16 disWide;
    s16 disHigh;
    s16 disRectCol;
    s16 disRectRow;
    s16 disRectWide;
    s16 disRectHigh;
    u8 cameraIndex;
} SLACurrentImageSize_t;
```

Sends the capture size and display size for the camera specified by **cameraIndex**. Sent in response to [SLAGetParameters_t\(\)](#).

17XX/4000 Display Size

The display width, height and rectangle will be 0 if the specified camera is not being displayed.

The display rectangle indicates the location in a multi (PiP and 2-Up) display where the specified camera is drawn. It will be all 0's if display is not in multi mode.

See also [SLAGetImageSize_t](#)

Message ID 0x4E

Byte Offset	Name	Description
4-5	capWide	Width of captured image
6-7	capHigh	Height of captured image
8-9	disWide	Width of display for the specified camera. 0 if not displaying. NOTE: Network size if displaying to Network plus Analog/HDMI/HDSDI.

10-11	disHigh	Height of display for the specified camera. 0 if not displaying
12-13	disRectCol	Display rectangle column offset
14-15	disRectRow	Display rectangle row offset
16-17	disRectWide	Display rectangle width
18-19	disRectHigh	Display rectangle heigh
20	cameraIndex	<u>Camera Index</u>

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentLensParameters_t Struct Reference

[Focus](#)

Description

Deprecated, use [SLALensParameters_t](#)

```
typedef struct {
    u8 lensType;
    u8 afMetricRegionSize;
    u8 zoomTrackFocus;
    u8 autofocusMethod;
    u16 autofocusRateAdjust;
    u8 autofocusChangePct;
    u8 zoomSpeed;
    u8 focusSpeed;
    u8 comPortNum;
    u8 colRoiFrac;
    u8 rowRoiFrac;
    u8 wideRoiFrac;
    u8 highRoiFrac;
} SLACurrentLensParameters\_t;
```

Deprecated, use [SLALensParameters_t](#)

Current lens parameters – format identical to [SLASetLensParameters_t](#).

Sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x6F

Byte Offset	Name	Description
4	lensType	Lens type. See SLASetLensParameters_t .
5	afMetricRegionSize	AutoFocus Metric Region Size % of screen (0 to 100)
6	zoomTrackFocus	Zoom Track Focus (0= disabled, 1= enabled)
7	autofocusMethod	Autofocus method (0=scan far to near, 1=seek)
8-9	autofocusRateAdjust	Autofocus Rate Adjust (0 to 255) 100 = default speed of motion
10	autofocusChangePct	Autofocus Change Percent (0 to 255) in 1/10ths of a percent. 40 (default) = 4%

11	zoomSpeed	Zoom speed (0=use default, 1=min 255=max)
12	focusSpeed	Focus speed (0=use default, 1=min 255=max)
13	comPortNum	Serial port number for lens control see <u>SLASetPortConfiguration_t</u>
14	colRoiFrac	Row ROI for focus stats. Row Offset in % of image width (255=100%); Full image(default) = 0
15	rowRoiFrac	Col ROI for focus stats. Column Offset in % of image height (255=100%); Full image(default) = 0
16	wideRoiFrac	High ROI for focus stats. Height in % of image height (255=100%); Full image(default) = 255
17	highRoiFrac	Wide ROI for focus stats. Width in % of image width (255=100%); Full image(default) = 255

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) [Camera Configuration](#)

[List of all members](#)

SLALensParameters_t Struct

Reference

[Focus](#)

Description

Lens parameters - control the lens.

```
typedef struct {
    u8 cameraIndex;
    u8 zoomTrackFocus;
    u32 reserved1;
    u16 zoomSpeed;
    u16 focusSpeed;
    u16 reserved2;
    u8 port;
    SVPLenString\_t filename;
} SLALensParameters\_t;
```

See [SLALensMode_t](#) for lens control and [SLAFocusParameters_t](#) for focus measurement control information. See [SLACustomAutoFocusParameters_t](#) for autofocus control information.

Focus metrics are reported using [SLAFocusStats_t](#).

See also

[SLASetOverlayMode_t](#).graphics to enable the onscreen [Overlays](#) for the region of interest.

File Naming Syntax

- All Lens Control DLLs will follow the same naming schema: *filename.so*
- Filename shall contain no spaces.
- Filename (including the file extension) shall be no more than 128 characters.
- The File shall be placed in the lens folder in the same directory as the VideoTrack application.
- Paths are *not* required when specifying the file name.
 - GOOD: MyLens02_03_04.so
 - BAD: MyLens02 03 04.so
 - BAD: ./MyLens02_03_04.so

Message ID 0xB1

Byte Offset	Name	Description
4	cameraIndex	<u>Camera Index</u>
5	zoomTrackFocus	Zoom Track Focus (0= disabled, 1= enabled). Stops when focus metric reaches peak threshold.
6-9	reserved1	<i>Reserved</i>
10-11	zoomSpeed	Zoom speed (0=use default, 1=min 65535=max. Real ranges are based on specific lens, driver should limit to valid values)
12-13	focusSpeed	Focus speed (0=use default, 1=min 65535=max. Real ranges are based on specific lens, driver should limit to valid values)
14-15	reserved2	Reserved
16	port	<u>Port Identifier</u> . Serial port ID for lens control. See <u>SLASetPortConfiguration_t</u>
17	filename.len	string length
18-...	filename.str	Name of the Lens Control DLL file.

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

[List of all members](#)

SLACurrentLensStatus_t Struct Reference

[Focus](#)

Description

Current lens focus and zoom position.

```
typedef struct {
    u16 focusPos;
    u16 zoomPos;
    u8 focusLimits;
    u8 zoomLimits;
    u8 camIdx;
    u8 customAFOn;
} SLACurrentLensStatus_t;
```

Sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x6D

Byte Offset	Name	Description												
4-5	focusPos	Focus position – motor pulse counts												
6-7	zoomPos	Zoom position – motor pulse counts												
8	focusLimits	focus mechanical and optical limit switch												
		<table><tr><th>Bit</th><th>Description</th></tr><tr><td>0</td><td>far mechanical limit</td></tr><tr><td>1</td><td>near mechanical limit</td></tr><tr><td>2</td><td>near optical limit</td></tr><tr><td>3</td><td>far optical limit</td></tr><tr><td>4:7</td><td>not used</td></tr></table>	Bit	Description	0	far mechanical limit	1	near mechanical limit	2	near optical limit	3	far optical limit	4:7	not used
		Bit	Description											
		0	far mechanical limit											
		1	near mechanical limit											
		2	near optical limit											
		3	far optical limit											
4:7	not used													
		zoom mechanical and optical limit switch												
		<table><tr><th>Bit</th><th>Description</th></tr><tr><td>0</td><td>narrow mechanical limit</td></tr><tr><td></td><td></td></tr></table>	Bit	Description	0	narrow mechanical limit								
		Bit	Description											
		0	narrow mechanical limit											

9	zoomLimits	1	wide mechanical limit
		2	wide optical limit
		3	narrow optical limit
		4:7	not used
10	camIdx	Camera index	
11	customAFOn	1 if custom autofocus is currently running, 0 otherwise.	

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentMetadataFrameValues_t

Struct Reference

[KLV Metadata](#)

Description

Current metadata frame values – format nearly identical to [SLASetMetadataFrameValues_t](#).

```
typedef struct {
    s32 frameCenterLat;
    s32 frameCenterLon;
    u16 frameCenterEl;
    u16 frameWidth;
    u32 slantRange;
    u8 userSuppliedFlags;
    s32 targetLat;
    s32 targetLon;
    u16 targetEl;
    u8 targetTrackGateHeight;
    u8 targetTrackGateWidth;
    s16 offsetCornerLat1;
    s16 offsetCornerLon1;
    s16 offsetCornerLat2;
    s16 offsetCornerLon2;
    s16 offsetCornerLat3;
    s16 offsetCornerLon3;
    s16 offsetCornerLat4;
    s16 offsetCornerLon4;
    u16 displayId;
} SLACurrentMetadataFrameValues\_t;
```

Sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x8C

Byte Offset	Name	Description
4-7	frameCenterLat	Frame center latitude (valid data bit = 0)
8-11	frameCenterLon	Frame center longitude (valid data bit = 1)
12-13	frameCenterEl	Frame center elevation (valid data bit = 2)
14-15	frameWidth	Target width (valid data bit = 3)
16-19	slantRange	Slant range (valid data bit = 4)
20	userSuppliedFlags	User-supplied metadata flags. See SLASetMetadataFrameValues_t .

21-24	targetLat	Target location specification mode
25-28	targetLon	Target location specification mode
29-30	targetEI	Target location specification mode
31	targetTrackGateHeight	Target track gate height
32	targetTrackGateWidth	Target track gate width
33-34	offsetCornerLat1	Offset corner latitude point 1
35-36	offsetCornerLon1	Offset corner longitude point 1
37-38	offsetCornerLat2	Offset corner latitude point 2
39-40	offsetCornerLon2	Offset corner longitude point 2
41-42	offsetCornerLat3	Offset corner latitude point 3
43-44	offsetCornerLon3	Offset corner longitude point 3
45-46	offsetCornerLat4	Offset corner latitude point 4
47-48	offsetCornerLon4	Offset corner longitude point 4
49-50	displayId	<u>Network Display ID</u> 4000/17XX: (0x0002 = Net0, 0x0080 = Net1) 41XX: (0x0002 = Net0, 0x0080 = Net1, 0x0010 = Net2)

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) [Camera Configuration](#)

[List of all members](#)

SLASetMetadataFrameValues_t

Struct Reference

[KLV Metadata](#)

Description

Sets new KLV metadata frame data values.

```
typedef struct {
    u16 validDataMask;
    s32 frameCenterLat;
    s32 frameCenterLon;
    u16 frameCenterEl;
    u16 frameWidth;
    u32 slantRange;
    u8 userSuppliedFlags;
    s32 targetLat;
    s32 targetLon;
    u16 targetEl;
    u8 targetTrackGateHeight;
    u8 targetTrackGateWidth;
    s16 offsetCornerLat1;
    s16 offsetCornerLon1;
    s16 offsetCornerLat2;
    s16 offsetCornerLon2;
    s16 offsetCornerLat3;
    s16 offsetCornerLon3;
    s16 offsetCornerLat4;
    s16 offsetCornerLon4;
    u16 displayId;
} SLASetMetadataFrameValues\_t;
```

Value are output as MPEG2-TS ([SLASetEthernetDisplayParameters_t](#)) and as part of the EXIF header in [SLASetSnapshot_t](#).

Set the rate at which these are transmitted using [SLASetMetadataRate_t](#).

The source of several data elements may be either automatically generated from other KLV data elements, or supplied by the user. With the user-supplied metadata flags (*userSuppliedFlags*) set to 0, *frameCenterLat*, *frameCenterLon*, *targetLat*, *targetLon*, *targetEl*, *targetTrackGateHeight*, *targetTrackGateWidth*, and offset corners are calculated using a flat-earth model utilizing platform angles and position, sensor field of view, and pixel track positions ([SLASetMetadataValues_t](#)).

When using user supplied metadata, make sure the appropriate value is set for the selector in [SLATagSourceSelector_t](#).

User-supplied metadata flags

Bits	Description
0	0 = calculate frame center lat/long from other metadata 1 = user-supplied frame center lat/long
1-2	Target location specification mode – 0 = calculate target information from other metadata 1 = user-supplied target metadata in MISB format 2 = user-supplied target row, column, height, width; target metadata calculated using these values. This mode provides a mechanism so that an encode-only system may be informed about target location in pixel coordinates, while still calculating MISB fields. 3 = Reserved
3	0 = calculate 4-corner metadata values 1 = user-supplied 4-corner metadata values
4	0 = default. 1 = user-supplied frame center elevation used in calculation of lat-lon.
5	0 = default. 1 = Use the Object Localization Service to calculate values.
6-7	Reserved , set to 0

Message ID 0x15

Byte Offset	Name	Description				
4-5	validDataMask	Valid data bit mask (see column). Update corresponding data element when bit value = 1.				
6-9	frameCenterLat	Frame center latitude (valid data bit = 0)				
10-13	frameCenterLon	Frame center longitude (valid data bit = 1)				
14-15	frameCenterEl	Frame center elevation (valid data bit = 2)				
16-17	frameWidth	Target width (valid data bit = 3)				
18-21	slantRange	Slant range (valid data bit = 4)				
22	userSuppliedFlags	User-supplied metadata flags				
23-26	targetLat	Target location specification mode				
		<table><tr><th>Mode</th><th>Use</th></tr><tr><td>0</td><td>field ignored</td></tr></table>	Mode	Use	0	field ignored
		Mode	Use			
0	field ignored					

		<table><tr><td>1</td><td>Target location latitude signed 32-bit integer in MISB format</td></tr><tr><td>2</td><td>Target location row unsigned 32-bit integer</td></tr></table>	1	Target location latitude signed 32-bit integer in MISB format	2	Target location row unsigned 32-bit integer				
1	Target location latitude signed 32-bit integer in MISB format									
2	Target location row unsigned 32-bit integer									
27-30	targetLon	Target location specification mode <table><tr><th>Mode</th><th>Use</th></tr><tr><td>0</td><td>field ignored</td></tr><tr><td>1</td><td>Target location longitude signed 32-bit integer in MISB format</td></tr><tr><td>2</td><td>Target location column unsigned 32-bit integer</td></tr></table>	Mode	Use	0	field ignored	1	Target location longitude signed 32-bit integer in MISB format	2	Target location column unsigned 32-bit integer
Mode	Use									
0	field ignored									
1	Target location longitude signed 32-bit integer in MISB format									
2	Target location column unsigned 32-bit integer									
31-32	targetEl	Target location specification mode <table><tr><th>Mode</th><th>Use</th></tr><tr><td>0,2</td><td>field ignored</td></tr><tr><td>1</td><td>Target location elevation</td></tr></table>	Mode	Use	0,2	field ignored	1	Target location elevation		
Mode	Use									
0,2	field ignored									
1	Target location elevation									
33	targetTrackGateHeight	Target track gate height								
34	targetTrackGateWidth	Target track gate width								
35-36	offsetCornerLat1	Offset corner latitude point 1								
37-38	offsetCornerLon1	Offset corner longitude point 1								
39-40	offsetCornerLat2	Offset corner latitude point 2								
41-42	offsetCornerLon2	Offset corner longitude point 2								
43-44	offsetCornerLat3	Offset corner latitude point 3								
45-46	offsetCornerLon3	Offset corner longitude point 3								
47-48	offsetCornerLat4	Offset corner latitude point 4								
49-50	offsetCornerLon4	Offset corner longitude point 4								
51-52	displayId	<u>Network Display ID</u> 4000/17XX: (0x0002 = Net0, 0x0080 = Net1, 0x0082 = both) 41XX: (0x0002 = Net0, 0x0080 = Net1, 0x0010 = Net2, 0x0092 = all)								

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentMetadataRate_t Struct Reference

[KLV Metadata](#)

Description

Current metadata rate – format similar to [SLASetMetadataRate_t](#).

```
typedef struct {
    u8 index;
    u8 frameStep;
    u16 displayId;
} SLACurrentMetadataRate_t;
```

Sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x8D

Byte Offset	Name	Description
4	index	Metadata element index – corresponds to Bit number in SLASetMetadataRate_t . enables
5	frameStep	Frame step at which to send the specified data values; 0 = sending disabled, 1 = send each frame, etc.
6-7	displayId	Network Display ID 4000/17XX: (0x0002 = Net0, 0x0080 = Net1) 41XX: (0x0002 = Net0, 0x0080 = Net1, 0x0010 = Net2)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetMetadataRate_t Struct Reference

[KLV Metadata](#)

Description

Deprecated, use [SLATagDataRate_t](#)

```
typedef struct {
    u64 enables;
    u8 frameStep;
    u16 displayId;
} SLASetMetadataRate_t;
```

Deprecated, use [SLATagDataRate_t](#)

Set the frame step rates at which KLV metadata is output.

Bit mask data enables

Bit	Tag	Field	Bit	Tag	Field
0	2	UTC time	26	48 / 6	Security: Releasing instructions
1	3	Mission ID	27	48 / 12	Security : Object country coding method
2	5	Platform heading angle	28	48 / 13	Security : Object country
3	6	Platform pitch angle	29	48 / 22	Security : Metadata version
4	7	Platform roll angle	30	65	UAS local set version
5	10	Platform designation	31	94	Motion imagery core identifier
6	11	Image source sensor	32	4	Platform tail number
7	13	Sensor latitude	33	26	Offset corner latitude point 1 (1)
8	14	Sensor longitude	34	27	Offset corner longitude point 1 (1)
9	15	Sensor true altitude MSL	35	28	Offset corner latitude point 2 (1)
10	16	Sensor horizontal field of view	36	29	Offset corner longitude point 2 (1)
11	17	Sensor vertical field of view	37	30	Offset corner latitude point 3 (1)
12	18	Sensor relative azimuth angle	38	31	Offset corner longitude point 3 (1)

13	19	Sensor relative elevation angle	39	32	Offset corner latitude point 4 (1)
14	20	Sensor relative roll angle	40	33	Offset corner longitude point 4 (1)
15	23	Frame center latitude	41	40	Target location latitude (2)
16	24	Frame center longitude	42	41	Target location longitude (2)
17	25	Frame center elevation MSL	43	42	Target location elevation (2)
18	22	Target width	44	43	Target track gate width (2)
19	21	Slant range	45	44	Target track gate height (2)
20	12	Image coordinate system	46	45	Target error estimate CE90
21	48 / 1	Security: Classification	47	46	Target error estimate LE90
22	48 / 2	Security : Classifying country coding method	48	47	Generic flag data 01
23	48 / 3	Security : Classifying country	49	59	Platform call sign
24	48 / 4	Security : SCI / SHI information	50	74	VMTI LDS targets (3)
25	48 / 5	Security : Caveats	51	100	Application Specific Metadata
–	–	–	52	75	Sensor true altitude HAE
–	–	–	53	78	Frame center elevation HAE

1. Values are calculated from platform angles, sensor angles, sensor position, frame center position, and slant range that are supplied by [SLASetMetadataValues_t](#) and [SLASetMetadataFrameValues_t](#).
2. Values are calculated from values supplied by [SLASetMetadataValues_t](#) and [SLASetMetadataFrameValues_t](#), and by internal tracking position and box size. Target elevation is assumed to be the same as frame center elevation, and target latitude and longitude are calculated in a manner similar to the four corners. Track gate width and height are in pixels, derived directly from target box size.
3. Values are calculated from internal tracking information. Multiple targets (either user-designated or automatically-generated) are indicated. Pixel coordinates are in 'display' coordinates suitable for directly rendering over streamed digital video.

Note

Per MISB STD 0601, version identifier fields for the UAS Local Data Set, Security Metadata Local Set, and Motion Imagery Track Metadata Local Set are emitted with elements of each local set. See also MISB ST 0902 for guidelines on setting the frame step for different fields.

Setting the frameStep (rate) to zero may make the resulting KLV metadata non compliant.

Message ID 0x62

Byte Offset	Name	Description
4-11	enables	Set rate bit mask. Update the rate for the corresponding data element when bit value = 1
12	frameStep	Frame step at which to send the specified data values; 0 = disable sending KLV metadata, 1 = send each frame, etc.
13-14	displayId	<u>Network Display ID</u> 4000/17XX: (0x0002 = Net0, 0x0080 = Net1, 0x0082 = both)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentMetadataValues_t Struct Reference

[KLV Metadata](#)

Description

Current metadata values – format nearly identical to [SLASetMetadataValues_t](#).

```
typedef struct {
    u64 utcTime;
    u16 heading;
    s16 pitch;
    s16 roll;
    s32 lat;
    s32 lon;
    u16 alt;
    u16 hfov;
    u16 vfov;
    u32 az;
    s32 el;
    u32 sensorRoll;
    u16 displayId;
} SLACurrentMetadataValues_t;
```

Sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x8B

Byte Offset	Name	Description
4-11	utcTime	UTC time (valid data bit = 0)
12-13	heading	Platform heading angle (valid data bit = 1)
14-15	pitch	Platform pitch angle signed (valid data bit = 2)
16-17	roll	Platform roll angle signed (valid data bit = 3)
18-21	lat	Sensor latitude (valid data bit = 4)
22-25	lon	Sensor longitude (valid data bit = 5)
26-27	alt	Sensor altitude (valid data bit = 6)
28-29	hfov	Sensor horizontal field of view (valid data bit = 7)
30-31	vfov	Sensor vertical field of view (valid data bit = 8)
32-35	az	Sensor relative azimuth angle (valid data bit = 9)

36-39	el	Sensor relative elevation angle (valid data bit = 10)
40-43	sensorRoll	Sensor relative roll angle (valid data bit = 11)
44-45	displayId	<u>Network Display ID</u> 4000/17XX: (0x0002 = Net0, 0x0080 = Net1) 41XX: (0x0002 = Net0, 0x0080 = Net1, 0x0010 = Net2)

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetMetadataValues_t Struct

Reference

[KLV Metadata](#)

Description

Set new KLV metadata data values.

```
typedef struct {
    u16 validDataMask;
    u64 utcTime;
    u16 heading;
    s16 pitch;
    s16 roll;
    s32 lat;
    s32 lon;
    u16 alt;
    u16 hfov;
    u16 vfov;
    u32 az;
    s32 el;
    u32 sensorRoll;
    u16 displayId;
} SLASetMetadataValues_t;
```

Value are output as MPEG2-TS ([SLASetEthernetDisplayParameters_t](#)) and as part of the EXIF header in [SLASetSnapShot_t](#).

Set the rate at which these are transmitted using [SLASetMetadataRate_t](#).

Message ID 0x13

Byte Offset	Name	Description
4-5	validDataMask	Valid data bit mask. Update corresponding data element when bit value = 1.
6-13	utcTime	UTC time (valid data bit = 0)
14-15	heading	Platform heading angle (valid data bit = 1)
16-17	pitch	Platform pitch angle (valid data bit = 2)
18-19	roll	Platform roll angle signed (valid data bit = 3)
20-23	lat	Sensor latitude (valid data bit = 4)
24-27	lon	Sensor longitude (valid data bit = 5)

28-29	alt	Sensor altitude (valid data bit = 6)
30-31	hfov	Sensor horizontal field of view (valid data bit = 7)
32-33	vfov	Sensor vertical field of view (valid data bit = 8)
34-37	az	Sensor relative azimuth angle (valid data bit = 9)
38-41	el	Sensor relative elevation angle (valid data bit = 10)
42-45	sensorRoll	Sensor relative roll angle (valid data bit = 11)
46-47	displayId	<u>Network Display ID</u> 4000/17XX: (0x0002 = Net0, 0x0080 = Net1, 0x0082 = both) 41XX: (0x0002 = Net0, 0x0080 = Net1, 0x0010 = Net2, 0x0092 = all)

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentMultipleAlignment_t

Struct Reference

[Blending](#)

Description

Deprecated, use [SLABlendAlign_t](#)

```
typedef struct {
    u8 nAlignments;
    struct {
        u8 vertical;
        u8 horizontal;
        u8 rotate;
        u8 zoom;
        u8 hzoom;
    } alignment[5];
} SLACurrentMultipleAlignment\_t;
```

Deprecated, use [SLABlendAlign_t](#)

Return the current values of all 5 alignment sets – format identical to [SLASetMultipleAlignment_t](#).

Sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x75

Byte Offset	Name	Description
4	nAlignments	Number of alignments present
varies	vertical	Vertical offset in pixels
varies	horizontal	Horizontal offset in pixels
varies	rotate	Rotation. See SLASetBlendParameters_t
varies	zoom	Zoom. See SLASetBlendParameters_t
varies	hzoom	Horizontal zoom scale. See SLASetBlendParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentNetworkList_t Struct

Reference

[List of all members](#)

[Network](#)

Description

List of all network interfaces controller names.

```
typedef struct {
    u8 numIds;
    struct {
        SVPEndString\_t iface;
    } iface[10];
} SLACurrentNetworkList\_t;
```

Sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x67

Byte Offset	Name	Description
4	numIds	Number of Network Interface Controllers (NICs)
varies	iface	interface name

Sightline Intelligence

Command, Control, and Script

API v3.11

[List of all members](#)

SLACurrentNetworkParameters_t

Struct Reference

[Network](#)

Description

Current network parameters – format identical to [SLASetNetworkParameters_t](#).

```
typedef struct {
    u8 mode;
    u32be ipAddr;
    u32be subnet;
    u32be gateway;
    u16be c2replyPort;
    u16be telemetryReplyPort;
    u8 modes;
    u8 index;
    u16be telemetryListenPort;
    u16be telemetryListenPort2;
    SVPLenString\_t hostName;
} SLACurrentNetworkParameters\_t;
```

Sent in response to [SLAGetNetworkParameters_t](#).

Message ID 0x49

Byte Offset	Name	Description
4	mode	0 = Use DHCP, 1 = Use specified Static IP address
5-8	ipAddr	IP address, if mode is static; 0 = No change
9-12	subnet	Subnet mask, if mode is static, 0 = No change
13-16	gateway	Gateway, if mode is static, 0 = No change
17-18	c2replyPort	Command and control reply port
19-20	telemetryReplyPort	<i>Reserved</i>
21	modes	<i>Reserved</i>
22	index	<i>Reserved</i>
23-24	telemetryListenPort	C2C command listen port, 0 indicates no change, default is 14001.
25-26	telemetryListenPort2	C2C command listen port2, 0 indicates no change, default is 14003.

27	hostName.len	string length
28-...	hostName.str	Host Name returned by Discover protocol.

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentOverlayMode_t Struct

Reference

[Overlays](#)

Description

Current overlay generation mode – format identical to [SLASetOverlayMode_t](#).

```
typedef struct {
    u8 primaryReticle;
    u8 secondaryReticle;
    u16 graphics;
    u8 mtiColor;
    u8 mtiSelectableColor;
    u8 cameraIndex;
    u8 selectedReticle;
    u8 personReticle;
    u8 cursorReticle;
    u8 lineThickness;
    u8 fontScale;
    u8 fontID;
    u8 extraLabels;
    u8 modernMode;
    u8 mtiReticle;
    u8 mtiLabelAdv;
    u8 mtiMultiColor;
    u8 detCounterColor;
    u8 mtiDisplayLabelLimit;
    u8 detCounterPos;
    u8 mtiLabelDeconflict;
} SLACurrentOverlayMode\_t;
```

Sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x42

Byte Offset	Name	Description
4	primaryReticle	SLASetOverlayMode_t
5	secondaryReticle	
6-7	graphics	
8	mtiColor	
9	mtiSelectableColor	
10	cameraIndex	

11	selectedReticle	
12	personReticle	
13	cursorReticle	
14	lineThickness	
15	fontScale	
16	fontID	
17	extraLabels	
18	modernMode	
19	mtiReticle	
20	mtiLabelAdv	
21	mtiMultiColor	
22	detCounterColor	
23	mtiDisplayLabelLimit	
24	detCounterPos	
25	mtiLabelDeconflict	

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetOverlayMode_t Struct Reference

[Overlays](#)

Description

Control display of overlay graphics.

```
typedef struct {
    u8 primaryReticle;
    u8 secondaryReticle;
    u16 graphics;
    u8 mtiColor;
    u8 mtiSelectableColor;
    u8 cameraIndex;
    u8 selectedReticle;
    u8 personReticle;
    u8 cursorReticle;
    u8 lineThickness;
    u8 fontScale;
    u8 fontID;
    u8 extraLabels;
    u8 modernMode;
    u8 mtiReticle;
    u8 mtiLabelAdv;
    u8 mtiMultiColor;
    u8 detCounterColor;
    u8 mtiDisplayLabelLimit;
    u8 detCounterPos;
    u8 mtiLabelDeconflict;
} SLASetOverlayMode\_t;
```

Command is applied to the Command Camera, or to the camera specified by [SLASetOverlayMode_t](#).cameraIndex. See [Draw Order](#) for additional drawing details.

Track Color Modes

Value	Description
0	Off
1	White (default)
2	Black
3	Auto(white or black)
4	Rainbow

5	Red
6	Orange
7	Yellow
8	Green
9	Blue
10	Violet
11	Cyan
12	BlackWhite (Black + white for contrast)

Track Reticle Type

Value	Description
0	Box corners (default)
1	Cross
2	Circle
3	Duplex cross-hair
4	Modern range
5	Target dot
6-15	<i>Reserved</i>

Note: Rotated reticles only apply to Box Corners (Default) overlay.

Graphic Overlays Selection

Enable and disable generation of specific overlay graphics.

Bit	0 = do not show, 1 = show as overlay
0..1	<i>Reserved</i> = 0
2	Overlay for SLATrackingBoxPixelStats_t
3	Histogram - Capture histogram green, Display histogram blue. Display histogram not supported on 17XX, 4000.
4	Track index
5	Track motion trails
6	Auto focus metric and ROI for SLASetLensParameters_t
7	Registration ignore edges for SLARegistrationParameters_t
8	MTI track index
9	Logo – Enabling this bit will insert an overlay on the video of a file that has been loaded by the user as 'userLogo.png'. To load, place the file in the user\AppData\Roaming\SightLineApplications directory, and use the firmware upgrade application to put the file on the SLA hardware. If it is not 640X150 it will not be downscaled by 2.
10	Overlay for SLALandingAid_t (turn landing aid overlays on/off)

11	Landing keep out
12	Overlay for SLADrawObject_t (turn user overlay objects on/off)
13	Draw track reticle dashed when coasting
14..15	<i>Reserved</i>

Message ID 0x06

Byte Offset	Name	Description				
4	primaryReticle	Color and shape of primary track overlay. <table><tr><td>Bits 0..3</td><td>Primary <u>track color mode</u></td></tr><tr><td>Bits 4..7</td><td>Primary <u>track reticle type</u></td></tr></table>	Bits 0..3	Primary <u>track color mode</u>	Bits 4..7	Primary <u>track reticle type</u>
Bits 0..3	Primary <u>track color mode</u>					
Bits 4..7	Primary <u>track reticle type</u>					
5	secondaryReticle	Color and shape of secondary track overlays. <table><tr><td>Bits 0..3</td><td>Secondary <u>track color mode</u></td></tr><tr><td>Bits 4..7</td><td>Secondary <u>track reticle type</u></td></tr></table>	Bits 0..3	Secondary <u>track color mode</u>	Bits 4..7	Secondary <u>track reticle type</u>
Bits 0..3	Secondary <u>track color mode</u>					
Bits 4..7	Secondary <u>track reticle type</u>					
6-7	graphics	<u>Graphic overlays selection</u>				
8	mtiColor	Color for detected tracks. <table><tr><td>Bits 0..3</td><td>MTI <u>track color mode</u>, BlackWhite is not supported</td></tr><tr><td>Bits 4..7</td><td><i>Reserved</i></td></tr></table>	Bits 0..3	MTI <u>track color mode</u> , BlackWhite is not supported	Bits 4..7	<i>Reserved</i>
Bits 0..3	MTI <u>track color mode</u> , BlackWhite is not supported					
Bits 4..7	<i>Reserved</i>					
9	mtiSelectableColor	Color for tracks which may be transitioned to primary tracks. <table><tr><td>Bits 0..3</td><td>MTI selectable <u>track color mode</u>, BlackWhite is not supported</td></tr><tr><td>Bits 4..7</td><td><i>Reserved</i></td></tr></table>	Bits 0..3	MTI selectable <u>track color mode</u> , BlackWhite is not supported	Bits 4..7	<i>Reserved</i>
Bits 0..3	MTI selectable <u>track color mode</u> , BlackWhite is not supported					
Bits 4..7	<i>Reserved</i>					
10	cameraIndex	<u>Camera Index</u> 255 = all cameras. If not present, applies to all cameras.				
11	selectedReticle	Color and shape of selected track overlay. If not present secondaryReticle will be used for selected track. <table><tr><td>Bits 0..3</td><td>Selected <u>track color mode</u></td></tr><tr><td>Bits 4..7</td><td>Selected <u>track reticle type</u></td></tr></table>	Bits 0..3	Selected <u>track color mode</u>	Bits 4..7	Selected <u>track reticle type</u>
Bits 0..3	Selected <u>track color mode</u>					
Bits 4..7	Selected <u>track reticle type</u>					
12	personReticle	Shape of track overlay when in person tracking mode. If not present default person reticle will be used <table><tr><td>Bits 0..3</td><td>Person mode primary <u>track color mode</u>, default is cyan</td></tr><tr><td>Bits 4..7</td><td>Reserved - reticle used will match type set in Primary, Secondary, Selected</td></tr></table>	Bits 0..3	Person mode primary <u>track color mode</u> , default is cyan	Bits 4..7	Reserved - reticle used will match type set in Primary, Secondary, Selected
Bits 0..3	Person mode primary <u>track color mode</u> , default is cyan					
Bits 4..7	Reserved - reticle used will match type set in Primary, Secondary, Selected					
13	cursorReticle	Color of the cursor. If not present default cursor will be used <table><tr><td>Bits 0..3</td><td>Cursor <u>color mode</u>, default is white</td></tr><tr><td>Bits 4..7</td><td>Reserved - shape is standard cross-hair cursor</td></tr></table>	Bits 0..3	Cursor <u>color mode</u> , default is white	Bits 4..7	Reserved - shape is standard cross-hair cursor
Bits 0..3	Cursor <u>color mode</u> , default is white					
Bits 4..7	Reserved - shape is standard cross-hair cursor					

14	lineThickness	Thickness of lines used for reticles, cursors, etc. Default = 1.	
15	fontScale	Font scaling is used for all internal text rendering, for example this applies to drawing track index, focus metrics, track pixel stats, etc. This does not apply to user overlays. Scaling is calculated as fontScale/32. 32 is 100% scale, 0 is also treated as 100%. Default = 32.	
16	fontID	Font ID for non-user objects, such as track ID etc. follows the same <u>Font IDs as fonts used by the user for text strings</u> .	
17	extraLabels	Extra Labels for classifier	
		Bit 0	Turn on Classifier Track Labels and confidence.
		Bit 1	Turn on Classifier MTI Track labels and confidence.
		Bit 2	Turn on overlay that displays total objects by class.
		Bit 3	Limit MTI Track to labels only. Must enable Bit 1.
		Bit 4	Add Tracker Reticle indicators (AI assist and track mode).
17	extraLabels	Bit 7	Reduces long label and PiP label names to a maximum of 10 character by removing vowels starting from the right side.
18	modernMode	Set as 1 to make certain overlays have a more modern look. These overlays include the track index, track pixel stats, autofocus and PiP labels. When there is no parameter file, this will be enabled.	
19	mtiReticle	Shape of the MTI reticles	
		Bit 0	Ellipse (Default)
		Bit 1	Ellipse with letter for classification label
		Bit 2	Inverted triangle
		Bit 3	Inverted triangle with letter for classification label
		Bit 4	Rectangle
19	mtiReticle	Bit 5	Rectangle with letter for classification label
20	mtiLabelAdv	Mti index and confidence label position and display options	
		Bits 0..3	Label position. 0 = auto (Default), 1 = top right, 2 = bottom left, 3 = bottom right
		Bits 4..7	Limit number of classification labels displayed. 0 = no limit
21	mtiMultiColor	Mti index and confidence label multi-pallete color. 0 = none (use mtiColor), 1 = Palette A Borders, 2 = Palette B Borders, 3 = Drone Palette Borders, 4 = Classification ID Palette A Borders, 5 = Classification ID Palette B Borders, 6 = Palette A Full Color, 2 = Palette B Full Color, 3 = Drone Palette Full Color, 4 = Classification ID Palette A Full Color, 5 =	

		Classification ID Palette B Full Color
22	detCounterColor	Sets the color of the visible detection counter overlay. 0 = none (use mtiColor), 1 = Palette A Borders, 2 = Palette B Borders, 3 = Palette A Full Color, 4 = Palette B Full Color
23	mtiDisplayLabelLimit	Limits the number of detection labels displayed. Default = 255
24	detCounterPos	Change the position of the detection counter. Label position. 0 = top left (Default), 1 = top right, 2 = bottom left, 3 = bottom right
25	mtiLabelDeconflict	Mti label de-conflicting. 0 = none (do not de-conflict labels), 1 = auto (de-conflict along shortest axis overlap), 2 = horizontal (de-conflict labels horizontally), 3 = vertical (de-conflict labels vertically)

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentOverlayObjectParameters_t Struct Reference

[Overlays](#)

Description

Properties of a particular graphic overlay specified by its Object ID.

```
typedef struct {
    u8 type;
    u8 id;
    u8 flags;
    u8 staticObject;
    u16 a;
    u16 b;
    u16 c;
    u16 d;
    u8 color;
    SVPEndString\_t text;
} SLACurrentOverlayObjectParameters\_t;
```

Sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x6B

Byte Offset	Name	Description
4	type	Object type – see SLADrawObject_t.type
5	id	Object ID – unique index assigned at creation
6	flags	See description of propertyFlags within SLADrawObject_t
7	staticObject	Used for static overlays such as text, doesn't move with camera.
8-9	a	See description of object parameters within SLADrawObject_t
10-11	b	
12-13	c	
14-15	d	
16	color	See backgroundColor in SLADrawObject_t
17	text	Text string

Sightline Intelligence

Command, Control, and Script

API v3.11

[List of all members](#)

SLACurrentOverlayObjectsIds_t

Struct Reference

[Overlays](#)

Description

A list of used object ids as a 255 long bit array.

```
typedef struct {
    u64 id0;
    u64 id64;
    u64 id128;
    u64 id192;
} SLACurrentOverlayObjectsIds_t;
```

Sent in response to [SLAGetParameters_t\(\)](#).
Use this information to request the properties of a particular object ID. '1' means it is drawn, '0' means it is not.

Message ID 0x68

Byte Offset	Name	Description
4-11	id0	Object IDs 1 through 63
12-19	id64	Object IDs 64 through 127
20-27	id128	Object IDs 128 through 191
28-35	id192	Object IDs 192 through 255

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentPortConfiguration_t

Struct Reference

[Serial Port](#)

Description

Current I/O port configuration parameters – format identical to [SLASetPortConfiguration_t](#).

```
typedef struct {
    u8 port;
    u8 baud;
    u8 dataBits;
    u8 stopBits;
    u8 parity;
    u8 maxPacket;
    u8 maxDelay;
    u8 protocol;
    u16 inputPort;
    u32be udpDestinationAddr;
    u16 udpDestinationPort;
    u16 udpAttNavPort;
} SLACurrentPortConfiguration\_t;
```

Sent in response to [SLAGetPortConfiguration_t\(\)](#).

Message ID 0x53

Byte Offset	Name	Description
4	port	See SLASetPortConfiguration_t
5	baud	
6	dataBits	
7	stopBits	
8	parity	
9	maxPacket	
10	maxDelay	
11	protocol	
12-13	inputPort	
14-17	udpDestinationAddr	

18-19	udpDestinationPort	
20-21	udpAttNavPort	

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentRegistrationParameters_t

Struct Reference

Description

Deprecated, use [SLARegistrationParameters_t](#)

```
typedef struct {
    u16 maxTranslation;
    u8 maxRotation;
    u8 zoomRange;
    u8 left;
    u8 right;
    u8 top;
    u8 bottom;
    u8 cameraIndex;
} SLACurrentRegistrationParameters_t;
```

Deprecated, use [SLARegistrationParameters_t](#)

Current registration parameters – format identical to [SLASetRegistrationParameters_t](#)

Sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x45

Byte Offset	Name	Description
4-5	maxTranslation	See SLASetRegistrationParameters_t
6	maxRotation	
7	zoomRange	
8	left	
9	right	
10	top	
11	bottom	
12	cameraIndex	

Sightline Intelligence

Command, Control, and Script

API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) ■ [Camera Configuration](#)

[List of all members](#)

SLARegistrationParameters_t Struct Reference

[Stabilization](#)

Description

Parameters used in frame to frame image registration matching.

```
typedef struct {
    u8 cameraIndex;
    u16 maxTranslation;
    u8 maxRotation;
    u8 zoomRange;
    u16 left;
    u16 right;
    u16 top;
    u16 bottom;
    u8 updateRate;
    u8 flags;
} SLARegistrationParameters_t;
```

Registration matches frame to frame motion in offset, rotation and zoom. It is a critical first step for stabilization, tracking and detection. If poor performance in stabilization, tracking or detection is observed, it is often due to registration failures. The most common reasons for registration failure along with possible solutions are:

- Black edges in the captured image. Adjust ROI in [SLASetAcquisitionParameters_t](#), or use Ignore Edge Pixels below.
- Large amounts of motion blur in the image. Use a shorter integration time camera or mechanical dampening.
- Zooming in to where a moving foreground object takes up a large fraction of the screen. Zoom out.

Ignore Edge Pixels

Used for overlays or foreground objects that appear near the edge of the image. 0 is default. At least 1/4 of the smaller dimension of the image must be remaining. For NTSC, remaining non - ignored image must be at least 120x120. Use the 'ignore edge pixel band' to indicate that the registration algorithm should not include these pixels when determining the registration match. This can be used to compensate for effects of the optics such as vignetting or when there is an obstruction along an edge of the image. See [SLASetOverlayMode_t](#) to enable onscreen graphics to better select ignore edge bands.

To get values: [SLAGetParameters_t](#), id=0x9E, payload1=[CameraIndex]. Returns [SLARegistrationParameters_t](#).

Byte Offset	Name	Description						
4	cameraIndex	<u>Camera Index</u>						
5-6	maxTranslation	Maximum translation in pixels per frame. Default of 0 is equivalent to 120 for a 480 high image (1/4 of the image height). NOTE: If you know that the maximum frame to frame translation is small, you can get faster speed with a smaller value here. Be careful not to limit it too much!						
7	maxRotation	Maximum rotation range in degrees per frame: 0..10 (values larger than 10 clipped to 10). 5 is default.. NOTE: 5 is a good value for an airborne camera, it is rare to need more than 5 degrees per frame. 0 executes significantly faster and is a good value for a fixed mount camera that does not experience roll.						
8	zoomRange	Maximum zoom range in percent zoom per frame: 0..10. 0 is default. NOTE: This parameter should almost always be 0. Small values may help in limited cases to maintain registration and tracking while a camera is actively zooming. Enabling this parameter when a camera is not zooming will make registration run slower with less accuracy.						
9-10	left	Left band of edge pixels to ignore. (0 is default)						
11-12	right	Right band of edge pixels to ignore. (0 is default)						
13-14	top	Top band of edge pixels to ignore. (0 is default)						
15-16	bottom	Bottom band of edge pixels to ignore. (0 is default)						
17	updateRate	Percentage to update the model each frame. 0 to 100%. (10 is default) 100: recommended for moving cameras. (Stationary Camera Low Drift unchecked in Panel Plus) 10: for ground mounted cameras desiring low drift scene lock and Staring detection mode. (Stationary Camera Low Drift checked in Panel Plus)						
18	flags	Registration Flags. <table><tr><th>Bits</th><th>Description</th></tr><tr><td>0-2</td><td>Registration mode 0 = Default - software based registration 1 = Reserved 2 = Use passed in registration, software based registration will be skipped. (Only supported with Library. Not supported on 17XX, 4000) 3-7 = Reserved</td></tr><tr><td>3-7</td><td>Reserved</td></tr></table>	Bits	Description	0-2	Registration mode 0 = Default - software based registration 1 = Reserved 2 = Use passed in registration, software based registration will be skipped. (Only supported with Library. Not supported on 17XX, 4000) 3-7 = Reserved	3-7	Reserved
Bits	Description							
0-2	Registration mode 0 = Default - software based registration 1 = Reserved 2 = Use passed in registration, software based registration will be skipped. (Only supported with Library. Not supported on 17XX, 4000) 3-7 = Reserved							
3-7	Reserved							

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentSDCardDirectoryInfo_t

Struct Reference

[Recording](#)

Description

A list of file names in the current recording directory.

```
typedef struct {
    ul6 nFiles;
    ul6 startIndex;
    ul6 endIndex;
} SLACurrentSDCardDirectoryInfo_t;
```

Sent in response to [SLASetSDRecordingParameters_t](#) command.
See [SLACurrentSDCardRecordingStatus_t](#) to check if SD Card is present

```
sent:SetSDRecordingParameters 51,AC,0C,1E,00,00,00,01,00,00,00,00,00,00,3F
```

Based on the number of file names that need to be sent, multiple packets (groups) might be needed.
In the event that the number of bytes for the file names forces the packet to be greater than 127 bytes, then this message will be an "extended packet" and the byte offsets below will be shifted by +1.

Example SLASetSDRecordingParameters Packet

```
51,AC,[D0,05],59,23,00,00,00,22,00,01,02,18,18,15,... [extended length]
51,AC,D0,05,[59],23,00,00,00,22,00,01,02,18,18,15,... [message ID]
51,AC,D0,05,59,[23,00],00,00,22,00,01,02,18,18,15,... [number of files]
51,AC,D0,05,59,23,00,[00,00],22,00,01,02,18,18,15,... [index of first file name in this packet]
51,AC,D0,05,59,23,00,00,00,[22,00],01,02,18,18,15,... [index of last file name in this packet]
51,AC,D0,05,59,23,00,00,00,[22,00],01,02,18,18,15,... [index of last file name in this packet]
51,AC,D0,05,59,23,00,00,00,22,00,[01,02,18,18,15,...] [lengths of the file names that will follow]
```

Note
A maximum of 1024 file names will be returned.

Message ID 0x59

Byte Offset	Name	Description
4-5	nFiles	Total number of files

6-7	startIndex	Start index in this packet
8-9	endIndex	End index in this packet; after endIndex are $N = 1 + \text{endIndex} - \text{startIndex}$ filename lengths followed by N null-terminated filename strings.

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentSnapShot_t Struct Reference

[Recording](#)

Description

Current snapshot state.

```
typedef struct {
    u8 mode;
    u8 format;
    u8 source;
    u8 quality;
    u8 downsample;
    u32be ipAddr;
    u16be port;
    SVPLenString\_t userName;
    SVPLenString\_t password;
    u8 frameStep;
    u8 numFrames;
    SVPLenString\_t fileName;
    u8 secondaryFormat;
    u8 cameraIndex;
} SLACurrentSnapShot\_t;
```

Sent in response to [SLAGetParameters_t\(\)](#).

See [SLASetSnapShot_t](#)

Message ID 0x5D

Byte Offset	Name	Description
4	mode	Snapshot destination – 0 = FTP server, 1 = Micro SD
5	format	File format – 0 = JPEG, 1 = PNG, 2 = JPEG Big, 3 = SLRaw (16-bit grey, 10 bit bayer only), 4 = TIFF (16-bit grey or 8-bit grey) If source is set to BOTH, this is the file format for the capture image
6	source	snapshot source: 1 = captured image, 2 = displayed image, 3 = both (requires secondaryFormat to be set for displayed image)
7	quality	JPEG compression quality level 0 to 100
8	downsample	Down Sample – 1: none, 2: 2x2 downsample, 4: 4x4 downsample

9-12	ipAddr	IP network address of FTP server
13-14	port	FTP server command port (default 21)
15	userName.len	string length
16-...	userName.str	FTP server user name
varies	password.len	string length
varies	password.str	FTP server password
varies	frameStep	Frame step at which to take snapshots
varies	numFrames	Number of snapshots to take
varies	fileName.len	string length
varies	fileName.str	Base file name for snapshots – names will save to <fileName>_N where N is an incrementing count, starting at 0
varies	secondaryFormat	File format for display snap if source is set to BOTH – 0 = JPEG (default), 1 = PNG
varies	cameraIndex	<u>Camera Index</u>

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentStabilizationBias_t Struct Reference

Description

Deprecated, use [SLAStabilizationBias_t](#)

```
typedef struct {
    s8 biasCol;
    s8 biasRow;
    u8 autoBias;
    u8 updateRate;
    u8 cameraIndex;
} SLACurrentStabilizationBias\_t;
```

Deprecated, use [SLAStabilizationBias_t](#)

Return the current stabilization bias parameters – format identical to [SLASetStabilizationBias_t](#).

Sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x7A

Byte Offset	Name	Description
4	biasCol	See SLASetStabilizationBias_t
5	biasRow	
6	autoBias	
7	updateRate	
8	cameraIndex	

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAStabilizationBias_t Struct Reference

[Stabilization](#)

Description

Adjust the stabilization solution by adding a constant bias in pixel coordinate space to the current coordinates each frame.

```
typedef struct {
    u8 cameraIndex;
    s16 biasCol;
    s16 biasRow;
    u8 autoBias;
    u8 updateRate;
} SLAStabilizationBias_t;
```

Stabilization bias is used to feed forward user controlled camera motion so that stabilization does not 'fight' against camera pan and tilt. Set 'auto bias' mode to automatically prevent the system from stabilizing against constant motion. Some amount of motion lag will still be experienced in 'auto bias' mode. Manual bias offsets may be used together with 'auto bias' mode. Changes to the column and row bias values will be added to the 'auto bias' solution. In 'auto bias' mode, set column and row bias to 0 for 'auto bias' only. Bias offsets may need to be scaled up as the operational frame rate decreases.

NOTE: Applies to offset stabilization only, not rotation.

See also

- [SLAResetStabilizationParameters_t](#) to reset only the auto-bias filter.
- [SLASetStabilizationParameters_t](#) for other stabilization control.

To get values: [SLAGetParameters_t](#), id=0x9F, payload1=[CameraIndex]. Returns [SLAStabilizationBias_t](#).

Message ID 0x9F

Byte Offset	Name	Description
4	cameraIndex	Camera Index 255 = All cameras.
5-6	biasCol	Per frame column adjustment (bias) in pixels (s16 value, typically within +- 1/2 image size)
7-8	biasRow	Per frame row adjustment (bias) in pixels (s16 value, typically within +- 1/2 image size)

9	autoBias	1 = Enable auto bias (combined auto + manual bias) 0 = Disable auto bias(manual bias only)
10	updateRate	Auto bias update rate 0..255 (default = 50) (Internally divided by 500)

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentStabilizationParameters_t

Struct Reference

[Stabilization](#)

Description

Describes the current stabilization mode.

```
typedef struct {
    u8 mode;
    u8 rate;
    u8 translationLimit;
    u8 angleLimit;
    u8 cameraIndex;
    u8 maxStabOff;
    u8 edgeY;
    u8 edgeU;
    u8 edgeV;
} SLACurrentStabilizationParameters_t;
```

Sent in response to [SLAGetParameters_t\(\)](#).
Same as [SLASetStabilizationParameters_t](#) with exceptions noted below
See also [SLAStabilizationBias_t](#).

Message ID 0x41

Byte Offset	Name	Description				
4	mode	Same as: <u>SLASetStabilizationParameters_t</u> with the following extra bit used:				
		<table><tr><th>Bit</th><th>Description</th></tr><tr><td>3</td><td>0 = Auto bias disabled, 1 = Auto bias enabled</td></tr></table>	Bit	Description	3	0 = Auto bias disabled, 1 = Auto bias enabled
		Bit	Description			
3	0 = Auto bias disabled, 1 = Auto bias enabled					
5	rate					
6	translationLimit					
7	angleLimit					
8	cameraIndex					
9	maxStabOff					
10	edgeY					

11	edgeU	
12	edgeV	

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentSystemValue_t Struct

Reference

[General](#)

Description

Current SystemValue – format identical to [SLASetSystemValue_t](#).

```
typedef struct {
    u32 systemValue;
    u32 setMode;
    s32 value0;
    s32 value1;
    s32 value2;
    s32 value3;
} SLACurrentSystemValue_t;
```

Sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x93

Byte Offset	Name	Description
4-7	systemValue	See SLASetSystemValue_t .
8-11	setMode	
12-15	value0	
16-19	value1	
20-23	value2	
24-27	value3	

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetSystemValue_t Struct

Reference

[General](#)

Description

Configure Special System Operational Values or Modes.

```
typedef struct {
    u32 systemValue;
    u32 setMode;
    s32 value0;
    s32 value1;
    s32 value2;
    s32 value3;
} SLASetSystemValue_t;
```

This function is used to customize or fine tune the system behavior by setting SystemValue. Most of these SystemValues are internally used by SightLine and users usually do not/should not need to use this function. Setting incorrect values may cause the system to crash. **value0** must be present for all SystemValue types. Some SystemValues may use additional arguments **value1..value3**.

System Value Type

systemValue	Name	Persist	Description
0	Reserved	-	-
1	Demo Mode	No	Sending this command puts the system into Demo Mode . value0 = 1 to enter demo mode value1 to value3 are ignored, set to 0. Restart system to get out of demo mode.
2	Reserved	-	-
3	Focus Metric	No	NOTE: Deprecated, use SLAFocusParameters_t
4	KLV Modes	Yes	value0: Deprecated value1: 0 = Use Platform Pitch Angle (Tag ID 6) & Platform Roll Angle (Tag ID 7) for calculation (default) 1 = Use Platform Pitch Angle Full (Tag ID 90) & Platform Roll Angle Full (Tag ID 91)

5	<i>Reserved</i>	-	-
6	<i>Reserved</i>	-	-
7	Laser Color	Yes	value0: UV color displayed for high bit laser camera display. LSB = U, MSB = V. Examples: 0xF040 = red (default), 0x1010 = green, 0x10F0 = teal, F0F0 = magenta See <u>SLASetAcquisitionParameters_t</u>
8	Gas Color	Yes	value0: UV color displayed for gas enhance/detect. LSB = U, MSB = V. 0x1010 = green (default), 0xF010 = orange, 0x10F0 = teal, 0xF0F0 = magenta, 0x8080 = no color value1: Add source * 256. 0: Show only difference, 128: add half of source intensity to difference, 255: add full source intensity to difference. NOTE: Starting in 3.7, use <i>detBitDepth</i> in <u>SLASetAdvancedDetectionParameters_t</u> to select 8 or 16 bit input image See <u>SLASetDetectionParameters_t</u>
9	CPU Frequency 0	Yes	4000 core 0-1 Only value0 (governor type): 0=default, 1=performance, 4=on demand value1: min MHz value2: max MHz, if min=max=0 then they are ignored value3: current MHz (read only) 17XX core 0-3 value0 (governor type): 0=default, 1=performance, 4=on demand value1: min MHz (valid options 1200, 1600 (default)) value2: max MHz (valid options 1200, 1600 (default)), if min=max=0 then they are ignored value3: current MHz (read only)
10	NUC Reset Pixel	No	Reset the gain and offset in the nuc table for a specific pixel value0: Camera index value1: Column of the pixel to be reset value2: Row of the pixel to be reset
11	CPU Frequency 2	Yes	4000 core 2-3 Only value0 (governor type): 0=default, 1=performance, 4=on demand value1: min MHz value2: max MHz, if min=max=0 then they are ignored value3: current MHz (read only)
12	<i>Reserved</i>	-	-
13	Linux Traffic Control	No	value0: Maximum rate in kilobits per second. value1: Burst in bytes. value2: MTU in bytes. Note: 17XX: If MTU is changed the video and communications may drop for 7-8 seconds as network is restarted Send 0 as any value to reset to No Traffic Control. Note: Wait at least 1 second between consecutive sending of this mode
			Split Cam 0 video which contains two video streams on alternating frames into Cam 0 and Cam 1 based on the value in one pixel. Requires setting Cam 1 as

14	Split Cam	Yes	Duplicate of Cam 0. value0: 0=disabled (default), 1=pixel value <> Threshold determines Cam 0/1, 2=alternating (for test) value1: Column of pixel to compare to threshold (from upper left) value2: Row of pixel to compare to threshold (from upper left) value3: Threshold 0->128 (default), Pixel(col,row) < Threshold -> cam 0, Pixel(col,row) >= Threshold -> cam 1
15	I2C Register Set	No	value0: i2c bus value1: device address on the i2c bus value2: 16bit register address followed by 16 bit value (as 32bits total). Example: 0x01342200 is address 0x0134 and value 0x2200 value3: number of bytes to write (either 3 or 4). 2 bytes for the address and 1 or 2 bytes for value to device.
16	System Up Time	No	Returns the system up time. value0 System uptime in minutes value1 Seconds portion of system uptime
17	Debug	No	<i>Internal use only</i>
18	Customer Use	Yes	This system value is to be used by customers to execute custom functionality in a LUA script or similar.
19	Customer Use 2	Yes	This system value is to be used by customers to execute custom functionality in a LUA script or similar.
20	User Warning Destination	Yes	value0: Destination(s) of user warning messages (see SLAUserWarningMessage_t), bit ORed value (1 << Port-ID, (see SLACommandPassThrough_t destPort), default is 0x04 (1<<2, Ethernet C&C reply port).
21	HTTP Debug Port	yes	HTTP debug port number: value0: TCP port number used as HTTP port, 0=disable server (default), 9090 is recommended. 80 should be avoided unless the default HTTP server is disabled on the system. To access the server from a Web Browser: IP-address:9090 (ex. 192.168.1.3:9090). value1 to value3: reserved and must be 0. A reboot is required to change the port number, disabling server works immediately.
22	I2C configuration	Yes	4xxx Only (advanced) value0: Change the i2c bus clock speed. Check Persistence, save parameters and reboot for this change to take effect. 0 = set i2c bus 1 and 3 to 400Khz (default) 0x0300 = set i2c bus 1 to 1MHz 0x03000000 = set i1c bus 3 to 1MHz 0x03000300 = set i2c bus 1 and 3 to 1Mhz value3 (4110 Only): Change the i2c bus clock speed. Check Persistence, save parameters and reboot for this change to take effect. 0 = set i2c bus 14 400Khz (default)

			0x030000 = set i2c bus 14 to 1MHz Note: Some camera I/F boards won't work at 1MHz. Please contact Technical Support for more details.
23	OGL Conf	Yes	value0: Max number of characters in a User Font to be displayed. Characters in User Fonts are cached in memory so that they can be rendered quickly. This value configures the max number of characters to be cached for a single User Font. Square of this number is the actual max number of characters to be cached, default is 16 (0..15 is also treated as 16) which results in max 256 characters. Some languages may require more than 256 characters, and you will need to increase value0 in that case. Need to check Persistence, save parameter, then reboot for the change to take effect. value1 to value3: reserved and must be 0.
24	Net1 Zoom10	Yes	value0: Network 1 zoom level x 1024, if 0 use camera zoom as normal. Only valid when same camera to both network 0 and network 1.
25	GPU Temp Workaround	Yes	4000 Only The GPU on the 4000 may experience faults at core temperatures between about 7°C (45°F) and 20°C (68°F) This feature works around this issue by using the CPU instead of the GPU for rendering in a specified temperature range value0: high-temperature - if core temperature <= this value in C, then use CPU, this feature is disabled if 0 value1: low-temperature - if temperature >= this value in C, then use CPU, ignored if 0 value2: settle time in seconds before switching back to GPU, if 0 then the internal default is used value3: lower 16 bits = skip rendering in CPU mode when >=2, 2=skips every other frame, 3=skips 2 out of 3 frames, etc bits 20..23 = number of warmup threads (max 4). Warmup thread help to warm up 4000 to get out of the critical temperature zone. Other bits must be 0.
26	Classifier	Yes	Each 32 bit value is per camera [0..3] bits 0..7 (Reporting Type) 0 = Simplified Report (default), 1 = Full Report. Simple report shows only drone or background, full report includes background, drone, fixed wing drone, or vehicle. bits 8..15 (Chip Padding) 0 = Default, 1 = No Padding, 2 = 2X Padding. Other values ignored. bits 16..23 (Rate8) 0 = Default to 75, 1 = minimum rate, 255 = maximum rate, 75 = default. bits 24..31 reserved.
27	No Video Timer	Yes	Control the no video screen shown when a camera is unplugged or incorrectly configured so that it can't capture video. value0: Frames per second to send the no video screen. Default and minimum 1 frame per second, maximum 30 frames per second. NOTE: VLC display may require at least 6 frames per second.

			value1: Contents of no video screen. 0 = default X screen with text, 1 = Black screen, 2 = Splash screen (see EAN-Digital-Overlay-Graphics.pdf .)
28	Track Box Size Reporting	Yes	valueN: Track box size reporting for Camera index N in drone mode. 0 = Default, 1 = Average of estimates, 2 = Current estimate
29	Auto restart VideoTrack	Yes	value0: 0=don't restart VideoTrack, 1=always restart, 2=(default) only when terminated abnormally.
30	Power Save	No	17XX only value0: 1=enter power save mode, restart when a character is received on the debug serial port, ttyMXC1.
31	Blob Control	Yes	Suppresses clumping of blobs into a larger blob: values: 0(Default) = Clump , 1 = Do not clump (suppress). values are per camera i.e. Value0 controls camera0 etc.
32	Bitrate Warnings	Yes	Provides user warnings when bitrate deviates from target bitrate beyond thresholds. value0: Net0, Bits 0..7, Seconds per warning when bitrate exceeds threshold. Rounds to even values. (0 = No Warnings) value0: Net0, Bits 8..31, Reserved value1: Net0, Bits 0..15, Lower bitrate warning threshold percent (0-100%). The warning is triggered. value1: Net0, Bits 15..31, Upper bitrate warning threshold percent (100+%). Bitrate values above this trigger the warning. value2: Net1, Bits 0..7, Seconds per warning when bitrate exceeds threshold. Rounds to even values. (0 = No Warnings) value2: Net1, Bits 8..31, Reserved value3: Net1, Bits 0..15, Lower bitrate warning threshold percent (0-100%). The warning is triggered. value3: Net1, Bits 15..31, Upper bitrate warning threshold percent (100+%). Bitrate values above this trigger the warning.
33	Max Image Queue	Yes	Sets the maximum number of images that can be queued in the system for each camera. More than this number of images in the queue will result in frames being dropped. **NOTE** may have unintended consequences, please contact Technical Support for more details. value0: Max images in queue. 0 to 3. 0 defaults to 3 images
34	Burst Mode Snapshot	Yes	Configure burst mode snapshot settings. Check Persistence, save parameters and reboot for this change to take effect. See also SLADoSnapShot_t value0: The width of burst snapshot in pixels. Setting to 0 resets to the default internal value. value1: The height of burst snapshot in pixels. Setting to 0 resets to the default internal value. value2: Max number of snapshots. The number is limited by the free memory on the system and it can be increased by reducing the width/height of the image buffer. Setting to 0 resets to the default internal value.

			value3: Reserved, must be 0
35	App Bits Testing	No	Sets a reduced set of App Bits for testing functionality with different configurations. value0: App Bits - Must be a subset of app-bits as reported by <u>SLAVersionNumber_t</u>
36	MISB Compliance	Yes	Adjustments for KLV values to be MISB compliant. value0: Time standard that has been used in the input. 0 = MISP Time. 1 = GPS Time. 2 = UTC Time. 3 = TAI Time. value1: Number of leap seconds to add. We will use this value and the time standard set in value 0 to calculate the offset from the UTC time set in <u>SLASetMetadataValues_t</u> and therefore calculate the correct MISP time. If already accounting for leap seconds use 0, otherwise use the correct number of leap seconds which is 37 as of December 2024. value2: VMTI pixel number calculations comply with MISB 0903.6. These calculations are applied to the centroid, bounding box top left and bounding box bottom right. Value 0 = does not comply so pixel number = (row * frame width) + col + 1. Value 1 = complies so pixel number = ((row - 1) * frame width) + col value3: Reserved
37	Disable Convert	Yes	Turn off convert processing - NOTE: must also disable display and registration, only for detection modes that don't require convert (eg. 16 bit blob detect) value0: 0: enable convert, 1: disable convert
38	Laser Frame Control	Yes	Controls for cameras that provide laser data frames along with regular camera frames value0: 0: 16bit threshold value, 1: LSB = U, MSB = V. 0xF040 = red (default), 0x1010 = green, 0xF010 = orange, 0x10F0 = teal, 0xF0F0 = magenta, 0x8080 = no color, 2: MTI overlay 0=off, 1=on
39	Zero Conf	Yes	Turn on Discover via ZeroConf value0: 0: Disable ZeroConf/Avahi on the board, 1: Enable ZeroConf/Avahi on the board
40	AUD Bits	Yes	Add AUD bits at the end of every frame. The bits are 0x00 0x00 0x00 0x01 0x09 0x10 value0: 0: Disable adding AUD bits into every frame, 1: Enable adding AUD bits into every frame
41	Enable Classifier NPU	Yes	Enable the NPU (neural processing unit) for classifications. This is off by default to allow the NPU resources to be used for full frame AI detections. value0: 0: Disable NPU classifications, 1: Enable NPU classifications
42	System clock	Yes	Sets the system clock when no active NTP. value0: 0: None, 1: Use the UTC time set in <u>SLASetMetadataValues_t</u> for Network 0 when this system value message is received with a value of 1, 2: Set the system clock whenever the UTC time is changed via <u>SLASetMetaDataValues_t</u> on Network 0 or when this message is received.

44	Network Additions	Yes	Provides additional network settings. value0: 0: disable mDNS, 1: enable mDNS (4100/17XX only) value1: The TTL value to be used for multicast video. Valid values are 1-255. 0 will result in default of 64
45	NITF Control	Yes	STANAG 4545 Compliance for NITF Metadata. value0: 0: Sensor ID Type for EO/IR Frame sensor format only. Follows STDI-0002-v1.2 (ACFTB TRE) 0:(default) VMFR 1: IHFR. 2: IMFR 3: ILFR ...10: VFFR value1: 0: Sensor ID. Follows STDI-0002-v1.2 (ACFTB TRE) 0:(default) CA265 1: APG-73. 2: AIP 3: ASARS1 ...22:TSAR
others	<i>Reserved</i>	-	-

Message ID 0x92

Byte Offset	Name	Description						
4-7	systemValue	<u>System Value Type</u>						
8-11	setMode	Persistence:<table><tr><th>Bit</th><th>Description</th></tr><tr><td>0</td><td>0 = value will not be saved to parameter file. 1 = the value will be saved to parameter file</td></tr><tr><td>other bits</td><td><i>Reserved and must be cleared to 0</i></td></tr></table>	Bit	Description	0	0 = value will not be saved to parameter file. 1 = the value will be saved to parameter file	other bits	<i>Reserved and must be cleared to 0</i>
Bit	Description							
0	0 = value will not be saved to parameter file. 1 = the value will be saved to parameter file							
other bits	<i>Reserved and must be cleared to 0</i>							
12-15	value0	Argument for systemValue. May depend on the platform and type, see <u>System Value Type</u> above.						
16-19	value1	Optional additional argument. Set to 0 if unused.						
20-23	value2	Optional additional argument. Set to 0 if unused.						
24-27	value3	Optional additional argument. Set to 0 if unused.						

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentTrackingParameters_t

Struct Reference

[Tracking](#)

Description

Current tracking parameters – format identical to [SLASetTrackingParameters_t](#).

```
typedef struct {
    u8 objectSize;
    u8 mode;
    u8 mode2;
    u8 maxMisses;
    u16 nearVal;
    u8 objectHeight;
    u8 cameraIndex;
    u8 zoomSmoothing;
    u8 rollSmoothing;
    u8 maxTracks;
    u8 maxPauseTime;
    u16 maxMissesExt;
    u8 modeFlags;
} SLACurrentTrackingParameters\_t;
```

Sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x44

Byte Offset	Name	Description
4	objectSize	See SLASetTrackingParameters_t
5	mode	
6	mode2	
7	maxMisses	
8-9	nearVal	
10	objectHeight	
11	cameraIndex	
12	zoomSmoothing	
13	rollSmoothing	
14	maxTracks	

15	maxPauseTime	
16-17	maxMissesExt	
18	modeFlags	

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetTrackingParameters_t Struct Reference

[Tracking](#)

Description

Set parameters used by tracking module.

```
typedef struct {
    u8 objectSize;
    u8 mode;
    u8 mode2;
    u8 maxMisses;
    u16 nearVal;
    u8 objectHeight;
    u8 cameraIndex;
    u8 zoomSmoothing;
    u8 rollSmoothing;
    u8 maxTracks;
    u8 maxPauseTime;
    u16 maxMissesExt;
    u8 modeFlags;
} SLASetTrackingParameters_t;
```

These parameters are applied to the specified camera. Changing the tracking mode, track size, or some of the other flags while tracking will cause the track to restart.

Message ID 0x0C

Byte Offset	Name	Description				
4	objectSize	Size of object, in pixels, to track. (for user designated tracking). if objectHeight==0 then this objectSize is both width and height, otherwise only width.				
		Tracking modes				
		<table><tr><th>Bits</th><th>Description</th></tr><tr><td></td><td>Tracking Mode 0 = no change - This only impacts the Tracking Mode not any of the other flags that are part of the mode or mode2 bytes. 1 = Stationary mode – Used to track non-moving object (e.g. door, window, </td></tr></table>	Bits	Description		Tracking Mode 0 = no change - This only impacts the Tracking Mode not any of the other flags that are part of the mode or mode2 bytes. 1 = Stationary mode – Used to track non-moving object (e.g. door, window,
		Bits	Description			
	Tracking Mode 0 = no change - This only impacts the Tracking Mode not any of the other flags that are part of the mode or mode2 bytes. 1 = Stationary mode – Used to track non-moving object (e.g. door, window,					

5	mode	0..3	building, etc.). Disables moving target detection. • 2 = Vehicle mode – Used to track moving objects. Works best with relatively constant velocity objects such as a car. • 3 = Person mode – Used to track moving objects like people which have more erratic motion than vehicles. • 4 = Scene mode – Uses frame-to-frame registration to determine position of target. May work better than Stationary Mode for low-contrast non moving targets. • 5 = <i>Reserved</i> • 6 = Static mode – Fixed location non-moving track box used for <u>SLATrackingBoxPixelStats_t</u> • 7 = No Registration – Deprecated Use Drone or Vehicle mode now, even in cases when it is difficult to accurately estimate frame-to-frame registration. • 8 = Drone mode – Erratically moving quadcopters, for example. • 9..14 = <i>Reserved</i> • 15 = Reserved – Internal Sightline Use.
		4	High noise compensation. • 0 = off (default) • 1 = on – improves tracking in very high noise situations, but processing time increases
		5	Acquisition assist • 0 = off (default) • 1 = on – assists initialization of the track box size and location. Impacts user designated targets in all tracking modes as well as targets auto designated from detections.
		6	Intelligent assist. • 0 = off (default) • 1 = on – tracker will look for turning tracks and if detected will attempt a reacquisition. The new track will be followed for a period of time before replacing the current track. Requires Acquisition Assist to be enabled
		7	Reduced search. • 0 = off (default). Tracker searches a slightly larger area including the estimated position and the last position • 1 = on – Tracker will only search in the estimated position
		Tracking modes	
		Bits	Description
		0	Run Classifier – Deprecated . See <u>SLAClassifierParameters_t</u>

6	mode2	1	Mode Switch Without Reset – When switching tracking modes while a track is active the track will be completely reset. Set this bit to switch between vehicle and person mode without resetting the track. Not valid for switching between any other modes.
		2	Zoom Scale – scale the track box with the estimated zoom from registration. Default is not set.
		3	AI Track Assist 0 = off (default) 1 = on – An enhancement to Intelligent Assist which uses AI MTI coordinates to improve track resizing.
		4..7	*Reserved*
7	maxMisses	Maximum number of frames to keep looking for a non-found object before stopping a track. (Default is 45 frames or 1.5 seconds.) Controls how long a track can be off screen or obscured(eg.behind a tree) before the track will give up. 0 = no change. Valid range is 15-255.	
8-9	nearVal	'radius of engagement' used with SLAStartTracking_t modes. This is the area around an existing target that interactions can take place.Large values allow greater tolerance in selecting tracks.	
10	objectHeight	Height of object, in pixels, to track. (for user designated tracking) if value is 0, uses objectSize above.	
11	cameraIndex	Camera Index	
12	zoomSmoothing	Percentage to smooth zoom estimates coming from registration. 0 to 100%. Default is 5%.	
13	rollSmoothing	Percentage to smooth rotation estimates coming from registration. 0 to 100%. Default is 5%.	
14	maxTracks	Maximum number of tracks, including those from Drone, Vehicle, and Person detection modes. Default 5, min 1, max 20. Values outside the max/min are set to the default.	
15	maxPauseTime	The maximum number of seconds to store queued frames. Valid ranges 0-10 seconds or 0-20 seconds depending on the platform and the number of cameras configured.	
16-17	maxMissesExt	Extended Maximum number of frames to keep looking for a non-found object. Valid range is 256 - 65535. (Will not be used if maxMisses is set)	
18	modeFlags	Tracking mode flags	
		Bits	Description
		0	Track Specific Mode New 3.11 0 = off – all tracks use the same track mode 1 = on – tracks will start with the Tracking Mode set in byte 5, but previously started tracks will use keep their original track mode
			Auto Track Mode New 3.11

		1	0 = off (default) 1 = on – When a track starts it is passed to a classifier which determines if it should use Person, Vehicle, or Drone tracking mode. Requires Track Specific Mode to be on.
		2	Color Track Mode New 3.11 0 = off (default) 1 = on – Color components of the track aid in tracking. Color inputs only (G8 and G16 not supported).

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentVideoEnhancementParameters_t Struct Reference

[Enhancement](#)

Description

Current video enhancement parameters – format identical to [SLASetVideoEnhancementParameters_t](#).

```
typedef struct {
    u8 sharpenEnhance;
    u8 alphaBlend;
    u8 enhanceParam;
    u8 denoiseRate;
    u8 cameraIndex;
    u8 flags;
    u8 histAveRate8;
    u8 histMaxPctBin;
    u16 roiRow;
    u16 roiCol;
    u16 roiHigh;
    u16 roiWide;
    u8 deconvSigma;
    u8 gaussianBlur;
    u8 lapMinDiff;
    u8 colorEnhance;
    u8 brightness;
    u8 contrast;
    u8 scintillationMode;
    u8 sharpenRadius;
    SVPSignedData\_t customKernel;
    u8 normalizeKernel;
} SLACurrentVideoEnhancementParameters\_t;
```

Sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x4A

Byte Offset	Name	Description
4	sharpenEnhance	see SLASetVideoEnhancementParameters_t
5	alphaBlend	
6	enhanceParam	
7	denoiseRate	
8	cameraIndex	
9	flags	

10	histAveRate8	
11	histMaxPctBin	
12-13	roiRow	
14-15	roiCol	
16-17	roiHigh	
18-19	roiWide	
20	deconvSigma	
21	gaussianBlur	
22	lapMinDiff	
23	colorEnhance	
24	brightness	
25	contrast	
26	scintillationMode	
27	sharpenRadius	
28	customKernel.len	Number of signed 8-bit integers
29-...	customKernel.values	
varies	normalizeKernel	

Sightline Intelligence

Command, Control, and Script

API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) [Camera Configuration](#)

[List of all members](#)

SLASetVideoEnhancementParameters_t Struct Reference

[Enhancement](#)

Description

Set up enhancement of the display video.

```
typedef struct {
    u8 sharpenEnhance;
    u8 alphaBlend;
    u8 enhanceParam;
    u8 denoiseRate;
    u8 cameraIndex;
    u8 flags;
    u8 histAveRate8;
    u8 histMaxPctBin;
    u16 roiRow;
    u16 roiCol;
    u16 roiHigh;
    u16 roiWide;
    u8 deconvSigma;
    u8 gaussianBlur;
    u8 lapMinDiff;
    u8 colorEnhance;
    u8 brightness;
    u8 contrast;
    u8 scintillationMode;
    u8 sharpenRadius;
    SVPSignedData\_t customKernel;
    u8 normalizeKernel;
} SLASetVideoEnhancementParameters\_t;
```

Under control of the **denoiseRate**, a running average of frames is optionally calculated. The denoised frame is then passed through one of several filters to enhance contrast and optionally sharpened to emphasize fine detail.

NOTE: CLAHE 9 and 10 bit modes are only valid for high bit depth cameras. Selecting normal CLAHE for a high bit depth camera converts the video to 8 bits first then runs CLAHE. 9 bit CLAHE is slower than 8 bit, 10 bit is slower than 9 bit.

NOTE: [SLASetADCPParameters_t](#) -> **saturation** is used in Histogram Equalization and CLAHE 9 and 10 bit modes to control the range of grey values which go into the histogram to be equalized.

- 255 => 100 % min / max based (allows you to see detail on very bright or very dark objects)
- 0 => 100 % auto gain spread based (high contrast image, may be saturated in some areas)
- 128 => even mix of the two methods

NOTE: Applied to the Command Camera.

Byte Offset	Name	Description										
4	sharpenEnhance	Contrast enhancement mode and sharpening control. <table><tr><th>Bits</th><th>Description</th></tr><tr><td>0-3</td><td>Contrast enhancement mode 0 = None 1 = CLAHE 2 = LAP 3 = CLAHE 9 Bit (for >8 bit cameras) 4 = CLAHE 10 Bit (for >8 bit cameras) 5 = Histogram Equalization 6 = Gamma Correction (17XX/4000 only) 7 = LAP 16 Bit (for 16 bit cameras) 8 = Histogram Equalize into a Gamma curve</td></tr><tr><td>4-7</td><td>Sharpening: 0 (none) - 15 (max)</td></tr></table>	Bits	Description	0-3	Contrast enhancement mode 0 = None 1 = CLAHE 2 = LAP 3 = CLAHE 9 Bit (for >8 bit cameras) 4 = CLAHE 10 Bit (for >8 bit cameras) 5 = Histogram Equalization 6 = Gamma Correction (17XX/4000 only) 7 = LAP 16 Bit (for 16 bit cameras) 8 = Histogram Equalize into a Gamma curve	4-7	Sharpening: 0 (none) - 15 (max)				
		Bits	Description									
		0-3	Contrast enhancement mode 0 = None 1 = CLAHE 2 = LAP 3 = CLAHE 9 Bit (for >8 bit cameras) 4 = CLAHE 10 Bit (for >8 bit cameras) 5 = Histogram Equalization 6 = Gamma Correction (17XX/4000 only) 7 = LAP 16 Bit (for 16 bit cameras) 8 = Histogram Equalize into a Gamma curve									
4-7	Sharpening: 0 (none) - 15 (max)											
5	alphaBlend	Alpha blending (source with filtered) (0-255, 255 default, 200 nominal). NOTE: Forced to 255 for CLAHE 9 Bit and CLAHE 10 Bit modes. <table><tr><th>Example</th><th>Description</th></tr><tr><td>0</td><td>use input frame only</td></tr><tr><td>128</td><td>50/50 mix</td></tr><tr><td>255</td><td>use filtered frame only (default)</td></tr></table>	Example	Description	0	use input frame only	128	50/50 mix	255	use filtered frame only (default)		
		Example	Description									
		0	use input frame only									
		128	50/50 mix									
255	use filtered frame only (default)											
6	enhanceParam	Contrast enhancement strength parameter (0-127, bit 7 <i>reserved</i>) <table><tr><th>Contrast Enhancement Mode</th><th>Description</th></tr><tr><td>None</td><td>Not used</td></tr><tr><td>CLAHE, CLAHE 9 Bit, CLAHE 10 Bit</td><td>Contrast limit (typical 25)</td></tr><tr><td>LAP, LAP16</td><td>Width of high pass kernel, clipped to 0-18 (typical 10)</td></tr><tr><td>Gamma Correction, Histogram Equalization + Gamma</td><td>Gamma value clipped to integer 3-28 (typical 10; 10=1.0, 8=0.8, 22=2.2)</td></tr></table>	Contrast Enhancement Mode	Description	None	Not used	CLAHE, CLAHE 9 Bit, CLAHE 10 Bit	Contrast limit (typical 25)	LAP, LAP16	Width of high pass kernel, clipped to 0-18 (typical 10)	Gamma Correction, Histogram Equalization + Gamma	Gamma value clipped to integer 3-28 (typical 10; 10=1.0, 8=0.8, 22=2.2)
		Contrast Enhancement Mode	Description									
		None	Not used									
		CLAHE, CLAHE 9 Bit, CLAHE 10 Bit	Contrast limit (typical 25)									
		LAP, LAP16	Width of high pass kernel, clipped to 0-18 (typical 10)									
Gamma Correction, Histogram Equalization + Gamma	Gamma value clipped to integer 3-28 (typical 10; 10=1.0, 8=0.8, 22=2.2)											
7	denoiseRate	Denoising coefficient by running average (0-255, 0 default) <table><tr><th>Example</th><th>Description</th></tr><tr><td>0</td><td>No denoising (default)</td></tr><tr><td>128</td><td>Half from new image, half from running average.</td></tr><tr><td>255</td><td>Maximum averaging</td></tr></table>	Example	Description	0	No denoising (default)	128	Half from new image, half from running average.	255	Maximum averaging		
		Example	Description									
		0	No denoising (default)									
		128	Half from new image, half from running average.									
255	Maximum averaging											

8	cameraIndex	Camera Index										
9	flags	Default 0										
		<table><tr><th>Bit</th><th>Description</th></tr><tr><td>0</td><td>Turn on Aerial De-Noise Motion Mask to keep from blurring moving objects. For moving camera platforms. New 3.1</td></tr><tr><td>1</td><td>Feature Based Histogram Equalization - emphasizes higher spatial frequency areas of the image</td></tr><tr><td>2</td><td>Square Root Histogram Equalization - reduces over-emphasis on a single grey-level</td></tr><tr><td>4</td><td>Turn on Staring De-Noise Motion Mask to keep from blurring moving objects. For fixed cameras or step-stare ground PTZ cameras. New 3.2</td></tr></table>	Bit	Description	0	Turn on Aerial De-Noise Motion Mask to keep from blurring moving objects. For moving camera platforms. New 3.1	1	Feature Based Histogram Equalization - emphasizes higher spatial frequency areas of the image	2	Square Root Histogram Equalization - reduces over-emphasis on a single grey-level	4	Turn on Staring De-Noise Motion Mask to keep from blurring moving objects. For fixed cameras or step-stare ground PTZ cameras. New 3.2
		Bit	Description									
		0	Turn on Aerial De-Noise Motion Mask to keep from blurring moving objects. For moving camera platforms. New 3.1									
		1	Feature Based Histogram Equalization - emphasizes higher spatial frequency areas of the image									
		2	Square Root Histogram Equalization - reduces over-emphasis on a single grey-level									
4	Turn on Staring De-Noise Motion Mask to keep from blurring moving objects. For fixed cameras or step-stare ground PTZ cameras. New 3.2											
NOTE Aerial and Staring De-Noise Motion Mask can not be used at the same time as detection using detIdx 1 from SLASetDetectionParameters_t .												
10	histAveRate8	Time averaging rate for histogram in histogram equalization. 0 = no averaging, 255 = maximum averaging. Hist = NewHist*(256-rate8)/256 + OldHist*rate8/256 New 3.2										
11	histMaxPctBin	Maximum percentage of counts that go into a single histogram bin in histogram equalization. New 3.2										
12-13	roiRow	Row coordinate of region of interest. New 3.3 Enhancement functions are restricted to the specified area of the source frame. Set roiHigh and roiWide to 0 to disable ROI processing. NOTE: When ROI is enabled and PiP is enabled, if the PiP index is the same as the main camera the PiP image will not have enhancements applied. New 3.7										
14-15	roiCol	Column coordinate of region of interest										
16-17	roiHigh	Height of region of interest										
18-19	roiWide	Width of region of interest										
20	deconvSigma	Reserved										
21	gaussianBlur	0 = don't gaussian blur, 1 = minimum blur, 6 = maximum blur New 3.4 . For value>0, Gaussian sigma = (value+2)*0.25 The valid range is 0-6. For 17xx, larger values take more processing time.										
22	lapMinDiff	In LAP processing, differences between image and blurred image below this value get set to 0. Prevents contour line artifacts in flat scene areas. Default = 0, Typical = 2. New 3.4 In Histogram Equalize + Gamma, acts as the enhanceParam for the Histogram Equalization portion of the algorithm. New 3.10										
23	colorEnhance	Color enhancement control. 0 = no color enhancement (default), 255 = maximum color enhancement. New 3.5										
24	brightness	Brightness control for CLAHE and HistEq. 128 = default, >128 = brighter, <128 = darker. New 3.6										
25	contrast	Contrast control for CLAHE and HistEq. 128 = default, >128 = more contrast, <128 =										

		less contrast New 3.6										
26	scintillationMode	Scintillation mitigation preset. Default 0. New 3.6 NOTE: enhanceParam, sharpenEnhance, alphaBlend, flags bits 0 or 4, and denoiseRate cannot be changed while using a scintillation mode other than manual. These values should be 0 when scintillation mode is not Manual to avoid a warning. <table><tr><th>Value</th><th>Description</th></tr><tr><td>0</td><td>Manual</td></tr><tr><td>1</td><td>Low Motion</td></tr><tr><td>2</td><td>High Motion</td></tr><tr><td>3</td><td>Infra-Red</td></tr></table>	Value	Description	0	Manual	1	Low Motion	2	High Motion	3	Infra-Red
Value	Description											
0	Manual											
1	Low Motion											
2	High Motion											
3	Infra-Red											
27	sharpenRadius	How large of features to sharpen. Options are 1, 2, and 3 pixels. Default 1. New 3.9										
28	customKernel.len	Number of signed 8-bit integers										
29-...	customKernel.values	Custom kernel to convolve the image with. Signed 8-bit Integers. Supported sizes are 3x3,5x5,7x7,9x9. Default 0. New 3.10										
varies	normalizeKernel	Whether to normalize the kernel, default is 0.										

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentVideoModeParameters_t

Struct Reference

Description

Deprecated, use [SLAVideoDisplay_t](#) and [SLAMultiDisplay_t](#).

```
typedef struct {
    u8 nCameras;
    u8 nDisplays;
    u8 displayMode;
    u8 displayDest;
    u8 cameraIndex0;
    u8 cameraIndex1;
    u8 cameraIndex2;
    u8 pipColor;
    u8 pipScale;
    u8 pipQuadrant;
    u8 displayIndex0;
    u8 displayIndex1;
    u8 displayIndex2;
    u8 displayIndex3;
    u8 analogDest;
    u8 hdmiDest;
    u8 net0Dest;
    u8 net1Dest;
    u8 HDSDDIDest;
    u8 hdmiRes;
    u8 hdsdiRes;
    u8 analogRes;
} SLACurrentVideoModeParameters_t;
```

Deprecated, use [SLAVideoDisplay_t](#) and [SLAMultiDisplay_t](#).

Current video mode parameters – format identical to [SLASetVideoMode_t](#)

Sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x4B

Byte Offset	Name	Description
4	nCameras	see SLASetVideoMode_t
5	nDisplays	
6	displayMode	
7	displayDest	

8	cameraIndex0	
9	cameraIndex1	
10	cameraIndex2	
11	pipColor	
12	pipScale	
13	pipQuadrant	
14	displayIndex0	
15	displayIndex1	
16	displayIndex2	
17	displayIndex3	
18	analogDest	
19	hdmiDest	
20	net0Dest	
21	net1Dest	
22	HDSDIDest	
23	hdmiRes	
24	hdsdiRes	
25	analogRes	

Sightline Intelligence Command, Control, and Script API v3.11

[Main Page](#)
[Related Pages](#)
[Modules](#)
[Classes](#)
[Camera Configuration](#)

List of all members

SLAVideoDisplay_t Struct Reference

Display

Description

Configure which cameras and resolutions to display.

```
typedef struct {
    u8 streamIdx;
    u8 source;
    u16 destination;
    u8 resolution;
    u8 flags;
} SLAVideoDisplay_t;
```

NOTE: When network video display is selected, configure network video settings (including resolution) with `SLASetEthernetVideoParameters_t`.

NOTE: Video Recording and Snap Shot are only available for cameras that are being processed for telemetry or display output.

See **Draw Order** for additional drawing details.

NOTE: Switching the display modes can be a time consuming operation and it may cause delays in other part of the system. For example, switching HDMI video modes may delay telemetry packets for a few seconds.

Display Resolutions

This table specifies support for display resolutions on 4000, 4100, and 17XX. Grey cells marked **N** are not possible. White cells marked **N** may be supported in future releases.

[illegible]

	640	512	p	30	N(3)	N	Y(1)	N(3)	N	Y(7)	Y(1)
	768	576	p	25	N	N	Y	N	N	N	Y
	768	576	i	50	N	N	Y(1)	N	N	N	Y(1)
	768	576	p	50	N	N	Y(1)	Y	N	N	Y(1)
720	800	600	p	60	Y	N	Y(1)	Y	N	N	Y(1)
	960	720	p	30	Y	N	Y	Y	N	N	Y
	1024	768	p	60(4)	Y(4)	N	Y(1)	Y	N	N	Y(1)
	1280	720	p	60/59.94/50	Y	Y(5)	Y	Y	Y(5)	Y(7)	Y
	1280	720	p	30	N(3)	N	Y	N(3)	N	Y(7)	Y
1080	1280	768	p	60	Y/N(8)	N	Y(1)	Y	N	N	Y(1)
	1280	960	p	30	N	N	Y	N	N	N	Y
	1280	1024	p	30	N(3)	N	Y(1)	N(3)	N	N	Y(1)
	1280	1024	p	50	N(3)	N	Y(1)	N(3)	N	Y(7)	Y(1)
	1440	1080	p	30	N	N	Y	N	N	N	Y
	1600	1200	p	60	Y	N	Y(1)	Y	N	N	Y(1)
	1920	1080	p	60/59.94/50	Y	Y(5)	Y	Y	Y(5)	N	Y
	1920	1080	p	30/29.97/25	Y	Y(5)	Y	Y	Y(5)	Y(7)	Y
	1920	1080	i	60/59.94/50	N	N	Y	N	N	N	Y
4K	1920	1200	p	60	Y/N(8)	N	Y(1)	N	N	N	N
	1920	1536	p	30	N(3)	N	Y(1)	N	N	N	N
	2560	1440	p	30	N	N	Y	N	N	N	N
	3840	2160	p	30/29.97/25	Y	N	Y	N	N	N	N

1. Supported for network stream with Out=In, not a selectable resolution
2. SDI part supports limited resolutions: 720p, 1080i, 1080p, 480i, 576i. See [SDI Camera Frame Rate Compatibility](#).
3. Possible future resolution support
4. 1024x768 HDMI output on 4000 is at **70** fps, 17XX and 4100 output 1024x768 at 60 fps
5. For SDI output on 4000, 4100, and 17XX, set **destination** to HDMI and attach the [SLA-HDMI-SDI](#) adapter board
6. For LVDS output on 17XX, limited resolutions are supported: 720p60/50/30, 1080p30/25, 640x512p60/30, 1280x1024p50. LVDS only supported on 1750-OEM.
7. For LVDS output on 17XX, lowest latency, turn off Network display and disable all processing.
8. This resolution is NOT supported on the 4100.

Display Resolution Examples for 17XX, 4100, and 4000

When camera, physical display and network display resolutions don't match, cropping, windowing and sometimes scaling are used as shown in the following examples.

Camera Resolution	Network Resolution	Physical Display(1)	Physical Resolution	17XX/4000/4100 Display	

				Network	Physical HDMI
1280x720	1920x1080	HDMI	1280x720	720 in 1080 with gray border	720
1280x720	1920x1080	HDMI	1920x1080	720 in 1080 with gray border	720 in 1080 with gray border
1920x1080	1280x720	HDMI	1280x720	720 cropped	720 cropped
1920x1080	1920x1080	HDMI	1280x720	1080	720 cropped
1920x1080	1280x720	HDMI	1920x1080	1080(3)	1080

1. HDSDI display behaves the same as HDMI, attach the [SLA-HDMI-SDI](#) adapter board
2. Network resolution is the maximum of the network and physical display resolution by default, but can be decoupled (see the flags byte, bit 1)

Multi Camera Display Combinations

The table below shows valid (white) and invalid (grey) combinations for display.

The system will warn and disallow setting invalid combinations. **A** is any camera (0, 1 or 2). **A**, **B**, and **C** are two different cameras.

In PiP or Blend configurations **A** is main PiP, **B** is secondary PiP, and **C** is non PiP camera.

Supported	Network 0	Network 1	Network 2	HDMI or HDSDI or LVDS(1,2)	Video Record	Description and Notes
Single and Dual Stream Configurations						
Yes	A	None	None	A	A	Single stream
Yes	A	A	None	A	Net0 or Net1	Single camera to 2 streams at different resolution, bitrate, etc (3)
Yes	A	B	None	A or B	Net0 and/or Net1	Dual stream
Yes	A	B	C	A or B or C	Net0 and/or Net1 and/or Net2	Triple stream
Yes	A	A	B	A or B	Net0 and/or Net1 and/or Net2	Dual camera to 3 streams, 2 of same at different resolution, bitrate, etc (3)
Yes	A	B	None	C	Net0 and/or Net1	3 cameras - 2 to Network, 1 to HDMI
No	A	A	A			Same camera to 3 streams - Can only output same to 2 streams at a time
No	A	B	A			Same camera to Net 0 and Net 2
No	B	A	A			Same camera to Net 1 and Net 2
No	A	None			B	Can only record a network stream (2)
No	A	B			C	Can only record a network stream (2)

Picture in Picture Configurations						
Yes	PiP	None	None	PiP	PiP	PiP to single stream
Yes	PiP	PiP	None	PiP	Net0 and/or Net1	PiP to 2 streams at different resolution, bitrate, etc (3)
Yes	PiP	A	None	A or PiP	Net0 and/or Net1	PiP to Net0, A to Net1
Yes	PiP	C	None	C or PiP	Net0 and/or Net1	PiP to Net0, C to Net1
Yes	PiP	A or B or C	A or B or C (4)	PiP or A or B or C (5)	Net0 and/or Net1 and/or Net1	PiP to Net0, individual cameras to Net1 and Net2
Yes	PiP	PiP	C	PiP or C	Net0 and/or Net1 and/or Net2	PiP to 2 streams at different resolution, bitrate, etc, and 3rd camera to Net2 (3)
No	PiP	None	None	A		Use PiP to Net0, A to Net1 if you want to show A to HDMI or HDSDI
No	A	PiP	None	PiP		PiP has to go to network 0 if you want other displays
No	PiP	A		None	B	Can only record a network stream (2)
No	PiP	PiP	A or B			PiP to 2 streams, one of PiP cameras to other stream not supported
No	PiP	PiP	PiP			PiP to 3 streams - Can only output same to 2 streams at a time
Blend Configurations - 4000/4100 Only						
Yes	Blend	None	None	Blend	Blend	Blend to single stream
Yes	Blend	Blend	None	Blend	Net0 or Net1	Blend to 2 streams at different resolution, bitrate, etc (3)
No	Blend	A or B or PiP (to any of these)				With Blend, other cameras can not be shown
No	Blend	Blend	Blend			Blend to 3 streams - Can only output same to 2 streams at a time

1. HDSDI is only through the [SLA-HDMI-SDI](#) adapter board. Select HDMI and HDSDI will be auto-detected if the adapter board is connected.
2. System only records configured network streams using the settings for that network stream
3. When sending the same input (camera, multi, or blend) to both networks Net1 is a copy of Net0 so if the resolution of Net1 is smaller than Net0 it will crop out the center. If Net1 resolution is larger than Net0 it will have a border.
[SLASetSystemValue_t](#) Net1 Zoom10 (24) can be used to set digital zoom independently on Net1.
4. Net2 can't be the same camera as Net1 in this case.
5. HDMI needs to be one of the same outputs as Net0 or Net1 or Net2

9	0x0200	LVDS - display processed 8bit gray image, color overlay graphics are also displayed in gray (1750 only)
10	0x0400	LVDS - Color output 16bit 422 (1750 only)
11	0x0800	LVDS - display processed 8bit gray image using upper bits. e.g. 0x0A00 (1750 only)
12	0x1000	UVC - USB Video Class (17xx and 41xx only)
others		<i>Reserved, must be 0</i>

Physical display resolution (**Analog, HDMI or HDS DI**).

Configure **network** display resolution with [**SLASetEthernetVideoParameters_t**](#).

NOTE: For 17XX/4000/4100, HDMI resolution will be as set. Network resolution will be the maximum of the specified Network and HDMI (if active) resolutions.

NOTE: For 17XX/4000/4100, see [Resolution Examples](#) above.

Value	Description
0	Default
1	NTSC Analog (Not supported on 17XX/4000/4100).
4	720p60 (3)
5	1080p30 (1,3)
6	1080p25 (1,3)
7	1080p60
8	720p50 (1,3)
9	1080p50 (1)
10	PAL Analog (Not supported on 17XX/4000/4100)
11	720p59.94 HDMI or HDS DI (17XX/4000/4100) (1,2)
12	1080p29.97 HDMI or HDS DI (17XX/4000/4100) (1,2)
13	1080p59.94 HDMI or HDS DI (17XX/4000/4100) (1,2)
14	1080i60 HDMI or HDS DI (Not supported on 17XX/4000/4100)
15	1080i59.94 HDMI or HDS DI (Not supported on 17XX/4000/4100)
16	1080i50 HDMI or HDS DI (Not supported on 17XX/4000/4100)
17	480P60 (17XX HDMI only) (1) - May need to set monitor aspect ratio
18	4Kp30 UHD 3840x2160 HDMI (4000/4100 only) (1)
19	4Kp25 UHD 3840x2160 HDMI (4000/4100 only) (1)
20	4Kp29.97 UHD 3840x2160 HDMI (4000/4100 only) (1)
21	600P60 800x600 HDMI (17XX/4000/4100) (1)
22	768P60 1024x768 HDMI (17XX/4000/4100) (1)
23	1200P60 1600x1200 HDMI (4000/4100) (1)
24	1280x768P60 1280x768 HDMI (17XX/4000) (1)

8

resolution

25	1920x1200P60 1920x1200 HDMI (4000) (1)
26	640x512P60 640x512 LVDS (17XX) (3)
27	640x512P30 640x512 LVDS (17XX) (3)
28	1280x1024P50 1280x1024 LVDS (17XX) (3)
29	720P30 1280x720 LVDS (17XX) (3)

1. Some monitors do not support this format
2. For SDI output on 4000, 4100, and 17XX, set **destination** to HDMI and attach the **SLA-HDMI-SDI** adapter board
3. For LVDS (if available) output on 17XX, resolutions 720p60, 720p50, 720p30, 1080p30, 1080p25, 640x512P60, 640x512P30, and 1280x1024p50 are available

Bits	Description
0	<i>Reserved</i>
1	0 = force network video resolution to match HDMI resolution (default), 1 = network resolution not forced to match HDMI resolution, resolutions can differ
1-7	<i>Reserved</i>

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAMultiDisplay_t Struct Reference

[Display](#)

Description

Configure multi-camera display options such as Picture-In-Picture.

```
typedef struct {
    u8 displayMode;
    u8 cameraIndex0;
    u8 cameraIndex1;
    u8 reserved1;
    u8 pipLabel;
    u8 pipColor;
    u8 pipScale;
    u8 pipQuadrant;
    u8 multiPipZoom;
} SLAMultiDisplay t;
```

See [Draw Order](#) for additional drawing details.

Display Mode

Value	Display mode
0	One UP (1-UP) (default)
1	Picture In Picture
2-6	<i>Reserved</i>
7	No Display
8-15	<i>Reserved</i>

Display Modes

Color of PiP window border

Value	Color	Value	Color
0x00 - 0x0F	No change	0x18	Light green

0x10	White	0x19	Green
0x11	Black	0x1A	Dark green
0x12	Light gray	0x1B	Red
0x13	Gray	0x1C	Orange
0x14	Dark gray	0x1D	Yellow
0x15	Light blue	0x1E	Transparent
0x16	Blue	0x1F	Automatic
0x17	Dark blue	Other values	No change

Picture-In-Picture (PiP) Thumbnail Scale & Options

Value	Description
0	Do not change
1	1/4 screen size thumbnail of full screen or full size of PiP source, the smaller of the two.
2	3/8 screen size thumbnail of full screen or full size of PiP source, the smaller of the two.
3	1/2 screen size thumbnail of full screen or full size of PiP source, the smaller of the two.
4	1/4 screen size zoom on primary track (see SLAStartTracking_t)
5	3/8 screen size zoom on primary track
6	1/2 screen size zoom on primary track
7	zoom on all tracks (see SLAStartTracking_t)
8	zoom on MTI/detection tracks 5 (see SLASetDetectionParameters_t)
9	zoom on MTI/detection tracks 10
10	zoom on MTI/detection track 8 (corners + top + bottom + left + right)

Picture-in-Picture Scale

Zoom on all tracks & Zoom on Detection

Note

- Overlays such as reticles or user drawn objects (see [SLADrawOverlay_t](#)) will not be scaled and rendered in the PiP.
- When using PiP options 7-10 above, scale is pre-determined and not set by user.

Multi Picture-In-Picture (PiP) Zoom Options

Value	Description

0	Zoom 2X - Zoom the area around the track center by 2x (default)
1	Track Fit - Scale track size to PiP window
2	Track Fit 2X - Scale track size to PiP window and zoom by 2x
3	Zoom 3X - Zoom the area around the track center by 3x
4	Zoom 4X - Zoom the area around the track center by 4x

Note

- Only applies when pipScale is 7 to 10
- In 2X, 3X and 4X Zoom modes, the track and/or detection size isn't taken into account. Instead a region around the center of the track/detection is selected and scaled by either 2x,3x,4x to fill up the size of the PiP Window. Any user digital zoom is included in the scaling, but only digital zoom in, zooming out will not make the PiP use less zoom.
- in Track Fit modes, the track/detection size is scaled to fit the PiP area. Digital zoom has the same effect as above.

Screen Quadrant for PiP Display

Thumbnails will be displayed along top, bottom, left, or right side of the image.

Value	Description
0	Top Right (Right for all tracks or mti/detection tracks)
1	Bottom Right (Bottom for all tracks or mti/detection tracks)
2	Bottom Left (Left for all tracks or mti/detection tracks)
3	Top Left (Top for all tracks or mti/detection tracks)

Picture-in-Picture Quadrant

Label for PiP Display

Value	Description
0	Track Index
1	Classification Type
2	None
3	Track Index + Class
4	Track Index + Class + Conf

Message ID 0xA5

Byte Offset	Name	Description
4	displayMode	<u>Display Mode</u>

5	cameraIndex0	<u>Camera Index</u> Main camera for PiP
6	cameraIndex1	<u>Camera Index</u> PiP camera
7	reserved1	<i>Reserved</i>
8	pipLabel	<u>PiP Label</u> , applies to pipScale 7 to 10
9	pipColor	<u>Color of PiP</u> window border, default is red.
10	pipScale	<u>Scale</u> of Picture-in-picture thumbnail
11	pipQuadrant	<u>Screen quadrant</u> in which to render PiP window
12	multiPipZoom	<u>Zoom</u> of Picture-in-picture thumbnail, applies to pipScale 7 to 10

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentVideoParameters_t Struct Reference

[Capture](#)

Description

Current video parameters – format identical to [SLASetVideoParameters_t](#).

```
typedef struct {
    u8 autoChop;
    u8 chopTop;
    u8 chopBottom;
    u8 chopLeft;
    u8 chopRight;
    u8 deinterlace;
    u8 autoReset;
    u8 cameraIndex;
} SLACurrentVideoParameters\_t;
```

Sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x46

Byte Offset	Name	Description
4	autoChop	see SLASetVideoParameters_t
5	chopTop	
6	chopBottom	
7	chopLeft	
8	chopRight	
9	deinterlace	
10	autoReset	
11	cameraIndex	

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetVideoParameters_t Struct Reference

[Capture](#)

Description

Set parameters used by video input module.

```
typedef struct {
    u8 autoChop;
    u8 chopTop;
    u8 chopBottom;
    u8 chopLeft;
    u8 chopRight;
    u8 deinterlace;
    u8 autoReset;
    u8 cameraIndex;
} SLASetVideoParameters_t;
```

- NOTE:** Applied to the Command Camera.
- NOTE:** Chop is only recommended for Analog (NTSC or PAL) cameras in firmware versions 3.5 and below. Otherwise, use Region of Interest (Valid Row, etc) in [SLASetAcquisitionParameters_t](#) to remove black edges from an image.

Many cameras produce images with black pixels along one or more edges. It is important to remove these pixels as the hard edge transition can cause frame - to - frame registration to fail. The edge pixels are removed by either setting automatic detection mode (byte 4) or manually specifying top, bottom, left and right (bytes 5, 6, 7, 8) edge pixels to remove. For a known camera, manually specifying edge pixels is the most reliable option. If you see black edges in a moving stabilized image, that is an indication that edge pixel removal is not set up correctly. Number of pixels to remove are rounded down to the nearest 8.

Message ID 0x10

Byte Offset	Name	Description
4	autoChop	0 = Manual chop – removed specified edge pixels (default), 1 = automatically detect boundary pixels to remove NOTE: Auto and Manual Chop are not recommended. Use Acquisition Region of Interest in SLASetAcquisitionParameters_t .
5	chopTop	Top pixels to remove (values 8 to 64)

6	chopBottom	Bottom pixels to remove (values 8 to 64)
7	chopLeft	Left pixels to remove (values 8 to 128)
8	chopRight	Right pixels to remove (values 8 to 128)
9	deinterlace	0 = no deinterlacing, 1 = apply digital deinterlacing (default)
10	autoReset	0 = never automatically reset video decoder, 1 = automatically reset video decoder when failed frame synchronization loss is detected
11	cameraIndex	<u>Camera Index</u>

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

[List of all members](#)

SLACursorOnTarget_t Struct

Reference

[KLV Metadata](#)

Description

Cursor on Target (CoT) CoT packet contents are converted from KLV Metadata inputs.

```
typedef struct {
    u16 mode;
    u32be ipAddr;
    u16be port;
    u16 rate;
    u16 displayId;
} SLACursorOnTarget_t;
```

CoT packets requires KLV to be streaming ([SLAStreamingControl_t](#))

Message ID 0xB0

Byte Offset	Name	Description
4-5	mode	Reserved
6-9	ipAddr	IPv4 address of the receiver
10-11	port	destination network port
12-13	rate	rate at which packets will be generated (frames)
14-15	displayId	Network Display ID 4000/17XX: (0x0002 = Net0, 0x0080 = Net1, 0x0082 = both) 41XX: (0x0002 = Net0, 0x0080 = Net1, 0x0010 = Net2, 0x0092 = all)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACustomAIDetect_t Struct Reference

[Detection](#)

Description

Custom AI Object Detector Model.

```
typedef struct {
    u8 cameraIndex;
    SVPLenString\_t fileName;
} SLACustomAIDetect\_t;
```

Used to update the default ai object detector model with a customer specific model. File name length is restricted to 256 characters.

NOTE: 17XX and 4100 Only

Message ID 0xBA

Byte Offset	Name	Description
4	cameraIndex	Camera Index
5	fileName.len	string length
6-...	fileName.str	Filename for custom classifier (example: drone.slaod)

Sightline Intelligence

Command, Control, and Script

API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) [Camera Configuration](#)

[List of all members](#)

SLACustomAutoFocusParameters_t

Struct Reference

[Focus](#)

Description

Set Custom Autofocus Parameters.

```
typedef struct {
    u8 cameraIdx;
    u8 method;
    u16 calibStart;
    u16 calibEnd;
    u8 frameOffsetMult8;
    u8 calibFlags;
    u16 reserved1;
    u16 rateAdjust;
    u8 changePercent10ths;
    u16 reserved2;
    u8 nInitialSteps;
    u16 FineSpeed;
    u16 MidSpeed;
    u16 CoarseSpeed;
    u8 CoarseThresh;
    u8 FineThresh;
    u8 CdownMax;
    u16 reserved3;
    SVPLenString\_t fileName;
    SVPLenData\_t data;
} SLACustomAutoFocusParameters\_t;
```

Autofocus Method

Autofocus algorithms move the focus motor while recording focus metrics and focus positions. Algorithms have different strategies for when and how fast to move the motor. When autofocus terminates, the lens will move to the focus position that had the highest focus metric. Note that autofocus methods work best when the focus metric changes in the search area.

Scan

This is based on the "Rule-Based Search" autofocus algorithm. Start by moving the focus towards the far focus limit for some number of steps. Begin search in mid phase by focusing near. Switches to fine phase when the metric rapidly increases. Switches back to mid phase when the metric decreases. Switches to coarse phase when the current metric is far below the maximum found metric.

Terminating conditions: When the motor arrives at its most near focus position, when coarse phase begins if a fine phase

had been performed, or when the metric is neither increasing nor decreasing.

Seek

This is based on the hill-climbing autofocus algorithm. Searches near current focus. Start by moving the focus in the far direction until the focus metric changes more than the value set in autofocusChangePct.

- If the focus metric is increasing: Keep moving in this direction until the focus metric starts to decrease by the value set in autofocusChangePct, e.g., moving up the hill then down the hill.
- If the focus metric is decreasing: Change directions until the focus metric increases and then decreases by the value set in autofocusChangePct.

Message ID 0xB5

Byte Offset	Name	Description										
4	cameraldx	Lens type. See SLASetLensParameters_t .										
5	method	The autofocus algorithm to use. 0 = Scan and 1 = Seek.										
6-7	calibStart	Frame Offset Calibration: focus start position. Set this to the farther focus value. Trace Curve creation: zoom start position. Set this to the wider zoom position.										
8-9	calibEnd	Frame Offset Calibration: focus end position. Set this to the nearer focus value. Trace Curve creation: zoom end position. Set this to the narrower zoom position.										
10	frameOffsetMult8	The lag between when a new focus position is requested and when the focus position is received from the lens. Unit is eighths of a step, roughly corresponding to frames.										
11	calibFlags	Starts or stops calibration, and specifies which calibration to perform. If both are selected, the calibration will not be performed.<table><tr><th>Bit</th><th>Option</th></tr><tr><td>0</td><td>Start Calibration</td></tr><tr><td>1</td><td>Stop Calibration</td></tr><tr><td>2</td><td>Calculate frame offset</td></tr><tr><td>3</td><td>Create a track curve for keeping focus through zoom</td></tr></table>	Bit	Option	0	Start Calibration	1	Stop Calibration	2	Calculate frame offset	3	Create a track curve for keeping focus through zoom
Bit	Option											
0	Start Calibration											
1	Stop Calibration											
2	Calculate frame offset											
3	Create a track curve for keeping focus through zoom											
12-13	reserved1	<i>Reserved</i>										
14-15	rateAdjust	Seek: Adjusts the automatically calculated speed of the lens. The value range is 0 to 255, where 100 is the default speed of motion. Scan: adjusts the focus speed to be slower as you zoom in. A higher number corresponds to slower speeds as you zoom in. 255 = decrease the speed in exact proportion to the remaining zoom range; 0 = do not change speed based on zoom level.										
16	changePercent10ths	Seek: Autofocus Change Percent Adjusts the sensitivity to focus metric change in 10ths of a percent. The value range is 0 to 255 with a default of 40 (equivalent to 4% change). When using the Seek AutoFocus algorithm (see "method" above), this value is the percentage change in the focus metric that is considered significant enough to indicate that the focus metric is going in the right (or wrong) direction.										

17-18	reserved2	<i>Reserved</i>
19	nInitialSteps	Scan: The number of steps the algorithm takes to move to an initial (focused far) position before beginning the focus search. During the initial step, the lens will move at the speed specified as the coarse speed (5 is the default). A small value (3-5) is typically sufficient to move to an area that encompasses the focused position.
20-21	FineSpeed	Scan: The lens focus speed when in fine phase. The likelihood of being near max focus is high.
22-23	MidSpeed	Scan: The lens focus speed at the start of the algorithm and when in mid phase. It is unlikely the lens is close to being focused.
24-25	CoarseSpeed	Scan: The lens focus speed when in coarse phase. It is very unlikely the lens is focused. Coarse speed will only be used if: 1) the current metric is far below the highest metric seen (see coarseThresh), and 2) the algorithm never entered fine phase (ie, it is unlikely a focus peak was already found)
26	CoarseThresh	Scan: The ratio (the numerator out of 255) of the current focus metric to the highest focus metric seen required to switch to coarse phase. The default is 63, for a threshold value of $63/255 = 0.25$. A higher number increases sensitivity since a smaller difference is required to switch to coarse phase. Increase this value to make the algorithm end earlier.
27	FineThresh	Scan: Ratio (the numerator out of 255) of change from the previous step's metric to the current metric required to switch to fine phase. The default is 51, for a threshold value of $51/255 = 0.20$. A higher number decreases sensitivity since the difference between the current and previous metric must be high to switch to fine phase. Increase this value to increase the speed of the algorithm, especially in low-contrast scenes.
28	CdownMax	Scan: The number of consecutive decreases considered a "true" decrease. If in fine phase, CdownMax consecutive decreases in the focus metric will trigger a switch to mid phase.
29-30	reserved3	<i>Reserved</i>
31	fileName.len	string length
32-...	fileName.str	Filename field. Used when saving and loading track curves for tracking focus through zoom.
varies	data.len	string length
varies	data.str	Customer-defined data. It is not used by the autofocus object by default. This can be used for adding custom parameters to user-created autofocus algorithms.

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACustomClassifier_t Struct Reference

[Classification](#)

Description

Custom Classifier.

```
typedef struct {
    u8 cameraIndex;
    SVPLenString_t fileName;
} SLACustomClassifier_t;
```

Used to update the default classifier model with a customer specific model. File name length is restricted to 256 characters.

NOTE: To remove a custom classifier and revert to the built-in classifier, send this message with an empty filename (length 0).

NOTE: 17XX and 4000 Only

Message ID 0xA7

Byte Offset	Name	Description
4	cameraIndex	Camera Index
5	fileName.len	string length
6-...	fileName.str	Filename for custom classifier (example: classifier_params.cls)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACustomResponse_t Struct Reference

[General](#)

Description

A custom response that a customer can send from an overlay DLL.

```
typedef struct {
    SVPLenData_t customData;
} SLACustomResponse_t;
```

Can also be used to send data into an overlay DLL See vtcallbacks_sample for usage.

Message ID 0xBC

Byte Offset	Name	Description
4	customData.len	string length
5-...	customData.str	Customer defined data. Max length 255

Sightline Intelligence

Command, Control, and Script

API v3.11

SLADeadPixel_t Struct Reference

[Non Uniformity Correction](#)

Description

Modify Dead Pixel List Manually.

```
typedef struct {
    u8 cameraIndex;
    u8 mode;
    u16 a;
    u16 b;
    u8 c;
    u8 d;
} SLADeadPixel_t;
```

Used to indicate a specific pixel is dead. This pixel is then added to the current dead pixel table and replaced along with other dead pixels in dead pixel table.

Mode Based Parameters

Parameter	a	b	c	d
Add Dead Pixel	Column (x)	Row (y)	Update Map	–
Remove Dead Pixel	Column (x)	Row (y)	Update Map	–
Dynamic Dead Pixel Detection	– Kernel Size	Maxium Pixel Difference	–	–

Update Map

This parameter is used to tell Videotrack whether it should expect more pixels to modify. If this is set Videotrack will not update the Dead Pixel mapping and the replacements will not be reflected in the steamed image BUT the pixels will still be modified and when the mapping is updated will be reflected in the streamed image. This parameter is used when a list is being programmatically sent to Videotrack.

E.g. Sending 3 pixels to add to the dead list. (Pixel 0, c = 1) (Pixel 1, c = 1) (Pixel 2, c = 0) When pixel 2 is received the Dead Pixel mapping will be updated and all 3 pixels will now be replaced in the streamed image.

Value	Description
0	Update Dead Pixel mapping. Use this on the last pixel being modified

1	Do not update Dead Pixel mapping. Use this if you are sending a list of pixels and are not on the last pixel.
---	---

Kernel Size

8-bit (0-255) number that determines the width and height around a pixel that is used to during the Dynamic Detection Algorithm. This is used when calculating the Pixel difference. A larger number means the pixel is compared against more surrounding pixels. If a 0 is used here the algorithm will default to 15.

Maximum Pixel Difference

This parameter is used to determine whether an individual pixel is considered "Dead". If a 0 is used here the algorithm will use a variable automatic number based on the surrounding pixels in the same height and width as the kernel size.

Message ID 0xA8

Byte Offset	Name	Description								
4	cameraIndex	<u>Camera Index</u>								
5	mode	Mode of operation								
		<table><tr><th>Value</th><th>Description</th></tr><tr><td>0</td><td>Add Dead Pixel</td></tr><tr><td>1</td><td>Remove Dead Pixel</td></tr><tr><td>2</td><td>Dynamic Dead Pixel Detection</td></tr></table>	Value	Description	0	Add Dead Pixel	1	Remove Dead Pixel	2	Dynamic Dead Pixel Detection
		Value	Description							
		0	Add Dead Pixel							
		1	Remove Dead Pixel							
2	Dynamic Dead Pixel Detection									
6-7	a	<u>Mode Based Parameters</u>								
8-9	b	<u>Mode Based Parameters</u>								
10	c	<u>Mode Based Parameters</u>								
11	d	<u>Mode Based Parameters</u>								

Sightline Intelligence

Command, Control, and Script

API v3.11

SLADeadPixelStats_t Struct

Reference

[Non Uniformity Correction](#)

[List of all members](#)

Description

Statistics from Dead Pixel Replacement calibration process.

```
typedef struct {
    u8 cameraIndex;
    s32 nDead;
    u32 nGainLo;
    u32 nGainHi;
    u32 nAvgLo;
    u32 nAvgHi;
    u32 nOffLo;
    u32 nOffHi;
    u32 nDevHi;
} SLADeadPixelStats_t;
```

Message ID 0xA1

Byte Offset	Name	Description
4	cameraIndex	Camera Index
5-8	nDead	Number of Dead Pixels Detected
9-12	nGainLo	Number of pixels with gain less than the minimum dead pixel gain
13-16	nGainHi	Number of pixels with gain greater than the maximum dead pixel gain
17-20	nAvgLo	Number of pixels with average value less than the minimum input pixel value
21-24	nAvgHi	Number of pixels with average value greater than the maximum input pixel value
25-28	nOffLo	Number of pixels with offset less than the minimum dead pixel offset
29-32	nOffHi	Number of pixels with offset greater than the maximum dead pixel offset
33-36	nDevHi	Number of pixels with standard deviation greater than the maximum dead pixel standard deviation

Sightline Intelligence

Command, Control, and Script

API v3.11

[List of all members](#)

SLADecoderParameters_t Struct

Reference

[Capture](#)

Description

Configure format of Onboard decoder.

```
typedef struct {
    u32be ipAddr;
    u16 port;
    SVPLenString\_t customStr;
    u8 cameraIndex;
} SLADecoderParameters\_t;
```

Set the IP address and port number where the video will be received and decoded. Only one IP Camera can be connected at a time.

Message ID 0x99

Byte Offset	Name	Description
4-7	ipAddr	IpAddress where the system will look for the encoded video. This is typically the IP address of the board itself, but could be another address under certain cases, such as multicast video.
8-9	port	Port where the system will look for the encoded video.
10	customStr.len	string length
11-...	customStr.str	Custom string - if non null replaces use of ipAddr and port This can be a stream description like: udb://@224.10.10.10:15004 Or the name of a recorded .ts video file like: \filename.ts . The leading "\" is replaced with the microSD card path or the full path can be specified See EAN-Decoder.pdf
varies	cameraIndex	Camera Index

Sightline Intelligence

Command, Control, and Script

API v3.11

SLADesignateSelectedTrackPrimary_t Struct Reference

[Tracking](#)

Description

Command the system to designate the selected track as primary.

```
typedef struct {  
    u8 cameraIndex;  
} SLADesignateSelectedTrackPrimary_t;
```

Message ID 0x32

Byte Offset	Name	Description
4	cameraIndex	Camera Index

Sightline Intelligence

Command, Control, and Script

API v3.11

SLADetailedTimingMessage_t Struct Reference

[General](#)

Description

Packet is sent from the SLA hardware to report DSP and ARM detailed timing measurements.

```
typedef struct {
    u8 reserved0;
    u8 reserved1;
    SVPEndString\_t data;
} SLADetailedTimingMessage\_t;
```

Use [SLASystemStatusMode_t](#) to configure this message.

NOTE: This packet is sent only to Ethernet Command and Control Reply Port, and never sent to a Serial port.

Message ID 0x88

Byte Offset	Name	Description
4	reserved0	<i>Reserved</i>
5	reserved1	<i>Reserved</i>
6	data	Text characters for the message itself, comprised of one or more lines delimited by '\n' character (terminating null character may or may not be present)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLADigitalVideoParserParameters_t

Struct Reference

[Capture](#)

Description

Set parameters for the digital video SightLine protocol parser.

```
typedef struct {
    u8 cameraIndex;
    u8 enabled;
    u8 flags;
    u8 oneMessagePerImage;
    u8 usePixelValue;
} SLADigitalVideoParserParameters_t;
```

This function provides the ability to manipulate the video frames to encode SightLine protocol messages on the **two lines** immediately following the valid video window defined in [SLASetAcquisitionParameters_t](#). The first pixel for encoding in the Luma (Y) buffer is either:

- ((Valid_ROI_Row)+(ROI_Height))*(Image_Width), if bit 1 of **flags** is set to false, (first line after ROI window)
- (Image_Height)*(Image_Width), if bit 1 of **flags** is set to true (first line after acquired window).

The maximum number of pixels for encoding is Image_Width*2 in either case.

Performance Notes:

- If flags Bit 0 is zero, parsing may take more time, as the entire buffer is scanned for SLA protocol message(s).
- If oneMessagePerImage is zero, parsing may take more time, as more SLA protocol messages are looked for after any successful parsing.
- Setting usePixelValue to 0 or 2 may be required in the *unlikely* case that the exact value of the pixel cannot be guaranteed.

Also sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x91

Byte Offset	Name	Description
4	cameraIndex	Camera Index

5	enabled	Enable parsing (0 = disable, 1 = enable). Default is disable.											
6	flags	<table><tr><th>Bit</th><th>Description</th></tr><tr><td>0</td><td>SLA protocol message is expected to start at first pixel (0 = false, 1 = true) default is true</td></tr><tr><td>1</td><td>SLA protocol message is encoded beginning at first line right after Acquired window. (1 = true, 0 = false: the message is embedded right after the ROI window, as defined in <u>SLASetAcquisitionParameters_t</u>).</td></tr><tr><td>2</td><td>SLA protocol message is encoded at last line of frame or acquired window. (0 = false. 1 = true : the message is embedded at the last line, depending on bit 1 above, of the ROI window or image (overrides bit 1)).</td></tr></table>	Bit	Description	0	SLA protocol message is expected to start at first pixel (0 = false, 1 = true) default is true	1	SLA protocol message is encoded beginning at first line right after Acquired window. (1 = true, 0 = false: the message is embedded right after the ROI window, as defined in <u>SLASetAcquisitionParameters_t</u>).	2	SLA protocol message is encoded at last line of frame or acquired window. (0 = false. 1 = true : the message is embedded at the last line, depending on bit 1 above, of the ROI window or image (overrides bit 1)).			
Bit	Description												
0	SLA protocol message is expected to start at first pixel (0 = false, 1 = true) default is true												
1	SLA protocol message is encoded beginning at first line right after Acquired window. (1 = true, 0 = false: the message is embedded right after the ROI window, as defined in <u>SLASetAcquisitionParameters_t</u>).												
2	SLA protocol message is encoded at last line of frame or acquired window. (0 = false. 1 = true : the message is embedded at the last line, depending on bit 1 above, of the ROI window or image (overrides bit 1)).												
7	oneMessagePerImage	0 - Multiple protocol message may be present in the frame 1 - a single SLA protocol message is expected per frame											
8	usePixelValue	<table><tr><th colspan="2">Interpret pixel byte value.</th></tr><tr><th>Value</th><th>Description</th></tr><tr><td>0</td><td>Each image byte will be interpreted as a bit in LSB to MSB order. Any value less than 128 translates to 0, any value 128 and above translates to 1</td></tr><tr><td>1 (default)</td><td>Each image byte will be interpreted as a byte</td></tr><tr><td>2</td><td>Each image byte will be interpreted as a bit in MSB to LSB order. Any value less than 128 translates to 0, any value 128 and above translates to 1</td></tr></table>	Interpret pixel byte value.		Value	Description	0	Each image byte will be interpreted as a bit in LSB to MSB order. Any value less than 128 translates to 0, any value 128 and above translates to 1	1 (default)	Each image byte will be interpreted as a byte	2	Each image byte will be interpreted as a bit in MSB to LSB order. Any value less than 128 translates to 0, any value 128 and above translates to 1	
Interpret pixel byte value.													
Value	Description												
0	Each image byte will be interpreted as a bit in LSB to MSB order. Any value less than 128 translates to 0, any value 128 and above translates to 1												
1 (default)	Each image byte will be interpreted as a byte												
2	Each image byte will be interpreted as a bit in MSB to LSB order. Any value less than 128 translates to 0, any value 128 and above translates to 1												

Sightline Intelligence

Command, Control, and Script

API v3.11

SLADirectoryStatisticsReply_t Struct Reference

[Recording](#)

Description

Sent in response to a [SLASetSDRecordingParameters_t](#).

```
typedef struct {
    u32 totalSize;
    u32 spaceUsed;
    u32 totalSizeUpper;
    u32 spaceUsedUpper;
} SLADirectoryStatisticsReply_t;
```

getDirectory request. See [SLACurrentSDCardDirectoryInfo_t](#) to get the list of files.

Message ID 0x79

Byte Offset	Name	Description
4-7	totalSize	Maximum disk size in Kbytes
8-11	spaceUsed	Total space used in Kbytes
12-15	totalSizeUpper	Maximum disk size in Kbytes, Upper 32 bits for large sizes.
16-19	spaceUsedUpper	Total space used in Kbytes, Upper 32 bits for large sizes.

Sightline Intelligence

Command, Control, and Script

API v3.11

SLADisplayAngle_t Struct Reference

[Display](#)

Description

Set display rotation angle settings for a single camera or for all cameras See also [SLASetDisplayParameters_t](#).

```
typedef struct {
    u8 cameraIndex;
    u16 rotationDegrees;
    u16 rotationLimit;
    u8 decayRate;
} SLADisplayAngle\_t;
```

Use rotationLimit and decayRate for smooth rotation transitions.

Also sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x5C

Byte Offset	Name	Description
4	cameraIndex	Camera Index 255 = all
5-6	rotationDegrees	Rotation angle in degrees (0..360) * 128
7-8	rotationLimit	Rotation rate limit in degrees (0..360) * 128 per frame, (default 256).
9	decayRate	Rotation decay rate per frame, 0(0%) to 255(100%) (default 50).

Sightline Intelligence

Command, Control, and Script

API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) ■ [Camera Configuration](#)

[List of all members](#)

SLADoDetectSnapShot_t Struct Reference

[Recording](#) | [Detection](#)

Description

Execute a track/detection snapshot to the MicroSD card or an external FTP server.

```
typedef struct {
    u8 cameraIndex;
    u8 mode;
    u8 sizeType;
    u16 snapSize;
    u8 maxPerFrame;
    u8 minFramesBetween;
    u8 flags;
    u8 reserved1;
    SVPLenString\_t baseName;
} SLADoDetectSnapShot\_t;
```

Use [SLASetSnapShot_t](#) first to specify the FTP server, login, and other relevant parameters.

See [File Recording](#) for additional details on file naming, etc.

Notes:

1. These small snapshots may also be referred to as **chips**.
2. These source setting in [SLASetSnapShot_t](#) must be the source image.
3. The downsample setting in [SLASetSnapShot_t](#) is not valid for detection snapshots.
4. The chips must be at least 32x32 and the size must be a multiple of 16.
5. Auto foldering as described in [SLADoSnapShot_t](#) will always be used in this mode since the number of snapshots can grow very quickly. The default max per folder is 2000 files.
6. Files will be saved in the format <baseName>_C_NNNN_TRK.jpg where C is the camera index and NNNN is the incrementing counter and TRK is the track index. Note that detection track index values start at 10 and go to 109. (or 20-109 with Advanced Tracking Appbit)
7. Use the flags byte to insert an 8 byte utc timestamp instead of the NNNN number into the filename. This number corresponds to the timestamp accessible through telemetry and klv.
8. Caution should be used when running with scaling as it can reduce frame rates as well as how many snapshots you can record per second.
9. Raw 16-bit snapshots are not available, any 16-bit input will have an AGC applied before output.
10. The only valid settings for format in [SLASetSnapShot_t](#) is JPEG or PNG.

11. This feature is not available in [Demo Mode](#).

Message ID 0xAB

Byte Offset	Name	Description								
4	cameraIndex	<u>Camera Index</u>								
5	mode	Detection SnapShot Mode <table><tr><th>Bits</th><th>Description</th></tr><tr><td>0</td><td>SnapShot Detections - Includes Aerial MTI, Staring MTI, Motion MTI, Anomaly, Maritime, Radiometric, and Blob detect modes.</td></tr><tr><td>1</td><td>SnapShot Tracks - Includes all tracking modes as well as Vehicle and Drone detect modes.</td></tr><tr><td>2-7</td><td>*Reserved*</td></tr></table>	Bits	Description	0	SnapShot Detections - Includes Aerial MTI, Staring MTI, Motion MTI, Anomaly, Maritime, Radiometric, and Blob detect modes.	1	SnapShot Tracks - Includes all tracking modes as well as Vehicle and Drone detect modes.	2-7	*Reserved*
Bits	Description									
0	SnapShot Detections - Includes Aerial MTI, Staring MTI, Motion MTI, Anomaly, Maritime, Radiometric, and Blob detect modes.									
1	SnapShot Tracks - Includes all tracking modes as well as Vehicle and Drone detect modes.									
2-7	*Reserved*									
6	sizeType	Detection SnapShot Sizing <table><tr><th>Values</th><th>Description</th></tr><tr><td>0</td><td>Fixed Size - Square centered on detection of <u>snapSize</u> is used.</td></tr><tr><td>1</td><td>Scaled - Detection is scaled to <u>snapSize</u>.</td></tr><tr><td>2</td><td>Auto - snapSize is ignored and the detection size is used.</td></tr></table>	Values	Description	0	Fixed Size - Square centered on detection of <u>snapSize</u> is used.	1	Scaled - Detection is scaled to <u>snapSize</u> .	2	Auto - snapSize is ignored and the detection size is used.
Values	Description									
0	Fixed Size - Square centered on detection of <u>snapSize</u> is used.									
1	Scaled - Detection is scaled to <u>snapSize</u> .									
2	Auto - snapSize is ignored and the detection size is used.									
7-8	snapSize	Detection snap shot size. See <u>sizeType</u> . Minimum and default size is 32. Maximum size is 512. Size must be a multiple of 16.								
9	maxPerFrame	Maximum number of detections to snap per frame. Default value is 1, valid range is 1-10. Setting this number too high will result in lost snapshots.								
10	minFramesBetween	Minimum number of frames between when you snapshot a detection and when you snapshot it again. Default value is 1, valid range is 1-255.								
11	flags	Detection SnapShot Mode <table><tr><th>Bits</th><th>Description</th></tr><tr><td>0</td><td>Use 8-byte UTC time instead of NNNN for the filename. Same timestamp that shows up in telemetry messages.</td></tr><tr><td>1-7</td><td>*Reserved*</td></tr></table>	Bits	Description	0	Use 8-byte UTC time instead of NNNN for the filename. Same timestamp that shows up in telemetry messages.	1-7	*Reserved*		
Bits	Description									
0	Use 8-byte UTC time instead of NNNN for the filename. Same timestamp that shows up in telemetry messages.									
1-7	*Reserved*									
12	reserved1	<i>Reserved</i>								
13	baseName.len	string length								
14-...	baseName.str	Base file name of saved files. See <u>File Recording</u> for how this is used to construct file names.								

Sightline Intelligence

Command, Control, and Script

API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) ■ [Camera Configuration](#)

[List of all members](#)

SLADoSnapShot_t Struct Reference

[Recording](#)

Description

Execute an image snapshot.

```
typedef struct {
    u8 frameStep;
    u8 numFrames;
    SVPLenString\_t fileName;
    u8 snapAllCameras;
    u8 shouldScan;
    u8 autoFolder;
    u16 maxFiles;
    u8 flags;
} SLADoSnapShot\_t;
```

Use [SLASetSnapShot_t](#) first to specify the FTP server, login, and other relevant parameters.

See [File Recording](#) for additional details on file naming, etc.

Notes:

1. Snapshot applies to the **Command Camera**.
2. Snapshots are truncated to a multiple of 128 pixels wide and a multiple of 8 pixels high.
3. When saving a snapshot of the **Displayed Image**:
 - If the Command Camera is not displayed, then no snapshot will be taken.
 - If the Command Camera is displayed to multiple displays(say, Analog and Network0 with 1280x720 resolution), then the snapshot will be taken from the largest of them(so it will be 1280x720 in size).
 - If the Command Camera is used for a multi-camera operation(e.g. PiP, Blend, etc.), then the snapshot will be taken but the resulting filename may have different camIdx number(if PiP or 2 - Up, it is the Main camera index, if Blend, it is the Fixed camera index).
 - If the video is being displayed to both Network0 and Network1, then the result will be the same size as Network0. This applies to 2 and 3 above.
 - If a multi-camera operation is displayed, and also a single camera is displayed to Network1 as 1-up at the same time, then it is unpredictable which image will be taken for snapshot.

NOTE: Capture snapshot is not available in [Demo Mode](#).

Message ID 0x60

Byte Offset	Name	Description						
4	frameStep	Frame Step – step between frames (e.g. 2 = every other)						
5	numFrames	Number of snapshots to take (1 to 254), 255 = continuous, 0 = Stop; Ignored if Snap All Cameras is used.						
6	fileName.len	string length						
7-...	fileName.str	Base file name of saved files. See File Recording for how this is used to construct file names.						
varies	snapAllCameras	Bit mask of Cameras to Snap (for example, use 0x5 to snap cameras 0 and 2). 0 = snap the command Camera. For multi camera snaps, only a single frame snap-shot is allowed. NOTE: Do not use a file name ending in a digit when snapAllCameras is used. NOTE: Snapshot settings are per camera and must be configured for each camera using SLASetSnapShot_t						
varies	shouldScan	When using file auto-numbering, begin file numbering after highest-numbered existing filename. This requires a scan of the contents on the SD card once at startup and again if the filename is changed or if the autoFolder state is changed. If this is not set each time the filename or the autoFolder state is changed the auto-numbering will restart at 0.						
varies	autoFolder	1 = create a new folder every maxFiles, 0 = don't create new folders NOTE: Does not work for FTP recording, or recording to non SD card or USB devices.						
varies	maxFiles	Max files per folder (2000 is default, 20000 is max). NOTE: maxFiles is the maximum number of files per folder, the folder may auto increment sooner under certain conditions such as snapshots being taken too fast.						
varies	flags	SnapShot Flags<table><tr><th>Bits</th><th>Description</th></tr><tr><td>0</td><td>Enable burst mode. In this mode up to 120 snapshots at 1920x1080 (or smaller) can be recorded to either an SD card or USB device without dropping frames. 4000 only</td></tr><tr><td>1-7</td><td>*Reserved*</td></tr></table>	Bits	Description	0	Enable burst mode. In this mode up to 120 snapshots at 1920x1080 (or smaller) can be recorded to either an SD card or USB device without dropping frames. 4000 only	1-7	*Reserved*
Bits	Description							
0	Enable burst mode. In this mode up to 120 snapshots at 1920x1080 (or smaller) can be recorded to either an SD card or USB device without dropping frames. 4000 only							
1-7	*Reserved*							

Sightline Intelligence

Command, Control, and Script

API v3.11

SLADrawObject_t Struct Reference

Description

Deprecated, use [SLADrawOverlay_t](#)

```
typedef struct {
    u8 objId;
    u8 action;
    u8 propertyFlags;
    u8 type;
    u16 a;
    u16 b;
    u16 c;
    u16 d;
    u8 backgroundColor;
    SVPEndString\_t text;
} SLADrawObject\_t;
```

Deprecated, use [SLADrawOverlay_t](#)

Object Types

Value	Description
0	Circle
1	Rectangle
2	Line
3	<i>Reserved</i>
4	Filled Circle
5	Filled Rectangle
6	Text Extended
7	<i>Reserved</i>
8	KLV Field
9 to 254	<i>Reserved</i>
255	Error or Object Not Found

Object Type Specific Parameters

Object Type	a	b	c	d
Circle	Center X	Center Y	Radius	–
Filled Circle	Center X	Center Y	Radius	–
Rectangle	Upper left corner X	Upper left corner Y	Width	Height
Filled rectangle	Upper left corner X	Upper left corner Y	Width	Height
Line	End point 1 X	End point 1 Y	End point 2 X	End point 2 Y
<u>Text Extended</u>	Upper left X	Upper left Y	<u>H Scaling</u> <u>V Scaling</u>	<u>Font ID</u> <u>Spacing</u>
<u>KLK Field</u>	Upper left X	Upper left Y	<u>Scaling, Font ID</u>	<u>Field, Format</u>

H Scaling

Horizontal scale shifted left 5 (e.g. 32 means no scaling. 0 is also interpreted as no scaling). Not used for font ID 5 and higher.

V Scaling

Vertical scale shifted left 5. Not used for font ID 5 and higher.

Spacing

Pixels between a font to the next font. If 0, then uses the font's default spacing. Not used for font ID 5 and higher.

Text Extended Object Type

Parameter	Bits	Description
c	0-7	Horizontal scaling is value of (Bits 0-7)/32, 0 = no scaling
c	8-15	Vertical scaling is value of (Bits 8-15)/32, 0 = no scaling
d	0-4	Font ID 0 = Courier, 1 = Courier bold, 5 = Arial, 6 = Arial Bold, 7 = Verdana, 8 = Verdana Bold, 9 = Calibri, 10 = Calibri Bold
d	5	Enable fixed width – applicable to fonts with ID 5..10
d	6	Enable outline effect – applicable when background color set to transparent
d	7	Enable shadow effect – applicable when background color set to transparent
d	8-15	Pixels between a font to the next font; 0 = font default spacing

KLK Field Object Type

Parameter	Bits	Description
c	0-7	Horizontal and Vertical scaling is value of (Bits 0-7)/32, 0 = no scaling
c	8-12	Font ID 0 = Courier, 1 = Courier bold, 5 = Arial, 6 = Arial Bold, 7 = Verdana, 8 = Verdana Bold, 9 = Calibri, 10 = Calibri Bold
c	13	Enable fixed width – applicable to fonts with ID 5..10

c	14	Enable outline effect – applicable when background color set to transparent
c	15	Enable shadow effect – applicable when background color set to transparent
d	0-7	KLV Field
d	8-15	KLV Format

KLV Field Options

Value	Field Name	KLV Format
0	UTC Time	Time
1	Mission ID	N/A (Send 0)
2	Platform Heading	Angle
3	Platform Pitch	Angle
4	Platform Roll	Angle
7	Sensor Latitude	Position
8	Sensor Longitude	Position
9	Sensor Altitude	Distance
10	Sensor HFOV	Angle
11	Sensor VFOV	Angle
12	Sensor Relative Azimuth Angle	Angle
13	Sensor Relative Elevation Angle	Angle
14	Sensor Relative Roll Angle	Angle
15	Frame Center Latitude	Position
16	Frame Center Longitude	Position
17	Frame Center Elevation	Distance
19	Slant Range	Distance
20	Target Location Elevation	Distance
21	Target Location Latitude	Position
22	Target Location Longitude	Position

KLV Format Types

Type	Values
Time	0 = YYYY-MM-DD hh:mm:ss, 1 = DDMonthYYYY, 2 = HH:MM:SS, 3 = Month DD YYYY, 4 = MM/DD/YY, 5 = MM/DD/YYYY, 6 = DD HHMMZ MON YY (DTG), 7 = DDHHMMZMONYY (DTG - No Spaces)
Angle	0 = Degrees, 1 = Radians, 2 = Milliradians
Distance	0 = Meters, 1 = Feet
Position	0 = MGRS-10 (1 Meter), 1 = Decimal Degrees, 2 = Degrees Minutes Seconds, 3 = MGRS-8 (10 Meter)

Color To Number Mapping

Value	Color	Value	Color
0	White	8	Light Green
1	Black	9	Green
2	Light Gray	10	Dark Green
3	Gray	11	Red
4	Dark Gray	12	Orange
5	Light Blue	13	Yellow
6	Blue	14	Transparent
7	Dark Blue	15	Automatic

Message ID 0x3B

Byte Offset	Name	Description																					
4	objId	Unique object ID																					
		<table><tr><th>ID</th><th>Description - Maximum number of unique objects is 128</th></tr><tr><td>0</td><td>Used for destroying all objects when Action==Destroy</td></tr><tr><td>1 - 199</td><td>Unique ID for each object created</td></tr><tr><td>200 - 255</td><td>Reserved</td></tr></table>	ID	Description - Maximum number of unique objects is 128	0	Used for destroying all objects when Action==Destroy	1 - 199	Unique ID for each object created	200 - 255	Reserved													
		ID	Description - Maximum number of unique objects is 128																				
		0	Used for destroying all objects when Action==Destroy																				
		1 - 199	Unique ID for each object created																				
200 - 255	Reserved																						
5	action	Action to perform 0 = Destroy specified object ID 1 = Create new object with specified object ID																					
6	propertyFlags	<table><tr><th>Bits</th><th colspan="2">Description</th></tr><tr><td rowspan="4">0 - 3</td><td>1</td><td>Coordinates are specified in source coordinate space and object is drawn at the specified location in the source image</td></tr><tr><td>2</td><td>Coordinates are specified in display coordinate space and object is moved along with scene motion – equivalent to scene mode tracking with a custom overlay</td></tr><tr><td>4</td><td>Static overlays that do not move (such as text), coordinates are specified in display coordinate space (default)</td></tr><tr><td>other</td><td>Reserved</td></tr><tr><td>4 - 5</td><td colspan="2">Reserved</td></tr><tr><td>6</td><td colspan="2">0 – don't save this object, 1 – save this object to parameter file (only valid if action is set to create)</td></tr><tr><td>7</td><td colspan="2">0 – coordinate origin (0,0) is at the center of the image, 1 – coordinate origin is at the upper left corner of the image</td></tr></table>	Bits	Description		0 - 3	1	Coordinates are specified in source coordinate space and object is drawn at the specified location in the source image	2	Coordinates are specified in display coordinate space and object is moved along with scene motion – equivalent to scene mode tracking with a custom overlay	4	Static overlays that do not move (such as text), coordinates are specified in display coordinate space (default)	other	Reserved	4 - 5	Reserved		6	0 – don't save this object, 1 – save this object to parameter file (only valid if action is set to create)		7	0 – coordinate origin (0,0) is at the center of the image, 1 – coordinate origin is at the upper left corner of the image	
		Bits	Description																				
		0 - 3	1	Coordinates are specified in source coordinate space and object is drawn at the specified location in the source image																			
			2	Coordinates are specified in display coordinate space and object is moved along with scene motion – equivalent to scene mode tracking with a custom overlay																			
			4	Static overlays that do not move (such as text), coordinates are specified in display coordinate space (default)																			
			other	Reserved																			
		4 - 5	Reserved																				
		6	0 – don't save this object, 1 – save this object to parameter file (only valid if action is set to create)																				
7	0 – coordinate origin (0,0) is at the center of the image, 1 – coordinate origin is at the upper left corner of the image																						

7	type	<u>Object Types</u>						
8-9	a	<u>Object Type Specific Parameters</u>						
10-11	b	<u>Object Type Specific Parameters</u>						
12-13	c	<u>Object Type Specific Parameters</u>						
14-15	d	<u>Object Type Specific Parameters</u>						
16	backgroundColor	<u>Object Color</u>						
		<table><tr><th>Bits</th><th>Description</th></tr><tr><td>0-3</td><td>Background color: for vertical and horizontal lines, the background color is drawn as a shadow; use transparent (14) to disable drawing of shadow</td></tr><tr><td>4-7</td><td>Foreground color: for filled objects "Automatic" will result in object being filled with a partially transparent middle gray value which can be used as a background for drawing text to ensure the text is always visible. For non filled objects it will be either white or black depending on the scene content. Note: See action byte for setting transparency level for filled objects. Note: Transparent is not supported for text foreground</td></tr></table>	Bits	Description	0-3	Background color: for vertical and horizontal lines, the background color is drawn as a shadow; use transparent (14) to disable drawing of shadow	4-7	Foreground color: for filled objects " Automatic " will result in object being filled with a partially transparent middle gray value which can be used as a background for drawing text to ensure the text is always visible. For non filled objects it will be either white or black depending on the scene content. Note: See action byte for setting transparency level for filled objects. Note: Transparent is not supported for text foreground
		Bits	Description					
		0-3	Background color: for vertical and horizontal lines, the background color is drawn as a shadow; use transparent (14) to disable drawing of shadow					
4-7	Foreground color: for filled objects " Automatic " will result in object being filled with a partially transparent middle gray value which can be used as a background for drawing text to ensure the text is always visible. For non filled objects it will be either white or black depending on the scene content. Note: See action byte for setting transparency level for filled objects. Note: Transparent is not supported for text foreground							
Note: Using the same foreground and background color for text including automatic will result in unreadable text.								
17	text	Text and KLV objects only: text string (64 characters max) For KLV objects use "%s" for time, mission ID, latitude, and longitudes and "%f" for all others. Example "Time: %s" Example "Slant Range: %05.0f (ft)" Example "HFOV: %2.1f (deg)" Example "Sensor Position: %s"						

Sightline Intelligence

Command, Control, and Script

API v3.11

SLADrawOverlay_t Struct Reference

[Overlays](#)

Description

Draw user specified overlay graphics on the screen.

```
typedef struct {
    u8 cameraIndex;
    u8 objId;
    u8 action;
    u8 propertyFlags;
    u8 type;
    u16 a;
    u16 b;
    u16 c;
    u16 d;
    u8 backgroundColor;
    SVPLenString\_t text;
    u16 e;
    u16 f;
} SLADrawOverlay\_t;
```

NOTE: This replaces [SLADrawObject_t](#)

NOTE: To get all the object ids of current drawn objects use [SLAGetParameters_t\(DrawObject\)](#) with an object id of 0. To get the current object information use [SLAGetParameters_t\(DrawOverlay\)](#).

NOTE: Applied to the Command Camera or the camera index specified.

See [Draw Order](#) for additional drawing details.

Object Types

Value	Description
0	Circle
1	Rectangle
2	Line
3	<i>Reserved</i>
4	Filled Circle
5	Filled Rectangle
6	Text Extended

7	<i>Reserved</i>
8	KLV Field
9	Cross
10	Square
11	Filled Square
12	KLV Widget
13 to 254	<i>Reserved</i>
255	Error or Object Not Found

Object Type Specific Parameters

Object Type	a	b	c	d	e	f
Circle	Center X	Center Y	Radius	Start angle (for arc)	<u>Line Thickness</u>	End angle (for arc)
Filled Circle	Center X	Center Y	Radius	Start angle (for arc)	<u>Line Thickness</u>	–
Rectangle	Upper left corner X	Upper left corner Y	Width	Height	<u>Line Thickness</u>	Angle
Filled rectangle	Upper left corner X	Upper left corner Y	Width	Height	<u>Line Thickness</u>	Angle
Line	End point 1 X	End point 1 Y	End point 2 X	End point 2 Y	<u>Line Thickness</u>	-
<u>Text Extended</u>	Upper left X	Upper left Y	<u>H Scaling</u> <u>V Scaling</u>	<u>Font ID</u> <u>Spacing</u>	–	-
<u>KLV Field</u>	Upper left X	Upper left Y	<u>Scaling, Font</u>	<u>Field, Format</u>	–	-
Cross	Center X	Center Y	Width	—	<u>Line Thickness</u>	-
Square	Center X	Center Y	Width/Height	Angle	<u>Line Thickness</u>	-
Filled Square	Center X	Center Y	Width/Height	Angle	<u>Line Thickness</u>	-
<u>KLV Widget</u>	Center at row	Center at col	Width	<u>Field, Type</u>	-	-

H/V Scaling

Horizontal/Vertical scale shifted left 5 (e.g. 32 means no scaling (= 100%). 0 is also interpreted as no scaling).
 Using a scale other than 100% may result in blurred/smeared text drawing.

Spacing

Pixels between a character to the next character. If 0, then uses the font's default spacing. For font ID 0 .. 10, this specifies the width of drawn fonts. For UserFont0..UserFont15, this specifies the extra space (padding) between fonts.

Line Thickness (New 3.4)

Thickness of line strokes in pixels. 0 is same as 1.

Text Extended Object Type

Parameter	Bits	Description
c	0-7	Horizontal scaling is value of (Bits 0-7)/32, 0 = no scaling
c	8-15	Vertical scaling is value of (Bits 8-15)/32, 0 = no scaling
d	0-4	Font ID 0 = Courier (Old Default), 1 = Courier bold, 5 = Arial, 6 = Arial Bold, 7 = Verdana, 8 = Verdana Bold, 9 = Calibri, 10 = Calibri Bold (17XX/4000 only) 16..31 = UserFont0..UserFont15 – SightLine provides default user fonts but any of them can be replaced by SLAUserFont_t . The new default is 16, which SightLine currently sets as Roboto Mono.
d	5	Enable fixed width – applicable to fonts with ID 5 .. 10, draw fonts with a predefined fixed width.
d	6	Enable outline effect – applicable when background color set to transparent
d	7	Enable shadow effect – applicable when background color set to transparent
d	8-15	Pixels between a font to the next font; 0 = font default spacing
e	0-15	Ignored, must be 0
f	0-15	Ignored, must be 0

KLV Field Object Type

Parameter	Bits	Description
c	0-7	Horizontal and Vertical scaling is value of (Bits 0-7)/32, 0 = no scaling
c	8-12	Font ID 0 = Courier, 1 = Courier bold, 5 = Arial, 6 = Arial Bold, 7 = Verdana, 8 = Verdana Bold, 9 = Calibri, 10 = Calibri Bold (17XX/4000 only) 16..31 = UserFont0..UserFont15 – SightLine provides default user fonts but any of them can be replaced by SLAUserFont_t . New 3.4
c	13	Enable fixed width – applicable to fonts with ID 5 .. 10, draw fonts with a predefined fixed width.
c	14	Enable outline effect – applicable when background color set to transparent
c	15	Enable shadow effect – applicable when background color set to transparent
d	0-7	KLV Field
d	8-15	KLV Format
e	0-15	Ignored, must be 0
f	0-15	Ignored, must be 0

KLV Field Options

Value	Field Name	KLV Format
0	UTC Time	Time
1	Mission ID	N/A (Send 0)
2	Platform Heading	Angle
3	Platform Pitch	Angle
4	Platform Roll	Angle
5	Platform Pitch Full	Angle
6	Platform Roll Full	Angle
7	Sensor Latitude	Position
8	Sensor Longitude	Position
9	Sensor Altitude	Distance
10	Sensor HFOV	Angle
11	Sensor VFOV	Angle
12	Sensor Relative Azimuth Angle	Angle
13	Sensor Relative Elevation Angle	Angle
14	Sensor Relative Roll Angle	Angle
15	Frame Center Latitude	Position
16	Frame Center Longitude	Position
17	Frame Center Elevation	Distance
19	Slant Range	Distance
20	Target Location Elevation	Distance
21	Target Location Latitude	Position
22	Target Location Longitude	Position

KLV Format Types

Type	Values
Time	0 = YYYY-MM-DD hh:mm:ss, 1 = DDMonthYYYY, 2 = HH:MM:SS, 3 = Month DD YYYY, 4 = MM/DD/YY, 5 = MM/DD/YYYY, 6 = DD HHMMZ MON YY (DTG), 7 = DDHHMMZMONYY (DTG - No Spaces)
Angle	0 = Degrees, 1 = Radians, 2 = Milliradians
Distance	0 = Meters, 1 = Feet
Position	0 = MGRS-10 (1 Meter), 1 = Decimal Degrees, 2 = Degrees Minutes Seconds, 3 = MGRS-8 (10 Meter

KLV Widget

Parameter	Bits	Description
c	0-7	Field with 0 = Heading sensor relative Azimuth angle

c	8-15	KLV Widget with 0 = Heading arrow based on the KLV platform heading and sensor relative azimuth angle.
---	------	--

Color To Number Mapping

Value	Color	Value	Color
0	White	8	Light Green
1	Black	9	Green
2	Light Gray	10	Dark Green
3	Gray	11	Red
4	Dark Gray	12	Orange
5	Light Blue	13	Yellow
6	Blue	14	Transparent for Background; Turquoise for Foreground
7	Dark Blue	15	<u>Automatic</u>

Automatic Color on 17XX/4000

Using automatic-color is not recommended on 17XX/4000 and future systems. It may be deprecated in the future on 17XX/4000. Due to the implementation of graphics drawing, using automatic-color may slow down processing speed.

NOTE: If automatic color is specified for a User Font (UserFont0 .. UserFont15), then it is ignored and treated as Transparent.

NOTE: Also, if automatic color is specified for a line or other shape, then line thickness is ignored and treated as 1 pixel thick.

Message ID 0x9C

Byte Offset	Name	Description									
4	cameraIndex	<u>Camera Index</u>									
5	objId	Unique object ID - Maximum number of unique objects is 128<table><tr><th>ID</th><th>Description</th></tr><tr><td>0</td><td>Used for destroying all objects when Action==Destroy</td></tr><tr><td>1 - 199</td><td>Unique ID for each object created</td></tr><tr><td>200 - 255</td><td>Reserved</td></tr></table>	ID	Description	0	Used for destroying all objects when Action==Destroy	1 - 199	Unique ID for each object created	200 - 255	Reserved	
ID	Description										
0	Used for destroying all objects when Action==Destroy										
1 - 199	Unique ID for each object created										
200 - 255	Reserved										
6	action	<table><tr><th>Bits</th><th>Description</th></tr><tr><td rowspan="2">0</td><td>0 Destroy specified object ID</td></tr><tr><td>1 Create new object with specified object ID</td></tr><tr><td>1-2</td><td>Reserved</td></tr><tr><td></td><td></td></tr></table>	Bits	Description	0	0 Destroy specified object ID	1 Create new object with specified object ID	1-2	Reserved		
Bits	Description										
0	0 Destroy specified object ID										
	1 Create new object with specified object ID										
1-2	Reserved										

		3-7	Transparency for filled circle, filled rectangle, and filled square. 0 = No transparency ... 31 = fully transparent. New 3.0																				
7	propertyFlags	<table><tr><th>Bits</th><th colspan="2">Description</th></tr><tr><td rowspan="4">0 - 3</td><td>1</td><td>Coordinates are specified in source coordinate space and object is drawn at the specified location in the source image</td></tr><tr><td>2</td><td>Coordinates are specified in display coordinate space and object is moved along with scene motion – equivalent to scene mode tracking with a custom overlay</td></tr><tr><td>4</td><td>Static overlays that do not move (such as text), coordinates are specified in display coordinate space (default)</td></tr><tr><td>other</td><td><i>Reserved</i></td></tr><tr><td>4 - 5</td><td colspan="2"><i>Reserved</i></td></tr><tr><td>6</td><td colspan="2">0 – don't save this object, 1 – save this object (only valid if action is set to create)</td></tr><tr><td>7</td><td colspan="2">0 – coordinate origin (0,0) is at the center of the image, 1 – coordinate origin is at the upper left corner of the image</td></tr></table>	Bits	Description		0 - 3	1	Coordinates are specified in source coordinate space and object is drawn at the specified location in the source image	2	Coordinates are specified in display coordinate space and object is moved along with scene motion – equivalent to scene mode tracking with a custom overlay	4	Static overlays that do not move (such as text), coordinates are specified in display coordinate space (default)	other	<i>Reserved</i>	4 - 5	<i>Reserved</i>		6	0 – don't save this object, 1 – save this object (only valid if action is set to create)		7	0 – coordinate origin (0,0) is at the center of the image, 1 – coordinate origin is at the upper left corner of the image	
Bits	Description																						
0 - 3	1	Coordinates are specified in source coordinate space and object is drawn at the specified location in the source image																					
	2	Coordinates are specified in display coordinate space and object is moved along with scene motion – equivalent to scene mode tracking with a custom overlay																					
	4	Static overlays that do not move (such as text), coordinates are specified in display coordinate space (default)																					
	other	<i>Reserved</i>																					
4 - 5	<i>Reserved</i>																						
6	0 – don't save this object, 1 – save this object (only valid if action is set to create)																						
7	0 – coordinate origin (0,0) is at the center of the image, 1 – coordinate origin is at the upper left corner of the image																						
8	type	<u>Object Types</u>																					
9-10	a	<u>Object Type Specific Parameters</u>																					
11-12	b	<u>Object Type Specific Parameters</u>																					
13-14	c	<u>Object Type Specific Parameters</u>																					
15-16	d	<u>Object Type Specific Parameters</u>																					
17	backgroundColor	<table><tr><th>Bits</th><th colspan="2">Description</th></tr><tr><td>0-3</td><td colspan="2">Background color: for vertical and horizontal lines, the background color is drawn as a shadow; use transparent (14) to disable drawing of shadow</td></tr><tr><td>4-7</td><td colspan="2">Foreground color: for filled objects "Automatic" will result in object being filled with a partially transparent middle gray value which can be used as a background for drawing text to ensure the text is always visible. For non filled objects it will be either white or black depending on the scene content. Note: See action byte for setting transparency level for filled objects. Note: Transparent is not supported for text foreground</td></tr></table> Note: Using the same foreground and background color for text including automatic will result in unreadable text.		Bits	Description		0-3	Background color: for vertical and horizontal lines, the background color is drawn as a shadow; use transparent (14) to disable drawing of shadow		4-7	Foreground color: for filled objects " Automatic " will result in object being filled with a partially transparent middle gray value which can be used as a background for drawing text to ensure the text is always visible. For non filled objects it will be either white or black depending on the scene content. Note: See action byte for setting transparency level for filled objects. Note: Transparent is not supported for text foreground												
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18	text.len	string length																					
		Text and KLV objects only: text string (64 characters max) For KLV objects use "%s" for time, mission ID, latitude, and longitudes and "%f" for all others. Example "Time: %s" Example "Slant Range: %05.0f (ft)" Example "HFOV: %2.1f (deg)"																					

19-...	text.str	Example "Sensor Position: %s" Character Encoding: Depending on the font type, different character encodings are used: - Courier(default), Courier bold: ASCII (7bit, 0x20..0x7F) - Arial..Calibri Bold: ISO 8859-1 (8bit, 0x20..0xff, ASCII + additional characters such as European characters) - UserFont0..UserFont15: UTF-8
varies	e	<u>(Optional) Object Type Specific Parameters (2bytes)</u> New 3.4
varies	f	<u>(Optional) Object Type Specific Parameters (2bytes)</u> New 3.8

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAExternalProgram_t Struct Reference

[Lua Script Functions](#)

Description

Configure external programs – Set names of external applications to be run on the SLA hardware.

```
typedef struct {
    u8 programType;
    SVPLenString_t filename0;
    SVPLenString_t filename1;
    SVPLenString_t filename2;
} SLAExternalProgram_t;
```

Up to three (3) external programs (DLLs or LUA scripts) may be loaded at run time. External programs must exist in the same directory as VideoTrack.

Note

- Script files should end with '.lua', DLLs should end with '.DLL' or '.so'.
- filename2 for DLL is *reserved* for internal use.

To get programs currently loaded: [SLAGetParameters_t](#), id=0x8F, payload0=[programType]. Returns [SLAExternalProgram_t](#).

Message ID 0x8F

Byte Offset	Name	Description
4	programType	Type of external program 0 = DLL 1 = Lua script
5	filename0.len	string length
6-...	filename0.str	Name of the first external program
varies	filename1.len	string length
varies	filename1.str	Name of the second external program
varies	filename2.len	string length

varies	filename2.str	Name of the third external program (<i>Reserved</i> for internal use)
--------	---------------	--

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

[List of all members](#)

SLAFocusParameters_t Struct

Reference

[Focus](#)

Description

Focus parameters – control the focus measurement.

```
typedef struct {
    u8 cameraIndex;
    u8 focusMetric;
    u32 focusDivider;
    u32 varianceDivider;
    u8 colRoiFrac;
    u8 rowRoiFrac;
    u8 wideRoiFrac;
    u8 highRoiFrac;
} SLAFocusParameters_t;
```

See [SLALensParameters_t](#) and [SLALensMode_t](#) for lens auto focus control information.

See [SLAFocusStats_t](#) for more information on the calculation of the **focusMetric**.

Message ID 0xB3

Byte Offset	Name	Description																		
4	cameraIndex	Camera Index																		
5	focusMetric	Metric used in the focus calculation returned in SLAFocusStats_t.<table><tr><th>value0</th><th>Name</th><th>Description</th></tr><tr><td>4</td><td>Sobel</td><td>Variance of Sobel</td></tr><tr><td>5</td><td>Brenner</td><td>Variance of Brenner (default)</td></tr><tr><td>6</td><td>Sobel8</td><td>Variance of Sobel with 256 scale for 16 bit images</td></tr><tr><td>7</td><td>Brenner8</td><td>Variance of Brenner with 256 scale for 16 bit images</td></tr><tr><td>8</td><td>Tenengrad Variance (TENV)</td><td>Square Root of variance of sum of squares of horizontal and vertical Sobel NOTE: Default may need larger focusDivider (eg. 256)</td></tr></table>	value0	Name	Description	4	Sobel	Variance of Sobel	5	Brenner	Variance of Brenner (default)	6	Sobel8	Variance of Sobel with 256 scale for 16 bit images	7	Brenner8	Variance of Brenner with 256 scale for 16 bit images	8	Tenengrad Variance (TENV)	Square Root of variance of sum of squares of horizontal and vertical Sobel NOTE: Default may need larger focusDivider (eg. 256)
value0	Name	Description																		
4	Sobel	Variance of Sobel																		
5	Brenner	Variance of Brenner (default)																		
6	Sobel8	Variance of Sobel with 256 scale for 16 bit images																		
7	Brenner8	Variance of Brenner with 256 scale for 16 bit images																		
8	Tenengrad Variance (TENV)	Square Root of variance of sum of squares of horizontal and vertical Sobel NOTE: Default may need larger focusDivider (eg. 256)																		

6-9	focusDivider	Divider for returned focus metric to prevent overflow, default 0 for no divider. See SLAFocusStats_t for more details.
10-13	varianceDivider	Divider for returned variance to prevent overflow, default 0 for no divider. See SLAFocusStats_t for more details.
14	colRoiFrac	Col ROI for focus stats. Column Offset in % of image height (255=100%); Full image = 0, Default = 64 (25%)
15	rowRoiFrac	Row ROI for focus stats. Row Offset in % of image width (255=100%); Full image = 0, Default = 64 (25%)
16	wideRoiFrac	High ROI for focus stats. Height in % of image height (255=100%); Full image = 255, Default = 128 (50%)
17	highRoiFrac	Wide ROI for focus stats. Width in % of image width (255=100%); Full image = 255, Default = 128 (50%)

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAFocusStats_t Struct Reference

[Telemetry](#) | [Focus](#)

Description

Periodic report of focus and brightness statistics.

```
typedef struct {
    u8 cameraIndex;
    u32 metric;
    u64 timeStamp;
    u32 frameId;
    u32 mean;
    u32 variance;
} SLAFocusStats_t;
```

Enable using [SLACoordinateReportingMode_t](#).
Set the region of interest with [SLASetLensParameters_t](#).

The focus metric is selected using [SLAFocusParameters_t](#). It can be one of the following:

Value	Method	Description
4	Sobel	Sum of the variance of the horizontal and vertical 3x3 sobel edge filters. Better than Laplacian at finding a peak at the focus, especially with noisy cameras.
5	Brenner	(default) - variance of the sum of the absolute values of horizontal and vertical 1x3 simple edge filters. Similar performance to Sobel observed in testing. It uses a little less processor load then Sobel to run.
6	Sobel8	Same as Sobel, but the metric and image variance are scaled up by 8 bits (*256) for 16-bit images.
7	Brenner8	Same as Brenner, but the metric and image variance are scaled up by 8 bits (*256) for 16-bit images.
8	Tenengrad Variance (TENV)	Square root of variance of the sum of squares of horizontal and vertical Sobel masks.

NOTE: **Sobel** and **Brenner** result scaling for 16 bit images was changed in 2.25.06 to fix overflows for high dynamic ranges. **Sobel8** and **Brenner8** match the pre-2.25.06 scaling for 16 bit images which is useful for lower dynamic range 16 bit images.

The mean and variance of the image give a **brightness** measure for control of aperture and exposure. These values are also useful to detect video that is dark, saturated or very low contrast.

NOTE: Telemetry output is not available in Demo Mode.

Message ID 0x55

Byte Offset	Name	Description
4	cameraIndex	<u>Camera Index</u>
5-8	metric	Calculated focus metric. Unsigned 32-bit integer. For 8 bit images, it is scaled up by 8 bits (*256). For 16 bit images in Sobel8 and Brenner8 modes, it is also scaled up by 8 bits (*256). For Tenengrad mode, it is scaled up by 256/Value1. In other modes it is not scaled. Changed 2.25
9-16	timeStamp	64-bit micro-second time at which this frame was captured
17-20	frameId	32-bit unsigned user controlled frame index – See <u>SLAFrameIndex_t</u>
21-24	mean	Mean (average) of the image pixel values scaled up by 8 bits (*256). Unsigned 32-bit integer
25-28	variance	Variance of the image pixel values. Unsigned 32-bit integer. For 8 bit images, it is scaled up by 8 bits (*256). For 16 bit images in Sobel8 and Brenner8 modes, it is also scaled up by 8 bits (*256). For Tenengrad mode, it is scaled up by 256/Value1. In other modes it is not scaled

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAFourAlignPoints_t Struct

Reference

[Blending](#)

Description

Used to assist in the alignment of two cameras for blending purposes.

```
typedef struct {
    u8 index;
    s16 LeftCameraPointACol;
    s16 LeftCameraPointARow;
    s16 RightCameraPointACol;
    s16 RightCameraPointARow;
    s16 LeftCameraPointBCol;
    s16 LeftCameraPointBRow;
    s16 RightCameraPointBCol;
    s16 RightCameraPointBRow;
    s16 LeftCameraPointCCol;
    s16 LeftCameraPointCRow;
    s16 RightCameraPointCCol;
    s16 RightCameraPointCRow;
    s16 LeftCameraPointDCol;
    s16 LeftCameraPointDRow;
    s16 RightCameraPointDCol;
    s16 RightCameraPointDRow;
} SLAFourAlignPoints_t;
```

Sets a group of 4 points on a video frame of one camera with the corresponding 4 points on the video frame of another camera. The point A in the left camera points at the same object as point A in the right camera, and so forth. Users should select 4 points close to the corner of the image for best results. The indexes do not interfere with the [0...4] indexes of the multiple alignment (SetMultipleAlignment_t) command.

Negative points will be set to 0. If all point values are set to 0 the cameras will be displayed on top of each other normally, without alignment.

See also [SLASetBlendParameters_t](#) and [SLABlendAlign_t](#).

Message ID 0x95

Byte Offset	Name	Description
4	index	index into table of alignment points [0..4]
5-6	LeftCameraPointACol	Column (x) coordinate of point A in the camera on the left of the 2-Up image.
7-8	LeftCameraPointARow	Row (y) coordinate of point A in the camera on the left of the 2-Up image.

9-10	RightCameraPointACol	Column (x) coordinate of point A in the camera on the right of the 2-Up image.
11-12	RightCameraPointARow	Row (y) coordinate of point A in the camera on the right of the 2-Up image.
13-14	LeftCameraPointBCol	Column (x) coordinate of point B in the camera on the left of the 2-Up image.
15-16	LeftCameraPointBRow	Row (y) coordinate of point B in the camera on the left of the 2-Up image.
17-18	RightCameraPointBCol	Column (x) coordinate of point B in the camera on the right of the 2-Up image.
19-20	RightCameraPointBRow	Row (y) coordinate of point B in the camera on the right of the 2-Up image.
21-22	LeftCameraPointCCol	Column (x) coordinate of point C in the camera on the left of the 2-Up image.
23-24	LeftCameraPointCRow	Row (y) coordinate of point C in the camera on the left of the 2-Up image.
25-26	RightCameraPointCCol	Column (x) coordinate of point C in the camera on the right of the 2-Up image.
27-28	RightCameraPointCRow	Row (y) coordinate of point C in the camera on the right of the 2-Up image.
29-30	LeftCameraPointDCol	Column (x) coordinate of point D in the camera on the left of the 2-Up image.
31-32	LeftCameraPointDRow	Row (y) coordinate of point D in the camera on the left of the 2-Up image.
33-34	RightCameraPointDCol	Column (x) coordinate of point D in the camera on the right of the 2-Up image.
35-36	RightCameraPointDRow	Row (y) coordinate of point D in the camera on the right of the 2-Up image.

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAFrameIndex_t Struct Reference

[General](#)

Description

User controlled frame index applied to specified camera index.

```
typedef struct {
    u8 cameraIndex;
    u32 index;
} SLAFrameIndex_t;
```

This frame index will be included in telemetry packets (position, positions, focus, landing, etc) if and only if the user also sets the flag to 'on' in the [SLACoordinateReportingMode_t](#).

NOTE: The user is responsible for changing the value per frame if so desired

Also sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x8A

Byte Offset	Name	Description
4	cameraIndex	Camera Index
5-8	index	Index to report in telemetry packets associated with cameraIndex

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAGetAcquisitionParameters_t

Struct Reference

Description

Return a [SLACurrentAcquisitionParameters_t](#) message.

```
typedef struct {
    u8 cameraIndex;
} SLAGetAcquisitionParameters\_t;
```

Message ID 0x38

Byte Offset	Name	Description
4	cameraIndex	Camera Index

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAGetADCParameters_t Struct Reference

Description

Return a [SLACurrentADCParameters_t](#) message.

```
typedef struct {
    u8 cameraIndex;
} SLAGetADCParameters_t;
```

Message ID *0x19*

Byte Offset	Name	Description
4	cameraIndex	Camera Index

Sightline Intelligence Command, Control, and Script API v3.11

[Main Page](#)[Related Pages](#)[Modules](#)[Classes](#) [Camera Configuration](#)

SLAGetBlendParameters_t Struct Reference

Description

Return a [SLACurrentBlendParameters_t](#) message.

```
typedef struct {  
} SLAGetBlendParameters t;
```

Message ID 0x30

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAGetDetectionParameters_t Struct Reference

[List of all members](#)

Description

Return a [SLACurrentDetectionParameters_t](#) message.

```
typedef struct {
    u8 cameraIndex;
    u8 detIdx;
} SLAGetDetectionParameters_t;
```

Message ID 0x2E

Byte Offset	Name	Description
4	cameraIndex	Camera Index
5	detIdx	Detection Index, 0 or 1. Optional, if left off, use the last detection index set

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAGetDisplayParameters_t Struct Reference

Description

Return a [SLACurrentDisplayParameters_t](#) message.

```
typedef struct {
    u8 cameraIndex;
} SLAGetDisplayParameters\_t;
```

Message ID 0x3A

Byte Offset	Name	Description
4	cameraIndex	Camera Index

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAGetEthernetDisplayParameters_t

Struct Reference

Description

Return a [SLACurrentEthernetDisplayParameters_t](#) message.

```
typedef struct {
    u16 displayId;
} SLAGetEthernetDisplayParameters_t;
```

Message ID 0x39

Byte Offset	Name	Description
4-5	displayId	Network Display ID 4000/17XX: (0x0002 = Net0, 0x0080 = Net1) 41XX: (0x0002 = Net0, 0x0080 = Net1, 0x0010 = Net2)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAGetEthernetVideoParameters_t

Struct Reference

Description

Return a [SLACurrentEthernetVideoParameters_t](#) message.

```
typedef struct {
    u16 displayId;
} SLAGetEthernetVideoParameters\_t;
```

Message ID 0x1B

Byte Offset	Name	Description
4-5	displayId	Network Display ID 4000/17XX: (0x0002 = Net0, 0x0080 = Net1) 41XX: (0x0002 = Net0, 0x0080 = Net1, 0x0010 = Net2)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAGetH264Parameters_t Struct Reference

Description

Return a [SLACurrentH264Parameters_t](#) message.

```
typedef struct {
    u16 displayId;
} SLAGetH264Parameters\_t;
```

Message ID 0x24

Byte Offset	Name	Description
4-5	displayId	Network Display ID 4000/17XX: (0x0002 = Net0, 0x0080 = Net1) 41XX: (0x0002 = Net0, 0x0080 = Net1, 0x0010 = Net2)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAGetHardwareID_t Struct Reference

[General](#)

Description

Return a [SLAGetHardwareID_t](#) message.

```
typedef struct {
    u64 hwID;
} SLAGetHardwareID_t;
```

This message is also sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x50

Byte Offset	Name	Description
4-11	hwID	64-bit hardware identifier is returned (ignored if used as a command)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAGetImageSize_t Struct Reference

[General](#)

Description

Return a [SLACurrentImageSize_t](#) message.

```
typedef struct {
    u8 cameraIndex;
} SLAGetImageSize\_t;
```

Message ID 0x31

Byte Offset	Name	Description
4	cameraIndex	Camera Index

Sightline Intelligence Command, Control, and Script API v3.11

[Main Page](#)[Related Pages](#)[Modules](#)[Classes](#) [Camera Configuration](#)

SLAGetNetworkList_t Struct Reference

Description

Return a [SLACurrentNetworkList_t](#) message.

```
typedef struct {  
} SLAGetNetworkList t;
```

Message ID 0x66

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAGetNetworkParameters_t Struct Reference

Description

Return a [SLACurrentNetworkParameters_t](#) message.

```
typedef struct {
    u8 index;
} SLAGetNetworkParameters_t;
```

Message ID 0x1D

Byte Offset	Name	Description
4	index	Network interface index – should be 0

Sightline Intelligence

Command, Control, and Script

API v3.11

[List of all members](#)

SLAGetOverlayMode_t Struct Reference

Description

Return a SLACurrentOverlay_t message.

```
typedef struct {
    u8 cameraIndex;
} SLAGetOverlayMode_t;
```

Message ID 0x07

Byte Offset	Name	Description
4	cameraIndex	Camera Index

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAGetPortConfiguration_t Struct Reference

Description

Return a [SLACurrentPortConfiguration_t](#) message.

```
typedef struct {
    u8 port;
} SLAGetPortConfiguration\_t;
```

Message ID 0x3F

Byte Offset	Name	Description
4	port	Port Identifier, see SLACommandPassThrough_t .destPort

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAGetRegistrationParameters_t

Struct Reference

Description

Deprecated, use [SLARegistrationParameters_t](#)

```
typedef struct {
    u8 cameraIndex;
} SLAGetRegistrationParameters\_t;
```

Deprecated, use [SLARegistrationParameters_t](#)

Return a [SLACurrentRegistrationParameters_t](#) message.

Message ID 0x0F

Byte Offset	Name	Description
4	cameraIndex	Camera Index

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAGetSnapShot_t Struct Reference

Description

Return a [SLACurrentSnapShot_t](#) message.

```
typedef struct {
    u8 cameraIndex;
} SLAGetSnapShot\_t;
```

Message ID 0x5F

Byte Offset	Name	Description
4	cameraIndex	Camera Index

Sightline Intelligence

Command, Control, and Script

API v3.11

[List of all members](#)

SLAGetStabilizationParameters_t

Struct Reference

Description

Return a [SLACurrentStabilizationParameters_t](#) message.

```
typedef struct {
    u8 cameraIndex;
} SLAGetStabilizationParameters\_t;
```

Message ID 0x03

Byte Offset	Name	Description
4	cameraIndex	Camera Index

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAGetTrackingParameters_t Struct Reference

[List of all members](#)

Description

Return a [SLACurrentTrackingParameters_t](#) message.

```
typedef struct {
    u8 cameraIndex;
} SLAGetTrackingParameters\_t;
```

Message ID 0x0D

Byte Offset	Name	Description
4	cameraIndex	Camera Index

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAGetVideoEnhancementParameters_t Struct Reference

Description

Return a [SLACurrentVideoEnhancementParameters_t](#) message.

```
typedef struct {
    u8 cameraIndex;
} SLAGetVideoEnhancementParameters\_t;
```

Message ID 0x22

Byte Offset	Name	Description
4	cameraIndex	Camera Index

Firmware Version: 3.11

Sightline Intelligence Command, Control, and Script API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) ■ [Camera Configuration](#)

SLAGetVideoMode_t Struct Reference

Description

Deprecated, use [SLAVideoDisplay_t](#) and [SLAMultiDisplay_t](#).

```
typedef struct {  
} SLAGetVideoMode t;
```

Deprecated, use [SLAVideoDisplay_t](#) and [SLAMultiDisplay_t](#).

Return a [SLACurrentVideoModeParameters_t](#) message

Message ID 0x20

Sightline Intelligence

Command, Control, and Script

API v3.11

[List of all members](#)

SLAGetVideoParameters_t Struct Reference

Description

Return a [SLACurrentVideoParameters_t](#) message.

```
typedef struct {
    u8 cameraIndex;
} SLAGetVideoParameters\_t;
```

Message ID 0x11

Byte Offset	Name	Description
4	cameraIndex	Camera Index

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAGPIO_t Struct Reference

[General](#)

Description

Write to and read from a GPIO.

```
typedef struct {
    u16 GpioNum;
    u8 Action;
    u8 Field;
    u8 Value;
} SLAGPIO_t;
```

The GPIO information comes from /sys/class/gpio/gpioXXX/direction and /sys/class/gpio/gpioXXX/value. In the table below, "state" refers to the contents of the GPIO's "value" file, and "direction" refers to the contents of its "direction" file. To set a GPIO pin state to high or low voltage, the GPIO must have direction="out".

Example Write: To configure a gpio to output high, set "Action" to 1. To change both the direction and state of the gpio, set bit 0 in "Field" to 1 and bit 1 in "Field" to 1. To change the direction to output, set bit 0 in "Value" to 0. To change the gpio state to high, set bit 1 of "Value" to 1.

Example Read: If a gpio pin is outputting a high value: "Action" will be 0 when this packet is received as a response to a GetParameters request. If successfully read, bits 0 and 1 of "Field" will both be 1. Bit 0 (direction) of "Value" will be 0 (out), and bit 1 (state) of "Value" will be 1 (high).

NOTE: Exercise caution when using "Write mode". Recommended only for use with customer-available GPIOs. Contact SightLine support for guidance on other GPIOs. See the ICD for your board for details.

Message ID 0xB6

Byte Offset	Name	Description
4-5	GpioNum	Software-defined number of the GPIO to query or write. eg: GpioNum= 452 when configuring "gpio452"
6	Action	0= read, 1=write. This will be 0 when this packet is a response to a GetParameters request. To configure the

		GPIO, set this to 1.												
7	Field	Write Mode (when Action = 1):<table><tr><th>Bit</th><th>Option</th></tr><tr><td>0: direction</td><td>0=do not change 1=use value in byte offset 8, bit 0</td></tr><tr><td>1: state</td><td>0=do not change 1=use value in byte offset 8, bit 1</td></tr></table> Read Mode (when Action = 0):<table><tr><th>Bit</th><th>Option</th></tr><tr><td>0: direction</td><td>0= was not read 1=direction value is in byte offset 8, bit 0</td></tr><tr><td>1: state</td><td>0=was not read 1= state value is in byte offset 8, bit 1</td></tr></table>	Bit	Option	0: direction	0=do not change 1=use value in byte offset 8, bit 0	1: state	0=do not change 1=use value in byte offset 8, bit 1	Bit	Option	0: direction	0= was not read 1=direction value is in byte offset 8, bit 0	1: state	0=was not read 1= state value is in byte offset 8, bit 1
Bit	Option													
0: direction	0=do not change 1=use value in byte offset 8, bit 0													
1: state	0=do not change 1=use value in byte offset 8, bit 1													
Bit	Option													
0: direction	0= was not read 1=direction value is in byte offset 8, bit 0													
1: state	0=was not read 1= state value is in byte offset 8, bit 1													
8	Value	<table><tr><th>Bit</th><th>Option</th></tr><tr><td>0: direction</td><td>0= "out" 1="in"</td></tr><tr><td>1: state</td><td>0= "0" 1= "1"</td></tr></table>	Bit	Option	0: direction	0= "out" 1="in"	1: state	0= "0" 1= "1"						
Bit	Option													
0: direction	0= "out" 1="in"													
1: state	0= "0" 1= "1"													

Sightline Intelligence Command, Control, and Script API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) [Camera Configuration](#)

[List of all members](#)

SLAI2CCommand_t Struct Reference

[Serial Port](#)

Description

Write or read I2C bus.

```
typedef struct {  
    u8 nread;  
    u8 bus;  
    u8 addr;  
    SVPLenData\_t data;  
} SLAI2CCommand\_t;
```

Response reports zero bytes read if there is a failure.

Example 1: Read the 2 byte part number from the tvp5150 on a 1500

Send Message: 51,AC,07,94,02,02,5D,01,80,FD

- 51,AC = header
- 07 = packet data length (don't count header or checksum)
- 94 = I2CCommand message ID
- 02 = number of bytes to read
- 02 = 1500 I2C bus number
- 5D = I2C address of tvp5150
- 01 = number of bytes to write (the address)
- 80 = register address for part number on tvp5150
- FD = checksum

Response Message: 51,AC,08,94,00,02,5D,02,51,50,44

- 51,50 = part number

Example 2: Read the 2 byte part number from the tvp5150 on a 3000 VIN0

(we write 1 byte - the 8 bit address 0x80, we then read 2 bytes from addresses 0x80,0x81)

Send Message: 51,AC,07,94,02,01,5D,01,80,75

Example 3: Write the green gain register (0x3056) on the Airborne Camera on a 1500

(we write 4 bytes - 2 address bytes and 2 data bytes; 16 bit address and 16 bit data)

Send Message: 51,AC,0A,94,00,02,10,04,30,56,00,55,29

Example 4: Write the green gain register (0x3056) on the Airborne Camera on a 3000 VIN1
Send Message: 51,AC,0A,94,00,04,10,04,30,56,00,55,A7

Message ID 0x94

Byte Offset	Name	Description																						
4	nread	0 : write >=1 : number of bytes to read																						
5	bus	I2C Bus																						
		<table><tr><th>Value</th><th>Description</th></tr><tr><td>1</td><td>3000-OEM I2C0 connected to VIN0 and VOUT</td></tr><tr><td>1</td><td>4000-OEM Camera 0 Video Input Connector</td></tr><tr><td>2</td><td>1500-OEM Digital Connector</td></tr><tr><td>3</td><td>4000-OEM Camera 1 Video Input Connector</td></tr><tr><td>4</td><td>3000-OEM I2C3 connected to VIN1 1750-OEM Camera 0 Video Input Connector</td></tr><tr><td>5</td><td>1750-OEM Camera 1 Video Input Connector</td></tr></table>	Value	Description	1	3000-OEM I2C0 connected to VIN0 and VOUT	1	4000-OEM Camera 0 Video Input Connector	2	1500-OEM Digital Connector	3	4000-OEM Camera 1 Video Input Connector	4	3000-OEM I2C3 connected to VIN1 1750-OEM Camera 0 Video Input Connector	5	1750-OEM Camera 1 Video Input Connector								
		Value	Description																					
		1	3000-OEM I2C0 connected to VIN0 and VOUT																					
		1	4000-OEM Camera 0 Video Input Connector																					
		2	1500-OEM Digital Connector																					
		3	4000-OEM Camera 1 Video Input Connector																					
		4	3000-OEM I2C3 connected to VIN1 1750-OEM Camera 0 Video Input Connector																					
5	1750-OEM Camera 1 Video Input Connector																							
6	addr	I2C address																						
		<table><tr><th>Value</th><th>Description</th></tr><tr><td>0x29</td><td>I2C to SPI converter on SLA-FPC-LI</td></tr><tr><td>0x2A</td><td>I2C to SPI adapter on SLA-3000-HDSDI-OUT board</td></tr><tr><td>0x2B</td><td>I2C to SPI on SLA-3000-HDSDI-IN Adapter</td></tr><tr><td>0x40</td><td>GPIO Expander on SLA-3000-TEST Board</td></tr><tr><td>0x48</td><td>LM75 Temperature Sensor on 3000-OEM</td></tr><tr><td>0x4C</td><td>HDMI receiver on 1500-OEM-HDMI and SLA-3000-HDMI</td></tr><tr><td>0x5D</td><td>1st TVP5150 analog video decoder on SLA-3000-AB</td></tr><tr><td>0x5C</td><td>2nd TVP5150 analog video decoder on SLA-3000-AB</td></tr><tr><td>0x5D</td><td>TVP5150 analog video decoder on 1500-OEM</td></tr><tr><td>0x68</td><td>Accelerometer on SLA-FPC-LI</td></tr></table>	Value	Description	0x29	I2C to SPI converter on SLA-FPC-LI	0x2A	I2C to SPI adapter on SLA-3000-HDSDI-OUT board	0x2B	I2C to SPI on SLA-3000-HDSDI-IN Adapter	0x40	GPIO Expander on SLA-3000-TEST Board	0x48	LM75 Temperature Sensor on 3000-OEM	0x4C	HDMI receiver on 1500-OEM-HDMI and SLA-3000-HDMI	0x5D	1st TVP5150 analog video decoder on SLA-3000-AB	0x5C	2nd TVP5150 analog video decoder on SLA-3000-AB	0x5D	TVP5150 analog video decoder on 1500-OEM	0x68	Accelerometer on SLA-FPC-LI
		Value	Description																					
		0x29	I2C to SPI converter on SLA-FPC-LI																					
		0x2A	I2C to SPI adapter on SLA-3000-HDSDI-OUT board																					
		0x2B	I2C to SPI on SLA-3000-HDSDI-IN Adapter																					
		0x40	GPIO Expander on SLA-3000-TEST Board																					
		0x48	LM75 Temperature Sensor on 3000-OEM																					
		0x4C	HDMI receiver on 1500-OEM-HDMI and SLA-3000-HDMI																					
		0x5D	1st TVP5150 analog video decoder on SLA-3000-AB																					
		0x5C	2nd TVP5150 analog video decoder on SLA-3000-AB																					
0x5D	TVP5150 analog video decoder on 1500-OEM																							
0x68	Accelerometer on SLA-FPC-LI																							
7	data.len	string length																						
8-...	data.str	Data buffer, length included																						

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAKlvClassFilters_t Struct Reference

[Classification](#)

Description

Set active Classification filters on KLV data.

```
typedef struct {
    u8 cameraIndex;
    u8 horizonFilter;
    s32 maxLat;
    s32 minLat;
    s32 maxLon;
    s32 minLon;
    u32 maxTargetWidth;
    u32 minTargetWidth;
    u32 maxTargetHeight;
    u32 minTargetHeight;
} SLAKlvClassFilters_t;
```

Message ID 0xC1

Byte Offset	Name	Description
4	cameraIndex	Camera Index
5	horizonFilter	0 = disabled/no horizon filtering. 1 = filter all objects above horizon. 2 = filter all objects below horizon.
6-9	maxLat	Filter all objects with a latitude greater than this value. Valid range: -90 to +90 degrees.
10-13	minLat	Filter all objects with a latitude lower than this value. Valid range: -90 to +90 degrees.
14-17	maxLon	Filter all objects with a longitude greater than this value. Valid range: -180 to +180 degrees.
18-21	minLon	Filter all objects with a longitude lower than this value. Valid range: -180 to +180 degrees.
22-25	maxTargetWidth	Filter all objects larger than this value. (meters)
26-29	minTargetWidth	Filter all objects smaller than this value. (meters)
30-33	maxTargetHeight	Filter all objects larger than this value. (meters)

34-37	minTargetHeight	Filter all objects smaller than this value. (meters)
-------	-----------------	--

Firmware Version: 3.11

Sightline Intelligence Command, Control, and Script API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) [Camera Configuration](#)

[List of all members](#)

SLALandingAid_t Struct Reference

[Landing Aid](#)

Description

Control landing aid detection.

```
typedef struct {  
    u8 mode;  
    u16 camHFovDeg8;  
    u32 blackTargetSize16;  
    u32 whiteTargetSize16;  
    u8 matchThresh;  
    u8 reserved0;  
    u8 reserved1;  
    u8 keepOutMode;  
    u8 reserved2;  
    u32 keepOutRadius16;  
    u16 ctrlParam0;  
    u16 ctrlParam1;  
    u16 ctrlParam2;  
    u16 ctrlParam3;  
    u8 cameraIndex;  
} SLALandingAid\_t;
```

SightLine's patented Landing Aid detection is capable of finding, tracking, and measuring the distance and angle to a SightLine specific target.

The target can be printed or painted or constructed of lights or heat elements. An overlay graphic can be enabled to help visualize the Landing Aid Detection Pattern – see [SLASetOverlayMode_t](#).graphics. The target size measurements are taken from circle center to circle center in the printed target as shown below.

Landing Aid Target Size

It is important to pick units for the target size that will not overflow the distance value returned in [SLALandingPosition_t](#): .e.g. if the target size is specified in meters, the maximum distance possible to report is 65536 meters. If the units are specified in cm, the maximum distance is 65536 cm = 655 meters.

If the white target size is larger than the black target size the system will look for a target that has a large white target on a black background (border) and a smaller black on white target.

This message is also sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x81

Byte Offset	Name	Description
4	mode	Landing Aid mode; 0 = off, 1 = run
5-6	camHFovDeg8	Camera horizontal field of view in degrees * 256 (default 30*256 → 30 degrees), 0 = no change
7-10	blackTargetSize16	Black target size * 65536 (default 0.175*65536 → 0.175 meters = 17.5 cm), 0 = no change
11-14	whiteTargetSize16	White target size * 65536 (default 0.00875*65536 → 0.00875 meters = 8.75 mm), 0 = no change
15	matchThresh	Match threshold, 0 to 100 (default 50) 0 = no change
16	reserved0	<i>Reserved</i>
17	reserved1	<i>Reserved</i>
18	keepOutMode	Landing Aid Keep Out Mode; 0 = off, 1 = run
19	reserved2	<i>Reserved</i>
20-23	keepOutRadius16	Keep out radius * 65536 (default 0.35f*65536 → 0.35 meters = 35 cm); the location of this zone is centered at the detected target location
24-25	ctrlParam0	Control parameter 0 – passed straight through to SLALandingPosition_t telemetry to be used for vehicle control; Final scaling and use is left up to the end user.
26-27	ctrlParam1	Control parameter 1 – passed straight through to SLALandingPosition_t telemetry to be used for vehicle control; Final scaling and use is left up to the end user.

28-29	ctrlParam2	Control parameter 2 – passed straight through to <u>SLALandingPosition_t</u> telemetry to be used for vehicle control; Final scaling and use is left up to the end user.
30-31	ctrlParam3	Control parameter 3 – passed straight through to <u>SLALandingPosition_t</u> telemetry to be used for vehicle control; Final scaling and use is left up to the end user.
32	cameraIndex	<u>Camera Index</u>

Firmware Version: 3.11

Sightline Intelligence Command, Control, and Script API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) [Camera Configuration](#)

[List of all members](#)

SLALandingPosition_t Struct Reference

[Landing Aid](#) | [Telemetry](#)

Description

Periodic report of landing aid location, orientation, distance and match score.

```
typedef struct {  
    u8 cameraIndex;  
    s16 col;  
    s16 row;  
    u16 angleDeg7;  
    u32 distance16;  
    u8 confidence;  
    u16 camHFovDeg8;  
    s16 capWide;  
    s16 capHigh;  
    u16 ctrlParam0;  
    u16 ctrlParam1;  
    u16 ctrlParam2;  
    u16 ctrlParam3;  
    u8 keepOutState;  
    u8 keepOutConfidence;  
    u16 keepOutSz;  
    u32 keepOutDist16;  
    u64 timeStamp;  
    u32 frameId;  
} SLALandingPosition\_t;
```

The position and angle are reported in camera source (unstabilized) pixel coordinates.

The confidence score indicates system confidence between 0 (low) and 100 (high). The distance to target units match those specified for black and white target size in [SLASetLandingAid_t](#).

The interval at which Landing Position is reported is controlled by [SLACoordinateReportingMode_t](#).

For reporting to multiple destinations see [SLASetTelemetryDestination_t](#).

NOTE: Telemetry output is not available in [Demo Mode](#).

See [SLALandingAid_t](#) for configuration.

Landing Aid Image

Message ID 0x83

Byte Offset	Name	Description						
4	cameraIndex	<u>Camera Index</u>						
5-6	col	Center of landing aid column in camera source coordinates						
7-8	row	Center of landing aid row in camera source coordinates						
9-10	angleDeg7	Landing aid angle (counter clockwise) in the source image (0..360 degrees) * 128						
11-14	distance16	Distance to landing aid target * 65536 (units match those for the target size specified <u>SLALandingAid_t</u>)						
15	confidence	Landing aid match confidence (0..100)						
16-17	camHFovDeg8	Camera horizontal field of view in degrees * 256 (default 30*256 → 30 degrees), 0 = no change						
18-19	capWide	Capture image width in pixels						
20-21	capHigh	Capture image height in pixels						
22-23	ctrlParam0	Control parameter 0, range is 0 to 65535 – final scaling and use is left up to the end user						
24-25	ctrlParam1	Control parameter 1, range is 0 to 65535 – final scaling and use is left up to the end user						
26-27	ctrlParam2	Control parameter 2, range is 0 to 65535 – final scaling and use is left up to the end user						
28-29	ctrlParam3	Control parameter 3, range is 0 to 65535 – final scaling and use is left up to the end user						
30	keepOutState	Keep Out State <table><tr><th>Bit</th><th>Description</th></tr><tr><td>0-6</td><td><i>Reserved</i></td></tr><tr><td>7</td><td>1 = detected, 0 = not detected</td></tr></table>	Bit	Description	0-6	<i>Reserved</i>	7	1 = detected, 0 = not detected
Bit	Description							
0-6	<i>Reserved</i>							
7	1 = detected, 0 = not detected							
31	keepOutConfidence	Keep Out Confidence (0..100) – confidence of the detection within the keep out zone						
32-33	keepOutSz	Keep Out Size – size of the detected object in pixels						
34-37	keepOutDist16	Keep Out Detect Distance * 65536 – distance from center of landing pattern to the detection in units that match those for the target size specification in <u>SLALandingAid_t</u>						
38-45	timeStamp	Timestamp, if auxiliary telemetry is enabled – see <u>SLACoordinateReportingMode_t</u>						
46-49	frameId	Frame ID, if auxiliary telemetry is enabled – see <u>SLACoordinateReportingMode_t</u>						

Sightline Intelligence

Command, Control, and Script

API v3.11

SLALensCommand_t Struct Reference

[List of all members](#)

[Focus](#)

Description

Perform a lens control command.

```
typedef struct {
    u8 cameraIndex;
    u8 commandId;
    SVPLenData_t data;
} SLALensCommand_t;
```

If any data needs to be sent along with a command, this data will be in the bytes following the LensCommand. Changing the lens position will stop autofocus (see [SLASetLensParameters_t](#)).

Message ID 0xB2

Byte Offset	Name	Description			
4	cameraIndex	Camera Index			
5	commandId	Value	Description	Data bytes	Data
		0, 1	Reserved	–	–
		2	Reset Lens Mechanism	0	–
		3	Request Lens Status	0	–
		4	Zoom in Wide Direction	0	–
		5	Zoom in Narrow Direction	0	–
		6	Stop Zoom Motion	0	–
		7	Focus in Far Direction	0	–
		8	Focus in Near Direction	0	–
		9	Stop Focus Motion	0	–

		10	One Push Auto Focus	0	–
		11	<i>Reserved</i>	0	–
		12	Lens NUC (IR camera)	1	0 = no shutter, 1 = shutter
		13	Set Zoom Position	2	Motor pulse counts
		14	Set Focus Position	2	Motor pulse counts
		15	Command Passthrough	Variable	Command to send to Lens
		128 & Above	<i>Reserved</i> for customer defined modes.	2	customer defined
6	data.len	string length			
7-...	data.str	Data bytes			

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLALensRanges_t Struct Reference

[Focus](#)

Description

Lens focus and zoom ranges.

```
typedef struct {
    u8 camIdx;
    u16 focusMin;
    u16 focusMax;
    u16 zoomMin;
    u16 zoomMax;
} SLALensRanges_t;
```

Sent in response to [SLAGetParameters_t\(\)](#).
These values come from the GetFocusRange and GetZoomRange subclass functions in the Lens DLL.

Message ID 0xB4

Byte Offset	Name	Description
4	camIdx	Camera index
5-6	focusMin	Minimum Focus position – motor pulse counts
7-8	focusMax	Maximum Focus position – motor pulse counts
9-10	zoomMin	Minimum Zoom position – motor pulse counts
11-12	zoomMax	Maximum Zoom position – motor pulse counts

Sightline Intelligence

Command, Control, and Script

API v3.11

[List of all members](#)

SLALogoParameters_t Struct

Reference

[Overlays](#)

Description

Used to set the opacity of the logo (watermark) and its placement relative to the bottom right corner of the output frame.

```
typedef struct {
    u8 cameraIndex;
    u8 logoOpacity;
    u16 OffsetX;
    u16 OffsetY;
} SLALogoParameters_t;
```

Message ID 0x9B

Byte Offset	Name	Description
4	cameraIndex	Camera Index
5	logoOpacity	0 means the logo will not be shown, 255 means no background blending.
6-7	OffsetX	Offset in columns from the lower right corner of the frame.
8-9	OffsetY	Offset in rows from the lower right corner of the frame.

Sightline Intelligence

Command, Control, and Script

API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) [Camera Configuration](#)

[List of all members](#)

SLAMetadataStaticValues_t Struct Reference

[KLV Metadata](#)

Description

Sets new KLV metadata values.

```
typedef struct {  
    u8 type;  
    SVPLenString\_t value;  
    u16 displayId;  
} SLAMetadataStaticValues\_t;
```

Value are output as MPEG2-TS ([SLASetEthernetDisplayParameters_t](#)) and as part of the EXIF header in [SLASetSnapShot_t](#).

Set the rate at which these are transmitted using [SLASetMetadataRate_t](#).

For non-string values, data encoding is big-endian as defined by MISB 0603. For example, to encode **Target Error Estimate CE90** which is specified as a uint16 value, 'Identifier string length' should be set to 2, byte 6 to the most significant byte of the value, and byte 7 to the least significant byte of value.

This message is also sent in response to [SLAGetParameters_t\(\)](#).

Special Notes

Sensor Field of View (Type 18)

Replaces [SLASetMetadataValues_t](#) *Sensor horizontal field of view* (valid data bit = 7) and *Sensor vertical field of view* (valid data bit = 8).

Set the validDataMask for bits 7 & 8 to 0 in [SLASetMetadataValues_t](#)

Example: 51 AC 0A 14 12 04 9C CD 17 D9 00 02 0D

Sensor true altitude HAE (Type 19)

Set the validDataMask for bit 52 in [SLASetMetadataRate_t](#).

Sensor frame center elevation HAE (Type 20)

Set the validDataMask for bit 53 in [SLASetMetadataRate_t](#).

Target Info (Type 21)

Replaces [SLASetMetadataFrameValues_t](#) (targetTrackGateHeight, targetTrackGateWidth, etc.) which allowed user to mauually specify target location in pixels

Using this command will cause the Slant Range (MISB TAG 21) and Target Width (TAG 22) to be autocalculated.

USE:

```
sub type = 21 <15>
value = {
  sub len = 9
  u16 Frame Center Elevation
  u16 target Width - in pixels which is then convert to meters. must be > 0 or no target will
    calculated
  u16 Origin Row - offset is to center of the target relative to the display coordinates
  u16 Origin Col - offset is to center of the target relative to the display coordinates
  u8 mode - see below
}
```

Target Info Mode

bit	description
0	enable / disable frame center lat/long = target position lat/long
2	Enable Auto/Manual modes that would normally defined in 0x15 userSuppliedFlags in SLASetMetadataFrameValues_t

EXAMPLE: 51 AC 0F 14 15 09 f3 34 37 00 f0 00 40 10 01 00 02 A3

Security Metadata (TAG 48)

In addition to the security tags below, Version (TAG 48/22) is set internally to value of 0x000A and sent at the default rate.

See also

[MISB ST 0102.10 Table 2](#) for data sizes and values.

Message ID 0x14

Byte Offset	Name	Description																					
		Static element identifier																					
		<table><tr><th>Value</th><th>Identifier type</th><th>MISB TAG</th></tr><tr><td>0</td><td>Mission Identifier</td><td>3</td></tr><tr><td>1</td><td>Platform Designation</td><td>10</td></tr><tr><td>2</td><td>Image Source Sensor</td><td>11</td></tr><tr><td>3</td><td>Image Coordinate System</td><td>12</td></tr><tr><td>4</td><td>Security: Classification (1 byte)</td><td>48 - 1</td></tr><tr><td>5</td><td>Security: Classifying country coding method (1 byte)</td><td>48 - 2</td></tr></table>	Value	Identifier type	MISB TAG	0	Mission Identifier	3	1	Platform Designation	10	2	Image Source Sensor	11	3	Image Coordinate System	12	4	Security: Classification (1 byte)	48 - 1	5	Security: Classifying country coding method (1 byte)	48 - 2
Value	Identifier type	MISB TAG																					
0	Mission Identifier	3																					
1	Platform Designation	10																					
2	Image Source Sensor	11																					
3	Image Coordinate System	12																					
4	Security: Classification (1 byte)	48 - 1																					
5	Security: Classifying country coding method (1 byte)	48 - 2																					

4	type	6	Security: Classifying country	48 - 3
		7	Security: SCI / SHI information	48 - 4
		8	Security: Caveats	48 - 5
		9	Security: Releasing Instructions	48 - 6
		10	Security: Object Country Coding Method (1 byte)	48 - 2
		11	Security: Object Country Code	48 - 13
		12	Motion Imagery Core Identifier	94
		13	Platform Tail Number	4
		14	Target Error Estimate CE90	45
		15	Target Error Estimate LE90	46
		16	Generic Flag Data 01	47
		17	Platform Call Sign	59
		18	<u>Sensor Field of View</u>	16
		19	<u>Sensor true altitude HAE</u>	
		20	<u>Frame center elevation HAE</u>	25
		21	<u>Target Info</u>	NA
		22 - 255	<i>Reserved</i>	NA
5	value.len	string length		
6-...	value.str	Identifier string - maximum string length is 127 characters		
varies	displayId	<u>Network Display ID</u> 4000/17XX: (0x0002 = Net0, 0x0080 = Net1, 0x0082 = both) 41XX: (0x0002 = Net0, 0x0080 = Net1, 0x0010 = Net2, 0x0092 = all)		

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAModifyTrackIndex_t Struct

Reference

[Tracking](#)

Description

Modify a particular track by its index (stop or designate as primary).

```
typedef struct {
    u8 trackIndex;
    u8 flags;
    u8 cameraIndex;
    u16 width;
    u16 height;
} SLAModifyTrackIndex_t;
```

Applied to the Command Camera.

PiP Locations

Picture In Picture Indices

Message ID 0x17



Byte Offset	Name	Description																												
4	trackIndex	0..253 = By Index, 254 = Selected, 255 = Primary 0..9 = PiP index if using flags = 10 or 12																												
5	flags	<table><tr><th>Flags</th><th>Result</th></tr><tr><td>0</td><td>Stop Track</td></tr><tr><td>1</td><td>Make Primary</td></tr><tr><td>2</td><td>Reinitialize: run acquisition assist logic to optimize track box location and size</td></tr><tr><td>3</td><td>Coast Mode Off</td></tr><tr><td>4</td><td>Coast Mode On - Freeze Track Model Updates</td></tr><tr><td>5</td><td>Coast Mode On - Freeze Search and Updates (use estimated velocity to report position)</td></tr><tr><td>6</td><td>Coast Mode On - Freeze Propagation</td></tr><tr><td>7</td><td>Reserved</td></tr><tr><td>8</td><td>Adjust track size - don't run acquisition assist</td></tr><tr><td>9</td><td>Adjust track size - run acquisition assist if enabled</td></tr><tr><td>10</td><td>Make Primary track from a PiP location. If this flag is set the trackIndex referenes a PiP location. PiP is setup using SLAVideoDisplay_t or SLASetVideoMode_t.</td></tr><tr><td>11</td><td>Similar to #1 above, but Make Secondary</td></tr><tr><td>12</td><td>Similar to #10 above, but make secondary instead of primary.</td></tr></table>	Flags	Result	0	Stop Track	1	Make Primary	2	Reinitialize: run acquisition assist logic to optimize track box location and size	3	Coast Mode Off	4	Coast Mode On - Freeze Track Model Updates	5	Coast Mode On - Freeze Search and Updates (use estimated velocity to report position)	6	Coast Mode On - Freeze Propagation	7	Reserved	8	Adjust track size - don't run acquisition assist	9	Adjust track size - run acquisition assist if enabled	10	Make Primary track from a PiP location. If this flag is set the trackIndex referenes a PiP location . PiP is setup using SLAVideoDisplay_t or SLASetVideoMode_t .	11	Similar to #1 above, but Make Secondary	12	Similar to #10 above, but make secondary instead of primary.
		Flags	Result																											
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		3	Coast Mode Off																											
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		11	Similar to #1 above, but Make Secondary																											
12	Similar to #10 above, but make secondary instead of primary.																													
6	cameraIndex	Camera Index																												
7-8	width	New track width of object in pixels. (Only applies to flags Adjust Track Size)																												
9-10	height	New track height of object in pixels. (Only applies to flags Adjust Track Size)																												

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAModifyTracking_t Struct Reference

[Tracking](#)

Description

NOTE: Newer functionality in [SLAStartTracking_t](#).

```
typedef struct {
    u16 col;
    u16 row;
    u8 flags;
    u8 width;
    u8 height;
    u8 cameraIndex;
} SLAModifyTracking_t;
```

Command the system to modify tracking: start a track, stop a track, designate a track as primary, nudge a track, etc.

Column and Row coordinates correspond to the pixel coordinate within a frame of video in display(stabilized) coordinates. The origin is in the upper left corner of the image with values increasing down and to the right. Users can set the maximum number of simultaneously engaged tracks in [SLASetTrackingParameters_t](#). See the Target Tracking Guide for more details about tracking multiple targets. To control specifying tracks 'near' existing tracks, see [SLASetTrackingParameters_t.nearVal](#). See [SLACurrentImageSize_t](#) for more information about capture image size. See also [SLACommandCamera_t](#).

Modify Mode

Value	Description
0	Show Cursor only. If in this mode Display Coordinate and Source Coordinate bits are ignored.
1	Kill any existing targets and then designate a new primary target at the cursor.
2	Designate a primary target at the cursor.
3	If there is a track near the coordinates, move track to coordinates and make secondary.
4	If there is a track near the coordinates, move track to coordinates and make secondary. Otherwise, add a new secondary track at coordinate location.
5	Designate track near coordinates as primary target. Also supports starting primary tracks from detections.
6	Designate track near coordinates as primary target if there is one. If not, add new track at coordinates and make primary. Also supports starting primary tracks from detections.

7	If there is a track near coordinates, move track to location of coordinates and designate as primary. Otherwise, add a new track at coordinates and make primary.
8	If there is a track near coordinates, move track to location of coordinates and designate as primary. Otherwise, kill all existing tracks and add a new primary track at location of coordinates.
9	Kill track near coordinates.
10	Kill all tracks, but the primary track. (Coordinates are ignored).
11	Reserved
12	Designate track near coordinates as secondary target. Also supports starting secondary tracks from detections.
13	Designate track near coordinates as secondary target if there is one. If not, add new track at coordinates and make secondary. Also supports starting secondary tracks from detections.

Message ID 0x05

Byte Offset	Name	Description								
4-5	col	Track column coordinate								
6-7	row	Track row coordinate								
8	flags	Coordinate space flag and modification mode. Coordinate space bits – OR these with any modify mode. NOTE: Do not set both Display and Source coordinates.								
		<table><tr><th>Bits</th><th>Description</th></tr><tr><td>7</td><td>Display Coordinates – Rotate Zoom Modifier. OR this flag (set bit 7 to 1) with any of the modify modes to indicate coordinates are rotated and zoomed with the display. By default, set this flag (bit 7 = 1) for coordinates in the display image space.</td></tr><tr><td>6</td><td>Source Coordinates – OR this flag (set bit 6 to 1) with any of the modify modes to indicate that the coordinates specified are in the source image. By default, don't set this flag (bit 6 = 0) for coordinates in the display image space.</td></tr><tr><td>0-5</td><td>Modify Mode</td></tr></table>	Bits	Description	7	Display Coordinates – Rotate Zoom Modifier. OR this flag (set bit 7 to 1) with any of the modify modes to indicate coordinates are rotated and zoomed with the display. By default, set this flag (bit 7 = 1) for coordinates in the display image space.	6	Source Coordinates – OR this flag (set bit 6 to 1) with any of the modify modes to indicate that the coordinates specified are in the source image. By default, don't set this flag (bit 6 = 0) for coordinates in the display image space.	0-5	Modify Mode
		Bits	Description							
		7	Display Coordinates – Rotate Zoom Modifier. OR this flag (set bit 7 to 1) with any of the modify modes to indicate coordinates are rotated and zoomed with the display. By default, set this flag (bit 7 = 1) for coordinates in the display image space.							
		6	Source Coordinates – OR this flag (set bit 6 to 1) with any of the modify modes to indicate that the coordinates specified are in the source image. By default, don't set this flag (bit 6 = 0) for coordinates in the display image space.							
0-5	Modify Mode									
9	width	width of object in pixels to track. (for user designated tracking) If width is not included or zero the sizes specified in SLASetTrackingParameters_t will be used. If width is not included or zero the height parameter will be ignored and values specified in SLASetTrackingParameters_t will be used. NOTE: In static track mode, if width or height is 255, use full screen for track box pixel stats.								
10	height	height of object in pixels to track. (for user designated tracking) If height is not included or zero and width is non zero the width will be used resulting in a square track box. If height and width are not included or both zero the sizes specified in SLASetTrackingParameters_t will be used.								
11	cameraIndex	Camera Index								

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAStartTracking_t Struct Reference

[Tracking](#)

Description

Command the system to start a track, stop a track, designate a track as primary, nudge a track, etc.

```
typedef struct {
    u8 cameraIndex;
    u16 col;
    u16 row;
    u16 width;
    u16 height;
    u8 flags;
    u16 nearVal;
    u8 userTrackId;
    u64 framePTS;
} SLAStartTracking_t;
```

Column and Row coordinates correspond to the pixel coordinate within a frame of video in display(stabilized) coordinates. The origin is in the upper left corner of the image with values increasing down and to the right. Users can set the maximum number of simultaneously engaged tracks in [SLASetTrackingParameters_t](#). See the Target Tracking Guide for more details about tracking multiple targets. See [SLACurrentImageSize_t](#) for more information about capture image size.

Modify Mode	
Value	Description
0	Show Cursor only. If in this mode Display Coordinate and Source Coordinate bits are ignored.
1	Kill any existing targets and then designate a new primary target at the cursor.
2	Designate a primary target at the cursor.
3	If there is a track near the coordinates, move track to coordinates and make secondary.
4	If there is a track near the coordinates, move track to coordinates and make secondary. Otherwise, add a new secondary track at coordinate location.
5	Designate track near coordinates as primary target. Also supports starting primary tracks from detections.
6	Designate track near coordinates as primary target if there is one. If not, add new track at coordinates and make primary. Also supports starting primary tracks from detections.
7	If there is a track near coordinates, move track to location of coordinates and designate as primary. Otherwise, add a new track at coordinates and make primary.

8	If there is a track near coordinates, move track to location of coordinates and designate as primary. Otherwise, kill all existing tracks and add a new primary track at location of coordinates.
9	Kill track near coordinates.
10	Kill all tracks, but the primary track. (Coordinates are ignored).
11	Reserved
12	Designate track near coordinates as secondary target. Also supports starting secondary tracks from detections.
13	Designate track near coordinates as secondary target if there is one. If not, add new track at coordinates and make secondary. Also supports starting secondary tracks from detections.

Message ID 0x08

Byte Offset	Name	Description								
4	cameraIndex	<u>Camera Index</u> If not a valid value, applies to <u>SLACommandCamera_t</u> .								
5-6	col	Track column coordinate								
7-8	row	Track row coordinate								
9-10	width	Width of object in pixels to track. (for user designated tracking) Maximum track box width is 500 pixels for all modes except static NOTE: In static track mode, if width or height is 255, use full screen for track box pixel stats.								
11-12	height	Height of object in pixels to track. (for user designated tracking) Maximum track box height is 500 pixels for all modes except static								
13	flags	Coordinate space flag and modification mode. Coordinate space bits – OR these with any modify mode. NOTE: Do not set both Display and Source coordinates.								
		<table><tr><th>Bits</th><th>Description</th></tr><tr><td>0-5</td><td>Modify Mode</td></tr><tr><td>6</td><td>Source Coordinates – OR this flag (set bit 6 to 1) with any of the modify modes to indicate that the coordinates specified are in the source image. By default, don't set this flag (bit 6 = 0) for coordinates in the display image space.</td></tr><tr><td>7</td><td>Display Coordinates – Rotate Zoom Modifier. OR this flag (set bit 7 to 1) with any of the modify modes to indicate coordinates are rotated and zoomed with the display. By default, set this flag (bit 7 = 1) for coordinates in the display image space.</td></tr></table>	Bits	Description	0-5	Modify Mode	6	Source Coordinates – OR this flag (set bit 6 to 1) with any of the modify modes to indicate that the coordinates specified are in the source image. By default, don't set this flag (bit 6 = 0) for coordinates in the display image space.	7	Display Coordinates – Rotate Zoom Modifier. OR this flag (set bit 7 to 1) with any of the modify modes to indicate coordinates are rotated and zoomed with the display. By default, set this flag (bit 7 = 1) for coordinates in the display image space.
		Bits	Description							
		0-5	Modify Mode							
		6	Source Coordinates – OR this flag (set bit 6 to 1) with any of the modify modes to indicate that the coordinates specified are in the source image. By default, don't set this flag (bit 6 = 0) for coordinates in the display image space.							
7	Display Coordinates – Rotate Zoom Modifier. OR this flag (set bit 7 to 1) with any of the modify modes to indicate coordinates are rotated and zoomed with the display. By default, set this flag (bit 7 = 1) for coordinates in the display image space.									
14-15	nearVal	'radius of engagement' used with some modes. This is the area around an existing target that interactions can take place. Large values allow greater tolerance in selecting tracks. (default = 0 = use value from <u>SLASetTrackingParameters_t</u>)								
16	userTrackId	User managed ID for this track. (0 for no ID)								
17-24	framePTS	Timestamp of the frame in the queue to act on (<u>SLASetTrackingParameters_t</u> : maxPauseTime > 0). If the timestamp is found in the queue, the system will track up to the most recent frame,								

		then continue to track as normal. This is the MISB Precision Time Stamp.
--	--	--

		NOTE: Set to 0 for normal track.
--	--	---

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLANoise3D_t Struct Reference

[Non Uniformity Correction](#)

Description

Statistics from 3D Noise calculation.

```
typedef struct {
    u8 cameraIndex;
    u16 SIG_t_8;
    u16 SIG_v_8;
    u16 SIG_h_8;
    u16 SIG_vh_8;
    u16 SIG_tv_8;
    u16 SIG_th_8;
    u16 SIG_tvh_8;
    u16 Noise_temporal_8;
} SLANoise3D_t;
```

All values are scaled up by 256.0

Message ID 0xAF

Byte Offset	Name	Description
4	cameraIndex	<u>Camera Index</u>
5-6	SIG_t_8	Standard Deviation of frame averages
7-8	SIG_v_8	Standard Deviation of 2D row slices
9-10	SIG_h_8	Standard Deviation of 2D column slices
11-12	SIG_vh_8	Standard Deviation of all average pixel values over time
13-14	SIG_tv_8	Standard Deviation of all row averages
15-16	SIG_th_8	Standard Deviation of all column averages
17-18	SIG_tvh_8	Standard Deviation of all individual pixel values
19-20	Noise_temporal_8	Average Standard Deviation of pixel value over time

Sightline Intelligence

Command, Control, and Script

API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) [Camera Configuration](#)

[List of all members](#)

SLANucParameters_t Struct

Reference

[Non Uniformity Correction](#)

Description

Set Non Uniformity Correction (NUC) and Dead Pixel Removal (DPR) parameters.

```
typedef struct {
    u8 cameraIndex;
    u8 nucShow;
    u8 nucRunMode;
    u8 numFrames;
    u16 minDeadGain;
    u16 maxDeadGain;
    u16 minDeadVal;
    u16 maxDeadVal;
    s32 minDeadOff;
    s32 maxDeadOff;
    u32 maxStdDevDead;
    s32 maxNumDead;
    u8 deadReplace;
    u8 numReplace;
    u8 deadFilter;
    s16 deadFilterThresh;
    u8 destripeAmount;
    u8 destripeSections;
    SVPLenString\_t nucName;
} SLANucParameters\_t;
```

Works with 16-bit or 14-bit gray scale capability. The maximum image size for NUC and DPR is limited to:

- 17XX: 1920x1440
- 4000: 2560x1440

Adding Frames

Adds the number of frames to the NUC calculation. Frames continue to be added until the calibration method { 1Pt NUC, 2PT NUC, Shutter Flattening } is run.

1 Point Correction

Frames can be added using the Add Frames nucMode, or can be added when command is called by setting numFrames > 0. When using numFrames with 1Pt Calibration any frames in queue will be ignored (see Clear Frames).

2 Point Correction

When using nucRunMode = Calculate 2 point correction, numFrames must be set to 0 . Frames must be added first using the Add Frames nucMode.

Shutter Flatten

Frames can be added using the Add Frames nucMode, or can be added when command is called by setting numFrames > 0.

Message ID 0x35

Byte Offset	Name	Description																												
4	cameraIndex	<u>Camera Index</u>																												
5	nucShow	Display mode																												
		<table><tr><th>Value</th><th>Description</th></tr><tr><td>0</td><td>Show uncorrected captured video</td></tr><tr><td>1</td><td>Apply both NUC and DPR to video</td></tr><tr><td>2</td><td>Apply only NUC to video</td></tr><tr><td>3</td><td>Apply only DPR to video</td></tr><tr><td>4</td><td>Display Gain Calculations in the form of an image</td></tr><tr><td>5</td><td>Show Offset Calculations in the form of an image</td></tr><tr><td>6</td><td>Show Dead Pixels in the form of an image</td></tr></table>	Value	Description	0	Show uncorrected captured video	1	Apply both NUC and DPR to video	2	Apply only NUC to video	3	Apply only DPR to video	4	Display Gain Calculations in the form of an image	5	Show Offset Calculations in the form of an image	6	Show Dead Pixels in the form of an image												
		Value	Description																											
		0	Show uncorrected captured video																											
		1	Apply both NUC and DPR to video																											
		2	Apply only NUC to video																											
		3	Apply only DPR to video																											
		4	Display Gain Calculations in the form of an image																											
		5	Show Offset Calculations in the form of an image																											
6	Show Dead Pixels in the form of an image																													
6	nucRunMode	Calc NUC, Calc DPR, Add Frames																												
		<table><tr><th>Value</th><th>Description</th></tr><tr><td>0</td><td>Do Nothing</td></tr><tr><td>1</td><td>Add Frames</td></tr><tr><td>2</td><td>Clear Frames that have already been added</td></tr><tr><td>3</td><td>Reset Gain Calculations</td></tr><tr><td>4</td><td>Reset Offset Calculations</td></tr><tr><td>5</td><td>Reset Dead Pixel Calculation</td></tr><tr><td>6</td><td>Reset all calculations (Gain, Offset, and Dead Pixel)</td></tr><tr><td>7</td><td>Calculate 1 point correction</td></tr><tr><td>8</td><td>Calculate 2 point correction</td></tr><tr><td>9</td><td>Calculate Dead Pixels and Replacement Pixels</td></tr><tr><td>10</td><td>Calculate only Replacement Pixels for DPR</td></tr><tr><td>11</td><td>Use an Automatic Detection algorithm for Dead Pixels in the current frame</td></tr><tr><td>12</td><td>Run Shutter Flattening mode.</td></tr></table>	Value	Description	0	Do Nothing	1	Add Frames	2	Clear Frames that have already been added	3	Reset Gain Calculations	4	Reset Offset Calculations	5	Reset Dead Pixel Calculation	6	Reset all calculations (Gain, Offset, and Dead Pixel)	7	Calculate 1 point correction	8	Calculate 2 point correction	9	Calculate Dead Pixels and Replacement Pixels	10	Calculate only Replacement Pixels for DPR	11	Use an Automatic Detection algorithm for Dead Pixels in the current frame	12	Run Shutter Flattening mode.
		Value	Description																											
		0	Do Nothing																											
		1	Add Frames																											
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		3	Reset Gain Calculations																											
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		5	Reset Dead Pixel Calculation																											
		6	Reset all calculations (Gain, Offset, and Dead Pixel)																											
		7	Calculate 1 point correction																											
		8	Calculate 2 point correction																											
		9	Calculate Dead Pixels and Replacement Pixels																											
		10	Calculate only Replacement Pixels for DPR																											
		11	Use an Automatic Detection algorithm for Dead Pixels in the current frame																											
12	Run Shutter Flattening mode.																													

		13	Calculate 3D Noise Stats.
7	numFrames	Number of Frames to add to the NUC calculation (0 to 255) NOTE: When not used set to 0	
8-9	minDeadGain	Minimum Gain limit for DPR (0% to 999%) NOTE: Only Used in Calculate Dead Pixel Mode (nucRunMode=9), otherwise set to 0	
10-11	maxDeadGain	Maximum Gain limit for DPR (0% to 999%) NOTE: Only Used in Calculate Dead Pixel Mode (nucRunMode=9), otherwise set to 0	
12-13	minDeadVal	Minimum input pixel value limit for DPR (0 to 65535) NOTE: Only Used in Calculate Dead Pixel Mode (nucRunMode=9), otherwise set to 0	
14-15	maxDeadVal	Maximum input pixel value limit for DPR (0 to 65535) NOTE: Only Used in Calculate Dead Pixel Mode (nucRunMode=9), otherwise set to 0	
16-19	minDeadOff	Minimum Offset limit for DPR (-999999 to 999999) NOTE: Only Used in Calculate Dead Pixel Mode(nucRunMode=9), otherwise set to 0	
20-23	maxDeadOff	Maximum Offset limit for DPR (-999999 to 999999) NOTE: Only Used in Calculate Dead Pixel Mode (nucRunMode=9), otherwise set to 0	
24-27	maxStdDevDead	Maximum Standard Deviation allowed in Dead Pixel Replacement NOTE: Only Used in Calculate Dead Pixel Mode (nucRunMode=9), otherwise set to 0* NOTE: Max value is 65535. If a number that is larger is supplied, 65535 will be used instead	
28-31	maxNumDead	Maximum number of dead pixels allowed NOTE: Only Used in Calculate Dead Pixel Mode (nucRunMode=9), otherwise set to 0	
32	deadReplace	Type of dead pixel replacement	
		Value	Description
		0	Replace dead pixels with their nearest neighbor (default)
		1	Replace dead pixels with the average of the numReplace nearest neighboring pixels
2	Replace dead pixels with the median of the numReplace nearest neighboring pixels		
33	numReplace	Number of neighboring pixels to use for average and median dead pixel replacement 1 to 8, default 5	
34	deadFilter	Type of dead pixel filtering	
		Value	Description
		0	None (default)
		1	MinMax - Replace pixels more than threshold above max or threshold below min of 3x3 neighbors.
2	NearFar - Replace pixels that are min/max of a 3x3 if near is more than a threshold above far .		
255	Ignore this parameter and deadFilterThresh.		
35-36	deadFilterThresh	Threshold used in dead pixel filtering. 0 to 255. Default 64.	

37	destripeAmount	Amount to destripe the image: 0=none, 255=aggressive. This can be used when it isn't possible to perform an updated NUC and the image has developed stripes over time. Default 0
38	destripeSections	Number of sections to break image into for destripe (more = fewer vertical artifacts but more horizontal) Default 1
39	nucName.len	string length
40-...	nucName.str	Name of NUC to add the frames to when creating multiple NUCs at once.

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLANudgeTrackingCoordinate_t

Struct Reference

[Tracking](#)

Description

Adjust the primary track's coordinates by adding a nudge in pixel coordinate space to the current tracking coordinates.

```
typedef struct {
    s8 offsetCol;
    s8 offsetRow;
    u8 rotate;
    u8 cameraIndex;
    s16 offsetZoom10;
    u8 trackIndex;
    s16 offsetColExt;
    s16 offsetRowExt;
} SLANudgeTrackingCoordinate_t;
```

NOTE: Nudging the track results in the track re-training so should be done with caution, for example nudging right before an occlusion will likely result in track loss. In this case track statistics are maintained, so this is less aggressive than making a new track at the new location/szie.

NOTE: Nudging while the track is in a coasting state will result in a nudge of the estimated position, and will not cause the track to re-train. A larger nudge value may be required while coasting to see an impact.

Message ID 0x0A

Byte Offset	Name	Description
4	offsetCol	-128..127: Column adjustment (2's complement signed 8-bit integer)
5	offsetRow	-128..127: Row adjustment (2's complement signed 8-bit integer)
6	rotate	0: Do not rotate command with display, 1: Rotate command with display
7	cameraIndex	Camera Index
8-9	offsetZoom10	Used to adjust the track box size up or down. -1024 to 1024 (this corresponds to zoom scale of -1.0 to 1.0). NOTE The values sent are integrated per frame, and added to an original zoom of 1.0 (no zoom). As an example if you send three messages of 0.01, 0.01, 0.01 the end zoom will be 1.03.

		NOTE Because track box sizes are integers small adjustments may have no impact. NOTE Minimum integrated zoom scale is 0.1 (10%) and maximum is 10.0 (1000%).
10	trackIndex	0..252 = By Index, 253 = All (only applicable to offsetZoom), 254 = Selected, 255 = Primary If not included will default to primary only.
11-12	offsetColExt	-32,768..32,767: Column adjustment (2's complement signed 16-bit integer). If set, will use over 8-bit offsetCol.
13-14	offsetRowExt	-32,768..32,767: Column adjustment (2's complement signed 16-bit integer). If set, will use over 8-bit offsetRow.

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAReadWriteNuc_t Struct Reference

[Non Uniformity Correction](#)

[List of all members](#)

Description

Used to save or load NUC/DPR tables as well as set a table to be applied on startup.

```
typedef struct {
    u8 cameraIndex;
    u8 reserved;
    u8 nucReadWriteMode;
    SVPLenString\_t fileName;
    SVPLenString\_t secondaryFileName;
    u8 interpolationRatio;
} SLAReadWriteNuc\_t;
```

When saving tables, the file extension {`.nuc`, `.dead`} will be appended to the end of the filename provided; *filename.nuc* will be treated the same as *filename*. Files are stored on the microSD card.

Note

To clear the default table, send a zero length `fileName` with `0x30` or `0x40` ***nucReadWriteMode***.

See also

Special note in [SLAGetParameters_t](#) regarding getting the ReadWriteNuc params.

Message ID 0x36

Byte Offset	Name	Description												
4	cameraIndex	Camera Index												
5	reserved	<i>Reserved</i>												
		Function that will be performed on the filename <table><tr><th>Bits</th><th>Value</th><th>Description</th></tr><tr><td rowspan="4">0 - 3</td><td>0</td><td>Do Nothing</td></tr><tr><td>1</td><td>Save NUC Table</td></tr><tr><td>2</td><td>Load NUC Table</td></tr><tr><td>3</td><td>Save Dead Table</td></tr></table>	Bits	Value	Description	0 - 3	0	Do Nothing	1	Save NUC Table	2	Load NUC Table	3	Save Dead Table
Bits	Value	Description												
0 - 3	0	Do Nothing												
	1	Save NUC Table												
	2	Load NUC Table												
	3	Save Dead Table												

6	nucReadWriteMode		4	Load Dead Table
			5	Load Interpolated NUC Table between two saved NUC Tables
		6	Load NUC Table with shutter flatten applied from secondary NUC Table	
			7	Save Shutter Flatten difference as NUC table
			8 - 15	*Reserved*
		4 - 7	0	Do Nothing
			1	Set NUC Table to be loaded as default on startup
			2	Set Dead Table to be loaded as default on startup
			3	Clear NUC Table to be loaded, so that no table is loaded on startup. fileName is ignored in this mode.
			4	Clear Dead Table to be loaded, so that no table is loaded on startup. fileName is ignored in this mode.
			5 - 15	*Reserved*
7	fileName.len	string length		
8-...	fileName.str	Filename for NUC/Dead table to be acted on. See File Recording NOTE: A blank filename will not be accepted unless nucReadWriteMode is 0x30 or 0x40.		
varies	secondaryFileName.len	string length		
varies	secondaryFileName.str	Filename for secondary NUC table when interpolating between two NUCs.		
varies	interpolationRatio	What ratio to use when interpolating between primary and secondary. 0 = fully primary NUC table, 255 = fully secondary NUC table		

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAResetStabilizationParameters_t

Struct Reference

[Stabilization](#)

Description

Reset the internal motion smoothing filters that control video stabilization.

```
typedef struct {
    u8 resetType;
    u8 cameraIndex;
} SLAResetStabilizationParameters_t;
```

Re-center the current video frame in the field of view.

Message ID 0x04

Byte Offset	Name	Description	
4	resetType	Value	Description
		0	Reset all filters (default)
		1	Reset only the display filter
		2	Reset only the auto bias filter
		3-255	Reserved
5	cameraIndex	<u>Camera Index</u> If not present or 255, command applies to all cameras.	

Sightline Intelligence

Command, Control, and Script

API v3.11

[Main Page](#)[Related Pages](#)[Modules](#)[Classes](#) [Camera Configuration](#)

SLASaveParameters_t Struct Reference

[General](#)

Description

Commits current parameters to flash.

```
typedef struct {  
} SLASaveParameters t;
```

Parameters are then loaded when system restarts. See [SLAResetAllParameters_t](#) for different options to reset parameters. Parameter files can also be modified using SSH or FTP or similar.

NOTE: Do not repeat sending this command within 1 second.

NOTE: Save parameters is not available in [Demo Mode](#).

Message ID 0x25

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASendToBTS_t Struct Reference

[General](#)

Description

Start sending data to BTS from SLA.

```
typedef struct {
    u8 mode;
    u32be ipAddr;
    u16be port;
    u16 rate;
    u16 displayId;
} SLASendToBTS_t;
```

Message ID 0xBE

Byte Offset	Name	Description
4	mode	Enable or disable
5-8	ipAddr	IPv4 address of the receiver
9-10	port	destination network port
11-12	rate	rate at which packets will be generated (frames)
13-14	displayId	<u>Network Display ID</u> 4000/17XX: (0x0002 = Net0, 0x0080 = Net1, 0x0082 = both) 41XX: (0x0002 = Net0, 0x0080 = Net1, 0x0010 = Net2, 0x0092 = all)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASendTraceStr_t Struct Reference

[General](#)

Description

Log commands and telemetry in human-readable format.

```
typedef struct {
    SVPEndString_t str;
} SLASendTraceStr_t;
```

Message ID 0x5A

Byte Offset	Name	Description
4	str	Line of trace data

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetDisplayAdjustments_t Struct Reference

[Display](#)

Description

Set the parameters for fine-tuned adjustments of the display.

```
typedef struct {
    u16 ratio8;
    s16 pan;
    s16 tilt;
    u8 cameraIndex;
} SLASetDisplayAdjustments_t;
```

Message ID 0x2A

Byte Offset	Name	Description
4-5	ratio8	Ratio of secondary to primary imager zoom levels multiplied by 256
6-7	pan	Display pan column offset in pixels.
8-9	tilt	Display tilt row offset in pixels.
10	cameraIndex	Camera Index

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetKlvData_t Struct Reference

[List of all members](#)

[KLV Metadata](#)

Description

The KLV blob data is constructed by user to be sent with MPEG2-TS stream (see [SLASetEthernetDisplayParameters_t](#)).

```
typedef struct {
    u16 displayId;
    SVPEndData\_t klvData;
} SLASetKlvData\_t;
```

The KLV data will be sent along with the next H.264 frame. When using this feature, you should disable the transmission of the built-in KLV metadata by setting [SLASetMetadataRate_t](#).frameStep to 0.

example: 51 ac <len> 61 00 82 <klv blob> <chk sum> len := 1 + 2 + <klvblob length> + 1

KLV data payload immediately follows [Network Display ID](#).

Message ID 0x61

Byte Offset	Name	Description
4-5	displayId	Network Display ID 4000/17XX: (0x0002 = Net0, 0x0080 = Net1, 0x0082 = both) 41XX: (0x0002 = Net0, 0x0080 = Net1, 0x0010 = Net2, 0x0092 = all)
6	klvData	KLV payload data

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetLensMode_t Struct Reference

[Focus](#)

Description

Perform camera lens control.

```
typedef struct {
    u8 lensMode;
    u16 data;
} SLASetLensMode_t;
```

If any data needs to be sent along with a command, this data will be in the bytes following the LensMode. Changing the lens position will stop autofocus (see [SLASetLensParameters_t](#)).

Message ID 0x6C

Byte Offset	Name	Description			
4	lensMode	Value	Description	Data bytes	Data
		0, 1	Reserved	–	–
		2	Reset Lens Mechanism	0	–
		3	Request Lens Status	0	–
		4	Zoom in Wide Direction	0	–
		5	Zoom in Narrow Direction	0	–
		6	Stop Zoom Motion	0	–
		7	Focus in Far Direction	0	–
		8	Focus in Near Direction	0	–
		9	Stop Focus Motion	0	–
		10	One Push Auto Focus	0	–
		11	Reserved	0	–
					0 = no shutter, 1 =

		12	Lens NUC (IR camera)	1	shutter
		13	Set Zoom Position	2	Motor pulse counts
		14	Set Focus Position	2	Motor pulse counts
		128 & Above	<i>Reserved</i> for customer defined modes.	2	customer defined
5-6	data	Data bytes			

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) [Camera Configuration](#)

[List of all members](#)

SLASetLensParameters_t Struct

Reference

[Focus](#)

Description

Deprecated, use [SLALensParameters_t](#)

```
typedef struct {
    u8 lensType;
    u8 afMetricRegionSize;
    u8 zoomTrackFocus;
    u8 autofocusMethod;
    u16 autofocusRateAdjust;
    u8 autofocusChangePct;
    u8 zoomSpeed;
    u8 focusSpeed;
    u8 comPortNum;
    u8 colRoiFrac;
    u8 rowRoiFrac;
    u8 wideRoiFrac;
    u8 highRoiFrac;
} SLASetLensParameters\_t;
```

Deprecated, use [SLALensParameters_t](#)

Set the current parameters for controlling the lens/focus.

Lens Type

A set of known lenses or cameras with integrated lenses for which command and control has been implemented. Assumes the lens is connected to a known serial port (e.g. Tamron SC001 is connected to Serial 1 on 1500-OEM), but can be overridden using the *comPortNum* parameter. This serial port will be automatically configured to the correct baud rate, etc. Command passthrough and similar commands will not be available. Corresponding serial port must be set to *Port not used* via [SLASetPortConfiguration_t](#).

See also

LUA lens driver [See example code page](#) as well

Auto Focus Method

Scan

Start by moving the focus to the far focus limit. Scan the full range to the near focus limit. Look for the peak focus metric over the entire range. This method may be problematic due to defocused imagery at the limits of the focus travel being noisy. Depending on the focus metric chosen, this noise can produce high metric values.

Seek

Searches near current focus. Start by moving the focus in the far direction until the focus metric changes more than the value set in autofocusChangePct.

- If the focus metric is increasing: Keep moving in this direction until the focus metric starts to decrease by the value set in autofocusChangePct, e.g., moving up the hill then down the hill.
- If the focus metric is decreasing: Change directions until the focus metric increases and then decreases by the value set in autofocusChangePct.

Set the focus metric algorithm using [SLASetSystemValue_t](#). Focus metrics are reported using [SLAFocusStats_t](#).

See also

[SLASetOverlayMode_t](#).graphics to enable the onscreen [Overlays](#) for the region of interest.

Message ID 0x6E

Byte Offset	Name	Description		
4	lensType			
		Value	Device	Serial Port
		0	None	N/A
		1	Tamron SC001	1
		2	Sony	2
		3	Hitachi	2
		128 and above	Custom Lens Driver	Any
5	afMetricRegionSize	Deprecated, use Roi controls below. AutoFocus Metric Region Size % of screen (0 to 100)		
6	zoomTrackFocus	Zoom Track Focus (0= disabled, 1= enabled). Stops when focus metric reaches peak threshold.		
7	autofocusMethod	Autofocus method (0=scan far to near, 1=seek, 128 and above reserved for customer use)		
8-9	autofocusRateAdjust	Autofocus Rate Adjust (0 to 65535) 100 = default speed of motion		
10	autofocusChangePct	Autofocus Change Percent (0 to 255) in 1/10ths of a percent. 40 (default) = 4% When using Seek in AutoFocus Method, this value (0 to 100) is the percentage of change in the focus metric that is considered significant enough to indicate that the focus metric is going in the right (or wrong) direction.		
11	zoomSpeed	Zoom speed (1-255 with 0 being the default)		
12	focusSpeed	Focus speed (1-255 with 0 being the default)		
13	comPortNum	Serial port number for lens control see SLASetPortConfiguration_t		

14	colRoiFrac	Row ROI for focus stats. Row Offset in % of image width (255=100%); Full image(default) = 0
15	rowRoiFrac	Col ROI for focus stats. Column Offset in % of image height (255=100%); Full image(default) = 0
16	wideRoiFrac	High ROI for focus stats. Height in % of image height (255=100%); Full image(default) = 255
17	highRoiFrac	Wide ROI for focus stats. Width in % of image width (255=100%); Full image(default) = 255

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLATagDataRate_t Struct Reference

[KLV Metadata](#)

Description

Set the output frequency of a KLV Tag or a range of tags.

```
typedef struct {
    u8 reserved1;
    u8 reserved2;
    u8 mode;
    u8 tagId1;
    u8 tagId2;
    u16 frameStep;
    u16 displayId;
} SLATagDataRate_t;
```

Rate will not be set for unsupported tags. See [SLATagData_t](#) for extending supported tags by adding new Tag and SubTag.

Note

Per MISB STD 0601, version identifier fields for the UAS Local Data Set, Security Metadata Local Set, and Motion Imagery Track Metadata Local Set are emitted with elements of each local set. See also MISB ST 0902 for guidelines on setting the frame step for different fields.

Setting the frameStep (rate) to zero may make the resulting KLV metadata non compliant with MISB standards.

Also sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x97

Byte Offset	Name	Description						
4	reserved1	Reserved						
5	reserved2	Reserved						
6	mode	Mode is used to set rate for a range of tags, ignored in the <u>SLAGetParameters_t()</u> reply						
		<table><tr><th>Value</th><th>Description</th></tr><tr><td>0</td><td>Setting the rate of a single KLV tag</td></tr><tr><td>1</td><td>Setting the rate of a range of KLV tags to the frameStep value</td></tr></table>	Value	Description	0	Setting the rate of a single KLV tag	1	Setting the rate of a range of KLV tags to the frameStep value
		Value	Description					
		0	Setting the rate of a single KLV tag					
1	Setting the rate of a range of KLV tags to the frameStep value							

7	tagId1	TagId to set or get the rate, If mode is 1 then this is the start Tag value of the range. TagId should corresponds to the tag values in the UAS Data link local Metadata set elements in STD-0601
8	tagId2	ignored if mode is 0 or this is a SLAGetParameters_t() . reply if mode is 1 then tagId2 represents the end tag in the range tagId1, tagId1 +1, ..., tagId2 will be set to the rate mentioned in the frameStep
9-10	frameStep	Frame step at which to send the specified tag values; 0 = disable sending KLV metadata, 1 = send each frame, etc. Per MISB STD 0601, version identifier fields for the UAS Local Data Set, Security Metadata Local Set, and Motion Imagery Track Metadata Local Set are emitted with elements of each local set. See also MISB ST 0902 for guidelines on setting the frame step for different fields.
11-12	displayId	<u>Network Display ID</u> 4000/17XX: (0x0002 = Net0, 0x0080 = Net1, 0x0082 = both) 41XX: (0x0002 = Net0, 0x0080 = Net1, 0x0010 = Net2, 0x0092 = all)

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetMultipleAlignment_t Struct Reference

[Blending](#)

Description

Deprecated, use [SLABlendAlign_t](#)

```
typedef struct {
    u8 nAlignments;
    struct {
        u8 vertical;
        u8 horizontal;
        u8 rotate;
        u8 zoom;
        u8 hzoom;
    } alignment[5];
} SLASetMultipleAlignment\_t;
```

Deprecated, use [SLABlendAlign_t](#)

Set the (5) alignment sets for a dual camera setup. See details for warp parameters in [SLASetBlendParameters_t](#). Values of unused sets to may be set to 0.

Message ID 0x74

Byte Offset	Name	Description
4	nAlignments	Number of alignments present
varies	vertical	Vertical offset in pixels.
varies	horizontal	Horizontal offset
varies	rotate	Rotation
varies	zoom	Zoom
varies	hzoom	Horizontal zoom scale

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetRegistrationParameters_t

Struct Reference

Description

Deprecated, use [SLARegistrationParameters_t](#)

```
typedef struct {
    u16 maxTranslation;
    u8 maxRotation;
    u8 zoomRange;
    u8 left;
    u8 right;
    u8 top;
    u8 bottom;
    u8 cameraIndex;
} SLASetRegistrationParameters_t;
```

Deprecated, use [SLARegistrationParameters_t](#)

Set parameters used by registration module.

NOTE: Applied to the Command Camera.

Ignore Edge Pixels

Used for overlays or foreground objects that appear near the edge of the image. 0 is default. At least 1/4 of the smaller dimension of the image must be remaining. For NTSC, remaining non - ignored image must be at least 120x120. Use the 'ignore edge pixel band' to indicate that the registration algorithm should not include these pixels when determining the registration match. This can be used to compensate for effects of the optics such as vignetting or when there is an obstruction along an edge of the image. See [SLASetOverlayMode_t](#) to enable onscreen graphics to better select ignore edge bands.

Message ID 0x0E

Byte Offset	Name	Description
4-5	maxTranslation	Maximum translation in pixels. Default of 0 is equivalent to 120 for a 480 high image (1/4 of the image height)
		Maximum rotation range in degrees per frame: 0..10 (values larger than 10 clipped to 10). 5 is

6	maxRotation	default.
7	zoomRange	Maximum zoom range in percent zoom per frame: 0..10. 0 is default. NOTE: This parameter should almost always be 0. Small values may help in limited cases to maintain registration and tracking while a camera is actively zooming. Enabling this parameter when a camera is not zooming will make registration run slower and be less accurate.
8	left	Left band of edge pixels to ignore. 0..255 (0 is default)
9	right	Right band of edge pixels to ignore. 0..255 (0 is default)
10	top	Top band of edge pixels to ignore. 0..255 (0 is default)
11	bottom	Bottom band of edge pixels to ignore. 0..255 (0 is default)
12	cameraIndex	<u>Camera Index</u>

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetSnapShot_t Struct Reference

[Recording](#)

Description

Set up parameters for image snapshot.

```
typedef struct {
    u8 mode;
    u8 format;
    u8 source;
    u8 quality;
    u8 downsample;
    u32be ipAddr;
    u16be port;
    SVPLenString\_t userName;
    SVPLenString\_t password;
    u8 secondaryFormat;
    u8 cameraIndex;
} SLASetSnapShot\_t;
```

For viewing a list of existing files or recording video, see [SLASetSDRecordingParameters_t](#). Image is recorded from the Camera set with [SLACommandCamera_t](#). When Image source is set to Capture, image contents and size will be the same as the raw image from the camera. For example, if using an Analog NTSC Video Source, the image size will be 640x480. If using a digital image from a Sony FCB-EH6300 the image will be 1280x720. When Image Source is set to Display Image, then the image will be a 640x480 image with all the overlay graphics, digital zoom, and other image processing applied. See [File Recording](#) for additional details.

NOTE: Applied to the Command Camera.

File Format

JPEG/JPG

A form of lossy compression defined by the Joint Photographic Experts Group (JPEG).

- The Quality value affects the size and quality/loss of the JPEG.
- SnapShot images are created with EXIF metadata headers populated from the following MISB fields:

MISB data element	EXIF tag	To set, use...
Sensor Latitude	GPS Latitude	SLASetMetadataValues_t

Sensor Longitude	GPS Longitude	<u>SLASetMetadataValues_t</u>
Sensor Altitude	GPS Altitude	<u>SLASetMetadataValues_t</u>
UTC Time	GPS Time	<u>SLASetMetadataValues_t</u>
Platform Indicated Air Speed	GPS Speed	<u>SLATagData_t</u>
Platform Heading	GPS Track Direction	<u>SLATagData_t</u>
Image Source Sensor	Model	<u>SLAMetadataStaticValues_t</u>
Slant Range	SubjectDistance	<u>SLATagData_t</u> (or automatically calculated)

- Compatibility with Exif is only in the sense that we use Exif tags, we do not comply with any specific Exif version.
- JPEG snap shots of high bit depth **Capture** (source) images are converted to 8-bit images before jpeg compression. Use PNG, SLRaw or TIFF to save full bit depth images.

PNG

PNG is a lossless compression standard that works with all image types.

- The Quality value affects the size of the PNG and time to generate it. A Quality value of 80 is recommended, this gives the best file size and performance.
- When saving a 14-bit grayscale capture source, the PNG will be 16-bits.
- When working with 16-bit PNG files, Microsoft Windows Paint is NOT recommended. Use SightLine PNGViewer or ImageJ.
- Downsample is not supported for PNG snapshots.
- Non-Uniformity Correction and Dead Pixel Removal are optionally applied **before** taking a capture snapshot. Dynamic Dead Filter is **not** applied before taking a capture snapshot.

JPEG Big

Same as JPEG except allows a larger snapshot to be taken of capture (source) image.

- Image processing is halted while a *JPEG Big* snapshot is taken.
- See [SLASetAcquisitionParameters_t](#) to configure the capture for JPEG Big snapshots

SLRaw

SightLine raw image standard for high (>8) bit depth grayscale and color **Capture** snapshots only.

- Non-Uniformity Correction and Dead Pixel Removal are optionally applied **before** taking a capture snapshot. Dynamic Dead Filter is **not** applied before taking a capture snapshot.
- Downsample and quality are not supported for SLRaw snapshots.
- See [EAN-Infrared-Temperature](#) for details on the SLRaw image structure.
- The *PngViewer* SLAExamples program has example code that reads an SLRaw image file in *SLRawImage.cpp*.

TIFF

TIFF format captures uncompressed images with a standard tiff wrapper for grayscale **Capture** snapshots only.

- Downsample is not supported for TIFF snapshots.
- Non-Uniformity Correction and Dead Pixel Removal are optionally applied **before** taking a capture snapshot. Dynamic

Dead Filter is **not** applied before taking a capture snapshot.

Metadata Format

- **Single KML file:** Produces a single KML file for all the snapshots taken with the same filename. NOTE: This does not work over FTP
- **Multiple KML files:** Produces a KML for for each individual snapshot taken.
- **KLV files:** Produces a file for each snapshot taken which contains a binary KLV blob, equivalent to what you would find in the MPEG-2 TS.

NOTE: Capture snapshot is not available in **Demo Mode**.

Message ID 0x5E

Byte Offset	Name	Description																																	
4	mode	Snapshot destination – 0 = FTP server, 1 = Micro SD (default), 2 = USB																																	
5	format	Image file and Metadata file formats																																	
		<table><tr><th>Bits</th><th colspan="2">Description</th></tr><tr><td rowspan="7">0-3</td><td colspan="2">File formats:</td></tr><tr><th>Value</th><th>Description</th></tr><tr><td>0</td><td>JPEG (8 bit compressed)</td></tr><tr><td>1</td><td>PNG (lossless 8/16 bit)</td></tr><tr><td>2</td><td>JPEG Big (Capture only)</td></tr><tr><td>3</td><td>SLRaw (>8 bit Capture only)</td></tr><tr><td>4</td><td>TIFF (Grayscale Capture only)</td></tr><tr><td rowspan="7">4-7</td><td colspan="2">Metadata formats:</td></tr><tr><th>Value</th><th>Description</th></tr><tr><td>0</td><td>No Metadata file</td></tr><tr><td>1</td><td>Single aggregate KML file</td></tr><tr><td>2</td><td>Multiple KML files</td></tr><tr><td>3</td><td>NITF file</td></tr><tr><td>4</td><td>KLV</td></tr></table>	Bits	Description		0-3	File formats:		Value	Description	0	JPEG (8 bit compressed)	1	PNG (lossless 8/16 bit)	2	JPEG Big (Capture only)	3	SLRaw (>8 bit Capture only)	4	TIFF (Grayscale Capture only)	4-7	Metadata formats:		Value	Description	0	No Metadata file	1	Single aggregate KML file	2	Multiple KML files	3	NITF file	4	KLV
		Bits	Description																																
		0-3	File formats:																																
			Value	Description																															
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4	TIFF (Grayscale Capture only)																																		
4-7	Metadata formats:																																		
	Value	Description																																	
	0	No Metadata file																																	
	1	Single aggregate KML file																																	
	2	Multiple KML files																																	
	3	NITF file																																	
	4	KLV																																	
When source is set to BOTH, the image format portion is only used for the captured image but metadata is used for both.																																			
6	source	snapshot source: 1 = captured image, can be 8 bit or high bit depth (>8 bit) (default) 2 = displayed image, always 8 bit color 3 = both captured and displayed images																																	

7	quality	JPEG: 0 (lowest quality) to 100 (highest quality) (default 80). >95 is not recommended, may truncate the image. PNG: 60 (maximum compression) to 100 (uncompressed) (default 80, recommended).
8	downsample	Down Sample (JPEG Only) 1: none (default) 2: 2x2 downsample 4: 4x4 downsample
9-12	ipAddr	IP network address of FTP server
13-14	port	Command and control reply port for FTP server (default 21)
15	userName.len	string length
16-...	userName.str	FTP server user name
varies	password.len	string length
varies	password.str	FTP server password
varies	secondaryFormat	Format for display snapshot if source is set to BOTH 0 = JPEG (8 bit compressed) (default) 2 = PNG (lossless 8/16 bit)
varies	cameraIndex	<u>Camera Index</u>

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetStabilizationBias_t Struct Reference

Description

Deprecated, use [SLAStabilizationBias_t](#)

```
typedef struct {
    s8 biasCol;
    s8 biasRow;
    u8 autoBias;
    u8 updateRate;
    u8 cameraIndex;
} SLASetStabilizationBias_t;
```

Deprecated, use [SLAStabilizationBias_t](#)

Adjust the stabilization solution by adding a constant bias in pixel coordinate space to the current coordinates each frame.

Stabilization bias is used to feed forward user controlled camera motion so that stabilization does not `fight` against camera pan and tilt. Set `auto bias` mode to automatically prevent the system from stabilizing against constant motion. Some amount of motion lag will still be experienced in `auto bias` mode. Manual bias offsets may be used together with `auto bias` mode. Changes to the column and row bias values will be added to the `auto bias` solution. In `auto bias` mode, set column and row bias to 0 for `auto bias` only. Bias offsets may need to be scaled to up as the operational frame rate decreases.

See also

[SLAResetStabilizationParameters_t](#) to reset only the auto-bias filter. Message ID 0x12

Byte Offset	Name	Description
4	biasCol	-128..127: Per frame column adjustment (bias) in pixels
5	biasRow	-128..127: Per frame row adjustment (bias) in pixels
6	autoBias	1 = Enable auto bias (combined auto + manual bias) 0 = Disable auto bias(manual bias only)
7	updateRate	Auto bias update rate 0..255 (default = 50) (Internally divided by 500)
8	cameraIndex	Camera Index 255 = All cameras.

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetVideoMode_t Struct Reference

Description

Deprecated, use [SLAVideoDisplay_t](#) and [SLAMultiDisplay_t](#).

```
typedef struct {
    u8 nCameras;
    u8 nDisplays;
    u8 displayMode;
    u8 displayDest;
    u8 cameraIndex0;
    u8 cameraIndex1;
    u8 cameraIndex2;
    u8 pipColor;
    u8 pipScale;
    u8 pipQuadrant;
    u8 displayIndex0;
    u8 displayIndex1;
    u8 displayIndex2;
    u8 displayIndex3;
    u8 analogDest;
    u8 hdmiDest;
    u8 net0Dest;
    u8 net1Dest;
    u8 HDSDIDest;
    u8 hdmiRes;
    u8 hdsdiRes;
    u8 analogRes;
} SLASetVideoMode_t;
```

Deprecated, use [SLAVideoDisplay_t](#) and [SLAMultiDisplay_t](#).

Message ID 0x1F

Byte Offset	Name	Description
4	nCameras	<i>Reserved</i> – set to 0
5	nDisplays	<i>Reserved</i> – set to 0
6	displayMode	Display Mode
7	displayDest	Display destination or resolution
8	cameraIndex0	Camera Index

9	cameraIndex1	<i>Reserved</i>
10	cameraIndex2	<i>Reserved</i>
11	pipColor	Color of PiP window border, default is red.
12	pipScale	Scale of Picture-in-picture thumbnail
13	pipQuadrant	Screen quadrant in which to render PiP window
14	displayIndex0	Camera Display Order index 0 (For multi camera display, Camera 0 is the main (PiP) or left image (Two Up). Camera 1 for multi-camera display is the PiP image or camera on the right in Two Up mode.)
15	displayIndex1	Camera Display Order index 1
16	displayIndex2	<i>Reserved</i>
17	displayIndex3	<i>Reserved</i>
18	analogDest	3000 only Display Image Mode for analog display
19	hdmiDest	3000 only Display Image Mode for HDMI display
20	net0Dest	3000 only Display Image Mode for Network 0 display
21	net1Dest	3000 only Display Image Mode for Network 1 display
22	HDSDIDest	3000 only Display Image Mode for HD-SDI display
23	hdmiRes	3000 only Display resolution for HDMI display (3000 only). Default is 1080p60.
24	hdsdiRes	3000 only Display resolution for HDSDI display (3000 only). Default is 1080p60.
25	analogRes	3000 only Display resolution for Analog display (3000 only). Default is NTSC.

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) [Camera Configuration](#)

[List of all members](#)

SLASetVMTI_t Struct Reference

[KLV Metadata](#)

Description

Allows targets from outside SightLine to be injected into the MPEG2-TS KLV stream.

```
typedef struct {
    u8 numTargets;
    struct {
        u8 targetID;
        u8 confidence;
        u16 col;
        u16 row;
        u16 wide;
        u16 high;
        u16 newTargetDetectionFlag;
    } targetPack[127];
    u16 displayId;
} SLASetVMTI\_t;
```

Targets received from an external source shall be merged with any detections generated by

[SLASetDetectionParameters_t](#) (0x2D) and are output as a single MISB/KLV Local Data Set Tag 74 (VMTI Local Data Set).

SightLine generated VMTI entries can be enabled/disabled with [SLASetAdvancedDetectionParameters_t](#) (0x76) *maxKlvTrks* argument.

Note

VMTI data generated automatically requires Full Telemetry ([SLAVersionNumber_t](#))

Target ID Number can be used to differentiate different "sources" of targets:

- 1 to 10 are tracks set manually or by Vehicle/Drone detection modes
- 11 to 101 are detections generated by modes such as Aerial, Motion, Staring, Maritime, Blob, etc.
- 128 to 255 are inserted with this function

Target center row and column are relative to upper left corner of the image.

Targets position and size are bound within the display frame.

Externally defined targets continue to be sent until a new [SLASetVMTI_t](#) message is received.

Example: Send SetVMTI with number of targets set to 0 (stops user set VMTI)

```
51, ac, 03, 84, 00, D0
```

Example: Send SetVMTI with 1 target at the center of 320x240 that is 40 pixels high and wide

51,AC,0F,84,01,01,62,40,01,F0,00,28,00,28,00,01,00,46

Message ID 0x84

Byte Offset	Name	Description
4	numTargets	Bits 0..6: number of tracks (up to 110 reported)
varies	targetID	Values 1 - 127. This will be transmitted as target ID 128 - 255. MISB 74.101.OEB.
varies	confidence	Target Confidence (0-100) MISB 74.101.5
varies	col	Target Centroid Column MISB 74.101.20
varies	row	Target Centroid Row MISB 74.101.19
varies	wide	Target width (pixels)
varies	high	Target height (pixels)
varies	newTargetDetectionFlag	The number of times this target has been detected MISB 74.101.6
varies	displayId	<u>Network Display ID</u> 4000/17XX: (0x0002 = Net0, 0x0080 = Net1, 0x0082 = both) 41XX: (0x0002 = Net0, 0x0080 = Net1, 0x0010 = Net2, 0x0092 = all)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAShiftSelectedTrack_t Struct Reference

[Tracking](#)

Description

Command the system to shift the selected track to the next track.

```
typedef struct {  
    u8 cameraIndex;  
} SLAShiftSelectedTrack_t;
```

Message ID 0x33

Byte Offset	Name	Description
4	cameraIndex	Camera Index

Sightline Intelligence

Command, Control, and Script

API v3.11

[List of all members](#)

SLAStopSelectedTrack_t Struct Reference

[Tracking](#)

Description

Command the system to stop the currently selected track.

```
typedef struct {
    u8 cameraIndex;
} SLAStopSelectedTrack_t;
```

Message ID 0x3C

Byte Offset	Name	Description
4	cameraIndex	Camera Index

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAStopTracking_t Struct Reference

[Tracking](#)

Description

Turn off all tracks.

```
typedef struct {
    u8 reserved1;
    u8 reserved2;
    u8 reserved3;
    u8 cameraIndex;
} SLAStopTracking_t;
```

Message ID 0x09

Byte Offset	Name	Description
4	reserved1	Reserved
5	reserved2	Reserved
6	reserved3	Reserved
7	cameraIndex	Camera Index

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAStreamingControl_t Struct Reference

[Compression](#) | [KLV Metadata](#)

Description

Control to Start and Stop Network Streaming.

```
typedef struct {  
    ul6 streamingControlMask;  
} SLAStreamingControl_t;
```

Application should previously have configured the corresponding network display via [SLASetEthernetDisplayParameters_t](#). Applications can start and stop streaming on both networks at the same time by enabling Streaming Control Mask 'Start' bits or 'Stop' bits for both the network displays. Streaming control is ignored if 'Start' bit and 'Stop' bit of that network is set to 1.

When an H.264 stream is started, an I-FRAME is generated.

Also sent in response to [SLAGetParameters_t\(\)](#).

Note

- KLV output depends on rates set by [SLATagDataRate_t](#)

Message ID 0x90

Byte Offset	Name	Description								
		Streaming control: Corresponding Start and Stop bits should not both be 1 or the entire command will be ignored								
		<table><tr><th>Bit</th><th>Description</th></tr><tr><td>0</td><td>1 = Start streaming on Net 0 (indicates streaming active in response), 0 = no effect on streaming</td></tr><tr><td>1</td><td>1 = Start streaming on Net 1 (indicates streaming active in response), 0 = no effect on streaming</td></tr><tr><td></td><td></td></tr></table>	Bit	Description	0	1 = Start streaming on Net 0 (indicates streaming active in response), 0 = no effect on streaming	1	1 = Start streaming on Net 1 (indicates streaming active in response), 0 = no effect on streaming		
Bit	Description									
0	1 = Start streaming on Net 0 (indicates streaming active in response), 0 = no effect on streaming									
1	1 = Start streaming on Net 1 (indicates streaming active in response), 0 = no effect on streaming									

4-5	streamingControlMask	2	1 = Start KLV streaming on Net 0, 0 = no effect on KLV streaming
		3	1 = Start KLV streaming on Net 1, 0 = no effect on KLV streaming
		4	1 = Start streaming on Net 2 (indicates streaming active in response), 0 = no effect on streaming
		5	1 = Start streaming on Net 3 (indicates streaming active in response), 0 = no effect on streaming
		6	1 = Start KLV streaming on Net 2, 0 = no effect on KLV streaming
		7	1 = Start KLV streaming on Net 3, 0 = no effect on KLV streaming
		8	1 = Stop streaming on Net 0 (indicates streaming stopped in response), 0 = no effect on streaming
		9	1 = Stop streaming on Net 1 (indicates streaming stopped in response), 0 = no effect on streaming
		10	1 = Stop KLV streaming on Net 0, 0 = no effect on KLV streaming
		11	1 = Stop KLV streaming on Net 1, 0 = no effect on KLV streaming
		12	1 = Stop streaming on Net 2 (indicates streaming stopped in response), 0 = no effect on streaming
		13	1 = Stop streaming on Net 3 (indicates streaming stopped in response), 0 = no effect on streaming
		14	1 = Stop KLV streaming on Net 2, 0 = no effect on KLV streaming
		15	1 = Stop KLV streaming on Net 3, 0 = no effect on KLV streaming

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASystemStatusMessage_t Struct

Reference

[General](#)

Description

Packet is sent from the SLA hardware to report diagnostic information such as CPU load and temperature, or the performance of software tools.

```
typedef struct {
    u64 errorFlags;
    s16 temperatureF;
    u8 load0;
    u8 load1;
    u8 load2;
    u8 load3;
    s16 temperatureC;
    u8 missedFrames0;
    u8 missedFrames1;
    u8 missedFrames2;
    u8 missedFrames3;
} SLASystemStatusMessage_t;
```

Use [SLASystemStatusMode_t](#) to configure this message.

This information can be used to measure and track performance of the system over time under different conditions.

NOTE: This packet is sent only to Ethernet Command and Control Reply Port, and never sent to a Serial port.

Message ID 0x87

Byte Offset	Name	Description
4-11	errorFlags	<i>Reserved</i>
12-13	temperatureF	Temperature in degrees F
14	load0	17XX/4000: Arm 0 load %; 4100: Core 2
15	load1	17XX/4000: Arm 0 load %; 4100: Core 3
16	load2	17XX/4000: Arm 0 load %; 4100: Core 4
17	load3	17XX/4000: Arm 0 load %; 4100: Core 5
18-19	temperatureC	Temperature in degrees C NEW 3.6
20	missedFrames0	Missed frames since last SystemStatusMessage call: Cam 0

21	missedFrames1	Missed frames since last SystemStatusMessage call: Cam 1
22	missedFrames2	Missed frames since last SystemStatusMessage call: Cam 2
23	missedFrames3	Missed frames since last SystemStatusMessage call: Cam 3

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASystemStatusMode_t Struct Reference

[List of all members](#)

[General](#)

Description

Enable / disable System Status Messages ([SLASystemStatusMessage_t](#)) and Detailed Timing Messages ([SLADetailedTimingMessage_t](#)) sent from the SLA hardware at 1 Hz.

```
typedef struct {
    u16 systemStatusBits;
    u32 systemDebugBits;
} SLASystemStatusMode_t;
```

Also sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x80

Byte Offset	Name	Description																
4-5	systemStatusBits	System status mode																
		<table><tr><th>Bit</th><th>Description</th></tr><tr><td>0</td><td>1 = Enable system status messages and missed frames reporting, 0 = disable system status messages and missed frames reporting</td></tr><tr><td>1-15</td><td>Reserved</td></tr></table>	Bit	Description	0	1 = Enable system status messages and missed frames reporting, 0 = disable system status messages and missed frames reporting	1-15	Reserved										
		Bit	Description															
		0	1 = Enable system status messages and missed frames reporting, 0 = disable system status messages and missed frames reporting															
1-15	Reserved																	
6-9	systemDebugBits	System debug flags																
		<table><tr><th>Bit</th><th>Description</th></tr><tr><td>0-9</td><td>Reserved</td></tr><tr><td>10</td><td>1 = Enable DSP and ARM timing measurement</td></tr><tr><td>11</td><td>Reserved</td></tr><tr><td>12</td><td>1 = Enable internal latency timing measurement</td></tr><tr><td>13</td><td>1 = Enable low level DSP detailed timing measurement</td></tr><tr><td></td><td></td></tr><tr><td></td><td></td></tr></table>	Bit	Description	0-9	Reserved	10	1 = Enable DSP and ARM timing measurement	11	Reserved	12	1 = Enable internal latency timing measurement	13	1 = Enable low level DSP detailed timing measurement				
		Bit	Description															
		0-9	Reserved															
		10	1 = Enable DSP and ARM timing measurement															
		11	Reserved															
		12	1 = Enable internal latency timing measurement															
		13	1 = Enable low level DSP detailed timing measurement															

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLATagData_t Struct Reference

[KLV Metadata](#)

Description

Set the value of any MISB Tag and subtag.

```
typedef struct {
    u8 reserved1;
    u8 reserved2;
    u8 tagId;
    u8 tagSubId;
    u16 reservedInternal;
    u16 displayId;
    SVPLenData\_t klvData;
} SLATagData\_t;
```

Used to set value for any KLV Tag specified in the MISB standard ST0601.xx.

This command can be used to set any custom value to a tag and subtag defined as per ST0601.xx. TagData can be used to extend the default supported tags by Sightline. Set the rate of the tag explicitly by calling [SLATagDataRate_t](#) after using TagData if the tag was not supported before. Also sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x96

Byte Offset	Name	Description
4	reserved1	<i>Reserved</i>
5	reserved2	<i>Reserved</i>
6	tagId	tagId corresponding to the tag key value as per ST0601.xx
7	tagSubId	tagSubId in case if tagId corresponds to a metaDataTag, For example "Security Local Set" with sub tag "Classifying country" will have tagId as 48 and tagSubId as 3. For regular tags tagSubId is ignored.
8-9	reservedInternal	reserved and should be set to zero, Used internally by video track
10-11	displayId	Network Display ID 4000/17XX: (0x0002 = Net0, 0x0080 = Net1, 0x0082 = both) 41XX: (0x0002 = Net0, 0x0080 = Net1, 0x0010 = Net2, 0x0092 = all)
12	klvData.len	string length

13-...	klvData.str	KLV payload data
--------	-------------	------------------

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLATagSourceSelector_t Struct Reference

[KLV Metadata](#)

Description

Select the source for a particular klv tag or a range of klv tags.

```
typedef struct {
    u8 reserved1;
    u8 reserved2;
    u8 mode;
    u8 tagId1;
    u8 tagId2;
    u16 selector;
    u16 displayId;
} SLATagSourceSelector_t;
```

Also sent in response to [SLAGetParameters_t](#) .

Message ID 0x98

Byte Offset	Name	Description						
4	reserved1	Reserved						
5	reserved2	Reserved						
6	mode	Used to set KLV source for either a single tag or range of tags. <i>ignored in the <u>SLAGetParameters_t</u> reply</i>						
		<table><tr><th>Value</th><th>Description</th></tr><tr><td>0</td><td>Setting the source of a single KLV tag</td></tr><tr><td>1</td><td>Setting the rate of a range of KLV tags to the selector given</td></tr></table>	Value	Description	0	Setting the source of a single KLV tag	1	Setting the rate of a range of KLV tags to the selector given
		Value	Description					
		0	Setting the source of a single KLV tag					
1	Setting the rate of a range of KLV tags to the selector given							
7	tagId1	TagId to set or get the selector, If mode is 1 then this is the start Tag value of the range. TagId should be set corresponding to the tags defined in UAS Datalink Local metadata set data elements in ST0601.xx						
8	tagId2	Ignored if mode is 0 if mode is 1 then tagId2 represents the end tag in the range tagId1, tagId1 +1, ..., tagId2 will be set to the selector given						

Source Selector identifies where values for tags come from. IDs are mutually exclusive.

Value	Source	Description
0x01	SLA protocol	KLV related SightLine commands (<u>SLASetMetadataValues_t</u> , <u>SLAMetadataStaticValues_t</u> , <u>SLASetMetadataFrameValues_t</u> and <u>SLATagData_t</u>)
0x02	NMEA	External NMEA strings (GGA, RMC, etc.)
0x04	VIDEOTRACK_INTERNAL	Internally generated by VideoTrack
0x08	KLV_BLOB	Externally generated KLV data is parsed for keys
0x10	VBI	Data is contained in the Vertical Blanking Interval of the input image. - used internally
0x20	TEST	For use with unit test, 3rd party apps. Internal any APP using SVP for KLV should set selector as 1
0x40	Other	<i>Reserved.</i> Other sources such as an autopilot, gimbal, or other device

Network Display ID
4000/17XX: (0x0002 = Net0, 0x0080 = Net1, 0x0082 = both)
41XX: (0x0002 = Net0, 0x0080 = Net1, 0x0010 = Net2, 0x0092 = all)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLATrackingBoxPixelStats_t Struct Reference

[Telemetry](#)

Description

Periodic report of pixel statistics within track boxes.

```
typedef struct {
    u8 cameraIndex;
    u8 numTracks;
    struct {
        u8 index;
        u16 mean;
        u16 max;
        u16 min;
    } stat[255];
    u64 timeStamp;
    u32 frameId;
} SLATrackingBoxPixelStats_t;
```

Enable with [SLACoordinateReportingMode_t](#).
This feature is available for manual tracks as well as radiometric and blob (**NEW 3.7**) detections.
Limited to 10 tracks.

NOTE: Telemetry output is not available in [Demo Mode](#).

Message ID 0x78

Byte Offset	Name	Description
4	cameraIndex	Camera Index
5	numTracks	Number of reported tracks
varies	index	Track index
varies	mean	Average value over tracking box area
varies	max	Maximum value over tracking box area
varies	min	Minimum value over tracking box area
varies	timeStamp	Timestamp, if auxiliary telemetry is enabled – see SLACoordinateReportingMode_t
varies	frameId	Frame ID, if auxiliary telemetry is enabled – see SLACoordinateReportingMode_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLATrackingMultiClass_t Struct Reference

[Telemetry](#) | [Tracking](#)

Description

Periodic report of all tracked multi classifier information.

```
typedef struct {
    u8 cameraIndex;
    u8 reserved1;
    u8 reserved2;
    u8 numTracks;
    struct {
        u8 index;
        u8 numSubClass;
        u8 baseClassLabel;
        u8 baseClassConf;
        u8 subClassLabel1;
        u8 subClassConf1;
        u8 subClassLabel2;
        u8 subClassConf2;
        u8 subClassLabel3;
        u8 subClassConf3;
        u8 subClassLabel4;
        u8 subClassConf4;
        u8 reserved3;
        u8 reserved4;
        u8 reserved5;
        u8 reserved6;
    } multiClass[127];
    u64 timeStamp;
    u32 frameId;
} SLATrackingMultiClass_t;
```

Enable with [SLACoordinateReportingMode_t](#).
For reporting telemetry to multiple destinations see [SLASetTelemetryDestination_t](#).

Message ID 0xBD

Byte Offset	Name	Description
4	cameraIndex	Camera Index
5	reserved1	Reserved
6	reserved2	Reserved

7	numTracks	Bits 0..6: number of tracks (up to 110 reported), Bit 7 indicates MTI tracks are present
varies	index	Track index N: 0-9 = Object tracker or Vehicle/Person/Drone detection; 10-109 = All other detection modes; With Advanced Tracking Appbit index 10-19 can become Tracks.
varies	numSubClass	Number of classifications valid 0-5
varies	baseClassLabel	Primary class label id assigned by the neural network classifier file
varies	baseClassConf	Primary class confidence of given label assigned by neural network classifier, from 0-100
varies	subClassLabel1	Subclass1 label id assigned by the neural network classifier file
varies	subClassConf1	Subclass1 Confidence of given label assigned by neural network classifier, from 0-100
varies	subClassLabel2	Subclass2 label id assigned by the neural network classifier file
varies	subClassConf2	Subclass2 Confidence of given label assigned by neural network classifier, from 0-100
varies	subClassLabel3	Subclass3 label id assigned by the neural network classifier file
varies	subClassConf3	Subclass3 Confidence of given label assigned by neural network classifier, from 0-100
varies	subClassLabel4	Subclass4 label id assigned by the neural network classifier file
varies	subClassConf4	Subclass4 Confidence of given label assigned by neural network classifier, from 0-100
varies	reserved3	<i>Reserved</i>
varies	reserved4	<i>Reserved</i>
varies	reserved5	<i>Reserved</i>
varies	reserved6	<i>Reserved</i>
varies	timeStamp	Timestamp, if auxiliary telemetry is enabled – see SLACoordinateReportingMode_t
varies	frameId	Frame ID, if auxiliary telemetry is enabled – see SLACoordinateReportingMode_t

Firmware Version: 3.11

Sightline Intelligence Command, Control, and Script API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) ■ [Camera Configuration](#)

[List of all members](#)

SLATrackingPosition_t Struct Reference

[Telemetry](#) | [Tracking](#)

Description

Periodic report of primary track's position, measured scene translation, rotation and scale, and stabilization display offset and rotation of a camera.

```
typedef struct {
    s16 trackingCol;
    s16 trackingRow;
    s16 sceneCol;
    s16 sceneRow;
    s16 offsetCol;
    s16 offsetRow;
    u8 trackingConfidence;
    u8 sceneConfidence;
    u16 rotation;
    u8 cameraIndex;
    u8 userTrackId;
    u8 sceneColFrac8;
    u8 sceneRowFrac8;
    s16 sceneAngle7;
    s16 sceneScale8;
    u64 timeStamp;
    u32 frameId;
} SLATrackingPosition\_t;
```

For reporting telemetry to multiple destinations see [SLASetTelemetryDestination_t](#). Multiple tracks (or MTI Tracks) result in a [SLATrackingPositions_t](#) message.

Tracking and scene confidence scores indicate the system's confidence between 0 (low) and 100 (high). Tracking position is in camera source (unstabilized) coordinates. Track column and row is the center of the track box. See also [SLASetTrackingParameters_t](#) and [SLAStartTracking_t](#).

Scene translation is the frame - to - frame offset, rotation, and scale accumulated over the number of frames since the last Tracking Position report. This interval is controlled by [SLACoordinateReportingMode_t](#).

Display offset is the translation applied to the current camera frame to create the current display frame. Positive values of offset indicate shifts down and to the right. In order to

render the tracking point of interest in display coordinates, with $(0, 0)$ being the top left pixel, use the following equation :

where (x_t, y_t) are reported tracking coordinates, (x_c, y_c) are the coordinates of the center pixel of the frame, (x_d, y_d) are reported display offsets, (Z_d) is the optional digital zoom factor, available in [SLASetDisplayParameters_t](#), and A is the reported screen rotation.

NOTE: If flip(mirror) is set in [SLASetDisplayParameters_t](#), apply the following before the above equation:

If flip left to right: $x_t = \text{captureWide} - x_t$ and $A = -A$

If flip top to bottom: $y_t = \text{captureHigh} - y_t$ and $A = -A$

The following code snippet shows how to correctly calculate fixed or floating point scene translation :

```
SLATrackingPosition_t *tp;

// Fixed point, scaled up by 8 bits
s32 sceneCol8 = ((s32)(tp->sceneCol << 16) | (tp->sceneColFrac8 << 8)) >> 8;
s32 sceneRow8 = ((s32)(tp->sceneRow << 16) | (tp->sceneRowFrac8 << 8)) >> 8;

// Floating point
f32 sceneColF = sceneCol8 / 256.0f;
f32 sceneRowF = sceneRow8 / 256.0f;
```

NOTE: In tracker only configuration only trackingCol, trackingRow, and trackingConfidence are populated with valid data.

NOTE: Telemetry output is not available in [Demo Mode](#).

Message ID 0x43

Byte Offset	Name	Description
4-5	trackingCol	Tracking column in camera source coordinates
6-7	trackingRow	Tracking row in camera source coordinates
8-9	sceneCol	Horizontal scene translation columns in pixels, positive indicates the scene has moved to the right (equivalent to panning left).
10-11	sceneRow	Vertical scene translation rows in pixels, positive indicates the scene has moved down (equivalent to panning up)
12-13	offsetCol	Display column offset
14-15	offsetRow	Display row offset
		Bits 0..6 Tracking confidence (0..100), Bit 7 Target coasting flag. It is important to use bit 7 to determine if the target is in a coasting state

16	trackingConfidence	because the confidence at which a track enters the coasting state may vary based on the tracking mode.
17	sceneConfidence	Scene measurement confidence (0..100)
18-19	rotation	Current display rotation (0..360 degrees) * 128
20	cameraIndex	<u>Camera Index</u>
21	userTrackId	User managed track ID.
22	sceneColFrac8	Horizontal scene translation 8 bit fractional pixel (combine with sceneCol)
23	sceneRowFrac8	Vertical scene translation 8 bit fractional pixel (combine with sceneRow).
24-25	sceneAngle7	Scene frame to frame rotation (-128 to 127 degrees)*128
26-27	sceneScale8	Scene frame to frame scale change factor * 256. 1.0*256 indicates no change
28-35	timeStamp	Timestamp, if auxiliary telemetry is enabled – see <u>SLACoordinateReportingMode_t</u>
36-39	frameId	Frame ID, if auxiliary telemetry is enabled – see <u>SLACoordinateReportingMode_t</u>

Firmware Version: 3.11

Sightline Intelligence Command, Control, and Script API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) [Camera Configuration](#)

[List of all members](#)

SLATrackingPositions_t Struct Reference

[Telemetry](#) | [Tracking](#)

Description

Periodic report of all tracked positions.

```
typedef struct {  
    u8 cameraIndex;  
    u8 numTracks;  
    struct {  
        u8 index;  
        s16 col;  
        s16 row;  
        s16 wide;  
        s16 high;  
        s16 velCol8;  
        s16 velRow8;  
        u8 confidence;  
        u8 flags;  
    } track[127];  
    u64 timeStamp;  
    u32 frameId;  
} SLATrackingPositions\_t;
```

Tracking and scene confidence scores indicates the system's confidence between 0 (low) and 100 (high). Tracking positions are in camera (unstabilized) coordinates. Enable with [SLACoordinateReportingMode_t](#). For reporting telemetry to multiple destinations see [SLASetTelemetryDestination_t](#).

Rendering Track Points

In order to render the tracking point of interest in display coordinates, with $(0, 0)$ being the top left pixel, use the following equation :

where (x_t, y_t) are reported tracking coordinates, (x_c, y_c) are the coordinates of the center pixel of the frame, (x_d, y_d) are reported display offsets, (Z_d) is the optional digital zoom factor, available in [SLACurrentDisplayParameters_t](#), and A is the

reported screen rotation. This information is contained in the [SLATrackingPosition_t](#) packet.

Primary Track

- This target will be affected by [SLANudgeTrackingCoordinate_t](#)
- Telemetry information is reported by [SLATrackingPosition_t](#)
- Reticle color and shape change using [SLASetOverlayMode_t](#)
- New user designated tracks will become primary [SLAStartTracking_t](#)
- Zoom to track will use Primary Target [SLASetDisplayParameters_t](#)

NOTE: In tracker only configuration only row, col, high, wide, confidence, and flags are populated with valid data, and only for one track.

NOTE: Telemetry output is not available in [Demo Mode](#).

Message ID 0x51

Byte Offset	Name	Description																
4	cameraIndex	<u>Camera Index</u>																
5	numTracks	Bits 0..6: number of tracks (up to 110 reported), Bit 7 indicates MTI tracks are present																
varies	index	Track index N: 0-9 = Object tracker or Vehicle/Person/Drone detection; 10-109 = All other detection modes; With Advanced Tracking Appbit index 10-19 can become Tracks.																
varies	col	Tracking column in camera coordinates																
varies	row	Tracking row in camera coordinates																
varies	wide	Track width																
varies	high	Track height																
varies	velCol8	Track horizontal velocity multiplied by 256 in pixels/frame – positive is to the right																
varies	velRow8	Track vertical velocity multiplied by 256 in pixels/frame – positive is down																
varies	confidence	Track Confidence (0-100) Bit 7 high indicates coasting of target (obstructed or out of FOV). It is important to use bit 7 to determine if the target is in a coasting state because the confidence at which a track enters the coasting state may vary based on the tracking mode.																
varies	flags	Track flags																
		<table><tr><th>Bit</th><th>Description</th></tr><tr><td>0</td><td>1 = Primary track, 0 = Not primary</td></tr><tr><td>1</td><td>1 = Selected track</td></tr><tr><td>2-3</td><td>0 = Track Off, 1 = Track Below Threshold, 2 = Track Off Screen, 3 = Track Active</td></tr><tr><td>4</td><td>Detection Index: 0 = detection 0, 1 = detection 1</td></tr><tr><td>5</td><td>MTI Detection Indicator: 0 = NC Track, 1 = MTI</td></tr><tr><td>6</td><td>1 = track is processing past frames (maxPauseTime must be > 0 in <u>SLASetTrackingParameters_t</u>)</td></tr><tr><td></td><td></td></tr></table>	Bit	Description	0	1 = Primary track, 0 = Not primary	1	1 = Selected track	2-3	0 = Track Off, 1 = Track Below Threshold, 2 = Track Off Screen, 3 = Track Active	4	Detection Index: 0 = detection 0, 1 = detection 1	5	MTI Detection Indicator: 0 = NC Track, 1 = MTI	6	1 = track is processing past frames (maxPauseTime must be > 0 in <u>SLASetTrackingParameters_t</u>)		
		Bit	Description															
		0	1 = Primary track, 0 = Not primary															
		1	1 = Selected track															
		2-3	0 = Track Off, 1 = Track Below Threshold, 2 = Track Off Screen, 3 = Track Active															
		4	Detection Index: 0 = detection 0, 1 = detection 1															
		5	MTI Detection Indicator: 0 = NC Track, 1 = MTI															
		6	1 = track is processing past frames (maxPauseTime must be > 0 in <u>SLASetTrackingParameters_t</u>)															

		7	1 = Coast mode on (freeze model update, freeze search and update, or freeze propagation), 0 = Coast mode off. Controlled via SLAModifyTrackIndex_t
varies	timeStamp	Timestamp, if auxiliary telemetry is enabled – see SLACoordinateReportingMode_t	
varies	frameId	Frame ID, if auxiliary telemetry is enabled – see SLACoordinateReportingMode_t	

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) [Camera Configuration](#)

[List of all members](#)

SLATrackingPositionsExtended_t

Struct Reference

[Telemetry](#) | [Tracking](#)

Description

Periodic report of all tracked positions, with classifier information.

```
typedef struct {
    u8 cameraIndex;
    u8 numTracks;
    struct {
        u8 index;
        s16 col;
        s16 row;
        s16 wide;
        s16 high;
        s16 velCol8;
        s16 velRow8;
        u8 confidence;
        u8 flags;
        u8 classifierLabel;
        u8 classifierConf;
        u8 userTrackId;
        u8 trackColFrac8;
        u8 trackRowFrac8;
        u8 flags1;
    } track[127];
    u64 timeStamp;
    u32 frameId;
} SLATrackingPositionsExtended\_t;
```

Tracking and scene confidence scores indicates the system's confidence between 0 (low) and 100 (high). Tracking positions are in camera (unstabilized) coordinates. Enable with [SLACoordinateReportingMode_t](#). For reporting telemetry to multiple destinations see [SLASetTelemetryDestination_t](#).

Rendering Track Points

In order to render the tracking point of interest in display coordinates, with (0, 0) being the top left pixel, use the following equation :

where (x_t, y_t) are reported tracking coordinates, (x_c, y_c) are the coordinates of the center pixel of the frame, (x_d, y_d) are reported display offsets, (Z_d) is the optional digital zoom factor, available in [SLACurrentDisplayParameters_t](#), and A is the reported screen rotation. This information is contained in the [SLATrackingPosition_t](#) packet.

Primary Track

- This target will be affected by [SLANudgeTrackingCoordinate_t](#)
- Telemetry information is reported by [SLATrackingPosition_t](#)
- Reticle color and shape change using [SLASetOverlayMode_t](#)
- New user designated tracks will become primary [SLAStartTracking_t](#)
- Zoom to track will use Primary Target [SLASetDisplayParameters_t](#)

NOTE: In tracker only configuration only row, col, high, wide, confidence, and flags are populated with valid data, and only for one track.

Classifier Labels

NOTE: The default SightLine classifier is for drone detection and tracking and contains types 0 to 3. Other types are only available if using a custom classifier, see [SLACustomClassifier_t](#).

Value	Label	Description
0	Background / Other	
1	Rotary Wing Drone	Drone/Rotary Wing Head Classification ID
2	Fixed Wing Drone	Drone/Fixed Wing Head Classification ID
3	Vehicle	Vehicle Head Classification ID
4	Person	Person Head Classification ID
5	Boat	Boat Head Classification (Deprecated - see Maritime)
6	Airplane	Airplane Head Classification
7	Vehicle Sedan	Vehicle Sub Classification Sedan <i>Type</i>
8	Vehicle Hatch	Vehicle Sub Classification Hatch <i>Type</i>
9	Vehicle Bus	Vehicle Sub Classification Bus <i>Type</i>
10	Vehicle Truck	Vehicle Sub Classification Truck <i>Type</i>
11	Vehicle Armored Fight Vehicle	Vehicle Sub Classification Armored Fight Vehicle <i>Type</i>
12	Vehicle Tank	Vehicle Sub Classification Tank <i>Type</i>
13	Vehicle Military Logistic Vehicle	Vehicle Sub Classification Military Logistic Vehicle <i>Type</i>
14	Vehicle Light Utility	Vehicle Sub Classification Light Utility <i>Type</i>
15	Vehicle Military	Vehicle Sub Classification Generic Military <i>Type</i>
16	Vehicle Civilian	Vehicle Sub Classification Generic Civilian <i>Type</i>
17	Vehicle Pickup	Vehicle Sub Classification Pickup <i>Type</i>
18	Vehicle Reserved 1	

19	Vehicle Reserved 2	
20	Vehicle Reserved 3	
21	Person Mil Uniform	Person Sub Classification Military <i>Uniform Type</i>
22	Person Civ Uniform	Person Sub Classification Non-military <i>Uniform Type</i>
23	Person Armed	Person Sub Classification Armed <i>Weapon Status</i>
24	Person Unarmed	Person Sub Classification Unarmed <i>Weapon Status</i>
25	Person No Surrender	Person Sub Classification No Surrender <i>Engagement State</i>
26	Person Surrender	Person Sub Classification Surrender <i>Engagement State</i>
27	Person Pose On Ground	Person Sub Classification On Ground <i>Pose State</i>
28	Person Pose Stand	Person Sub Classification Standing <i>Pose State</i>
29	Person Pose Intermediate	Person Sub Classification Intermediate <i>Pose State</i>
30	Person Military	Person Sub Classification Generic Military <i>Type</i>
31	Person Civilian	Person Sub Classification Generic Civilian <i>Type</i>
32	Person Reserved 0	
33	Person Reserved 1	
34	Person Reserved 2	
35	Person Reserved 3	
36	Person Reserved 4	
37	Person Reserved 5	
38	Maritime Vessel	Maritime Head Classification
39	Maritime Vessel Aircraft Carrier	Maritime Sub Classification Aircraft Carrier <i>Type</i>
40	Maritime Vessel Navy	Maritime Sub Classification Naval Vessel <i>Type</i>
41	Maritime Vessel Civilian	Maritime Sub Classification Civilian Vessel <i>Type</i>
42	Maritime Reserved 0	
43	Maritime Reserved 1	
44	Maritime Reserved 2	
45	Maritime Reserved 3	
46	Color Black	Sub Classification Black <i>Color</i> (Currently Vehicle Only)
47	Color White	Sub Classification White <i>Color</i> (Currently Vehicle Only)
48	Color Gray	Sub Classification Gray <i>Color</i> (Currently Vehicle Only)
49	Color Red	Sub Classification Red <i>Color</i> (Currently Vehicle Only)
50	Color Blue	Sub Classification Blue <i>Color</i> (Currently Vehicle Only)
51	Color Yellow	Sub Classification Yellow <i>Color</i> (Currently Vehicle Only)
52	Color Green	Sub Classification Green <i>Color</i> (Currently Vehicle Only)
53	Color Orange	Sub Classification Orange <i>Color</i> (Currently Unsupported)
54	Color Purple	Sub Classification Purple <i>Color</i> (Currently Unsupported)

55-59	Future Reserved	
60	Drone Multicopter	Sub Classification Multicopter <i>Type</i>
61	Drone Fixed Wing	Sub Classification Fixed Wing <i>Type</i>
62	Drone Reserved 0	
63	Drone Reserved 1	
100-199	User Types	
200-254	Reserved	
255	Unclassified	

NOTE: Telemetry output is not available in Demo Mode.

Message ID 0xA0

Byte Offset	Name	Description	
4	cameraIndex	<u>Camera Index</u>	
5	numTracks	Bits 0..6: number of tracks (up to 110 reported), Bit 7 indicates MTI tracks are present	
varies	index	Track index N: 0-9 = Object tracker or Vehicle/Person/Drone detection; 10-109 = All other detection modes; With Advanced Tracking Appbit index 10-19 can become Tracks.	
varies	col	Tracking column in camera coordinates	
varies	row	Tracking row in camera coordinates	
varies	wide	Track width	
varies	high	Track height	
varies	velCol8	Track horizontal velocity multiplied by 256 in pixels/frame – positive is to the right	
varies	velRow8	Track vertical velocity multiplied by 256 in pixels/frame – positive is down	
varies	confidence	Track Confidence (0-100) Bit 7 high indicates coasting of target (obstructed or out of FOV). It is important to use bit 7 to determine if the target is in a coasting state because the confidence at which a track enters the coasting state may vary based on the tracking mode	
varies	flags	Track flags	
		Bit	Description
		0	1 = Primary track, 0 = Not primary
		1	1 = Selected track
		2-3	0 = Track Off, 1 = Track Below Threshold, 2 = Track Off Screen, 3 = Track Active
		4	Detection Index: 0 = detection 0, 1 = detection 1
		5	MTI Detection Indicator: 0 = NC Track, 1 = MTI
		6	1 = track is processing past frames (maxPauseTime must be > 0 in <u>SLASetTrackingParameters_t</u>)
			1 = Coast mode on (freeze model update, freeze search and update, or freeze

		7	propagation), 0 = Coast mode off. Controlled via SLAModifyTrackIndex_t								
varies	classifierLabel	Label assigned by neural network classifier. Classifier must be enabled through SLASetTrackingParameters_t or SLASetDetectionParameters_t .									
varies	classifierConf	Confidence of given label assigned by neural network classifier, from 0-100									
varies	userTrackId	A user managed track ID. See SLAStartTracking_t ;									
varies	trackColFrac8	Tracking column in camera coordinates 8 bit fractional pixel (divide by 256 and add to col for subpixel resolution).									
varies	trackRowFrac8	Tracking row in camera coordinates 8 bit fractional pixel (divide by 256 and add to row for subpixel resolution).									
varies	flags1	Additional Track Flags. <table><tr><th>Bits</th><th>Description</th></tr><tr><td>0-3</td><td>Reserved</td></tr><tr><td>4</td><td>AI Assist: 0 = AI is not assisting track, 1 = AI is assisting track (see SLASetTrackingParameters_t mode2 bit 3)</td></tr><tr><td>5-7</td><td>Reserved</td></tr></table>		Bits	Description	0-3	Reserved	4	AI Assist: 0 = AI is not assisting track, 1 = AI is assisting track (see SLASetTrackingParameters_t mode2 bit 3)	5-7	Reserved
Bits	Description										
0-3	Reserved										
4	AI Assist: 0 = AI is not assisting track, 1 = AI is assisting track (see SLASetTrackingParameters_t mode2 bit 3)										
5-7	Reserved										
varies	timeStamp	Timestamp, if auxiliary telemetry is enabled – see SLACoordinateReportingMode_t									
varies	frameId	Frame ID, if auxiliary telemetry is enabled – see SLACoordinateReportingMode_t									

Sightline Intelligence

Command, Control, and Script

API v3.11

SLATrackTrails_t Struct Reference

[Overlays](#)

Description

Tracking Trail Information.

```
typedef struct {
    u8 cameraIndex;
    u16 flags;
    u16 tracksLen;
    u16 detectionLen;
} SLATrackTrails t;
```

Message ID 0x9D

Byte Offset	Name	Description										
4	cameraIndex	<u>Camera Index</u>										
5-6	flags	Flags to control the trails										
		<table><tr><th>Bits</th><th>Description</th></tr><tr><td>0</td><td>Track Trail Primary Only Enable</td></tr><tr><td>1</td><td>Track Trail All Tracks Enable</td></tr><tr><td>2</td><td>Detection Trail Enable</td></tr><tr><td>3-15</td><td>Reserved</td></tr></table>	Bits	Description	0	Track Trail Primary Only Enable	1	Track Trail All Tracks Enable	2	Detection Trail Enable	3-15	Reserved
		Bits	Description									
		0	Track Trail Primary Only Enable									
		1	Track Trail All Tracks Enable									
		2	Detection Trail Enable									
3-15	Reserved											
7-8	tracksLen	Track trail length										
9-10	detectionLen	Mti track trail length										

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAUsb3VisionFeature_t Struct

Reference

[Capture](#)

Description

Send commands to an attached USB3Vision camera or query feature value.

```
typedef struct {
    u8 cameraIndex;
    u8 type;
    SVPLenString_t name;
    SVPLenString_t value;
} SLAUsb3VisionFeature_t;
```

Returns a [SLAUsb3VisionFeature_t](#) message.

Feature Types

Value	Description
0	Integer
1	Double
2	String/Enumeration
3	Boolean
4	Command

Value Encoding

Values are encoded as variable length character strings. For boolean types, if the first character of the sent string is '0', the value is false, else the value is true. For integer and double types, if the sent string cannot be parsed, the value used is 0.

To query a feature value without setting it, leave the value string empty (length=0).

Examples:

Integer:

51 AC 11 A6 02 00 07 4F 66 66 73 65 74 58 04 31 30 30 30 74

Sets an integer feature "OffsetX" to the value 1000

Double:

51 AC 0D A6 02 01 04 47 61 69 6E 03 35 2E 30 27

Sets a double feature "Gain" to the value 5.0

String/Enumeration:

51 AC 11 A6 02 02 08 47 61 69 6E 41 75 74 6F 03 4F 66 66 D4

Sets an enumeration feature "GainAuto" to the enumeration entry with the label "Off"

Boolean:

51 AC 0E A6 02 03 07 43 65 6E 74 65 72 58 01 30 4A

Sets a boolean feature "CenterX" to "false"

Query Value (do not set):

51 AC 11 A6 02 00 0B 53 65 6E 73 6F 72 57 69 64 74 68 00 69

Queries an integer feature "SensorWidth"

51 AC 15 A6 02 00 0B 53 65 6E 73 6F 72 57 69 64 74 68 04 31 39 33 36 90

Example response, "SensorWidth" value is 1936

Send Command:

51 AC 11 A6 02 04 0B 44 65 76 69 63 65 52 65 73 65 74 00 91

Execute the command "DeviceReset"

Message ID 0xA6

Byte Offset	Name	Description
4	cameraIndex	Camera Index Index of USB3Vision camera
5	type	Feature Types
6	name.len	string length
7-...	name.str	Name of the feature
varies	value.len	string length
varies	value.str	Feature Value

Sightline Intelligence

Command, Control, and Script

API v3.11

[List of all members](#)

SLAUsbWebcamFeature_t Struct Reference

[Capture](#)

Description

Send commands to an attached V4L2-Controlled USB WebCamera, or query feature value.

```
typedef struct {
    u8 cameraIndex;
    u32 type;
    s32 value;
    u8 reset;
} SLAUsbWebcamFeature_t;
```

Returns a [SLAUsbWebcamFeature_t](#) message.

To query a feature value without setting it, set the value to -2^31 (0x80000000).

Message ID 0xB7

Byte Offset	Name	Description
4	cameraIndex	Camera Index Index of USB WebCamera
5-8	type	A number representing the V4L2 command index you wish to set or query
9-12	value	The value you wish to set the command to. Set to -2^31 (0x80000000) to get the value instead.
13	reset	Optional: Defaults to false. 1 for true.

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAUserFont_t Struct Reference

[Overlays](#)

Description

Set user font filename.

```
typedef struct {
    u8 userFontIndex;
    SVPLenString\_t fontFileName;
} SLAUserFont\_t;
```

Select which user specified font to be displayed/drawn **New 3.4**

To display user font:

- Copy the TrueType font file to **~/sl/bin/fonts (/home/slroot/sl/bin/fonts)** directory. A FTP client ([WinSCP](#), [FileZilla](#)) can be used to upload/copy the file to the SightLine system.
- Use [SLAUserFont_t](#) to assign the font file to any of UserFont0..UserFont15. **User Font dialog** in Panel+ can be used too (Panel+ -> Overlay -> User Font...).
- Use [SLADrawOverlay_t](#) to draw text string using the font. If no font file is set, then the SightLine provided default font is displayed.

Message ID 0xAE

Byte Offset	Name	Description
4	userFontIndex	Specifies which user font to set, 0..15 corresponds to UserFont0..UserFont15.
5	fontFileName.len	string length
6-...	fontFileName.str	TrueType font filename. The font file should have *.ttf or *.otf extension, and needs to be placed in /home/slroot/sl/bin/fonts directory. Specify a null string to clear the selection.

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAUserWarningLevel_t Struct

Reference

[List of all members](#)

[General](#)

Description

Enable / disable User Warning Messages ([SLAUserWarningMessage_t](#)) sent from SLA hardware.

```
typedef struct {
    ul6 warnLevel;
} SLAUserWarningLevel_t;
```

Also sent in response to [SLAGetParameters_t\(\)](#).

Message ID 0x7F

Byte Offset	Name	Description												
4-5	warnLevel	Warning level options												
		<table><tr><th>Bit</th><th>Description</th></tr><tr><td>0</td><td>Enable informative messages</td></tr><tr><td>1</td><td><i>Reserved</i></td></tr><tr><td>2</td><td>Enable warning messages</td></tr><tr><td>3</td><td>Enable critical messages</td></tr><tr><td>4-15</td><td><i>Reserved</i></td></tr></table>	Bit	Description	0	Enable informative messages	1	<i>Reserved</i>	2	Enable warning messages	3	Enable critical messages	4-15	<i>Reserved</i>
		Bit	Description											
		0	Enable informative messages											
		1	<i>Reserved</i>											
		2	Enable warning messages											
		3	Enable critical messages											
4-15	<i>Reserved</i>													

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAVMTIChips_t Struct Reference

[KLV Metadata](#)

Description

Configure KLV VMTI Chips.

```
typedef struct {
    u8 mode;
    u8 format;
    u8 sizeType;
    u16 sizeHint;
    u8 maxPerFrame;
    u8 minFramesBetween;
    u8 reserved0;
    u16 displayId;
} SLAVMTIChips_t;
```

Use [SLATagDataRate_t](#) first to ensure that VMTI data are being sent at some frame rate.

Notes:

1. According to [MISB standard ST 0903](#) the only valid formats for chips are JPEG and PNG. However additional vendor specific image options are available below. If choosing these other options understand that they do not meet the MISB standard.
2. Caution should be used when running with Scaled output as it can reduce frame rates.

Message ID 0xAD

Byte Offset	Name	Description	
4	mode	Bit	Description
		0	1 = Enable VMTI chips, 0 = Disable VMTI chips
		1-7	Reserved
		Chip format	
		Value	Description
		0	JPEG

5	format	1	PNG
		2	PNG16
		3	SLRaw16 (16-bit Grayscale)
		4	SLRaw8 (8-bit Grayscale).
		Note If the capture image is 16-bit and PNG16 or SLRaw16 is not selected then autogain will be used to convert data to 8-bit.	
6	sizeType	Chip Sizing	
		Values	Description
		0	Fixed Size - Square centered on vmti object of sizeHint is used.
		1	Scaled - VMTI object is scaled to sizeHint .
		2	Auto - sizeHint is ignored and the VMTI object size is used.
7-8	sizeHint	VMTI chip size hint. See sizeType . Default in 32, maximum is 512. JPEG and PNG: Minimum size is 32. Size must be a multiple of 16. SLRaw16 and SLRaw8: Minimum size is 8. Size must be a multiple of 8. 17XX/4000 Maximum KLV packet size per frame is 50000 bytes.	
9	maxPerFrame	Maximum number of VMTI chips to include per frame. Default value is 1, valid range is 1-10. Setting this number too high can result in reduced frame rate video as well as a more variable bit rate data stream.	
10	minFramesBetween	Minimum number of frames between when you created a chip for a detection and when you create a chip again. Default value is 1, valid range is 1-255.	
11	reserved0	<i>Reserved</i>	
12-13	displayId	Network Display ID 4000/17XX: (0x0002 = Net0, 0x0080 = Net1, 0x0082 = both) 41XX: (0x0002 = Net0, 0x0080 = Net1, 0x0010 = Net2, 0x0092 = all)	

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAVMTIFields_t Struct Reference

[KLV Metadata](#)

Description

Set active VMTI fields and Ontology Series Update Rate.

```
typedef struct {
    u16 displayId;
    u16 fields;
    u16 ontologySeriesRate;
} SLAVMTIFields_t;
```

Message ID 0xBF

Byte Offset	Name	Description								
4-5	displayId	<u>Network Display ID</u> 4000/17XX: (0x0002 = Net0, 0x0080 = Net1, 0x0082 = both) 41XX: (0x0002 = Net0, 0x0080 = Net1, 0x0010 = Net2, 0x0092 = all)								
6-7	fields	VMTI Fields (1 = enable, 0 = disable) <table><tr><th>Bit</th><th>Option</th></tr><tr><td>0</td><td>Offset Lat and Lon</td></tr><tr><td>1</td><td>Full Lat and Lon</td></tr><tr><td>2</td><td>Classification</td></tr></table>	Bit	Option	0	Offset Lat and Lon	1	Full Lat and Lon	2	Classification
Bit	Option									
0	Offset Lat and Lon									
1	Full Lat and Lon									
2	Classification									
8-9	ontologySeriesRate	rate at which the entire ontology series will be resent (seconds)								

Sightline Intelligence

Command, Control, and Script

API v3.11

SVPEndData_t Struct Reference

Description

Data block without specified length (determined using command length)

```
typedef struct {  
    u8 buf[2048];  
} SVPEndData_t;
```

Member Data Documentation

_buf

u8 SVPEndData_t::buf[2048]

Data buffer, length determined by command packet length.

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetStabilizationParameters_t Member List

This is the complete list of members for [SLASetStabilizationParameters_t](#), including all inherited members.

angleLimit (defined in SLASetStabilizationParameters_t)	SLASetStabilizationParameters_t
cameraIndex (defined in SLASetStabilizationParameters_t)	SLASetStabilizationParameters_t
edgeU (defined in SLASetStabilizationParameters_t)	SLASetStabilizationParameters_t
edgeV (defined in SLASetStabilizationParameters_t)	SLASetStabilizationParameters_t
edgeY (defined in SLASetStabilizationParameters_t)	SLASetStabilizationParameters_t
maxStabOff (defined in SLASetStabilizationParameters_t)	SLASetStabilizationParameters_t
mode (defined in SLASetStabilizationParameters_t)	SLASetStabilizationParameters_t
rate (defined in SLASetStabilizationParameters_t)	SLASetStabilizationParameters_t
translationLimit (defined in SLASetStabilizationParameters_t)	SLASetStabilizationParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

Stabilization

Classes

Classes

struct	SLASetStabilizationParameters_t
struct	SLAResetStabilizationParameters_t
struct	SLACurrentStabilizationParameters_t
struct	SLARegistrationParameters_t
struct	SLAStabilizationBias_t

Detailed Description

Stabilize moving video.

Remove camera movement, jitter and vibration.

Sightline Intelligence

Command, Control, and Script

API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) ■ [Camera Configuration](#)

Sightline IDD & Protocol Usage

How to use the IDD in combination with the [C++ API \(see the command and control page\)](#) provided by Sightline Intelligence.

Overview

- The IDD (Interface Design Document) describes the protocol used to communicate with VideoTrack. The structure and composition of all Sightline Video Protocol (SVP) packets as a part of this protocol are contained in the IDD.
- The [main page](#) of the IDD has information about the general structure of SVP packets and an overview of the API. It is important to go through this information when developing applications that interact with VideoTrack.
- In the IDD, the payloads of the SVP commands are represented as data structures bundling relevant members. Links to structures can be found from the Modules page and the Classes >> [Class List](#) page.

Modules

The [Modules](#) page contains a list of links each representing a group of related software functionality. Each group shows the list of SVP commands that are available for specific tasks. For example, the Overlay module shows a link to the [SLADrawOverlay_t](#) struct page that provides information for drawing shapes or text on top of the video stream.

LUA Functions and Message Identifiers

The [Modules](#) page also contains a link to SightLine's LUA scripting functions, and a link to the "Message Identifiers" page listing all possible SVP commands and their message IDs.

Classes

The [Class List](#) page is a list of all structures and their descriptions. This can be useful as a way to search for a SVP packet.

C++ API

It is not recommended to manually compose SVP packets byte by byte in C++ applications. Instead, construct SVP commands using the C++ API provided by SightLine. There are helper functions available in `slfip.h`, `slfip.cpp`, `slfiptypes.h`, `slfipport.h`, and `slfipport.cpp` found in downloadable [example code - see command and control page](#). For a comprehensive guide on generating, unpacking, and parsing these files, please refer to the [EAN-Command-Example-Code](#) document.

Searching the API for SVP-Building Functions

- Find the SVP command in the IDD, through the [Modules](#) or [Classes](#) pages.
- Find the message ID using the list of message IDs found in [Modules>>Message Identifiers](#). It will be a single hexadecimal byte.
- Search for the message ID byte in **slfiptypes.h** where the SVP packets are given enumerated identifiers. Note or copy the enum identifier.
- Search for that enum in **slflip.cpp** where it will be used in the function corresponding to that SVP packet.

Additional API Functions

There are useful functions and structs in the API, including copies of structures from the IDD, that can aid in packing or unpacking SVP commands and their parameters. Note this is not comprehensive. To use a struct in a custom C++ application if it doesn't already exist in the API, copy the struct exactly as it appears in the IDD into the header file. You will need to use the struct members' type information to calculate byte offsets.

How to use SVP-Building Functions

To craft an SVP packet, create a u8 (byte) buffer of the max length of an SVP packet (256). Input this as an argument to the SVP function. The buffer will be populated with the correct SVP packet bytes, which can then be written to the VideoTrack port directly.

How to Receive and Parse Replies from VideoTrack

In the API, functions prefixed with "Get" serve as commands directed to VideoTrack to acquire a response. Follow the steps outlined in the API documentation under "Searching the API for SVP-building functions" to construct the SVP command. If the ports are correctly configured, VideoTrack will respond with an SVP packet containing the requested information at the specified port.

When reading from the VideoTrack port, the received packets may not always match the expected reply. It is the responsibility of the custom C++ application to verify the message ID of any received packets against the message ID associated with the requested structure. Additionally, ensure data integrity by confirming the checksum. Examples demonstrating how to perform these checks can be found in the provided example code.

Sightline Intelligence

Command, Control, and Script

API v3.11

Sightline File Recording

This section consolidates information related to file recording (SnapShots, Video, NUC tables, etc.).

WARNING

Sightline Intelligence recommends that **all microSD** cards be **formatted** before being used with Sightline hardware. See [EAN-File-Recording.pdf](#).

Notes

- System may stall for several seconds while writing video file to microSD card when the **Stop Recording** command is issued.
- System may stall for several seconds when files are deleted.
- Current maximum disk space that can be used is 2.0 Terabytes.
 - If more than 2TB are used, system will report errors.
- The file name count resets to 0 on power up. Files will be overwritten if you use the same base file name after a power cycle unless you use the "shouldScan" option.
- When recording a video, file extension ".ts" is added to the file name. This video can then be played back in VLC or with Sightline sample [SLADecode \(see the command and control page\)](#).
- When performing a directory listing, if a file name is specified, it will be interpreted as a path.

File Naming

If file name ends with a **non-numeric** character, the file will save to <FileName>_NNNN.jpg (1500) or <FileName>_C_NNNN.jpg (3000/4000) where NNNN is an incrementing count, starting at 0 and maxing out at 4294967295 and C is the camera index (3000/4000 only). If file name ends with a numeric character (0-9), file will save to <FileName>.jpg. For values of NNNN less than 10,000 the file name count will pad 0's.

1500:

If you use FileName...	And then:	You should see...	Note
Hello	DoSnap (0x60)	Hello_0000.jpg	Count starts at 0
Hello	DoSnap (0x60)	Hello_0001.jpg	Count increments

World	DoSnap (0x60)	World_0000.jpg	Count restarts when FileName changes
Hello42	DoSnap (0x60)	Hello42.jpg	Count is not appended for FileName ending in a numeric character

3000/4000:

If you use FileName...	And then:	You should see...	Note
Hello	DoSnap (0x60)	Hello_2_0000.jpg	Count starts at 0
Hello	DoSnap (0x60)	Hello_2_0001.jpg	Count increments
World	DoSnap (0x60)	World_2_0000.jpg	Count restarts when FileName changes
Hello42	DoSnap (0x60)	Hello42.jpg	Count is not appended for FileName ending in a numeric character

(e.g. with Command Camera = 2)

Directories

- On the 1500, /mnt/mmcblk0p1/ is automatically prefixed to filename
- On the 3000/4000, /media/mmcblk0p1 is automatically prefixed to filename
- A (sub)directory may be specified in addition to a file name
- Directories are delimited by " / "
- if filename starts with " / " then no prefix is added and the file may be written to NAND FLASH.
 - NOTE: This is not recommended.
- ". ." (current directory) and ".. ." (parent directory) are valid path extensions
- standard file permissions are obeyed when attempting to write to directories
 - if an error is encountered, check that correct write permissions have been applied to the directory being written to

Examples:

If you use FileName...	And then:	You should see...	Note
filenametest	DoSnap (0x60)	/media/mmcblk0p1/filenamettest_0000.jpg	not directory name
filenametest/	DoSnap (0x60)	/media/mmcblk0p1/filenamettest/0000.jpg	assumes a directory already exists called <i>filenametest</i>
filenametest/myfilename	DoSnap	/media/mmcblk0p1/filenamettest/myfilename_0000.jpg	assumes a directory already exists called

	(0x60)		<i>filenametest</i>
./filenametest/myfilename	DoSnap (0x60)	/media/mmcblk0p1/filenamettest/myfilename_0000.jpg	assumes a directory already exists called <i>filenametest</i>

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

Sightline Protocol Changes

List of parameters/function changes for old releases.

Protocol Changes

Changes for 2.24

Type	Message	Description
MOD	SLASetAdvancedDetectionParameters_t	Added filtering debug byte. This can be used to help configure size and velocity thresholds for detection algorithms.
MOD	SLADrawObject_t , SLACurrentOverlayObjectParameters_t	Added ability to save to parameter file.
MOD	SLASetSnapShot_t	Added JPEG Big format.
MOD	SLASetAcquisitionParameters_t	Added Big Jpeg parameters: bigHigh, bigWide, bigVertFrontPorch, and bigHorzFrontPorch

Changes for 2.23

Type	Message	Description
MOD	SLASetOverlayMode_t	Added Note for SLA-3000 Blend and Pip display modes as showing custom overlays specifically for fixed (blend) and main(pip) images.
MOD	SLAVersionNumber_t	Added svn revision, build number, time and date.
ADD	SLAExternalProgram_t	Added ability to set and query external programs
MOD	SLASetAcquisitionParameters_t	Added InitAr0134CS and InitCameraLink flags
MOD	SLACurrentOverlayObjectsIds_t , SLACurrentOverlayObjectParameters_t	No longer labeled as NYI; fixed description of SLACurrentOverlayObjectParameters_t::moveWithCam to include details on coordinate origin
MOD	SLASetVideoEnhancementParameters_t , SLACurrentVideoEnhancementParameters_t	Changed color limit bit to reserved
ADD	SLASetSerialPassthrough_t	NMEA Parsing
	SLASetDetectionParameters_t ,	

MOD	<u>SLACurrentDetectionParameters_t</u> , <u>SLASetAdvancedDetectionParameters_t</u> , <u>SLACurrentAdvancedDetectionParameters_t</u> , <u>SLASetDetectionRegionOfInterestParameters_t</u> , <u>SLACurrentDetectionRegionOfInterestParameters_t</u>	Renamed all 'MovingTarget' packets to 'Detection'
ADD	<u>SLADigiVidParseParams_t</u>	Added ability to parse FIP messages from data added to the digital video input
MOD	<u>SLAUserWarningLevel_t</u> , <u>SLAUserWarningMessage_t</u>	Split user warnings out into a message that the user sends to set the level and a response from the system
ADD	<u>SLAFrameIndex_t</u>	Added way for user to set a specific frame number which can then be included in the telemetry data
MOD	<u>SLASystemStatusMode_t</u> , <u>SLASystemStatusMessage_t</u>	Split system status out into a message that the user sends to setup status messages and timing information and a response from the system
ADD	<u>SLADetailedTimingMessage_t</u>	New packet used to report detailed DSP and ARM timing data; previously data was sent as part of UserWarningLevel_t
MOD	<u>SLASetOverlayMode_t</u>	Added camera index as well as flag for disabling all overlay objects
MOD	<u>SLACoordinateReportingMode_t</u>	Added seconds flags byte with a new bit for enabling additional telemetry data to be included in all telemetry packets
ADD	User Program Communication Using FIP	Added dedicated port 14003 to VideoTrack for custom ARM applications to communicate with
MOD	<u>SLADoSnapShot_t</u>	Added byte for Snap All Cameras Mask
MOD	<u>SLASetSDRecordingParameters_t</u>	No longer supporting writing FiP commands to microSD Card; use network debug trace instead
ADD	<u>SLAStreamingControl_t</u>	Added Network Streaming Control to Start and Stop Network Streaming
MOD	<u>SLASetVideoMode_t</u>	Added 1080p25 output option for HD-SDI and HDMI
MOD	<u>SLASetDetectionParameters_t</u> , <u>SLASetAdvancedDetectionParameters_t</u>	Added details on Anomaly Detection and clarified which parameters applied to which algorithms
MOD	<u>SLASetDetectionParameters_t</u>	Extended packet to add additional mode bits, added Maritime Detect, Radiometric Detect and min/max temperature, and details on the debug flags
MOD	<u>SLASetStabilizationParameters_t</u>	Changed bit 7 of mode byte from PiP micro stabilization to reserved.
MOD	<u>SLAVersionNumber_t</u>	Added app bits for baseline tracking and telemetry
MOD	<u>SLACurrentConfiguration_t</u>	The type value of response packet for GetParameters / Get Configuration command changed from 0x01 to 0x8E to avoid confusion

Changes for 2.22

Type	Message	Description
MOD	<u>SLASetVideoMode_t</u>	Added 1080p59.94, 720p59.94, 1080p29.97, 1080i60, and 1080i59.94 output options for HD-SDI
MOD	<u>SLADrawObject_t</u>	Reserved Object IDs 200 - 255
ADD	<u>SLALandingAid_t</u> , <u>SLALandingPosition_t</u>	Utilize existing reserved parameters as keep out parameters for landing aid and landing position packets
MOD	<u>SLATrackingPositions_t</u>	Clarification that track velocities only apply to vehicle mode tracks.
MOD	<u>SLASetAcquisitionParameters_t</u>	Added Bit 11 to Flags parameter as Use Embedded Sync
MOD	<u>SLASetAcquisitionParameters_t</u>	Added TAU320, BT.656 documentation update only
MOD	<u>SLASetVideoMode_t</u>	Added 1080p50 and 720p50 output options for HDSDI and HDMI. Added 1080p30 output option for HDMI.
MOD	<u>SLACurrentAcquisitionParameters_t</u>	Added valid row, col, high, and wide parameters
ADD	<u>SLALandingAid_t</u>	Added landing aid detection
ADD	<u>SLACurrentOverlayObjectParameters_t</u> , <u>SLACurrentOverlayObjectsIds_t</u>	Get all parameters relating to an overlay object
MOD	<u>SLACurrentAcquisitionParameters_t</u>	Fixed description of current acquisition parameters.
MOD	<u>SLASetAdvancedDetectionParameters_t</u>	Downsample now applies to Vehicle MTI as well.
ADD	<u>SLASetAdvancedDetectionParameters_t</u>	Added hide overlap tracks parameter. Replaces byte 18 which was reserved.
MOD	<u>SLASetOverlayMode_t</u>	Moved display MTI indices from advanced moving target indication parameters to overlay mode. Added ability to set the color of MTI tracks and MTI selectable tracks.
MOD	<u>SLAModifyTracking_t</u>	Added ability to start objects tracks from SA or SV MTI under certain conditions.
MOD	<u>SLACurrentTrackingParameters_t</u>	Fixed errors in packet description.
MOD	<u>SLATrackingPositions_t</u>	Fixed errors in description of number of tracks and added clarification to track index description.
ADD	<u>SLASetVideoMode_t</u>	Add table for SLA-3000 describing display resolution.
ADD	Parameter File Format	Section describing parameter files
MOD	<u>SLACurrentVideoModeParameters_t</u>	Add HD-SDI Display Image
MOD	<u>SLASetEthernetDisplayParameters_t</u>	Clarification on Video Encoding and new codecs
MOD	<u>SLASetVideoMode_t</u>	Add HD-SDI Display Image
MOD	<u>SLASetTelemetryDestination_t</u>	Add flag bit to control saving to parameter file
MOD	<u>SLASetNetworkParameters_t</u>	Remove Last In First Out (LIFO) telemetry mode
MOD	<u>SLASetSerialPassthrough_t</u>	Fixed Protocol Type enumeration documentation. Fixed defaults for Max Length and Min Delay.

MOD	<u>SLASetTrackingParameters_t</u>	Added 'No Registration' tracking mode, byte 6 correctly marked as reserved
ADD	<u>SLAUserWarningLevel_t</u>	New message from board to user only.
MOD	<u>SLAModifyTrackIndex_t</u>	Added reinitialize option to message. Added option to specify primary or selected instead of specific index.
MOD	<u>SLASetTrackingParameters_t</u>	Added acquisition assist mode. Added intelligent assist mode.

Changes for 2.21

Type	Message	Description
MOD	<u>SLAFocusStats_t</u>	Added camera index
MOD	<u>SLASetAcquisitionParameters_t</u>	Add TAU Direct Connect and CameraLink Flags
MOD	<u>SLASetStabilizationParameters_t</u>	Add disable pan-tilt in zoom to track mode control
ADD	Analyze Render Synch	Mechanism to synchronize to distributed analyze and render systems.
MOD	<u>SLADoSnapShot_t</u>	Additional details regarding file naming.
MOD	<u>SLASetSnapShot_t</u>	Add Support for JPEG/PNG file format
MOD	<u>SLASetSnapShot_t</u>	Modified default recording to MicroSD card
FIX	Start Up Behavior	All systems send Version Number (0x40) on ready.
ADD	<u>SLAGetEthernetDisplayParameters_t</u>	Description – function was introduced earlier
MOD	<u>SLASetDetectionParameters_t</u>	Add frame step parameter
MOD	<u>SLASetDetectionParameters_t</u>	Added reset MTI flag.
MOD	<u>SLACoordinateReportingMode_t</u>	Added max number of tracking positions to report.
MOD	<u>SLASetAdvancedDetectionParameters_t</u>	Added down sample options.
MOD	<u>SLASetDetectionParameters_t</u>	Added suspicious score from advanced MTI message.
MOD	<u>SLACurrentDetectionParameters_t</u>	Changed from 2.20 to match <u>SLASetDetectionParameters_t</u> .
MOD	<u>SLASetAdvancedDetectionParameters_t</u>	Added parameters to control how many MTI tracks will be output in telemetry and KLV messages.

MOD	<u>SLASetLensMode_t</u>	Add in set focus and zoom positions
MOD	<u>SLASetLensParameters_t</u>	Add Zoom/Focus Speed
MOD	<u>SLACurrentLensParameters_t</u>	Add Zoom/Focus Speed
ADD	<u>SLACurrentStabilizationBias_t</u>	Return stabilization bias
ADD	<u>SLAAdvancedCaptureParameters_t</u>	Advanced analog capture control for image stretching
MOD	<u>SLASetTrackingParameters_t</u>	Object size = 255 will output pixel stats for full image
MOD	<u>SLASetMetadataValues_t</u> <u>SLACurrentMetadataValues_t</u> <u>SLAMetadataStaticValues_t</u> <u>SLASetMetadataFrameValues_t</u> <u>SLACurrentMetadataFrameValues_t</u> <u>SLASetMetadataRate_t</u> <u>SLASetKlvData_t</u>	Added additional parameter to support configuration of the independent KLV streams on SLA-3000.
ADD	<u>SLACurrentDetectionRegionOfInterestParameters_t</u>	Current Detection Region of Interest (0x7D) Added MTI region of interest parameters.
ADD	<u>SLASetVideoEnhancementParameters_t</u>	Added high bit depth CLAHE enhancement modes.
ADD	<u>SLASetStabilizationBias_t</u>	Added auto bias update rate
ADD	<u>SLAFocusStats_t</u>	Added focus statistics telemetry output.
MOD	<u>SLASetBlendParameters_t</u>	Added new blend modes for Fixed EO.
ADD	<u>SLADisplayAngle_t</u>	Set display angle for one or all cameras.

Changes for 2.20

Type	Message	Description
MOD	<u>SLASetAcquisitionParameters_t</u>	Add Frame Step Parameter.
MOD	<u>SLASetAcquisitionParameters_t</u>	Add InitPhoton to Flags in Generic Digital Interface
ADD	<u>SLADirectoryStatisticsReply_t</u>	Report back the total SD card space available (2.20.12)
MOD	<u>SLACurrentSDCardRecordingStatus_t</u>	Deprecated file size and time params (RESERVED)
MOD	<u>SLASetSDRecordingParameters_t</u>	Byte 6 sends <u>SLACurrentSDCardRecordingStatus_t</u> , Byte 7 can result in <u>SLADirectoryStatisticsReply_t</u>
MOD	<u>SLASetAcquisitionParameters_t</u>	Add InitCameraLinkLowSpeed Flag
MOD	<u>SLASetNetworkParameters_t</u>	Add NIC index
MOD	<u>SLASetTrackingParameters_t</u>	Added new parameter Near Value
ADD	<u>SLASetAcquisitionParameters_t</u>	New Error: Reference source not found
		Add 'Other Version' for hardware dependent version information.

MOD	<u>SLAVersionNumber_t</u>	e.g. FPGA version on SLA1500
MOD	<u>SLASetAcquisitionParameters_t</u>	Add SLA 1500 generic digital, add params for horizontal and vertical front porch, flags.
MOD	<u>SLATrackingBoxPixelStats_t</u>	Add Tracking Box Pixel Stats message for 14 bit digital camera mean,max,min
ADD	–	Add SLA-3000 specific protocol (beta)
MOD	<u>SLASetBlendParameters_t</u>	Current Blend Parameters (0x4D) Enable SD / HD image blending. Changed to warp the EO image. Added bit for alternate zoom control for HD. Returning EO/IR indices.
MOD	<u>SLASetDetectionParameters_t</u>	New MTI modes for small target detection.
NEW	<u>SLASetAdvancedDetectionParameters_t</u>	Beta advanced MTI control

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

Changes for 2.25

This section summaraizes commands that have been newly added or have changed in this release. Changes may include new parameters or additional values/functionality to existing parameters. For details, click on the link to jump to the command.

New Commands

These are commands that did not exist in previous release

Type	Message	ID
ADD	SLANucParameters_t	0x35
ADD	SLAReadWriteNuc_t	0x36
ADD	SLASetVMTI_t	0x84
ADD	SLAFourAlignPoints_t	0x95
ADD	SLATagData_t	0x96
ADD	SLATagDataRate_t	0x97
ADD	SLATagSourceSelector_t	0x98
ADD	SLADecoderParameters_t	0x99
ADD	SLALogoParameters_t	0x9B

Commands that have Changed

These commands have a new parameter, a parameter type has changes, may have a new mode (or similar), or a parameter name may have changed

Type	Message	ID
MOD	SLAModifyTracking_t	0x05
MOD	SLACoordinateReportingMode_t	0x0B
MOD	SLASetTrackingParameters_t	0x0C
MOD	SLASetNetworkParameters_t	0x1C
MOD	SLASetEthernetDisplayParameters_t	0x29

MOD	<u>SLASetDetectionParameters_t</u>	0x2D
MOD	<u>SLASetBlendParameters_t</u>	0x2F
MOD	<u>SLASetAcquisitionParameters_t</u>	0x37
MOD	<u>SLADrawObject_t</u>	0x3B
MOD	<u>SLACommandPassThrough_t</u>	0x3D
MOD	<u>SLACurrentNetworkParameters_t</u>	0x49
MOD	<u>SLACurrentVideoModeParameters_t</u>	0x4B
MOD	<u>SLACurrentBlendParameters_t</u>	0x4D
MOD	<u>SLACurrentAcquisitionParameters_t</u>	0x4F
MOD	<u>SLASetSnapShot_t</u>	0x5E
MOD	<u>SLADoSnapShot_t</u>	0x60
MOD	<u>SLASetKlvData_t</u>	0x61
MOD	<u>SLASetAdvancedDetectionParameters_t</u>	0x76
MOD	<u>SLALandingAid_t</u>	0x81
MOD	<u>SLACurrentConfiguration_t</u>	0x8E

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

Changes for 3.00

This section summaraizes commands that have been newly added or have changed in this release. Changes may include new parameters or additional values/functionality to existing parameters. Please follow links to IDD for complete details.

New Commands:

Type	Message	ID
ADD	SLATrackTrails_t	0x9D
ADD	SLAInternalCommand_t	0xA2
ADD	SLAInternalResponse_t	0xA3
ADD	SLAVideoDisplay_t	0xA4
ADD	SLAMultiDisplay_t	0xA5
ADD	SLASetUsb3VisionFeature_t	0xA6
ADD	SLASetBinaryClassifier_t	0xA7

Interfaces that have changed:

Type	Message	ID	Description
MOD	SLAModifyTracking_t	0x05	new modes
MOD	SLAStartTracking_t	0x08	new paramater userTrackID
MOD	SLASetTrackingParameters	0x0C	new parameter mode2
MOD	SLASetDisplayParameters_t	0x16	new parameter extendedZoom10
MOD	SLAModifyTrackIndex_t	0x17	new parameters width & height, new flags to adjust track size
MOD	SLASetDetectionParameters_t	0x2D	new parameter classSensitivity
MOD	SLASetEthernetDisplayParameters_t	0x29	added H.265 support
MOD	SLASetAcquisitionParameters_t	0x37	USB web cam support
MOD	SLAVersionNumber_t	0x40	new parameter v4AppBits
MOD	SLACurrentTrackingParameters_t	0x44	new parameter mode (was reserved)
MOD	SLACurrentDetectionParameters_t	0x54	new parameter classSensitivity
MOD	SLACurrentDisplayParameters_t	0x57	new parameters extendedZoom and zoomRate

MOD	<u>SLASetSystemValue_t</u>	0x92	New system values
MOD	<u>SLATrackingPositionsExtended_t</u>	0xA0	new parameters userTrackId trackColFrac8 trackRowFrac8

OBSOLETE

Type	Message	ID
DEL	SetRGB565Conversion	0x26
DEL	CurrentPrimaryTrackIndex	0x27
DEL	SetStitchParameters	0x2B
DEL	GetStitchParameters	0x2C
DEL	GetPrimaryTrackIndex	0x34
DEL	CurrentStitchParameters	0x4C
DEL	SetParameterBlock	0x69
DEL	ParameterBlock	0x6A
DEL	AnalyzeRenderSync	0x7E
DEL	SetGeneric	0x85
DEL	CurrentDecoderParameters	0x9A

Sightline Intelligence

Command, Control, and Script

API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) [Camera Configuration](#)

Changes for 3.01

This section summarizes commands that have been newly added or have changed in this release. Changes may include new parameters or additional values/functionality to existing parameters. For details, click on the link to jump to the command.

New Commands:

Type	Message	ID	Version
ADD	SLAUsb3VisionFeature_t	0xA6	3.01.00
ADD	SLABinaryClassifier_t	0xA7	3.01.00
ADD	SLADeadPixel_t	0xA8	3.01.01

Commands that have Changed:

Type	Message	ID	Version
MOD	SLASetOverlayMode_t	0x06	3.01.01
MOD	SLASetTrackingParameters_t	0x0C	3.01.01
MOD	SLASetVideoEnhancementParameters_t	0x21	3.01.00
MOD	SLASetAcquisitionParameters_t	0x37	3.01.00
MOD	SLASetPortConfiguration_t	0x3E	3.01.00
MOD	SLACurrentOverlayMode_t	0x42	3.01.01
MOD	SLATrackingPosition_t	0x42	3.01.00
MOD	SLACurrentVideoEnhancementParameters_t	0x4A	3.01.00
MOD	SLATrackingPositions_t	0x50	3.01.00
MOD	SLASetDigitalCameraParameters_t	0x70	3.01.00
MOD	SLACurrentDigitalCameraParameters_t	0x71	3.01.00
MOD	SLACurrentConfiguration_t	0x8E	3.01.00
MOD	SLAI2CCommand_t	0x94	3.01.00
MOD	SLADrawOverlay_t	0x9C	3.01.00
MOD	SLATrackingPositionsExtended_t	0x9F	3.01.00

MOD	<u>SLAVideoDisplay_t</u>	0xA3	3.01.00
MOD	<u>SLASetSystemValue_t</u>	0x92	3.01.01
MOD	<u>SLASetEthernetDisplayParameters_t</u>	0x29	3.02.00

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

Changes for 3.2

This section summaraizes commands that have been newly added or have changed in this release. Changes may include new parameters or additional values/functionality to existing parameters. For details, click on the link to jump to the command.

New Commands:

Type	Message	ID	Version
ADD	SLACustomClassifier_t	0xA7	3.2.2
ADD	SLAClassifierParameters_t	0xA9	3.2.2
ADD	SLASendScript_t	0xAA	3.2.2
ADD	SLADoDetectSnapShot_t	0xAB	3.2.2
ADD	SLAAncillaryTextMetadata_t	0xAC	3.2.2

Commands that have Changed:

Type	Message	ID	Version
MOD	SLAModifyTracking_t	0x05	3.2.2
MOD	SLAStartTracking_t	0x08	3.2.2
MOD	SLASetTrackingParameters_t	0x0C	3.2.2
MOD	SLAMetadataStaticValues_t	0x14	3.2.2
MOD	SLASetMetadataFrameValues_t	0x15	3.2.2
MOD	SLAGetEthernetVideoParameters_t	0x1B	3.2.2
MOD	SLASetSDRecordingParameters_t	0x1E	3.2.2
MOD	SLASetVideoEnhancementParameters_t	0x21	3.2.2
MOD	SLASetH264Parameters_t	0x23	3.2.2
MOD	SLAGetH264Parameters_t	0x24	3.2.2
MOD	SLASetEthernetDisplayParameters_t	0x29	3.2.2
MOD	SLASetBlendParameters_t	0x2F	3.2.2
MOD	SLANucParameters_t	0x35	3.2.2

MOD	<u>SLASetAcquisitionParameters_t</u>	0x37	3.2.2
MOD	<u>SLAGetEthernetDisplayParameters_t</u>	0x39	3.2.2
MOD	<u>SLAVersionNumber_t</u>	0x40	3.2.2
MOD	<u>SLACurrentVideoEnhancementParameters_t</u>	0x4A	3.2.2
MOD	<u>SLACurrentBlendParameters_t</u>	0x4D	3.2.2
MOD	<u>SLACurrentEthernetDisplayParameters_t</u>	0x52	3.2.2
MOD	<u>SLACurrentH264Parameters_t</u>	0x56	3.2.2
MOD	<u>SLASetSnapShot_t</u>	0x5E	3.2.2
MOD	<u>SLADoSnapShot_t</u>	0x60	3.2.2
MOD	<u>SLASetKlvData_t</u>	0x61	3.2.2
MOD	<u>SLASetMetadataRate_t</u>	0x62	3.2.2
MOD	<u>SLASetDigitalCameraParameters_t</u>	0x70	3.2.2
MOD	<u>SLACurrentDigitalCameraParameters_t</u>	0x71	3.2.2
MOD	<u>SLASetAdvancedDetectionParameters_t</u>	0x72	3.2.3
MOD	<u>SLASetVMTI_t</u>	0x84	3.2.2
MOD	<u>SLAAppendedMetadata_t</u>	0x89	3.2.2
MOD	<u>SLACurrentMetadataFrameValues_t</u>	0x8C	3.2.2
MOD	<u>SLACurrentMetadataRate_t</u>	0x8D	3.2.2
ADD	<u>SLASetSystemValue_t</u>	0x92	3.2.3
MOD	<u>SLATagData_t</u>	0x96	3.2.2
MOD	<u>SLATagDataRate_t</u>	0x97	3.2.2
MOD	<u>SLATagSourceSelector_t</u>	0x98	3.2.2
MOD	<u>SLAVideoDisplay_t</u>	0xA3	3.2.2

Sightline Intelligence

Command, Control, and Script

API v3.11

Changes for 3.3

This section summarizes commands that have been newly added or have changed in this release. Changes may include new parameters or additional values/functionality to existing parameters. For details, click on the link to jump to the command.

New Commands:

Type	Message	ID
ADD	SLAVMTIChips_t	0xAD

Commands that have Changed:

Type	Message	ID
MOD	SLASetStabilizationParameters_t	0x02
MOD	SLAStartTracking_t	0x08
MOD	SLASetDisplayParameters_t	0x16
MOD	SLASetSDRecordingParameters_t	0x1E
MOD	SLASetVideoEnhancementParameters_t	0x21
MOD	SLASetEthernetDisplayParameters_t	0x29
MOD	SLASetDetectionParameters_t	0x2D
MOD	SLANucParameters_t	0x35
MOD	SLACurrentVideoEnhancementParameters_t	0x4A
MOD	SLACurrentDetectionParameters_t	0x54
MOD	SLACurrentDisplayParameters_t	0x57
MOD	SLADoSnapShot_t	0x60
MOD	SLASetAdvancedDetectionParameters_t	0x76
MOD	SLACurrentAdvancedDetectionParameters_t	0x77
MOD	SLASystemStatusMode_t	0x80
MOD	SLAStreamingControl_t	0x90
MOD	SLAVideoDisplay_t	0xA3

MOD	SLAClassifierParameters_t	0xA9
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Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

Changes for 3.4

This section summarizes commands that have been newly added or have changed in this release. Changes may include new parameters or additional values/functionality to existing parameters. For details, click on the link to jump to the command.

New Commands:

Type	Message	ID	Version
ADD	SLAUserFont_t	0xAE	3.4.5
ADD	SLANoise3D_t	0xAF	3.4.5

Commands that have Changed:

Type	Message	ID	Version
MOD	SLASetOverlayMode_t	0x06	3.4.5
MOD	SLASetTrackingParameters_t	0x0C	3.4.5
MOD	SLAModifyTrackIndex_t	0x17	3.4.5
MOD	SLASetADCParameters_t	0x18	3.4.6
MOD	SLASetVideoEnhancementParameters_t	0x21	3.4.6
MOD	SLASetEthernetDisplayParameters_t	0x29	3.4.5
MOD	SLASetBlendParameters_t	0x2F	3.4.6
MOD	SLANucParameters_t	0x35	3.4.5
MOD	SLAReadWriteNuc_t	0x36	3.4.5
MOD	SLASetAcquisitionParameters_t	0x37	3.4.6
MOD	SLADrawObject_t	0x3B	3.4.5
MOD	SLACurrentOverlayMode_t	0x42	3.4.5
MOD	SLACurrentADCParameters_t	0x47	3.4.6
MOD	SLACurrentEthernetVideoParameters_t	0x48	3.4.5
MOD	SLACurrentVideoEnhancementParameters_t	0x4A	3.4.5
MOD	SLACurrentEthernetDisplayParameters_t	0x52	3.4.5

MOD	<u>SLACurrentH264Parameters_t</u>	0x56	3.4.5
MOD	<u>SLASetDigitalCameraParameters_t</u>	0x70	3.4.5
MOD	<u>SLACurrentDigitalCameraParameters_t</u>	0x71	3.4.5
MOD	<u>SLASetAdvancedDetectionParameters_t</u>	0x76	3.4.5
MOD	<u>SLACurrentAdvancedDetectionParameters_t</u>	0x77	3.4.5
MOD	<u>SLADrawOverlay_t</u>	0x9C	3.4.5
MOD	<u>SLARegistrationParameters_t</u>	0x9E	3.4.5
MOD	<u>SLADeadPixelStats_t</u>	0xA1	3.4.5
MOD	<u>SLAVideoDisplay_t</u>	0xA3	3.4.5

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

Changes for 3.5

This section summarizes commands that have been newly added or have changed in this release. Changes may include new parameters or additional values/functionality to existing parameters. For details, click on the link to jump to the command.

New Commands:

Type	Message	ID
ADD	SLACursorOnTarget_t	0xAF
ADD	SLALensParameters_t	0xB1
ADD	SLALensCommand_t	0xB2
ADD	SLAFocusParameters_t	0xB3

Commands that have Changed:

Type	Message	ID
MOD	SLASetStabilizationParameters_t	0x02
MOD	SLACoordinateReportingMode_t	0x0B
MOD	SLASetTrackingParameters_t	0x0C
MOD	SLASetDisplayParameters_t	0x16
MOD	SLASetVideoEnhancementParameters_t	0x21
MOD	SLASetH264Parameters_t	0x23
MOD	SLASetEthernetDisplayParameters_t	0x29
MOD	SLASetDetectionParameters_t	0x2D
MOD	SLASetBlendParameters_t	0x2F
MOD	SLANucParameters_t	0x35
MOD	SLAReadWriteNuc_t	0x36
MOD	SLASetAcquisitionParameters_t	0x37
MOD	SLADrawObject_t	0x3B
MOD	SLACurrentStabilizationParameters_t	0x41

MOD	<u>SLACurrentOverlayMode_t</u>	0x42
MOD	<u>SLACurrentVideoEnhancementParameters_t</u>	0x4A
MOD	<u>SLACurrentVideoModeParameters_t</u>	0x4B
MOD	<u>SLACurrentDetectionParameters_t</u>	0x54
MOD	<u>SLACurrentSDCardRecordingStatus_t</u>	0x58
MOD	<u>SLASetSnapShot_t</u>	0x5E
MOD	<u>SLASetMetadataRate_t</u>	0x62
MOD	<u>SLASetTelemetryDestination_t</u>	0x64
MOD	<u>SLACurrentLensStatus_t</u>	0x6D
MOD	<u>SLASetLensParameters_t</u>	0x6E
MOD	<u>SLACurrentLensParameters_t</u>	0x6F
MOD	<u>SLASetAdvancedDetectionParameters_t</u>	0x76
MOD	<u>SLACurrentAdvancedDetectionParameters_t</u>	0x77
MOD	<u>SLASetDetectionRegionOfInterestParameters_t</u>	0x7C
MOD	<u>SLACameraSwitch_t</u>	0x82
MOD	<u>SLADecoderParameters_t</u>	0x99
MOD	<u>SLADrawOverlay_t</u>	0x9C
MOD	<u>SLATrackTrails_t</u>	0x9D
MOD	<u>SLADeadPixel_t</u>	0xA8

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

Changes for 3.6

This section summarizes commands that have been newly added or have changed in this release. Changes may include new parameters or additional values/functionality to existing parameters. For details, click on the link to jump to the command.

New Commands:

Type	Message	ID
ADD	SLACustomAutoFocusParameters_t	0xB5
ADD	SLAGPIO_t	0xB6

Commands that have Changed:

Type	Message	ID
MOD	SLAResetAllParameters_t	0x01
MOD	SLASetOverlayMode_t	0x06
MOD	SLACoordinateReportingMode_t	0x0B
MOD	SLASetTrackingParameters_t	0x0C
MOD	SLASetMetadataValues_t	0x13
MOD	SLAMetadataStaticValues_t	0x14
MOD	SLASetMetadataFrameValues_t	0x15
MOD	SLASetADCPParameters_t	0x18
MOD	SLASetEthernetVideoParameters_t	0x18
MOD	SLAGetEthernetVideoParameters_t	0x1B
MOD	SLASetSDRecordingParameters_t	0x1E
MOD	SLASetVideoEnhancementParameters_t	0x21
MOD	SLASetH264Parameters_t	0x23
MOD	SLAGetH264Parameters_t	0x24
MOD	SLASetEthernetDisplayParameters_t	0x29
MOD	SLASetBlendParameters_t	0x2F

MOD	<u>SLASetAcquisitionParameters_t</u>	0x37
MOD	<u>SLAGetEthernetDisplayParameters_t</u>	0x39
MOD	<u>SLACommandPassThrough_t</u>	0x3D
MOD	<u>SLAVersionNumber_t</u>	0x40
MOD	<u>SLACurrentOverlayMode_t</u>	0x42
MOD	<u>SLACurrentTrackingParameters_t</u>	0x44
MOD	<u>SLACurrentEthernetVideoParameters_t</u>	0x48
MOD	<u>SLACurrentVideoEnhancementParameters_t</u>	0x4A
MOD	<u>SLACurrentEthernetDisplayParameters_t</u>	0x52
MOD	<u>SLACurrentSDCardRecordingStatus_t</u>	0x58
MOD	<u>SLASetKlvData_t</u>	0x61
MOD	<u>SLASetMetadataRate_t</u>	0x62
MOD	<u>SLACurrentLensStatus_t</u>	0x6D
MOD	<u>SLACurrentLensParameters_t</u>	0x6F
MOD	<u>SLASetDigitalCameraParameters_t</u>	0x70
MOD	<u>SLACurrentDigitalCameraParameters_t</u>	0x71
MOD	<u>SLASetAdvancedDetectionParameters_t</u>	0x76
MOD	<u>SLACurrentAdvancedDetectionParameters_t</u>	0x77
MOD	<u>SLASetDetectionRegionOfInterestParameters_t</u>	0x7C
MOD	<u>SLACurrentDetectionRegionOfInterestParameters_t</u>	0x7D
MOD	<u>SLASetVMTI_t</u>	0x84
MOD	<u>SLASystemStatusMessage_t</u>	0x87
MOD	<u>SLAAppendedMetadata_t</u>	0x89
MOD	<u>SLACurrentMetadataValues_t</u>	0x8B
MOD	<u>SLACurrentMetadataFrameValues_t</u>	0x8C
MOD	<u>SLACurrentMetadataRate_t</u>	0x8D
MOD	<u>SLAI2CCommand_t</u>	0x94
MOD	<u>SLATagData_t</u>	0x96
MOD	<u>SLATagDataRate_t</u>	0x97
MOD	<u>SLATagSourceSelector_t</u>	0x98
MOD	<u>SLARegistrationParameters_t</u>	0x9E
MOD	<u>SLAVideoDisplay_t</u>	0xA3
MOD	<u>SLAMultiDisplay_t</u>	0xA5
MOD	<u>SLAAncillaryTextMetadata_t</u>	0xAC
MOD	<u>SLAVMTIChips_t</u>	0xAD
MOD	<u>SLACursorOnTarget_t</u>	0xAF

MOD	SLALensParameters_t	0xB1
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Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) [Camera Configuration](#)

Changes for 3.7

This section summarizes commands that have been newly added or have changed in this release. Changes may include new parameters or additional values/functionality to existing parameters. For details, click on the link to jump to the command.

New Commands:

Type	Message	ID
ADD	SLAUsbWebcamFeature_t	0xB7
ADD	SLACreateDeviceOverlay_t	0xB8
ADD	SLABlendAlign_t	0xB9

Commands that have Changed:

Type	Message	ID
MOD	SLAResetAllParameters_t	0x01
MOD	SLANudgeTrackingCoordinate_t	0x0A
MOD	SLASetTrackingParameters_t	0x0C
MOD	SLAModifyTrackIndex_t	0x17
MOD	SLASetSDRecordingParameters_t	0x1E
MOD	SLASetH264Parameters_t	0x23
MOD	SLASetBlendParameters_t	0x2F
MOD	SLASetAcquisitionParameters_t	0x37
MOD	SLACurrentTrackingParameters_t	0x44
MOD	SLACurrentAcquisitionParameters_t	0x4F
MOD	SLACurrentH264Parameters_t	0x56
MOD	SLASetDigitalCameraParameters_t	0x70
MOD	SLACurrentDigitalCameraParameters_t	0x71
MOD	SLASystemStatusMessage_t	0x87
MOD	SLACurrentConfiguration_t	0x8E

Sightline Intelligence

Command, Control, and Script

API v3.11

Changes for 3.8

This section summarizes commands that have been newly added or have changed in this release. Changes may include new parameters or additional values/functionality to existing parameters. For details, click on the link to jump to the command.

New Commands:

Type	Message	ID
ADD	SLACustomAIDetect_t	0xBA
ADD	SLACameraCapabilities_t	0xBB

Commands that have Changed:

Type	Message	ID
MOD	SLASetOverlayMode_t	0x06
MOD	SLAStartTracking_t	0x08
MOD	SLANudgeTrackingCoordinate_t	0x0A
MOD	SLASetTrackingParameters_t	0x0C
MOD	SLASetADCParameters_t	0x18
MOD	SLASetVideoEnhancementParameters_t	0x21
MOD	SLASetH264Parameters_t	0x23
MOD	SLASetDetectionParameters_t	0x2D
MOD	SLAGetDetectionParameters_t	0x2E
MOD	SLASetAcquisitionParameters_t	0x37
MOD	SLACurrentOverlayMode_t	0x42
MOD	SLACurrentTrackingParameters_t	0x44
MOD	SLACurrentADCParameters_t	0x47
MOD	SLATrackingPositions_t	0x50
MOD	SLACurrentDetectionParameters_t	0x54
MOD	SLASetAdvancedDetectionParameters_t	0x76

MOD	<u>SLACurrentAdvancedDetectionParameters_t</u>	0x77
MOD	<u>SLASetDetectionRegionOfInterestParameters_t</u>	0x7C
MOD	<u>SLADecoderParameters_t</u>	0x99
MOD	<u>SLADrawOverlay_t</u>	0x9C
MOD	<u>SLATrackingPositionsExtended_t</u>	0x9F
MOD	<u>SLAVideoDisplay_t</u>	0xA3
MOD	<u>SLAVMTIChips_t</u>	0xAD

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) [Camera Configuration](#)

Changes for 3.9

This section summarizes commands that have been newly added or have changed in this release. Changes may include new parameters or additional values/functionality to existing parameters. For details, click on the link to jump to the command.

New Commands:

Type	Message	ID
ADD	SLACustomResponse_t	0xBC

Commands that have Changed:

Type	Message	ID
MOD	SLASetOverlayMode_t	0x06
MOD	SLASetVideoEnhancementParameters_t	0x21
MOD	SLANucParameters_t	0x35
MOD	SLAReadWriteNuc_t	0x36
MOD	SLAVersionNumber_t	0x40
MOD	SLACurrentOverlayMode_t	0x42
MOD	SLACurrentVideoEnhancementParameters_t	0x4A
MOD	SLATrackingPositions_t	0x50
MOD	SLACurrentSnapShot_t	0x5D
MOD	SLASetSnapShot_t	0x5E
MOD	SLASetDigitalCameraParameters_t	0x70
MOD	SLACurrentDigitalCameraParameters_t	0x71
MOD	SLASetAdvancedDetectionParameters_t	0x76
MOD	SLACurrentAdvancedDetectionParameters_t	0x77
MOD	SLATrackingPositionsExtended_t	0x9F

Sightline Intelligence

Command, Control, and Script

API v3.11

Changes for 3.10

This section summarizes commands that have been newly added or have changed in this release. Changes may include new parameters or additional values/functionality to existing parameters. For details, click on the link to jump to the command.

New Commands:

Type	Message	ID
ADD	SLASendToBTS_t	0xBE

Commands that have Changed:

Type	Message	ID
MOD	SLASetOverlayMode_t	0x06
MOD	SLASetTrackingParameters_t	0x0C
MOD	SLASetVideoEnhancementParameters_t	0x21
MOD	SLASetDetectionParameters_t	0x2D
MOD	SLANucParameters_t	0x35
MOD	SLASetAcquisitionParameters_t	0x37
MOD	SLACurrentOverlayMode_t	0x42
MOD	SLACurrentTrackingParameters_t	0x44
MOD	SLACurrentVideoEnhancementParameters_t	0x4A
MOD	SLACurrentDetectionParameters_t	0x54
MOD	SLASetAdvancedDetectionParameters_t	0x76
MOD	SLACurrentAdvancedDetectionParameters_t	0x77
MOD	SLATrackingPositionsExtended_t	0x9F
MOD	SLAVideoDisplay_t	0xA3
MOD	SLAClassifierParameters_t	0xA9

Sightline Intelligence

Command, Control, and Script

API v3.11

Changes for 3.11

This section summarizes commands that have been newly added or have changed in this release. Changes may include new parameters or additional values/functionality to existing parameters. For details, click on the link to jump to the command.

New Commands:

Type	Message	ID
ADD	SLACameraCalibration_t	0xC0
ADD	SLAKlvClassFilters_t	0xC1

Commands that have Changed:

Type	Message	ID
MOD	SLASetTrackingParameters_t	0x0C
MOD	SLASetDetectionParameters_t	0x2D
MOD	SLAReadWriteNuc_t	0x36
MOD	SLASetAcquisitionParameters_t	0x37
MOD	SLAVersionNumber_t	0x40
MOD	SLACurrentTrackingParameters_t	0x44
MOD	SLACurrentSnapShot_t	0x5D
MOD	SLASetSnapShot_t	0x5E
MOD	SLAGetSnapShot_t	0x5F
MOD	SLASetAdvancedDetectionParameters_t	0x76
MOD	SLACurrentAdvancedDetectionParameters_t	0x77
MOD	SLADoDetectSnapShot_t	0xAB

Sightline Intelligence

Command, Control, and Script

API v3.11

Blending

Classes

struct	SLASetBlendParameters_t
struct	SLACurrentBlendParameters_t
struct	SLASetMultipleAlignment_t
struct	SLACurrentMultipleAlignment_t
struct	SLAFourAlignPoints_t
struct	SLABlendAlign_t

Detailed Description

Two channel video blending.

Sightline Intelligence

Command, Control, and Script

API v3.11

Capture

Classes

Classes
struct SLASetVideoParameters_t
struct SLASetADCPParameters_t
struct SLASetAcquisitionParameters_t
struct SLACurrentVideoParameters_t
struct SLACurrentADCPParameters_t
struct SLACurrentAcquisitionParameters_t
struct SLASetDigitalCameraParameters_t
struct SLACurrentDigitalCameraParameters_t
struct SLAAdvancedCaptureParameters_t
struct SLACameraSwitch_t
struct SLADigitalVideoParserParameters_t
struct SLADecoderParameters_t
struct SLAUsb3VisionFeature_t
struct SLAUsbWebcamFeature_t
struct SLACreateDeviceOverlay_t
struct SLACameraCapabilities_t

Detailed Description

Camera acquisition controls.

Sightline Intelligence

Command, Control, and Script

API v3.11

Compression

[Classes](#)

Classes

struct	SLASetEthernetVideoParameters_t
struct	SLASetH264Parameters_t
struct	SLASetEthernetDisplayParameters_t
struct	SLACurrentEthernetVideoParameters_t
struct	SLACurrentEthernetDisplayParameters_t
struct	SLACurrentH264Parameters_t
struct	SLAStreamingControl_t

Detailed Description

Compression, encoding and network streaming of video.

SLA systems can compress video into H.264, MPEG4 and MJPEG and stream over ethernet.

Sightline Intelligence

Command, Control, and Script

API v3.11

Detection

Classes

Classes

struct	SLASetDetectionParameters_t
struct	SLACurrentDetectionParameters_t
struct	SLASetAdvancedDetectionParameters_t
struct	SLACurrentAdvancedDetectionParameters_t
struct	SLASetDetectionRegionOfInterestParameters_t
struct	SLACurrentDetectionRegionOfInterestParameters_t
struct	SLADoDetectSnapShot_t
struct	SLACustomAIDetect_t

Detailed Description

Automatic object detection - moving target, color anomaly, radiometric, marine.

See [SLASetDetectionParameters_t](#) for detailed description of the modes and application bit requirements.

- [SLASetStabilizationParameters_t](#) – Make sure you set Mode = 1 in addition to the other parameters
- [SLASetOverlayMode_t](#) – Enable Overlays so you can see the detection boxes
- [SLACoordinateReportingMode_t](#) – Enable reporting so the board sends back
SLACoordinateReportingMode_t::flags = 7
- [SLASetDetectionParameters_t](#) – Set the SLASetDetectionParameters_t::mode = 2 and change the
SLASetDetectionParameters_t::threshold = 5 (default). You can adjust the threshold down (1 = most sensitive and more false positives).
- [SLATrackingPositions_t](#) – Then look for these replies.

See [SLASetNetworkParameters_t](#) and [SLASetTelemetryDestination_t](#) to define where responses and telemetry will be sent. Querying the state of the hardware may be performed using [SLAGetParameters_t](#) which takes the 'setter' message ID as an input. In a few cases, more complicated messages have been implemented to get other types of results or status.

Sightline Intelligence

Command, Control, and Script

API v3.11

Display

Classes

struct	SLASetDisplayParameters_t
struct	SLASetDisplayAdjustments_t
struct	SLACurrentDisplayParameters_t
struct	SLADisplayAngle_t
struct	SLASetUserPalette_t
struct	SLACurrentUserPalette_t
struct	SLAVideoDisplay_t
struct	SLAMultiDisplay_t

Detailed Description

Video output display controls.

Draw Order

SightLine systems draw in the following order (later items are drawn on **top** of earlier items):

3000/4000 Single Camera:

1. Image
2. Track graphics (cursor, track box, ...)
3. Advanced overlays image (histogram, ...)
4. Graphic objects (circle, text, ...)

3000/4000 Multi Camera PiP or 2-Up:

1. **Main** image
2. Track graphics from the **Main** camera (cursor, track box, ...)
3. Advanced overlays from the **Main** camera (histogram, ...)
4. **Secondary** image (**PiP** or right image for **2-Up**)
5. Track graphics from the **Secondary** camera (cursor, track box, ...)

6. With Net 0 = **Multi PiP** and Net 1 = **Secondary** image: Graphic objects from the **Secondary** camera (circle, text, ...)
7. Graphic objects from the **Main** camera (circle, text, ...)

3000/4000 Blend:

1. Blended image
2. Track graphics from the **Fixed** camera (cursor, track box, ...)
3. Advanced overlays from the **Fixed** camera (histogram, ...)
4. Graphic objects from the **Fixed** camera (circle, text, ...)

1500:

1. Selected camera image
2. **PiP** image (if enabled, note graphics draw over PiP)
3. Track graphics from the displayed camera (cursor, track box, ...)
4. Advanced overlays from the displayed camera (histogram, ...)
5. Graphic objects from the displayed camera (circle, text, ...)

Sightline Intelligence

Command, Control, and Script

API v3.11

Enhancement

[Classes](#)

Classes

struct	SLASetVideoEnhancementParameters_t
struct	SLACurrentVideoEnhancementParameters_t
struct	SLACameraCalibration_t

Detailed Description

Enhancement of moving video.

Sightline Intelligence

Command, Control, and Script

API v3.11

Focus

[Classes](#)

Classes

struct	SLAFocusStats_t
struct	SLASetLensMode_t
struct	SLACurrentLensStatus_t
struct	SLASetLensParameters_t
struct	SLACurrentLensParameters_t
struct	SLALensParameters_t
struct	SLALensCommand_t
struct	SLAFocusParameters_t
struct	SLALensRanges_t
struct	SLACustomAutoFocusParameters_t

Detailed Description

Lens auto focus and brightness control.

Sightline Intelligence

Command, Control, and Script

API v3.11

General

[Classes](#)

Classes

struct	SLAGetVersionNumber_t
struct	SLAResetAllParameters_t
struct	SLASaveParameters_t
struct	SLAGetParameters_t
struct	SLAGetImageSize_t
struct	SLAVersionNumber_t
struct	SLACurrentImageSize_t
struct	SLAGetHardwareID_t
struct	SLASendTraceStr_t
struct	SLACommandCamera_t
struct	SLAUserWarningLevel_t
struct	SLASystemStatusMode_t
struct	SLAUserWarningMessage_t
struct	SLASystemStatusMessage_t
struct	SLADetailedTimingMessage_t
struct	SLAFrameIndex_t
struct	SLACurrentConfiguration_t
struct	SLASetSystemValue_t
struct	SLACurrentSystemValue_t
struct	SLAGPIO_t
struct	SLACustomResponse_t
struct	SLASendToBTS_t

Detailed Description

General system configuration - command camera, save parameters, user warnings, system profile etc.

Sightline Intelligence

Command, Control, and Script

API v3.11

KLV Metadata

[Classes](#)

Classes

struct	SLASetMetadataValues_t
struct	SLAMetadataStaticValues_t
struct	SLASetMetadataFrameValues_t
struct	SLASetKlvData_t
struct	SLASetMetadataRate_t
struct	SLASetVMTI_t
struct	SLAAppendedMetadata_t
struct	SLACurrentMetadataValues_t
struct	SLACurrentMetadataFrameValues_t
struct	SLACurrentMetadataRate_t
struct	SLAStreamingControl_t
struct	SLATagData_t
struct	SLATagDataRate_t
struct	SLATagSourceSelector_t
struct	SLAAncillaryTextMetadata_t
struct	SLAVMTIChips_t
struct	SLACursorOnTarget_t
struct	SLAVMTIFields_t

Detailed Description

Metadata embedding into network stream.

Metadata is generated in accordance with MISB standards 0102, 0601, 0603, 0604, and 0903. Selectable KLV elements may be chosen from a superset of the Motion Imagery Sensor Minimum Metadata Set defined in ST 0902. For conversion of values see Table 1 on page 16 of [ST 0601](#).

NOTE: KLV metadata is not available in [Demo Mode](#).

NOTE: Because packet processing and video output are asynchronous processes it is recommended that when setting KLV

data all KLV commands be placed into one UDP packet to ensure the data ends up in the same frame on the output.

Firmware Version: 3.11

Sightline Intelligence Command, Control, and Script API v3.11

Landing Aid

[Classes](#)

Classes	
struct	SLALandingAid_t
struct	SLALandingPosition_t

Detailed Description

Visual landing aid detection and tracking.

Sightline Intelligence

Command, Control, and Script

API v3.11

Network

[Classes](#)

Classes

struct	SLASetNetworkParameters_t
struct	SLACurrentNetworkParameters_t
struct	SLASetTelemetryDestination_t
struct	SLACurrentNetworkList_t

Detailed Description

Network communications and setup.

Sightline Intelligence

Command, Control, and Script

API v3.11

Non Uniformity Correction

[Classes](#)

Classes

struct	SLANucParameters_t
struct	SLAReadWriteNuc_t
struct	SLADeadPixelStats_t
struct	SLADeadPixel_t
struct	SLANoise3D_t

Detailed Description

Non Uniformity Correction and Dead Pixel Replacement.

Sightline Intelligence Command, Control, and Script API v3.11

Overlays

[Classes](#)

Classes

struct	SLASetOverlayMode_t
struct	SLACurrentOverlayMode_t
struct	SLACurrentOverlayObjectsIds_t
struct	SLACurrentOverlayObjectParameters_t
struct	SLALogoParameters_t
struct	SLADrawOverlay_t
struct	SLATrackTrails_t
struct	SLAUserFont_t

Detailed Description

Graphics overlays.

See [Draw Order](#) for additional overlay drawing details.

Sightline Intelligence

Command, Control, and Script

API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) [Camera Configuration](#)

[Classes](#)

Recording

Classes

struct	<u>SLASetSDRecordingParameters_t</u>
struct	<u>SLACurrentSDCardRecordingStatus_t</u>
struct	<u>SLACurrentSDCardDirectoryInfo_t</u>
struct	<u>SLACurrentSnapShot_t</u>
struct	<u>SLASetSnapShot_t</u>
struct	<u>SLADoSnapShot_t</u>
struct	<u>SLADirectoryStatisticsReply_t</u>
struct	<u>SLADoDetectSnapShot_t</u>

Detailed Description

Video recording and snapshots to FTP server, microSD card, or other media.

Recording is available as H.264,H.265, or MPEG4 for video streams.

Snapshots are available in compressed JPEG and uncompressed PNG image formats.

Metadata (KLV) can be recorded as KML or NITF.

Recording destinations can include a microSD card, USB drive, or other types of storage such as a RAM Drive or NFS. Each OEM has different hardware capabilities - please refer to <https://knowledge.sightlineintelligence.com/product-support-documentation/> for specific details.

see also [File Recording](#)

Sightline Intelligence Command, Control, and Script API v3.11

Serial Port

[Classes](#)

Classes

struct	SLACommandPassThrough_t
struct	SLASetPortConfiguration_t
struct	SLACurrentPortConfiguration_t
struct	SLAI2CCommand_t

Detailed Description

Serial port communications and setup.

Sightline Intelligence

Command, Control, and Script

API v3.11

Tracking

Classes

Classes

struct	SLAModifyTracking_t
struct	SLAStartTracking_t
struct	SLAStopTracking_t
struct	SLANudgeTrackingCoordinate_t
struct	SLASetTrackingParameters_t
struct	SLAModifyTrackIndex_t
struct	SLADesignateSelectedTrackPrimary_t
struct	SLAShiftSelectedTrack_t
struct	SLAStopSelectedTrack_t
struct	SLATrackingPosition_t
struct	SLACurrentTrackingParameters_t
struct	SLATrackingPositions_t
struct	SLATrackingPositionsExtended_t
struct	SLATrackingMultiClass_t

Detailed Description

User designated tracking of objects.

Sightline Intelligence

Command, Control, and Script

API v3.11

Classification

[Classes](#)

Classes

struct	SLACustomClassifier_t
struct	SLAClassifierParameters_t
struct	SLAKlvClassFilters_t

Detailed Description

AI Classifier.

Sightline Intelligence Command, Control, and Script API v3.11

SVPEndString_t Member List

This is the complete list of members for [SVPEndString_t](#), including all inherited members.

[str](#) [SVPEndString_t](#)

Sightline Intelligence Command, Control, and Script API v3.11

SVPLenString_t Member List

This is the complete list of members for [SVPLenString_t](#), including all inherited members.

- [len](#) [SVPLenString_t](#)
- [str](#) [SVPLenString_t](#)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetUserPalette_t Member List

This is the complete list of members for [SLASetUserPalette_t](#), including all inherited members.

palette (defined in SLASetUserPalette_t)	SLASetUserPalette_t	
u (defined in SLASetUserPalette_t)	SLASetUserPalette_t	
v (defined in SLASetUserPalette_t)	SLASetUserPalette_t	
y (defined in SLASetUserPalette_t)	SLASetUserPalette_t	

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentUserPalette_t Member List

This is the complete list of members for [SLACurrentUserPalette_t](#), including all inherited members.

palette (defined in SLACurrentUserPalette_t)	SLACurrentUserPalette_t
u (defined in SLACurrentUserPalette_t)	SLACurrentUserPalette_t
v (defined in SLACurrentUserPalette_t)	SLACurrentUserPalette_t
y (defined in SLACurrentUserPalette_t)	SLACurrentUserPalette_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SVPLenData_t Member List

This is the complete list of members for [SVPLenData_t](#), including all inherited members.

- [buf](#) [SVPLenData_t](#)
- [len](#) [SVPLenData_t](#)

Sightline Intelligence

Command, Control, and Script

API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) ■ [Camera Configuration](#)

SVPSignedData_t Member List

This is the complete list of members for [SVPSignedData_t](#), including all inherited members.

[buf](#) [SVPSignedData_t](#)
[len](#) [SVPSignedData_t](#)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAVersionNumber_t Member List

This is the complete list of members for [SLAVersionNumber_t](#), including all inherited members.

adapters (defined in SLAVersionNumber_t)	SLAVersionNumber_t
appBits (defined in SLAVersionNumber_t)	SLAVersionNumber_t
boardType (defined in SLAVersionNumber_t)	SLAVersionNumber_t
buildDate (defined in SLAVersionNumber_t)	SLAVersionNumber_t
buildTime (defined in SLAVersionNumber_t)	SLAVersionNumber_t
degreesC (defined in SLAVersionNumber_t)	SLAVersionNumber_t
degreesF (defined in SLAVersionNumber_t)	SLAVersionNumber_t
hwID (defined in SLAVersionNumber_t)	SLAVersionNumber_t
hwVersion (defined in SLAVersionNumber_t)	SLAVersionNumber_t
otherVersion (defined in SLAVersionNumber_t)	SLAVersionNumber_t
srcRevision (defined in SLAVersionNumber_t)	SLAVersionNumber_t
swBuild (defined in SLAVersionNumber_t)	SLAVersionNumber_t
swMajor (defined in SLAVersionNumber_t)	SLAVersionNumber_t
swMinor (defined in SLAVersionNumber_t)	SLAVersionNumber_t
swRelease (defined in SLAVersionNumber_t)	SLAVersionNumber_t
v4AppBits (defined in SLAVersionNumber_t)	SLAVersionNumber_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAGetParameters_t Member List

This is the complete list of members for [SLAGetParameters_t](#), including all inherited members.

id (defined in SLAGetParameters_t)	SLAGetParameters_t
payload0 (defined in SLAGetParameters_t)	SLAGetParameters_t
payload1 (defined in SLAGetParameters_t)	SLAGetParameters_t
payload2 (defined in SLAGetParameters_t)	SLAGetParameters_t
payload3 (defined in SLAGetParameters_t)	SLAGetParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetPortConfiguration_t Member List

This is the complete list of members for [SLASetPortConfiguration_t](#), including all inherited members.

baud (defined in SLASetPortConfiguration_t)	SLASetPortConfiguration_t
dataBits (defined in SLASetPortConfiguration_t)	SLASetPortConfiguration_t
inputPort (defined in SLASetPortConfiguration_t)	SLASetPortConfiguration_t
maxDelay (defined in SLASetPortConfiguration_t)	SLASetPortConfiguration_t
maxPacket (defined in SLASetPortConfiguration_t)	SLASetPortConfiguration_t
parity (defined in SLASetPortConfiguration_t)	SLASetPortConfiguration_t
port (defined in SLASetPortConfiguration_t)	SLASetPortConfiguration_t
protocol (defined in SLASetPortConfiguration_t)	SLASetPortConfiguration_t
stopBits (defined in SLASetPortConfiguration_t)	SLASetPortConfiguration_t
udpAttNavPort (defined in SLASetPortConfiguration_t)	SLASetPortConfiguration_t
udpDestinationAddr (defined in SLASetPortConfiguration_t)	SLASetPortConfiguration_t
udpDestinationPort (defined in SLASetPortConfiguration_t)	SLASetPortConfiguration_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetNetworkParameters_t Member List

This is the complete list of members for [SLASetNetworkParameters_t](#), including all inherited members.

c2replyPort (defined in SLASetNetworkParameters_t)	SLASetNetworkParameters_t
gateway (defined in SLASetNetworkParameters_t)	SLASetNetworkParameters_t
hostName (defined in SLASetNetworkParameters_t)	SLASetNetworkParameters_t
ipAddr (defined in SLASetNetworkParameters_t)	SLASetNetworkParameters_t
listenPort (defined in SLASetNetworkParameters_t)	SLASetNetworkParameters_t
listenPort2 (defined in SLASetNetworkParameters_t)	SLASetNetworkParameters_t
mode (defined in SLASetNetworkParameters_t)	SLASetNetworkParameters_t
reserved0 (defined in SLASetNetworkParameters_t)	SLASetNetworkParameters_t
reserved1 (defined in SLASetNetworkParameters_t)	SLASetNetworkParameters_t
reserved2 (defined in SLASetNetworkParameters_t)	SLASetNetworkParameters_t
subnet (defined in SLASetNetworkParameters_t)	SLASetNetworkParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAUserWarningMessage_t Member List

This is the complete list of members for [SLAUserWarningMessage_t](#), including all inherited members.

str (defined in SLAUserWarningMessage_t)	SLAUserWarningMessage_t
warnLevel (defined in SLAUserWarningMessage_t)	SLAUserWarningMessage_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetTelemetryDestination_t Member List

This is the complete list of members for [SLASetTelemetryDestination_t](#), including all inherited members.

cameraID (defined in SLASetTelemetryDestination_t)	SLASetTelemetryDestination_t
flags (defined in SLASetTelemetryDestination_t)	SLASetTelemetryDestination_t
ipAddr (defined in SLASetTelemetryDestination_t)	SLASetTelemetryDestination_t
mode (defined in SLASetTelemetryDestination_t)	SLASetTelemetryDestination_t
reserved (defined in SLASetTelemetryDestination_t)	SLASetTelemetryDestination_t
udpDestinationPort (defined in SLASetTelemetryDestination_t)	SLASetTelemetryDestination_t

Sightline Intelligence

Command, Control, and Script

API

v3.11

SLACommandCamera_t Member List

This is the complete list of members for [SLACommandCamera_t](#), including all inherited members.

cameraIndex (defined in [SLACommandCamera_t](#)) [SLACommandCamera_t](#)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentConfiguration_t Member List

This is the complete list of members for [SLACurrentConfiguration_t](#), including all inherited members.

cameraConfiguredBits (defined in SLACurrentConfiguration_t)	SLACurrentConfiguration_t
cameraConnectedBits (defined in SLACurrentConfiguration_t)	SLACurrentConfiguration_t
captureStateBits (defined in SLACurrentConfiguration_t)	SLACurrentConfiguration_t
displayPresentBits (defined in SLACurrentConfiguration_t)	SLACurrentConfiguration_t
maxCameras (defined in SLACurrentConfiguration_t)	SLACurrentConfiguration_t
maxProcessed (defined in SLACurrentConfiguration_t)	SLACurrentConfiguration_t
maxStreams (defined in SLACurrentConfiguration_t)	SLACurrentConfiguration_t
maxVirtCameras (defined in SLACurrentConfiguration_t)	SLACurrentConfiguration_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetEthernetDisplayParameters_t Member List

This is the complete list of members for [SLASetEthernetDisplayParameters_t](#), including all inherited members.

displayId (defined in SLASetEthernetDisplayParameters_t)	SLASetEthernetDisplayParameters_t
ipAddr (defined in SLASetEthernetDisplayParameters_t)	SLASetEthernetDisplayParameters_t
maxPacket (defined in SLASetEthernetDisplayParameters_t)	SLASetEthernetDisplayParameters_t
maxRawPacket (defined in SLASetEthernetDisplayParameters_t)	SLASetEthernetDisplayParameters_t
port (defined in SLASetEthernetDisplayParameters_t)	SLASetEthernetDisplayParameters_t
protocol (defined in SLASetEthernetDisplayParameters_t)	SLASetEthernetDisplayParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetSDRecordingParameters_t Member List

This is the complete list of members for [SLASetSDRecordingParameters_t](#), including all inherited members.

clearFlash (defined in SLASetSDRecordingParameters_t)	SLASetSDRecordingParameters_t
displayId (defined in SLASetSDRecordingParameters_t)	SLASetSDRecordingParameters_t
getDirectory (defined in SLASetSDRecordingParameters_t)	SLASetSDRecordingParameters_t
getStatus (defined in SLASetSDRecordingParameters_t)	SLASetSDRecordingParameters_t
lbl (defined in SLASetSDRecordingParameters_t)	SLASetSDRecordingParameters_t
recordMode (defined in SLASetSDRecordingParameters_t)	SLASetSDRecordingParameters_t
recordType (defined in SLASetSDRecordingParameters_t)	SLASetSDRecordingParameters_t
videoLength (defined in SLASetSDRecordingParameters_t)	SLASetSDRecordingParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentSDCardRecordingStatus_t Member List

This is the complete list of members for [SLACurrentSDCardRecordingStatus_t](#), including all inherited members.

displayId (defined in SLACurrentSDCardRecordingStatus_t)	SLACurrentSDCardRecordingStatus_t
length (defined in SLACurrentSDCardRecordingStatus_t)	SLACurrentSDCardRecordingStatus_t
space (defined in SLACurrentSDCardRecordingStatus_t)	SLACurrentSDCardRecordingStatus_t
state (defined in SLACurrentSDCardRecordingStatus_t)	SLACurrentSDCardRecordingStatus_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAResetAllParameters_t Member List

This is the complete list of members for [SLAResetAllParameters_t](#), including all inherited members.

resetType (defined in [SLAResetAllParameters_t](#)) [SLAResetAllParameters_t](#)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAAdvancedCaptureParameters_t Member List

This is the complete list of members for [SLAAdvancedCaptureParameters_t](#), including all inherited members.

horzControl (defined in SLAAdvancedCaptureParameters_t)	SLAAdvancedCaptureParameters_t
xstart (defined in SLAAdvancedCaptureParameters_t)	SLAAdvancedCaptureParameters_t
ystart (defined in SLAAdvancedCaptureParameters_t)	SLAAdvancedCaptureParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAAncillaryTextMetadata_t Member List

This is the complete list of members for [SLAAncillaryTextMetadata_t](#), including all inherited members.

creationTime (defined in SLAAncillaryTextMetadata_t)	SLAAncillaryTextMetadata_t
displayId (defined in SLAAncillaryTextMetadata_t)	SLAAncillaryTextMetadata_t
messageBody (defined in SLAAncillaryTextMetadata_t)	SLAAncillaryTextMetadata_t
originator (defined in SLAAncillaryTextMetadata_t)	SLAAncillaryTextMetadata_t
source (defined in SLAAncillaryTextMetadata_t)	SLAAncillaryTextMetadata_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAAppendedMetadata_t Member List

This is the complete list of members for [SLAAppendedMetadata_t](#), including all inherited members.

data (defined in SLAAppendedMetadata_t)	SLAAppendedMetadata_t
displayId (defined in SLAAppendedMetadata_t)	SLAAppendedMetadata_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLABlendAlign_t Member List

This is the complete list of members for [SLABlendAlign_t](#), including all inherited members.

horizontal (defined in SLABlendAlign_t)	SLABlendAlign_t
hzoom (defined in SLABlendAlign_t)	SLABlendAlign_t
index (defined in SLABlendAlign_t)	SLABlendAlign_t
rotate (defined in SLABlendAlign_t)	SLABlendAlign_t
vertical (defined in SLABlendAlign_t)	SLABlendAlign_t
zoom (defined in SLABlendAlign_t)	SLABlendAlign_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACameraCalibration_t Member List

This is the complete list of members for [SLACameraCalibration_t](#), including all inherited members.

cameraIndex (defined in SLACameraCalibration_t)	SLACameraCalibration_t
distortionCenterCol (defined in SLACameraCalibration_t)	SLACameraCalibration_t
distortionCenterRow (defined in SLACameraCalibration_t)	SLACameraCalibration_t
distortionCorrection (defined in SLACameraCalibration_t)	SLACameraCalibration_t
reserved (defined in SLACameraCalibration_t)	SLACameraCalibration_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACameraCapabilities_t Member List

This is the complete list of members for [SLACameraCapabilities_t](#), including all inherited members.

autoGainCapabilities (defined in SLACameraCapabilities_t)	SLACameraCapabilities_t
cameraIndex (defined in SLACameraCapabilities_t)	SLACameraCapabilities_t
capabilities (defined in SLACameraCapabilities_t)	SLACameraCapabilities_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACameraSwitch_t Member List

This is the complete list of members for [SLACameraSwitch_t](#), including all inherited members.

cameraIndex (defined in SLACameraSwitch_t)	SLACameraSwitch_t
cameraType (defined in SLACameraSwitch_t)	SLACameraSwitch_t
mode (defined in SLACameraSwitch_t)	SLACameraSwitch_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAClassifierParameters_t Member List

This is the complete list of members for [SLAClassifierParameters_t](#), including all inherited members.

cameraIndex (defined in SLAClassifierParameters_t)	SLAClassifierParameters_t
customFilters (defined in SLAClassifierParameters_t)	SLAClassifierParameters_t
detectionFlags (defined in SLAClassifierParameters_t)	SLAClassifierParameters_t
maxClassPerFrame (defined in SLAClassifierParameters_t)	SLAClassifierParameters_t
minClassDim (defined in SLAClassifierParameters_t)	SLAClassifierParameters_t
oodClassThresh (defined in SLAClassifierParameters_t)	SLAClassifierParameters_t
slaFilters (defined in SLAClassifierParameters_t)	SLAClassifierParameters_t
subClassThreshA (defined in SLAClassifierParameters_t)	SLAClassifierParameters_t
subClassThreshB (defined in SLAClassifierParameters_t)	SLAClassifierParameters_t
subClassThreshC (defined in SLAClassifierParameters_t)	SLAClassifierParameters_t
trackFlags (defined in SLAClassifierParameters_t)	SLAClassifierParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACommandPassThrough_t Member List

This is the complete list of members for [SLACommandPassThrough_t](#), including all inherited members.

data (defined in SLACommandPassThrough_t)	SLACommandPassThrough_t
destPort (defined in SLACommandPassThrough_t)	SLACommandPassThrough_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACoordinateReportingMode_t Member List

This is the complete list of members for [SLACoordinateReportingMode_t](#), including all inherited members.

cameraIndex (defined in SLACoordinateReportingMode_t)	SLACoordinateReportingMode_t
flags (defined in SLACoordinateReportingMode_t)	SLACoordinateReportingMode_t
framePeriod (defined in SLACoordinateReportingMode_t)	SLACoordinateReportingMode_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACreateDeviceOverlay_t Member List

This is the complete list of members for [SLACreateDeviceOverlay_t](#), including all inherited members.

clk (defined in SLACreateDeviceOverlay_t)	SLACreateDeviceOverlay_t
driverName (defined in SLACreateDeviceOverlay_t)	SLACreateDeviceOverlay_t
index (defined in SLACreateDeviceOverlay_t)	SLACreateDeviceOverlay_t
lanes (defined in SLACreateDeviceOverlay_t)	SLACreateDeviceOverlay_t
settle (defined in SLACreateDeviceOverlay_t)	SLACreateDeviceOverlay_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentAcquisitionParameters_t Member List

This is the complete list of members for [SLACurrentAcquisitionParameters_t](#), including all inherited members.

bigHigh (defined in SLACurrentAcquisitionParameters_t)	SLACurrentAcquisitionParameters_t
bigHorzFrontPorch (defined in SLACurrentAcquisitionParameters_t)	SLACurrentAcquisitionParameters_t
bigVertFrontPorch (defined in SLACurrentAcquisitionParameters_t)	SLACurrentAcquisitionParameters_t
bigWide (defined in SLACurrentAcquisitionParameters_t)	SLACurrentAcquisitionParameters_t
bitDepth (defined in SLACurrentAcquisitionParameters_t)	SLACurrentAcquisitionParameters_t
cameraIndex (defined in SLACurrentAcquisitionParameters_t)	SLACurrentAcquisitionParameters_t
cameraName (defined in SLACurrentAcquisitionParameters_t)	SLACurrentAcquisitionParameters_t
cameraType (defined in SLACurrentAcquisitionParameters_t)	SLACurrentAcquisitionParameters_t
camSwitchIndex (defined in SLACurrentAcquisitionParameters_t)	SLACurrentAcquisitionParameters_t
flags (defined in SLACurrentAcquisitionParameters_t)	SLACurrentAcquisitionParameters_t
frameStep (defined in SLACurrentAcquisitionParameters_t)	SLACurrentAcquisitionParameters_t
high (defined in SLACurrentAcquisitionParameters_t)	SLACurrentAcquisitionParameters_t
horzFrontPorch (defined in SLACurrentAcquisitionParameters_t)	SLACurrentAcquisitionParameters_t
numCamera (defined in SLACurrentAcquisitionParameters_t)	SLACurrentAcquisitionParameters_t
optArgs (defined in SLACurrentAcquisitionParameters_t)	SLACurrentAcquisitionParameters_t
reserved (defined in SLACurrentAcquisitionParameters_t)	SLACurrentAcquisitionParameters_t
validCol (defined in SLACurrentAcquisitionParameters_t)	SLACurrentAcquisitionParameters_t
validHigh (defined in SLACurrentAcquisitionParameters_t)	SLACurrentAcquisitionParameters_t
validRow (defined in SLACurrentAcquisitionParameters_t)	SLACurrentAcquisitionParameters_t
validWide (defined in SLACurrentAcquisitionParameters_t)	SLACurrentAcquisitionParameters_t
vertFrontPorch (defined in SLACurrentAcquisitionParameters_t)	SLACurrentAcquisitionParameters_t
wide (defined in SLACurrentAcquisitionParameters_t)	SLACurrentAcquisitionParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetAcquisitionParameters_t Member List

This is the complete list of members for [SLASetAcquisitionParameters_t](#), including all inherited members.

bigHigh (defined in SLASetAcquisitionParameters_t)	SLASetAcquisitionParameters_t
bigHorzFrontPorch (defined in SLASetAcquisitionParameters_t)	SLASetAcquisitionParameters_t
bigVertFrontPorch (defined in SLASetAcquisitionParameters_t)	SLASetAcquisitionParameters_t
bigWide (defined in SLASetAcquisitionParameters_t)	SLASetAcquisitionParameters_t
bitDepth (defined in SLASetAcquisitionParameters_t)	SLASetAcquisitionParameters_t
cameraIndex (defined in SLASetAcquisitionParameters_t)	SLASetAcquisitionParameters_t
cameraName (defined in SLASetAcquisitionParameters_t)	SLASetAcquisitionParameters_t
cameraType (defined in SLASetAcquisitionParameters_t)	SLASetAcquisitionParameters_t
camSwitchIndex (defined in SLASetAcquisitionParameters_t)	SLASetAcquisitionParameters_t
deviceId (defined in SLASetAcquisitionParameters_t)	SLASetAcquisitionParameters_t
flags (defined in SLASetAcquisitionParameters_t)	SLASetAcquisitionParameters_t
frameStep (defined in SLASetAcquisitionParameters_t)	SLASetAcquisitionParameters_t
high (defined in SLASetAcquisitionParameters_t)	SLASetAcquisitionParameters_t
horzFrontPorch (defined in SLASetAcquisitionParameters_t)	SLASetAcquisitionParameters_t
optArgs (defined in SLASetAcquisitionParameters_t)	SLASetAcquisitionParameters_t
portID (defined in SLASetAcquisitionParameters_t)	SLASetAcquisitionParameters_t
reserved (defined in SLASetAcquisitionParameters_t)	SLASetAcquisitionParameters_t
validCol (defined in SLASetAcquisitionParameters_t)	SLASetAcquisitionParameters_t
validHigh (defined in SLASetAcquisitionParameters_t)	SLASetAcquisitionParameters_t
validRow (defined in SLASetAcquisitionParameters_t)	SLASetAcquisitionParameters_t
validWide (defined in SLASetAcquisitionParameters_t)	SLASetAcquisitionParameters_t
vertFrontPorch (defined in SLASetAcquisitionParameters_t)	SLASetAcquisitionParameters_t
wide (defined in SLASetAcquisitionParameters_t)	SLASetAcquisitionParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentADCParameters_t Member List

This is the complete list of members for [SLACurrentADCParameters_t](#), including all inherited members.

brightness (defined in SLACurrentADCParameters_t)	SLACurrentADCParameters_t
cameraIndex (defined in SLACurrentADCParameters_t)	SLACurrentADCParameters_t
chroma1 (defined in SLACurrentADCParameters_t)	SLACurrentADCParameters_t
chroma2 (defined in SLACurrentADCParameters_t)	SLACurrentADCParameters_t
contrast (defined in SLACurrentADCParameters_t)	SLACurrentADCParameters_t
exposure (defined in SLACurrentADCParameters_t)	SLACurrentADCParameters_t
exposureFrac (defined in SLACurrentADCParameters_t)	SLACurrentADCParameters_t
hue (defined in SLACurrentADCParameters_t)	SLACurrentADCParameters_t
luma1 (defined in SLACurrentADCParameters_t)	SLACurrentADCParameters_t
luma2 (defined in SLACurrentADCParameters_t)	SLACurrentADCParameters_t
luma3 (defined in SLACurrentADCParameters_t)	SLACurrentADCParameters_t
mode (defined in SLACurrentADCParameters_t)	SLACurrentADCParameters_t
saturation (defined in SLACurrentADCParameters_t)	SLACurrentADCParameters_t
setMask (defined in SLACurrentADCParameters_t)	SLACurrentADCParameters_t
whiteBalanceCount (defined in SLACurrentADCParameters_t)	SLACurrentADCParameters_t
whiteBalanceExclusion (defined in SLACurrentADCParameters_t)	SLACurrentADCParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetADCParameters_t Member List

This is the complete list of members for [SLASetADCParameters_t](#), including all inherited members.

brightness (defined in SLASetADCParameters_t)	SLASetADCParameters_t
cameraIndex (defined in SLASetADCParameters_t)	SLASetADCParameters_t
chroma1 (defined in SLASetADCParameters_t)	SLASetADCParameters_t
chroma2 (defined in SLASetADCParameters_t)	SLASetADCParameters_t
contrast (defined in SLASetADCParameters_t)	SLASetADCParameters_t
exposure (defined in SLASetADCParameters_t)	SLASetADCParameters_t
exposureFrac (defined in SLASetADCParameters_t)	SLASetADCParameters_t
hue (defined in SLASetADCParameters_t)	SLASetADCParameters_t
luma1 (defined in SLASetADCParameters_t)	SLASetADCParameters_t
luma2 (defined in SLASetADCParameters_t)	SLASetADCParameters_t
luma3 (defined in SLASetADCParameters_t)	SLASetADCParameters_t
mode (defined in SLASetADCParameters_t)	SLASetADCParameters_t
saturation (defined in SLASetADCParameters_t)	SLASetADCParameters_t
setMask (defined in SLASetADCParameters_t)	SLASetADCParameters_t
whiteBalanceCount (defined in SLASetADCParameters_t)	SLASetADCParameters_t
whiteBalanceExclusion (defined in SLASetADCParameters_t)	SLASetADCParameters_t

Sightline Intelligence Command, Control, and Script API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) ■ [Camera Configuration](#)

SLACurrentAdvancedDetectionParameters_t Member List

This is the complete list of members for [SLACurrentAdvancedDetectionParameters_t](#), including all inherited members.

aiDetectIOU (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
bgEdgePenalty6 (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
bgResetAng (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
bgResetConf (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
bgResetFrames (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
bgResetOff (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
bgTimeConst (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
bgWarpAng (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
bgWarpConf (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
bgWarpFrames (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
bgWarpOff (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
blocksHigh (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
blocksWide (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
cameraIndex (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
closeFilter (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
debugFiltering (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
detBitDepth (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
detDirection (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
detIdx (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
detNearTrk (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
detUpdateRate (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
downsample (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
ellipsePadding (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
frameDepth (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t

hideOverlapTrks (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
maxDisplayTrks (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
maxHigh (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
maxKlvTrks (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
maxTelemTrks (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
maxTrackFrames (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
maxVel8 (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
maxWide (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
mergeDirTol (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
mergeRadius (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
minHigh (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
minVel8 (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
minWide (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
nFramesBack (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
reserved1 (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
surroundSize (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t
useRegistration (defined in SLACurrentAdvancedDetectionParameters_t)	SLACurrentAdvancedDetectionParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) [Camera Configuration](#)

SLASetAdvancedDetectionParameters_t Member List

This is the complete list of members for [SLASetAdvancedDetectionParameters_t](#), including all inherited members.

aiDetectIOU (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
bgEdgePenalty6 (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
bgResetAng (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
bgResetConf (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
bgResetFrames (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
bgResetOff (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
bgTimeConst (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
bgWarpAng (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
bgWarpConf (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
bgWarpFrames (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
bgWarpOff (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
blocksHigh (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
blocksWide (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
cameraIndex (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
closeFilter (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
debugFiltering (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
detBitDepth (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
detDirection (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
detIdx (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
detNearTrk (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
detUpdateRate (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
downsample (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
ellipsePadding (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
frameDepth (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
hideOverlapTrks (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
maxDisplayTrks (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
maxHigh (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
maxKlvTrks (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
maxTelemTrks (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
maxTrackFrames (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
maxVel8 (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
maxWide (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t

mergeDirTol (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
mergeRadius (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
minHigh (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
minVel8 (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
minWide (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
nFramesBack (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
reserved1 (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
surroundSize (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t
useRegistration (defined in SLASetAdvancedDetectionParameters_t)	SLASetAdvancedDetectionParameters_t

Firmware Version: 3.11

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentBlendParameters_t Member List

This is the complete list of members for [SLACurrentBlendParameters_t](#), including all inherited members.

absOffZoom (defined in SLACurrentBlendParameters_t)	SLACurrentBlendParameters_t
amt (defined in SLACurrentBlendParameters_t)	SLACurrentBlendParameters_t
coldEnd (defined in SLACurrentBlendParameters_t)	SLACurrentBlendParameters_t
down (defined in SLACurrentBlendParameters_t)	SLACurrentBlendParameters_t
fixedIndex (defined in SLACurrentBlendParameters_t)	SLACurrentBlendParameters_t
flags (defined in SLACurrentBlendParameters_t)	SLACurrentBlendParameters_t
hotStart (defined in SLACurrentBlendParameters_t)	SLACurrentBlendParameters_t
hue (defined in SLACurrentBlendParameters_t)	SLACurrentBlendParameters_t
hzoom (defined in SLACurrentBlendParameters_t)	SLACurrentBlendParameters_t
left (defined in SLACurrentBlendParameters_t)	SLACurrentBlendParameters_t
mode (defined in SLACurrentBlendParameters_t)	SLACurrentBlendParameters_t
presetAlignIndex (defined in SLACurrentBlendParameters_t)	SLACurrentBlendParameters_t
reserved (defined in SLACurrentBlendParameters_t)	SLACurrentBlendParameters_t
right (defined in SLACurrentBlendParameters_t)	SLACurrentBlendParameters_t
rotation (defined in SLACurrentBlendParameters_t)	SLACurrentBlendParameters_t
up (defined in SLACurrentBlendParameters_t)	SLACurrentBlendParameters_t
usePresetAlign (defined in SLACurrentBlendParameters_t)	SLACurrentBlendParameters_t
warpIndex (defined in SLACurrentBlendParameters_t)	SLACurrentBlendParameters_t
zoom (defined in SLACurrentBlendParameters_t)	SLACurrentBlendParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetBlendParameters_t Member List

This is the complete list of members for [SLASetBlendParameters_t](#), including all inherited members.

absOffZoom (defined in SLASetBlendParameters_t)	SLASetBlendParameters_t
amt (defined in SLASetBlendParameters_t)	SLASetBlendParameters_t
coldEnd (defined in SLASetBlendParameters_t)	SLASetBlendParameters_t
fixedIndex (defined in SLASetBlendParameters_t)	SLASetBlendParameters_t
flags (defined in SLASetBlendParameters_t)	SLASetBlendParameters_t
horizontal (defined in SLASetBlendParameters_t)	SLASetBlendParameters_t
hotStart (defined in SLASetBlendParameters_t)	SLASetBlendParameters_t
hue (defined in SLASetBlendParameters_t)	SLASetBlendParameters_t
hzoom (defined in SLASetBlendParameters_t)	SLASetBlendParameters_t
mode (defined in SLASetBlendParameters_t)	SLASetBlendParameters_t
presetAlignIndex (defined in SLASetBlendParameters_t)	SLASetBlendParameters_t
reserved (defined in SLASetBlendParameters_t)	SLASetBlendParameters_t
reset (defined in SLASetBlendParameters_t)	SLASetBlendParameters_t
rotation (defined in SLASetBlendParameters_t)	SLASetBlendParameters_t
usePresetAlign (defined in SLASetBlendParameters_t)	SLASetBlendParameters_t
vertical (defined in SLASetBlendParameters_t)	SLASetBlendParameters_t
warpIndex (defined in SLASetBlendParameters_t)	SLASetBlendParameters_t
zoom (defined in SLASetBlendParameters_t)	SLASetBlendParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentDetectionParameters_t Member List

This is the complete list of members for [SLACurrentDetectionParameters_t](#), including all inherited members.

cameraIndex (defined in SLACurrentDetectionParameters_t)	SLACurrentDetectionParameters_t
classSensitivity (defined in SLACurrentDetectionParameters_t)	SLACurrentDetectionParameters_t
debugCode (defined in SLACurrentDetectionParameters_t)	SLACurrentDetectionParameters_t
detIdx (defined in SLACurrentDetectionParameters_t)	SLACurrentDetectionParameters_t
enableAiMtd (defined in SLACurrentDetectionParameters_t)	SLACurrentDetectionParameters_t
manualThreshold (defined in SLACurrentDetectionParameters_t)	SLACurrentDetectionParameters_t
manualWatchFrames (defined in SLACurrentDetectionParameters_t)	SLACurrentDetectionParameters_t
maxTemperature (defined in SLACurrentDetectionParameters_t)	SLACurrentDetectionParameters_t
minTemperature (defined in SLACurrentDetectionParameters_t)	SLACurrentDetectionParameters_t
mode (defined in SLACurrentDetectionParameters_t)	SLACurrentDetectionParameters_t
modeUpperByte (defined in SLACurrentDetectionParameters_t)	SLACurrentDetectionParameters_t
suspScore (defined in SLACurrentDetectionParameters_t)	SLACurrentDetectionParameters_t
threshold (defined in SLACurrentDetectionParameters_t)	SLACurrentDetectionParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetDetectionParameters_t Member List

This is the complete list of members for [SLASetDetectionParameters_t](#), including all inherited members.

cameraIndex (defined in SLASetDetectionParameters_t)	SLASetDetectionParameters_t
classSensitivity (defined in SLASetDetectionParameters_t)	SLASetDetectionParameters_t
debugCode (defined in SLASetDetectionParameters_t)	SLASetDetectionParameters_t
detIdx (defined in SLASetDetectionParameters_t)	SLASetDetectionParameters_t
enableAiMtd (defined in SLASetDetectionParameters_t)	SLASetDetectionParameters_t
manualThreshold (defined in SLASetDetectionParameters_t)	SLASetDetectionParameters_t
manualWatchFrames (defined in SLASetDetectionParameters_t)	SLASetDetectionParameters_t
maxTemperature (defined in SLASetDetectionParameters_t)	SLASetDetectionParameters_t
minTemperature (defined in SLASetDetectionParameters_t)	SLASetDetectionParameters_t
mode (defined in SLASetDetectionParameters_t)	SLASetDetectionParameters_t
modeUpperByte (defined in SLASetDetectionParameters_t)	SLASetDetectionParameters_t
suspScore (defined in SLASetDetectionParameters_t)	SLASetDetectionParameters_t
threshold (defined in SLASetDetectionParameters_t)	SLASetDetectionParameters_t

Sightline Intelligence Command, Control, and Script API

v3.11

SLACurrentDetectionRegionOfInterestParameters_t Member List

This is the complete list of members for [SLACurrentDetectionRegionOfInterestParameters_t](#), including all inherited members.

cameraIndex (defined in SLACurrentDetectionRegionOfInterestParameters_t)	SLACurrentDetectionRegionOfInterestParameters_t
detectColUI (defined in SLACurrentDetectionRegionOfInterestParameters_t)	SLACurrentDetectionRegionOfInterestParameters_t
detectHeight (defined in SLACurrentDetectionRegionOfInterestParameters_t)	SLACurrentDetectionRegionOfInterestParameters_t
detectRowUI (defined in SLACurrentDetectionRegionOfInterestParameters_t)	SLACurrentDetectionRegionOfInterestParameters_t
detectWidth (defined in SLACurrentDetectionRegionOfInterestParameters_t)	SLACurrentDetectionRegionOfInterestParameters_t
flags (defined in SLACurrentDetectionRegionOfInterestParameters_t)	SLACurrentDetectionRegionOfInterestParameters_t
mask0 (defined in SLACurrentDetectionRegionOfInterestParameters_t)	SLACurrentDetectionRegionOfInterestParameters_t
mask1 (defined in SLACurrentDetectionRegionOfInterestParameters_t)	SLACurrentDetectionRegionOfInterestParameters_t
mask2 (defined in SLACurrentDetectionRegionOfInterestParameters_t)	SLACurrentDetectionRegionOfInterestParameters_t
mask3 (defined in SLACurrentDetectionRegionOfInterestParameters_t)	SLACurrentDetectionRegionOfInterestParameters_t
maskBlockHigh (defined in SLACurrentDetectionRegionOfInterestParameters_t)	SLACurrentDetectionRegionOfInterestParameters_t
maskBlockWide (defined in SLACurrentDetectionRegionOfInterestParameters_t)	SLACurrentDetectionRegionOfInterestParameters_t
searchColUI (defined in SLACurrentDetectionRegionOfInterestParameters_t)	SLACurrentDetectionRegionOfInterestParameters_t
searchHeight (defined in SLACurrentDetectionRegionOfInterestParameters_t)	SLACurrentDetectionRegionOfInterestParameters_t
searchRowUI (defined in SLACurrentDetectionRegionOfInterestParameters_t)	SLACurrentDetectionRegionOfInterestParameters_t
searchWidth (defined in SLACurrentDetectionRegionOfInterestParameters_t)	SLACurrentDetectionRegionOfInterestParameters_t

Sightline Intelligence Command, Control, and Script API v3.11

SLASetDetectionRegionOfInterestParameters_t Member List

This is the complete list of members for [SLASetDetectionRegionOfInterestParameters_t](#), including all inherited members.

cameraIndex (defined in SLASetDetectionRegionOfInterestParameters_t)	SLASetDetectionRegionOfInterestParameters_t
detectColUI (defined in SLASetDetectionRegionOfInterestParameters_t)	SLASetDetectionRegionOfInterestParameters_t
detectHeight (defined in SLASetDetectionRegionOfInterestParameters_t)	SLASetDetectionRegionOfInterestParameters_t
detectRowUI (defined in SLASetDetectionRegionOfInterestParameters_t)	SLASetDetectionRegionOfInterestParameters_t
detectWidth (defined in SLASetDetectionRegionOfInterestParameters_t)	SLASetDetectionRegionOfInterestParameters_t
flags (defined in SLASetDetectionRegionOfInterestParameters_t)	SLASetDetectionRegionOfInterestParameters_t
mask0 (defined in SLASetDetectionRegionOfInterestParameters_t)	SLASetDetectionRegionOfInterestParameters_t
mask1 (defined in SLASetDetectionRegionOfInterestParameters_t)	SLASetDetectionRegionOfInterestParameters_t
mask2 (defined in SLASetDetectionRegionOfInterestParameters_t)	SLASetDetectionRegionOfInterestParameters_t
mask3 (defined in SLASetDetectionRegionOfInterestParameters_t)	SLASetDetectionRegionOfInterestParameters_t
maskBlockHigh (defined in SLASetDetectionRegionOfInterestParameters_t)	SLASetDetectionRegionOfInterestParameters_t
maskBlockWide (defined in SLASetDetectionRegionOfInterestParameters_t)	SLASetDetectionRegionOfInterestParameters_t
searchColUI (defined in SLASetDetectionRegionOfInterestParameters_t)	SLASetDetectionRegionOfInterestParameters_t
searchHeight (defined in SLASetDetectionRegionOfInterestParameters_t)	SLASetDetectionRegionOfInterestParameters_t
searchRowUI (defined in SLASetDetectionRegionOfInterestParameters_t)	SLASetDetectionRegionOfInterestParameters_t
searchWidth (defined in SLASetDetectionRegionOfInterestParameters_t)	SLASetDetectionRegionOfInterestParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentDigitalCameraParameters_t Member List

This is the complete list of members for [SLACurrentDigitalCameraParameters_t](#), including all inherited members.

agHoldmax (defined in SLACurrentDigitalCameraParameters_t)	SLACurrentDigitalCameraParameters_t
agHoldmin (defined in SLACurrentDigitalCameraParameters_t)	SLACurrentDigitalCameraParameters_t
agRate8 (defined in SLACurrentDigitalCameraParameters_t)	SLACurrentDigitalCameraParameters_t
cameraIndex (defined in SLACurrentDigitalCameraParameters_t)	SLACurrentDigitalCameraParameters_t
colROI Pct (defined in SLACurrentDigitalCameraParameters_t)	SLACurrentDigitalCameraParameters_t
highBitDepthFlags (defined in SLACurrentDigitalCameraParameters_t)	SLACurrentDigitalCameraParameters_t
highROI Pct (defined in SLACurrentDigitalCameraParameters_t)	SLACurrentDigitalCameraParameters_t
maxExp (defined in SLACurrentDigitalCameraParameters_t)	SLACurrentDigitalCameraParameters_t
midpoint (defined in SLACurrentDigitalCameraParameters_t)	SLACurrentDigitalCameraParameters_t
minAGRange (defined in SLACurrentDigitalCameraParameters_t)	SLACurrentDigitalCameraParameters_t
minExp (defined in SLACurrentDigitalCameraParameters_t)	SLACurrentDigitalCameraParameters_t
mode (defined in SLACurrentDigitalCameraParameters_t)	SLACurrentDigitalCameraParameters_t
rejectBrightTail (defined in SLACurrentDigitalCameraParameters_t)	SLACurrentDigitalCameraParameters_t
rejectDarkTail (defined in SLACurrentDigitalCameraParameters_t)	SLACurrentDigitalCameraParameters_t
rowROI Pct (defined in SLACurrentDigitalCameraParameters_t)	SLACurrentDigitalCameraParameters_t
wideROI Pct (defined in SLACurrentDigitalCameraParameters_t)	SLACurrentDigitalCameraParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetDigitalCameraParameters_t Member List

This is the complete list of members for [SLASetDigitalCameraParameters_t](#), including all inherited members.

agHoldmax (defined in SLASetDigitalCameraParameters_t)	SLASetDigitalCameraParameters_t
agHoldmin (defined in SLASetDigitalCameraParameters_t)	SLASetDigitalCameraParameters_t
agRate8 (defined in SLASetDigitalCameraParameters_t)	SLASetDigitalCameraParameters_t
cameraIndex (defined in SLASetDigitalCameraParameters_t)	SLASetDigitalCameraParameters_t
colROIPct (defined in SLASetDigitalCameraParameters_t)	SLASetDigitalCameraParameters_t
highBitDepthFlags (defined in SLASetDigitalCameraParameters_t)	SLASetDigitalCameraParameters_t
highROIPct (defined in SLASetDigitalCameraParameters_t)	SLASetDigitalCameraParameters_t
maxExp (defined in SLASetDigitalCameraParameters_t)	SLASetDigitalCameraParameters_t
midpoint (defined in SLASetDigitalCameraParameters_t)	SLASetDigitalCameraParameters_t
minAGRange (defined in SLASetDigitalCameraParameters_t)	SLASetDigitalCameraParameters_t
minExp (defined in SLASetDigitalCameraParameters_t)	SLASetDigitalCameraParameters_t
mode (defined in SLASetDigitalCameraParameters_t)	SLASetDigitalCameraParameters_t
rejectBrightTail (defined in SLASetDigitalCameraParameters_t)	SLASetDigitalCameraParameters_t
rejectDarkTail (defined in SLASetDigitalCameraParameters_t)	SLASetDigitalCameraParameters_t
rowROIPct (defined in SLASetDigitalCameraParameters_t)	SLASetDigitalCameraParameters_t
wideROIPct (defined in SLASetDigitalCameraParameters_t)	SLASetDigitalCameraParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentDisplayParameters_t Member List

This is the complete list of members for [SLACurrentDisplayParameters_t](#), including all inherited members.

decayRate (defined in SLACurrentDisplayParameters_t)	SLACurrentDisplayParameters_t
extendedZoom10 (defined in SLACurrentDisplayParameters_t)	SLACurrentDisplayParameters_t
falseColorZTT (defined in SLACurrentDisplayParameters_t)	SLACurrentDisplayParameters_t
flipMode (defined in SLACurrentDisplayParameters_t)	SLACurrentDisplayParameters_t
index (defined in SLACurrentDisplayParameters_t)	SLACurrentDisplayParameters_t
panCol (defined in SLACurrentDisplayParameters_t)	SLACurrentDisplayParameters_t
rotationDegrees (defined in SLACurrentDisplayParameters_t)	SLACurrentDisplayParameters_t
rotationLimit (defined in SLACurrentDisplayParameters_t)	SLACurrentDisplayParameters_t
tiltRow (defined in SLACurrentDisplayParameters_t)	SLACurrentDisplayParameters_t
zoom (defined in SLACurrentDisplayParameters_t)	SLACurrentDisplayParameters_t
zoomRate (defined in SLACurrentDisplayParameters_t)	SLACurrentDisplayParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetDisplayParameters_t Member List

This is the complete list of members for [SLASetDisplayParameters_t](#), including all inherited members.

cameraIndex (defined in SLASetDisplayParameters_t)	SLASetDisplayParameters_t
decayRate (defined in SLASetDisplayParameters_t)	SLASetDisplayParameters_t
extendedZoom10 (defined in SLASetDisplayParameters_t)	SLASetDisplayParameters_t
falseColorZTT (defined in SLASetDisplayParameters_t)	SLASetDisplayParameters_t
flipMode (defined in SLASetDisplayParameters_t)	SLASetDisplayParameters_t
panCol (defined in SLASetDisplayParameters_t)	SLASetDisplayParameters_t
rotationDegrees (defined in SLASetDisplayParameters_t)	SLASetDisplayParameters_t
rotationLimit (defined in SLASetDisplayParameters_t)	SLASetDisplayParameters_t
tiltRow (defined in SLASetDisplayParameters_t)	SLASetDisplayParameters_t
zoom (defined in SLASetDisplayParameters_t)	SLASetDisplayParameters_t
zoomRate (defined in SLASetDisplayParameters_t)	SLASetDisplayParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentEthernetDisplayParameters_t Member List

This is the complete list of members for [SLACurrentEthernetDisplayParameters_t](#), including all inherited members.

displayId (defined in SLACurrentEthernetDisplayParameters_t)	SLACurrentEthernetDisplayParameters_t
ipAddr (defined in SLACurrentEthernetDisplayParameters_t)	SLACurrentEthernetDisplayParameters_t
maxPacket (defined in SLACurrentEthernetDisplayParameters_t)	SLACurrentEthernetDisplayParameters_t
maxRawPacket (defined in SLACurrentEthernetDisplayParameters_t)	SLACurrentEthernetDisplayParameters_t
port (defined in SLACurrentEthernetDisplayParameters_t)	SLACurrentEthernetDisplayParameters_t
protocol (defined in SLACurrentEthernetDisplayParameters_t)	SLACurrentEthernetDisplayParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentEthernetVideoParameters_t Member List

This is the complete list of members for [SLACurrentEthernetVideoParameters_t](#), including all inherited members.

customHigh (defined in SLACurrentEthernetVideoParameters_t)	SLACurrentEthernetVideoParameters_t
customWide (defined in SLACurrentEthernetVideoParameters_t)	SLACurrentEthernetVideoParameters_t
displayId (defined in SLACurrentEthernetVideoParameters_t)	SLACurrentEthernetVideoParameters_t
foveal (defined in SLACurrentEthernetVideoParameters_t)	SLACurrentEthernetVideoParameters_t
frameSize (defined in SLACurrentEthernetVideoParameters_t)	SLACurrentEthernetVideoParameters_t
frameStep (defined in SLACurrentEthernetVideoParameters_t)	SLACurrentEthernetVideoParameters_t
quality (defined in SLACurrentEthernetVideoParameters_t)	SLACurrentEthernetVideoParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetEthernetVideoParameters_t Member List

This is the complete list of members for [SLASetEthernetVideoParameters_t](#), including all inherited members.

customHigh (defined in SLASetEthernetVideoParameters_t)	SLASetEthernetVideoParameters_t
customWide (defined in SLASetEthernetVideoParameters_t)	SLASetEthernetVideoParameters_t
displayId (defined in SLASetEthernetVideoParameters_t)	SLASetEthernetVideoParameters_t
foveal (defined in SLASetEthernetVideoParameters_t)	SLASetEthernetVideoParameters_t
frameSize (defined in SLASetEthernetVideoParameters_t)	SLASetEthernetVideoParameters_t
frameStep (defined in SLASetEthernetVideoParameters_t)	SLASetEthernetVideoParameters_t
quality (defined in SLASetEthernetVideoParameters_t)	SLASetEthernetVideoParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentH264Parameters_t Member List

This is the complete list of members for [SLACurrentH264Parameters_t](#), including all inherited members.

airMbPeriod (defined in SLACurrentH264Parameters_t)	SLACurrentH264Parameters_t
displayId (defined in SLACurrentH264Parameters_t)	SLACurrentH264Parameters_t
flags (defined in SLACurrentH264Parameters_t)	SLACurrentH264Parameters_t
intraFrameInterval (defined in SLACurrentH264Parameters_t)	SLACurrentH264Parameters_t
lfDisableIdc (defined in SLACurrentH264Parameters_t)	SLACurrentH264Parameters_t
maxQp (defined in SLACurrentH264Parameters_t)	SLACurrentH264Parameters_t
minQp (defined in SLACurrentH264Parameters_t)	SLACurrentH264Parameters_t
sliceRefreshRowNumber (defined in SLACurrentH264Parameters_t)	SLACurrentH264Parameters_t
targetBitRate (defined in SLACurrentH264Parameters_t)	SLACurrentH264Parameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetH264Parameters_t Member List

This is the complete list of members for [SLASetH264Parameters_t](#), including all inherited members.

airMbPeriod (defined in SLASetH264Parameters_t)	SLASetH264Parameters_t
displayId (defined in SLASetH264Parameters_t)	SLASetH264Parameters_t
flags (defined in SLASetH264Parameters_t)	SLASetH264Parameters_t
intraFrameInterval (defined in SLASetH264Parameters_t)	SLASetH264Parameters_t
lfDisableIdc (defined in SLASetH264Parameters_t)	SLASetH264Parameters_t
maxQp (defined in SLASetH264Parameters_t)	SLASetH264Parameters_t
minQp (defined in SLASetH264Parameters_t)	SLASetH264Parameters_t
sliceRefreshRowNumber (defined in SLASetH264Parameters_t)	SLASetH264Parameters_t
targetBitRate (defined in SLASetH264Parameters_t)	SLASetH264Parameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentImageSize_t Member List

This is the complete list of members for [SLACurrentImageSize_t](#), including all inherited members.

cameraIndex (defined in SLACurrentImageSize_t)	SLACurrentImageSize_t
capHigh (defined in SLACurrentImageSize_t)	SLACurrentImageSize_t
capWide (defined in SLACurrentImageSize_t)	SLACurrentImageSize_t
disHigh (defined in SLACurrentImageSize_t)	SLACurrentImageSize_t
disRectCol (defined in SLACurrentImageSize_t)	SLACurrentImageSize_t
disRectHigh (defined in SLACurrentImageSize_t)	SLACurrentImageSize_t
disRectRow (defined in SLACurrentImageSize_t)	SLACurrentImageSize_t
disRectWide (defined in SLACurrentImageSize_t)	SLACurrentImageSize_t
disWide (defined in SLACurrentImageSize_t)	SLACurrentImageSize_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentLensParameters_t Member List

This is the complete list of members for [SLACurrentLensParameters_t](#), including all inherited members.

afMetricRegionSize (defined in SLACurrentLensParameters_t)	SLACurrentLensParameters_t
autofocusChangePct (defined in SLACurrentLensParameters_t)	SLACurrentLensParameters_t
autofocusMethod (defined in SLACurrentLensParameters_t)	SLACurrentLensParameters_t
autofocusRateAdjust (defined in SLACurrentLensParameters_t)	SLACurrentLensParameters_t
colRoiFrac (defined in SLACurrentLensParameters_t)	SLACurrentLensParameters_t
comPortNum (defined in SLACurrentLensParameters_t)	SLACurrentLensParameters_t
focusSpeed (defined in SLACurrentLensParameters_t)	SLACurrentLensParameters_t
highRoiFrac (defined in SLACurrentLensParameters_t)	SLACurrentLensParameters_t
lensType (defined in SLACurrentLensParameters_t)	SLACurrentLensParameters_t
rowRoiFrac (defined in SLACurrentLensParameters_t)	SLACurrentLensParameters_t
wideRoiFrac (defined in SLACurrentLensParameters_t)	SLACurrentLensParameters_t
zoomSpeed (defined in SLACurrentLensParameters_t)	SLACurrentLensParameters_t
zoomTrackFocus (defined in SLACurrentLensParameters_t)	SLACurrentLensParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLALensParameters_t Member List

This is the complete list of members for [SLALensParameters_t](#), including all inherited members.

cameraIndex (defined in SLALensParameters_t)	SLALensParameters_t
filename (defined in SLALensParameters_t)	SLALensParameters_t
focusSpeed (defined in SLALensParameters_t)	SLALensParameters_t
port (defined in SLALensParameters_t)	SLALensParameters_t
reserved1 (defined in SLALensParameters_t)	SLALensParameters_t
reserved2 (defined in SLALensParameters_t)	SLALensParameters_t
zoomSpeed (defined in SLALensParameters_t)	SLALensParameters_t
zoomTrackFocus (defined in SLALensParameters_t)	SLALensParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentLensStatus_t Member List

This is the complete list of members for [SLACurrentLensStatus_t](#), including all inherited members.

camIdx (defined in SLACurrentLensStatus_t)	SLACurrentLensStatus_t
customAFOn (defined in SLACurrentLensStatus_t)	SLACurrentLensStatus_t
focusLimits (defined in SLACurrentLensStatus_t)	SLACurrentLensStatus_t
focusPos (defined in SLACurrentLensStatus_t)	SLACurrentLensStatus_t
zoomLimits (defined in SLACurrentLensStatus_t)	SLACurrentLensStatus_t
zoomPos (defined in SLACurrentLensStatus_t)	SLACurrentLensStatus_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentMetadataFrameValues_t Member List

This is the complete list of members for [SLACurrentMetadataFrameValues_t](#), including all inherited members.

displayId (defined in SLACurrentMetadataFrameValues_t)	SLACurrentMetadataFrameValues_t
frameCenterEI (defined in SLACurrentMetadataFrameValues_t)	SLACurrentMetadataFrameValues_t
frameCenterLat (defined in SLACurrentMetadataFrameValues_t)	SLACurrentMetadataFrameValues_t
frameCenterLon (defined in SLACurrentMetadataFrameValues_t)	SLACurrentMetadataFrameValues_t
frameWidth (defined in SLACurrentMetadataFrameValues_t)	SLACurrentMetadataFrameValues_t
offsetCornerLat1 (defined in SLACurrentMetadataFrameValues_t)	SLACurrentMetadataFrameValues_t
offsetCornerLat2 (defined in SLACurrentMetadataFrameValues_t)	SLACurrentMetadataFrameValues_t
offsetCornerLat3 (defined in SLACurrentMetadataFrameValues_t)	SLACurrentMetadataFrameValues_t
offsetCornerLat4 (defined in SLACurrentMetadataFrameValues_t)	SLACurrentMetadataFrameValues_t
offsetCornerLon1 (defined in SLACurrentMetadataFrameValues_t)	SLACurrentMetadataFrameValues_t
offsetCornerLon2 (defined in SLACurrentMetadataFrameValues_t)	SLACurrentMetadataFrameValues_t
offsetCornerLon3 (defined in SLACurrentMetadataFrameValues_t)	SLACurrentMetadataFrameValues_t
offsetCornerLon4 (defined in SLACurrentMetadataFrameValues_t)	SLACurrentMetadataFrameValues_t
slantRange (defined in SLACurrentMetadataFrameValues_t)	SLACurrentMetadataFrameValues_t
targetEI (defined in SLACurrentMetadataFrameValues_t)	SLACurrentMetadataFrameValues_t
targetLat (defined in SLACurrentMetadataFrameValues_t)	SLACurrentMetadataFrameValues_t
targetLon (defined in SLACurrentMetadataFrameValues_t)	SLACurrentMetadataFrameValues_t
targetTrackGateHeight (defined in SLACurrentMetadataFrameValues_t)	SLACurrentMetadataFrameValues_t
targetTrackGateWidth (defined in SLACurrentMetadataFrameValues_t)	SLACurrentMetadataFrameValues_t
userSuppliedFlags (defined in SLACurrentMetadataFrameValues_t)	SLACurrentMetadataFrameValues_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetMetadataFrameValues_t Member List

This is the complete list of members for [SLASetMetadataFrameValues_t](#), including all inherited members.

displayId (defined in SLASetMetadataFrameValues_t)	SLASetMetadataFrameValues_t
frameCenterEI (defined in SLASetMetadataFrameValues_t)	SLASetMetadataFrameValues_t
frameCenterLat (defined in SLASetMetadataFrameValues_t)	SLASetMetadataFrameValues_t
frameCenterLon (defined in SLASetMetadataFrameValues_t)	SLASetMetadataFrameValues_t
frameWidth (defined in SLASetMetadataFrameValues_t)	SLASetMetadataFrameValues_t
offsetCornerLat1 (defined in SLASetMetadataFrameValues_t)	SLASetMetadataFrameValues_t
offsetCornerLat2 (defined in SLASetMetadataFrameValues_t)	SLASetMetadataFrameValues_t
offsetCornerLat3 (defined in SLASetMetadataFrameValues_t)	SLASetMetadataFrameValues_t
offsetCornerLat4 (defined in SLASetMetadataFrameValues_t)	SLASetMetadataFrameValues_t
offsetCornerLon1 (defined in SLASetMetadataFrameValues_t)	SLASetMetadataFrameValues_t
offsetCornerLon2 (defined in SLASetMetadataFrameValues_t)	SLASetMetadataFrameValues_t
offsetCornerLon3 (defined in SLASetMetadataFrameValues_t)	SLASetMetadataFrameValues_t
offsetCornerLon4 (defined in SLASetMetadataFrameValues_t)	SLASetMetadataFrameValues_t
slantRange (defined in SLASetMetadataFrameValues_t)	SLASetMetadataFrameValues_t
targetEI (defined in SLASetMetadataFrameValues_t)	SLASetMetadataFrameValues_t
targetLat (defined in SLASetMetadataFrameValues_t)	SLASetMetadataFrameValues_t
targetLon (defined in SLASetMetadataFrameValues_t)	SLASetMetadataFrameValues_t
targetTrackGateHeight (defined in SLASetMetadataFrameValues_t)	SLASetMetadataFrameValues_t
targetTrackGateWidth (defined in SLASetMetadataFrameValues_t)	SLASetMetadataFrameValues_t
userSuppliedFlags (defined in SLASetMetadataFrameValues_t)	SLASetMetadataFrameValues_t
validDataMask (defined in SLASetMetadataFrameValues_t)	SLASetMetadataFrameValues_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentMetadataRate_t Member List

This is the complete list of members for [SLACurrentMetadataRate_t](#), including all inherited members.

displayId (defined in SLACurrentMetadataRate_t)	SLACurrentMetadataRate_t
frameStep (defined in SLACurrentMetadataRate_t)	SLACurrentMetadataRate_t
index (defined in SLACurrentMetadataRate_t)	SLACurrentMetadataRate_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetMetadataRate_t Member List

This is the complete list of members for [SLASetMetadataRate_t](#), including all inherited members.

displayId (defined in SLASetMetadataRate_t)	SLASetMetadataRate_t
enables (defined in SLASetMetadataRate_t)	SLASetMetadataRate_t
frameStep (defined in SLASetMetadataRate_t)	SLASetMetadataRate_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentMetadataValues_t Member List

This is the complete list of members for [SLACurrentMetadataValues_t](#), including all inherited members.

alt (defined in SLACurrentMetadataValues_t)	SLACurrentMetadataValues_t
az (defined in SLACurrentMetadataValues_t)	SLACurrentMetadataValues_t
displayId (defined in SLACurrentMetadataValues_t)	SLACurrentMetadataValues_t
el (defined in SLACurrentMetadataValues_t)	SLACurrentMetadataValues_t
heading (defined in SLACurrentMetadataValues_t)	SLACurrentMetadataValues_t
hfov (defined in SLACurrentMetadataValues_t)	SLACurrentMetadataValues_t
lat (defined in SLACurrentMetadataValues_t)	SLACurrentMetadataValues_t
lon (defined in SLACurrentMetadataValues_t)	SLACurrentMetadataValues_t
pitch (defined in SLACurrentMetadataValues_t)	SLACurrentMetadataValues_t
roll (defined in SLACurrentMetadataValues_t)	SLACurrentMetadataValues_t
sensorRoll (defined in SLACurrentMetadataValues_t)	SLACurrentMetadataValues_t
utcTime (defined in SLACurrentMetadataValues_t)	SLACurrentMetadataValues_t
vfov (defined in SLACurrentMetadataValues_t)	SLACurrentMetadataValues_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetMetadataValues_t Member List

This is the complete list of members for [SLASetMetadataValues_t](#), including all inherited members.

alt (defined in SLASetMetadataValues_t)	SLASetMetadataValues_t
az (defined in SLASetMetadataValues_t)	SLASetMetadataValues_t
displayId (defined in SLASetMetadataValues_t)	SLASetMetadataValues_t
el (defined in SLASetMetadataValues_t)	SLASetMetadataValues_t
heading (defined in SLASetMetadataValues_t)	SLASetMetadataValues_t
hfov (defined in SLASetMetadataValues_t)	SLASetMetadataValues_t
lat (defined in SLASetMetadataValues_t)	SLASetMetadataValues_t
lon (defined in SLASetMetadataValues_t)	SLASetMetadataValues_t
pitch (defined in SLASetMetadataValues_t)	SLASetMetadataValues_t
roll (defined in SLASetMetadataValues_t)	SLASetMetadataValues_t
sensorRoll (defined in SLASetMetadataValues_t)	SLASetMetadataValues_t
utcTime (defined in SLASetMetadataValues_t)	SLASetMetadataValues_t
validDataMask (defined in SLASetMetadataValues_t)	SLASetMetadataValues_t
vfov (defined in SLASetMetadataValues_t)	SLASetMetadataValues_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentMultipleAlignment_t Member List

This is the complete list of members for [SLACurrentMultipleAlignment_t](#), including all inherited members.

alignment (defined in SLACurrentMultipleAlignment_t)	SLACurrentMultipleAlignment_t
horizontal (defined in SLACurrentMultipleAlignment_t)	SLACurrentMultipleAlignment_t
hzoom (defined in SLACurrentMultipleAlignment_t)	SLACurrentMultipleAlignment_t
nAlignments (defined in SLACurrentMultipleAlignment_t)	SLACurrentMultipleAlignment_t
rotate (defined in SLACurrentMultipleAlignment_t)	SLACurrentMultipleAlignment_t
vertical (defined in SLACurrentMultipleAlignment_t)	SLACurrentMultipleAlignment_t
zoom (defined in SLACurrentMultipleAlignment_t)	SLACurrentMultipleAlignment_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentNetworkList_t Member List

This is the complete list of members for [SLACurrentNetworkList_t](#), including all inherited members.

iface (defined in SLACurrentNetworkList_t)	SLACurrentNetworkList_t
iface (defined in SLACurrentNetworkList_t)	SLACurrentNetworkList_t
numIds (defined in SLACurrentNetworkList_t)	SLACurrentNetworkList_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentNetworkParameters_t Member List

This is the complete list of members for [SLACurrentNetworkParameters_t](#), including all inherited members.

c2replyPort (defined in SLACurrentNetworkParameters_t)	SLACurrentNetworkParameters_t
gateway (defined in SLACurrentNetworkParameters_t)	SLACurrentNetworkParameters_t
hostName (defined in SLACurrentNetworkParameters_t)	SLACurrentNetworkParameters_t
index (defined in SLACurrentNetworkParameters_t)	SLACurrentNetworkParameters_t
ipAddr (defined in SLACurrentNetworkParameters_t)	SLACurrentNetworkParameters_t
mode (defined in SLACurrentNetworkParameters_t)	SLACurrentNetworkParameters_t
modes (defined in SLACurrentNetworkParameters_t)	SLACurrentNetworkParameters_t
subnet (defined in SLACurrentNetworkParameters_t)	SLACurrentNetworkParameters_t
telemetryListenPort (defined in SLACurrentNetworkParameters_t)	SLACurrentNetworkParameters_t
telemetryListenPort2 (defined in SLACurrentNetworkParameters_t)	SLACurrentNetworkParameters_t
telemetryReplyPort (defined in SLACurrentNetworkParameters_t)	SLACurrentNetworkParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentOverlayMode_t Member List

This is the complete list of members for [SLACurrentOverlayMode_t](#), including all inherited members.

cameraIndex (defined in SLACurrentOverlayMode_t)	SLACurrentOverlayMode_t
cursorReticle (defined in SLACurrentOverlayMode_t)	SLACurrentOverlayMode_t
detCounterColor (defined in SLACurrentOverlayMode_t)	SLACurrentOverlayMode_t
detCounterPos (defined in SLACurrentOverlayMode_t)	SLACurrentOverlayMode_t
extraLabels (defined in SLACurrentOverlayMode_t)	SLACurrentOverlayMode_t
fontID (defined in SLACurrentOverlayMode_t)	SLACurrentOverlayMode_t
fontScale (defined in SLACurrentOverlayMode_t)	SLACurrentOverlayMode_t
graphics (defined in SLACurrentOverlayMode_t)	SLACurrentOverlayMode_t
lineThickness (defined in SLACurrentOverlayMode_t)	SLACurrentOverlayMode_t
modernMode (defined in SLACurrentOverlayMode_t)	SLACurrentOverlayMode_t
mtiColor (defined in SLACurrentOverlayMode_t)	SLACurrentOverlayMode_t
mtiDisplayLabelLimit (defined in SLACurrentOverlayMode_t)	SLACurrentOverlayMode_t
mtiLabelAdv (defined in SLACurrentOverlayMode_t)	SLACurrentOverlayMode_t
mtiLabelDeconflict (defined in SLACurrentOverlayMode_t)	SLACurrentOverlayMode_t
mtiMultiColor (defined in SLACurrentOverlayMode_t)	SLACurrentOverlayMode_t
mtiReticle (defined in SLACurrentOverlayMode_t)	SLACurrentOverlayMode_t
mtiSelectableColor (defined in SLACurrentOverlayMode_t)	SLACurrentOverlayMode_t
personReticle (defined in SLACurrentOverlayMode_t)	SLACurrentOverlayMode_t
primaryReticle (defined in SLACurrentOverlayMode_t)	SLACurrentOverlayMode_t
secondaryReticle (defined in SLACurrentOverlayMode_t)	SLACurrentOverlayMode_t
selectedReticle (defined in SLACurrentOverlayMode_t)	SLACurrentOverlayMode_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetOverlayMode_t Member List

This is the complete list of members for [SLASetOverlayMode_t](#), including all inherited members.

cameraIndex (defined in SLASetOverlayMode_t)	SLASetOverlayMode_t
cursorReticle (defined in SLASetOverlayMode_t)	SLASetOverlayMode_t
detCounterColor (defined in SLASetOverlayMode_t)	SLASetOverlayMode_t
detCounterPos (defined in SLASetOverlayMode_t)	SLASetOverlayMode_t
extraLabels (defined in SLASetOverlayMode_t)	SLASetOverlayMode_t
fontID (defined in SLASetOverlayMode_t)	SLASetOverlayMode_t
fontScale (defined in SLASetOverlayMode_t)	SLASetOverlayMode_t
graphics (defined in SLASetOverlayMode_t)	SLASetOverlayMode_t
lineThickness (defined in SLASetOverlayMode_t)	SLASetOverlayMode_t
modernMode (defined in SLASetOverlayMode_t)	SLASetOverlayMode_t
mtiColor (defined in SLASetOverlayMode_t)	SLASetOverlayMode_t
mtiDisplayLabelLimit (defined in SLASetOverlayMode_t)	SLASetOverlayMode_t
mtiLabelAdv (defined in SLASetOverlayMode_t)	SLASetOverlayMode_t
mtiLabelDeconflict (defined in SLASetOverlayMode_t)	SLASetOverlayMode_t
mtiMultiColor (defined in SLASetOverlayMode_t)	SLASetOverlayMode_t
mtiReticle (defined in SLASetOverlayMode_t)	SLASetOverlayMode_t
mtiSelectableColor (defined in SLASetOverlayMode_t)	SLASetOverlayMode_t
personReticle (defined in SLASetOverlayMode_t)	SLASetOverlayMode_t
primaryReticle (defined in SLASetOverlayMode_t)	SLASetOverlayMode_t
secondaryReticle (defined in SLASetOverlayMode_t)	SLASetOverlayMode_t
selectedReticle (defined in SLASetOverlayMode_t)	SLASetOverlayMode_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentOverlayObjectParameters_t Member List

This is the complete list of members for [SLACurrentOverlayObjectParameters_t](#), including all inherited members.

a (defined in SLACurrentOverlayObjectParameters_t)	SLACurrentOverlayObjectParameters_t
b (defined in SLACurrentOverlayObjectParameters_t)	SLACurrentOverlayObjectParameters_t
c (defined in SLACurrentOverlayObjectParameters_t)	SLACurrentOverlayObjectParameters_t
color (defined in SLACurrentOverlayObjectParameters_t)	SLACurrentOverlayObjectParameters_t
d (defined in SLACurrentOverlayObjectParameters_t)	SLACurrentOverlayObjectParameters_t
flags (defined in SLACurrentOverlayObjectParameters_t)	SLACurrentOverlayObjectParameters_t
id (defined in SLACurrentOverlayObjectParameters_t)	SLACurrentOverlayObjectParameters_t
staticObject (defined in SLACurrentOverlayObjectParameters_t)	SLACurrentOverlayObjectParameters_t
text (defined in SLACurrentOverlayObjectParameters_t)	SLACurrentOverlayObjectParameters_t
type (defined in SLACurrentOverlayObjectParameters_t)	SLACurrentOverlayObjectParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentOverlayObjectsIds_t Member List

This is the complete list of members for [SLACurrentOverlayObjectsIds_t](#), including all inherited members.

id0 (defined in SLACurrentOverlayObjectsIds_t)	SLACurrentOverlayObjectsIds_t
id128 (defined in SLACurrentOverlayObjectsIds_t)	SLACurrentOverlayObjectsIds_t
id192 (defined in SLACurrentOverlayObjectsIds_t)	SLACurrentOverlayObjectsIds_t
id64 (defined in SLACurrentOverlayObjectsIds_t)	SLACurrentOverlayObjectsIds_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentPortConfiguration_t Member List

This is the complete list of members for [SLACurrentPortConfiguration_t](#), including all inherited members.

baud (defined in SLACurrentPortConfiguration_t)	SLACurrentPortConfiguration_t
dataBits (defined in SLACurrentPortConfiguration_t)	SLACurrentPortConfiguration_t
inputPort (defined in SLACurrentPortConfiguration_t)	SLACurrentPortConfiguration_t
maxDelay (defined in SLACurrentPortConfiguration_t)	SLACurrentPortConfiguration_t
maxPacket (defined in SLACurrentPortConfiguration_t)	SLACurrentPortConfiguration_t
parity (defined in SLACurrentPortConfiguration_t)	SLACurrentPortConfiguration_t
port (defined in SLACurrentPortConfiguration_t)	SLACurrentPortConfiguration_t
protocol (defined in SLACurrentPortConfiguration_t)	SLACurrentPortConfiguration_t
stopBits (defined in SLACurrentPortConfiguration_t)	SLACurrentPortConfiguration_t
udpAttNavPort (defined in SLACurrentPortConfiguration_t)	SLACurrentPortConfiguration_t
udpDestinationAddr (defined in SLACurrentPortConfiguration_t)	SLACurrentPortConfiguration_t
udpDestinationPort (defined in SLACurrentPortConfiguration_t)	SLACurrentPortConfiguration_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentRegistrationParameters_t Member List

This is the complete list of members for [SLACurrentRegistrationParameters_t](#), including all inherited members.

bottom (defined in SLACurrentRegistrationParameters_t)	SLACurrentRegistrationParameters_t
cameraIndex (defined in SLACurrentRegistrationParameters_t)	SLACurrentRegistrationParameters_t
left (defined in SLACurrentRegistrationParameters_t)	SLACurrentRegistrationParameters_t
maxRotation (defined in SLACurrentRegistrationParameters_t)	SLACurrentRegistrationParameters_t
maxTranslation (defined in SLACurrentRegistrationParameters_t)	SLACurrentRegistrationParameters_t
right (defined in SLACurrentRegistrationParameters_t)	SLACurrentRegistrationParameters_t
top (defined in SLACurrentRegistrationParameters_t)	SLACurrentRegistrationParameters_t
zoomRange (defined in SLACurrentRegistrationParameters_t)	SLACurrentRegistrationParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLARegistrationParameters_t Member List

This is the complete list of members for [SLARegistrationParameters_t](#), including all inherited members.

bottom (defined in SLARegistrationParameters_t)	SLARegistrationParameters_t
cameraIndex (defined in SLARegistrationParameters_t)	SLARegistrationParameters_t
flags (defined in SLARegistrationParameters_t)	SLARegistrationParameters_t
left (defined in SLARegistrationParameters_t)	SLARegistrationParameters_t
maxRotation (defined in SLARegistrationParameters_t)	SLARegistrationParameters_t
maxTranslation (defined in SLARegistrationParameters_t)	SLARegistrationParameters_t
right (defined in SLARegistrationParameters_t)	SLARegistrationParameters_t
top (defined in SLARegistrationParameters_t)	SLARegistrationParameters_t
updateRate (defined in SLARegistrationParameters_t)	SLARegistrationParameters_t
zoomRange (defined in SLARegistrationParameters_t)	SLARegistrationParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentSDCardDirectoryInfo_t Member List

This is the complete list of members for [SLACurrentSDCardDirectoryInfo_t](#), including all inherited members.

endIndex (defined in SLACurrentSDCardDirectoryInfo_t)	SLACurrentSDCardDirectoryInfo_t
nFiles (defined in SLACurrentSDCardDirectoryInfo_t)	SLACurrentSDCardDirectoryInfo_t
startIndex (defined in SLACurrentSDCardDirectoryInfo_t)	SLACurrentSDCardDirectoryInfo_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentSnapShot_t Member List

This is the complete list of members for [SLACurrentSnapShot_t](#), including all inherited members.

cameraIndex (defined in SLACurrentSnapShot_t)	SLACurrentSnapShot_t
downsample (defined in SLACurrentSnapShot_t)	SLACurrentSnapShot_t
fileName (defined in SLACurrentSnapShot_t)	SLACurrentSnapShot_t
format (defined in SLACurrentSnapShot_t)	SLACurrentSnapShot_t
frameStep (defined in SLACurrentSnapShot_t)	SLACurrentSnapShot_t
ipAddr (defined in SLACurrentSnapShot_t)	SLACurrentSnapShot_t
mode (defined in SLACurrentSnapShot_t)	SLACurrentSnapShot_t
numFrames (defined in SLACurrentSnapShot_t)	SLACurrentSnapShot_t
password (defined in SLACurrentSnapShot_t)	SLACurrentSnapShot_t
port (defined in SLACurrentSnapShot_t)	SLACurrentSnapShot_t
quality (defined in SLACurrentSnapShot_t)	SLACurrentSnapShot_t
secondaryFormat (defined in SLACurrentSnapShot_t)	SLACurrentSnapShot_t
source (defined in SLACurrentSnapShot_t)	SLACurrentSnapShot_t
userName (defined in SLACurrentSnapShot_t)	SLACurrentSnapShot_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentStabilizationBias_t Member List

This is the complete list of members for [SLACurrentStabilizationBias_t](#), including all inherited members.

autoBias (defined in SLACurrentStabilizationBias_t)	SLACurrentStabilizationBias_t
biasCol (defined in SLACurrentStabilizationBias_t)	SLACurrentStabilizationBias_t
biasRow (defined in SLACurrentStabilizationBias_t)	SLACurrentStabilizationBias_t
cameraIndex (defined in SLACurrentStabilizationBias_t)	SLACurrentStabilizationBias_t
updateRate (defined in SLACurrentStabilizationBias_t)	SLACurrentStabilizationBias_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAStabilizationBias_t Member List

This is the complete list of members for [SLAStabilizationBias_t](#), including all inherited members.

autoBias (defined in SLAStabilizationBias_t)	SLAStabilizationBias_t
biasCol (defined in SLAStabilizationBias_t)	SLAStabilizationBias_t
biasRow (defined in SLAStabilizationBias_t)	SLAStabilizationBias_t
cameraIndex (defined in SLAStabilizationBias_t)	SLAStabilizationBias_t
updateRate (defined in SLAStabilizationBias_t)	SLAStabilizationBias_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentStabilizationParameters_t Member List

This is the complete list of members for [SLACurrentStabilizationParameters_t](#), including all inherited members.

angleLimit (defined in SLACurrentStabilizationParameters_t)	SLACurrentStabilizationParameters_t
cameraIndex (defined in SLACurrentStabilizationParameters_t)	SLACurrentStabilizationParameters_t
edgeU (defined in SLACurrentStabilizationParameters_t)	SLACurrentStabilizationParameters_t
edgeV (defined in SLACurrentStabilizationParameters_t)	SLACurrentStabilizationParameters_t
edgeY (defined in SLACurrentStabilizationParameters_t)	SLACurrentStabilizationParameters_t
maxStabOff (defined in SLACurrentStabilizationParameters_t)	SLACurrentStabilizationParameters_t
mode (defined in SLACurrentStabilizationParameters_t)	SLACurrentStabilizationParameters_t
rate (defined in SLACurrentStabilizationParameters_t)	SLACurrentStabilizationParameters_t
translationLimit (defined in SLACurrentStabilizationParameters_t)	SLACurrentStabilizationParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentSystemValue_t Member List

This is the complete list of members for [SLACurrentSystemValue_t](#), including all inherited members.

setMode (defined in SLACurrentSystemValue_t)	SLACurrentSystemValue_t
systemValue (defined in SLACurrentSystemValue_t)	SLACurrentSystemValue_t
value0 (defined in SLACurrentSystemValue_t)	SLACurrentSystemValue_t
value1 (defined in SLACurrentSystemValue_t)	SLACurrentSystemValue_t
value2 (defined in SLACurrentSystemValue_t)	SLACurrentSystemValue_t
value3 (defined in SLACurrentSystemValue_t)	SLACurrentSystemValue_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetSystemValue_t Member List

This is the complete list of members for [SLASetSystemValue_t](#), including all inherited members.

setMode (defined in SLASetSystemValue_t)	SLASetSystemValue_t
systemValue (defined in SLASetSystemValue_t)	SLASetSystemValue_t
value0 (defined in SLASetSystemValue_t)	SLASetSystemValue_t
value1 (defined in SLASetSystemValue_t)	SLASetSystemValue_t
value2 (defined in SLASetSystemValue_t)	SLASetSystemValue_t
value3 (defined in SLASetSystemValue_t)	SLASetSystemValue_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentTrackingParameters_t Member List

This is the complete list of members for [SLACurrentTrackingParameters_t](#), including all inherited members.

cameraIndex (defined in SLACurrentTrackingParameters_t)	SLACurrentTrackingParameters_t
maxMisses (defined in SLACurrentTrackingParameters_t)	SLACurrentTrackingParameters_t
maxMissesExt (defined in SLACurrentTrackingParameters_t)	SLACurrentTrackingParameters_t
maxPauseTime (defined in SLACurrentTrackingParameters_t)	SLACurrentTrackingParameters_t
maxTracks (defined in SLACurrentTrackingParameters_t)	SLACurrentTrackingParameters_t
mode (defined in SLACurrentTrackingParameters_t)	SLACurrentTrackingParameters_t
mode2 (defined in SLACurrentTrackingParameters_t)	SLACurrentTrackingParameters_t
modeFlags (defined in SLACurrentTrackingParameters_t)	SLACurrentTrackingParameters_t
nearVal (defined in SLACurrentTrackingParameters_t)	SLACurrentTrackingParameters_t
objectHeight (defined in SLACurrentTrackingParameters_t)	SLACurrentTrackingParameters_t
objectSize (defined in SLACurrentTrackingParameters_t)	SLACurrentTrackingParameters_t
rollSmoothing (defined in SLACurrentTrackingParameters_t)	SLACurrentTrackingParameters_t
zoomSmoothing (defined in SLACurrentTrackingParameters_t)	SLACurrentTrackingParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetTrackingParameters_t Member List

This is the complete list of members for [SLASetTrackingParameters_t](#), including all inherited members.

cameraIndex (defined in SLASetTrackingParameters_t)	SLASetTrackingParameters_t
maxMisses (defined in SLASetTrackingParameters_t)	SLASetTrackingParameters_t
maxMissesExt (defined in SLASetTrackingParameters_t)	SLASetTrackingParameters_t
maxPauseTime (defined in SLASetTrackingParameters_t)	SLASetTrackingParameters_t
maxTracks (defined in SLASetTrackingParameters_t)	SLASetTrackingParameters_t
mode (defined in SLASetTrackingParameters_t)	SLASetTrackingParameters_t
mode2 (defined in SLASetTrackingParameters_t)	SLASetTrackingParameters_t
modeFlags (defined in SLASetTrackingParameters_t)	SLASetTrackingParameters_t
nearVal (defined in SLASetTrackingParameters_t)	SLASetTrackingParameters_t
objectHeight (defined in SLASetTrackingParameters_t)	SLASetTrackingParameters_t
objectSize (defined in SLASetTrackingParameters_t)	SLASetTrackingParameters_t
rollSmoothing (defined in SLASetTrackingParameters_t)	SLASetTrackingParameters_t
zoomSmoothing (defined in SLASetTrackingParameters_t)	SLASetTrackingParameters_t

Sightline Intelligence Command, Control, and Script API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) ■ [Camera Configuration](#)

SLACurrentVideoEnhancementParameters_t Member List

This is the complete list of members for [SLACurrentVideoEnhancementParameters_t](#), including all inherited members.

alphaBlend (defined in SLACurrentVideoEnhancementParameters_t)	SLACurrentVideoEnhancementParameters_t
brightness (defined in SLACurrentVideoEnhancementParameters_t)	SLACurrentVideoEnhancementParameters_t
cameraIndex (defined in SLACurrentVideoEnhancementParameters_t)	SLACurrentVideoEnhancementParameters_t
colorEnhance (defined in SLACurrentVideoEnhancementParameters_t)	SLACurrentVideoEnhancementParameters_t
contrast (defined in SLACurrentVideoEnhancementParameters_t)	SLACurrentVideoEnhancementParameters_t
customKernel (defined in SLACurrentVideoEnhancementParameters_t)	SLACurrentVideoEnhancementParameters_t
deconvSigma (defined in SLACurrentVideoEnhancementParameters_t)	SLACurrentVideoEnhancementParameters_t
denoiseRate (defined in SLACurrentVideoEnhancementParameters_t)	SLACurrentVideoEnhancementParameters_t
enhanceParam (defined in SLACurrentVideoEnhancementParameters_t)	SLACurrentVideoEnhancementParameters_t
flags (defined in SLACurrentVideoEnhancementParameters_t)	SLACurrentVideoEnhancementParameters_t
gaussianBlur (defined in SLACurrentVideoEnhancementParameters_t)	SLACurrentVideoEnhancementParameters_t
histAveRate8 (defined in SLACurrentVideoEnhancementParameters_t)	SLACurrentVideoEnhancementParameters_t
histMaxPctBin (defined in SLACurrentVideoEnhancementParameters_t)	SLACurrentVideoEnhancementParameters_t
lapMinDiff (defined in SLACurrentVideoEnhancementParameters_t)	SLACurrentVideoEnhancementParameters_t
normalizeKernel (defined in SLACurrentVideoEnhancementParameters_t)	SLACurrentVideoEnhancementParameters_t
roiCol (defined in SLACurrentVideoEnhancementParameters_t)	SLACurrentVideoEnhancementParameters_t
roiHigh (defined in SLACurrentVideoEnhancementParameters_t)	SLACurrentVideoEnhancementParameters_t
roiRow (defined in SLACurrentVideoEnhancementParameters_t)	SLACurrentVideoEnhancementParameters_t
roiWide (defined in SLACurrentVideoEnhancementParameters_t)	SLACurrentVideoEnhancementParameters_t
scintillationMode (defined in SLACurrentVideoEnhancementParameters_t)	SLACurrentVideoEnhancementParameters_t
sharpenEnhance (defined in SLACurrentVideoEnhancementParameters_t)	SLACurrentVideoEnhancementParameters_t
sharpenRadius (defined in SLACurrentVideoEnhancementParameters_t)	SLACurrentVideoEnhancementParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetVideoEnhancementParameters_t Member List

This is the complete list of members for [SLASetVideoEnhancementParameters_t](#), including all inherited members.

alphaBlend (defined in SLASetVideoEnhancementParameters_t)	SLASetVideoEnhancementParameters_t
brightness (defined in SLASetVideoEnhancementParameters_t)	SLASetVideoEnhancementParameters_t
cameraIndex (defined in SLASetVideoEnhancementParameters_t)	SLASetVideoEnhancementParameters_t
colorEnhance (defined in SLASetVideoEnhancementParameters_t)	SLASetVideoEnhancementParameters_t
contrast (defined in SLASetVideoEnhancementParameters_t)	SLASetVideoEnhancementParameters_t
customKernel (defined in SLASetVideoEnhancementParameters_t)	SLASetVideoEnhancementParameters_t
deconvSigma (defined in SLASetVideoEnhancementParameters_t)	SLASetVideoEnhancementParameters_t
denoiseRate (defined in SLASetVideoEnhancementParameters_t)	SLASetVideoEnhancementParameters_t
enhanceParam (defined in SLASetVideoEnhancementParameters_t)	SLASetVideoEnhancementParameters_t
flags (defined in SLASetVideoEnhancementParameters_t)	SLASetVideoEnhancementParameters_t
gaussianBlur (defined in SLASetVideoEnhancementParameters_t)	SLASetVideoEnhancementParameters_t
histAveRate8 (defined in SLASetVideoEnhancementParameters_t)	SLASetVideoEnhancementParameters_t
histMaxPctBin (defined in SLASetVideoEnhancementParameters_t)	SLASetVideoEnhancementParameters_t
lapMinDiff (defined in SLASetVideoEnhancementParameters_t)	SLASetVideoEnhancementParameters_t
normalizeKernel (defined in SLASetVideoEnhancementParameters_t)	SLASetVideoEnhancementParameters_t
roiCol (defined in SLASetVideoEnhancementParameters_t)	SLASetVideoEnhancementParameters_t
roiHigh (defined in SLASetVideoEnhancementParameters_t)	SLASetVideoEnhancementParameters_t
roiRow (defined in SLASetVideoEnhancementParameters_t)	SLASetVideoEnhancementParameters_t
roiWide (defined in SLASetVideoEnhancementParameters_t)	SLASetVideoEnhancementParameters_t
scintillationMode (defined in SLASetVideoEnhancementParameters_t)	SLASetVideoEnhancementParameters_t
sharpenEnhance (defined in SLASetVideoEnhancementParameters_t)	SLASetVideoEnhancementParameters_t
sharpenRadius (defined in SLASetVideoEnhancementParameters_t)	SLASetVideoEnhancementParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentVideoModeParameters_t Member List

This is the complete list of members for [SLACurrentVideoModeParameters_t](#), including all inherited members.

analogDest (defined in SLACurrentVideoModeParameters_t)	SLACurrentVideoModeParameters_t
analogRes (defined in SLACurrentVideoModeParameters_t)	SLACurrentVideoModeParameters_t
cameraIndex0 (defined in SLACurrentVideoModeParameters_t)	SLACurrentVideoModeParameters_t
cameraIndex1 (defined in SLACurrentVideoModeParameters_t)	SLACurrentVideoModeParameters_t
cameraIndex2 (defined in SLACurrentVideoModeParameters_t)	SLACurrentVideoModeParameters_t
displayDest (defined in SLACurrentVideoModeParameters_t)	SLACurrentVideoModeParameters_t
displayIndex0 (defined in SLACurrentVideoModeParameters_t)	SLACurrentVideoModeParameters_t
displayIndex1 (defined in SLACurrentVideoModeParameters_t)	SLACurrentVideoModeParameters_t
displayIndex2 (defined in SLACurrentVideoModeParameters_t)	SLACurrentVideoModeParameters_t
displayIndex3 (defined in SLACurrentVideoModeParameters_t)	SLACurrentVideoModeParameters_t
displayMode (defined in SLACurrentVideoModeParameters_t)	SLACurrentVideoModeParameters_t
hdmiDest (defined in SLACurrentVideoModeParameters_t)	SLACurrentVideoModeParameters_t
hdmiRes (defined in SLACurrentVideoModeParameters_t)	SLACurrentVideoModeParameters_t
HDSDIDest (defined in SLACurrentVideoModeParameters_t)	SLACurrentVideoModeParameters_t
hdsdiRes (defined in SLACurrentVideoModeParameters_t)	SLACurrentVideoModeParameters_t
nCameras (defined in SLACurrentVideoModeParameters_t)	SLACurrentVideoModeParameters_t
nDisplays (defined in SLACurrentVideoModeParameters_t)	SLACurrentVideoModeParameters_t
net0Dest (defined in SLACurrentVideoModeParameters_t)	SLACurrentVideoModeParameters_t
net1Dest (defined in SLACurrentVideoModeParameters_t)	SLACurrentVideoModeParameters_t
pipColor (defined in SLACurrentVideoModeParameters_t)	SLACurrentVideoModeParameters_t
pipQuadrant (defined in SLACurrentVideoModeParameters_t)	SLACurrentVideoModeParameters_t
pipScale (defined in SLACurrentVideoModeParameters_t)	SLACurrentVideoModeParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAVideoDisplay_t Member List

This is the complete list of members for [SLAVideoDisplay_t](#), including all inherited members.

destination (defined in SLAVideoDisplay_t)	SLAVideoDisplay_t
flags (defined in SLAVideoDisplay_t)	SLAVideoDisplay_t
resolution (defined in SLAVideoDisplay_t)	SLAVideoDisplay_t
source (defined in SLAVideoDisplay_t)	SLAVideoDisplay_t
streamIdx (defined in SLAVideoDisplay_t)	SLAVideoDisplay_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAMultiDisplay_t Member List

This is the complete list of members for [SLAMultiDisplay_t](#), including all inherited members.

cameraIndex0 (defined in SLAMultiDisplay_t)	SLAMultiDisplay_t
cameraIndex1 (defined in SLAMultiDisplay_t)	SLAMultiDisplay_t
displayMode (defined in SLAMultiDisplay_t)	SLAMultiDisplay_t
multiPipZoom (defined in SLAMultiDisplay_t)	SLAMultiDisplay_t
pipColor (defined in SLAMultiDisplay_t)	SLAMultiDisplay_t
pipLabel (defined in SLAMultiDisplay_t)	SLAMultiDisplay_t
pipQuadrant (defined in SLAMultiDisplay_t)	SLAMultiDisplay_t
pipScale (defined in SLAMultiDisplay_t)	SLAMultiDisplay_t
reserved1 (defined in SLAMultiDisplay_t)	SLAMultiDisplay_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACurrentVideoParameters_t Member List

This is the complete list of members for [SLACurrentVideoParameters_t](#), including all inherited members.

autoChop (defined in SLACurrentVideoParameters_t)	SLACurrentVideoParameters_t
autoReset (defined in SLACurrentVideoParameters_t)	SLACurrentVideoParameters_t
cameraIndex (defined in SLACurrentVideoParameters_t)	SLACurrentVideoParameters_t
chopBottom (defined in SLACurrentVideoParameters_t)	SLACurrentVideoParameters_t
chopLeft (defined in SLACurrentVideoParameters_t)	SLACurrentVideoParameters_t
chopRight (defined in SLACurrentVideoParameters_t)	SLACurrentVideoParameters_t
chopTop (defined in SLACurrentVideoParameters_t)	SLACurrentVideoParameters_t
deinterlace (defined in SLACurrentVideoParameters_t)	SLACurrentVideoParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetVideoParameters_t Member List

This is the complete list of members for [SLASetVideoParameters_t](#), including all inherited members.

autoChop (defined in SLASetVideoParameters_t)	SLASetVideoParameters_t
autoReset (defined in SLASetVideoParameters_t)	SLASetVideoParameters_t
cameraIndex (defined in SLASetVideoParameters_t)	SLASetVideoParameters_t
chopBottom (defined in SLASetVideoParameters_t)	SLASetVideoParameters_t
chopLeft (defined in SLASetVideoParameters_t)	SLASetVideoParameters_t
chopRight (defined in SLASetVideoParameters_t)	SLASetVideoParameters_t
chopTop (defined in SLASetVideoParameters_t)	SLASetVideoParameters_t
deinterlace (defined in SLASetVideoParameters_t)	SLASetVideoParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACursorOnTarget_t Member List

This is the complete list of members for [SLACursorOnTarget_t](#), including all inherited members.

displayId (defined in SLACursorOnTarget_t)	SLACursorOnTarget_t
ipAddr (defined in SLACursorOnTarget_t)	SLACursorOnTarget_t
mode (defined in SLACursorOnTarget_t)	SLACursorOnTarget_t
port (defined in SLACursorOnTarget_t)	SLACursorOnTarget_t
rate (defined in SLACursorOnTarget_t)	SLACursorOnTarget_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACustomAIDetect_t Member List

This is the complete list of members for [SLACustomAIDetect_t](#), including all inherited members.

cameraIndex (defined in SLACustomAIDetect_t)	SLACustomAIDetect_t
fileName (defined in SLACustomAIDetect_t)	SLACustomAIDetect_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACustomAutoFocusParameters_t Member List

This is the complete list of members for [SLACustomAutoFocusParameters_t](#), including all inherited members.

calibEnd (defined in SLACustomAutoFocusParameters_t)	SLACustomAutoFocusParameters_t
calibFlags (defined in SLACustomAutoFocusParameters_t)	SLACustomAutoFocusParameters_t
calibStart (defined in SLACustomAutoFocusParameters_t)	SLACustomAutoFocusParameters_t
cameraldx (defined in SLACustomAutoFocusParameters_t)	SLACustomAutoFocusParameters_t
CdownMax (defined in SLACustomAutoFocusParameters_t)	SLACustomAutoFocusParameters_t
changePercent10ths (defined in SLACustomAutoFocusParameters_t)	SLACustomAutoFocusParameters_t
CoarseSpeed (defined in SLACustomAutoFocusParameters_t)	SLACustomAutoFocusParameters_t
CoarseThresh (defined in SLACustomAutoFocusParameters_t)	SLACustomAutoFocusParameters_t
data (defined in SLACustomAutoFocusParameters_t)	SLACustomAutoFocusParameters_t
fileName (defined in SLACustomAutoFocusParameters_t)	SLACustomAutoFocusParameters_t
FineSpeed (defined in SLACustomAutoFocusParameters_t)	SLACustomAutoFocusParameters_t
FineThresh (defined in SLACustomAutoFocusParameters_t)	SLACustomAutoFocusParameters_t
frameOffsetMult8 (defined in SLACustomAutoFocusParameters_t)	SLACustomAutoFocusParameters_t
method (defined in SLACustomAutoFocusParameters_t)	SLACustomAutoFocusParameters_t
MidSpeed (defined in SLACustomAutoFocusParameters_t)	SLACustomAutoFocusParameters_t
nInitialSteps (defined in SLACustomAutoFocusParameters_t)	SLACustomAutoFocusParameters_t
rateAdjust (defined in SLACustomAutoFocusParameters_t)	SLACustomAutoFocusParameters_t
reserved1 (defined in SLACustomAutoFocusParameters_t)	SLACustomAutoFocusParameters_t
reserved2 (defined in SLACustomAutoFocusParameters_t)	SLACustomAutoFocusParameters_t
reserved3 (defined in SLACustomAutoFocusParameters_t)	SLACustomAutoFocusParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLACustomClassifier_t Member List

This is the complete list of members for [SLACustomClassifier_t](#), including all inherited members.

cameraIndex	(defined in SLACustomClassifier_t)	SLACustomClassifier_t
fileName	(defined in SLACustomClassifier_t)	SLACustomClassifier_t

Sightline Intelligence

Command, Control, and Script

API

v3.11

SLACustomResponse_t Member List

This is the complete list of members for [SLACustomResponse_t](#), including all inherited members.

customData (defined in [SLACustomResponse_t](#)) [SLACustomResponse_t](#)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLADeadPixel_t Member List

This is the complete list of members for [SLADeadPixel_t](#), including all inherited members.

a (defined in SLADeadPixel_t)	SLADeadPixel_t	
b (defined in SLADeadPixel_t)	SLADeadPixel_t	
c (defined in SLADeadPixel_t)	SLADeadPixel_t	
cameraIndex (defined in SLADeadPixel_t)	SLADeadPixel_t	
d (defined in SLADeadPixel_t)	SLADeadPixel_t	
mode (defined in SLADeadPixel_t)	SLADeadPixel_t	

Sightline Intelligence

Command, Control, and Script

API v3.11

SLADeadPixelStats_t Member List

This is the complete list of members for [SLADeadPixelStats_t](#), including all inherited members.

cameraIndex (defined in SLADeadPixelStats_t)	SLADeadPixelStats_t
nAvgHi (defined in SLADeadPixelStats_t)	SLADeadPixelStats_t
nAvgLo (defined in SLADeadPixelStats_t)	SLADeadPixelStats_t
nDead (defined in SLADeadPixelStats_t)	SLADeadPixelStats_t
nDevHi (defined in SLADeadPixelStats_t)	SLADeadPixelStats_t
nGainHi (defined in SLADeadPixelStats_t)	SLADeadPixelStats_t
nGainLo (defined in SLADeadPixelStats_t)	SLADeadPixelStats_t
nOffHi (defined in SLADeadPixelStats_t)	SLADeadPixelStats_t
nOffLo (defined in SLADeadPixelStats_t)	SLADeadPixelStats_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLADecoderParameters_t Member List

This is the complete list of members for [SLADecoderParameters_t](#), including all inherited members.

cameraIndex (defined in SLADecoderParameters_t)	SLADecoderParameters_t
customStr (defined in SLADecoderParameters_t)	SLADecoderParameters_t
ipAddr (defined in SLADecoderParameters_t)	SLADecoderParameters_t
port (defined in SLADecoderParameters_t)	SLADecoderParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLADesignateSelectedTrackPrimary_t Member List

This is the complete list of members for [SLADesignateSelectedTrackPrimary_t](#), including all inherited members.

cameraIndex (defined in [SLADesignateSelectedTrackPrimary_t](#)) [SLADesignateSelectedTrackPrimary_t](#)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLADetailedTimingMessage_t Member List

This is the complete list of members for [SLADetailedTimingMessage_t](#), including all inherited members.

data (defined in SLADetailedTimingMessage_t)	SLADetailedTimingMessage_t
reserved0 (defined in SLADetailedTimingMessage_t)	SLADetailedTimingMessage_t
reserved1 (defined in SLADetailedTimingMessage_t)	SLADetailedTimingMessage_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLADigitalVideoParserParameters_t Member List

This is the complete list of members for [SLADigitalVideoParserParameters_t](#), including all inherited members.

cameraIndex (defined in SLADigitalVideoParserParameters_t)	SLADigitalVideoParserParameters_t
enabled (defined in SLADigitalVideoParserParameters_t)	SLADigitalVideoParserParameters_t
flags (defined in SLADigitalVideoParserParameters_t)	SLADigitalVideoParserParameters_t
oneMessagePerImage (defined in SLADigitalVideoParserParameters_t)	SLADigitalVideoParserParameters_t
usePixelValue (defined in SLADigitalVideoParserParameters_t)	SLADigitalVideoParserParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLADirectoryStatisticsReply_t Member List

This is the complete list of members for [SLADirectoryStatisticsReply_t](#), including all inherited members.

spaceUsed (defined in SLADirectoryStatisticsReply_t)	SLADirectoryStatisticsReply_t
spaceUsedUpper (defined in SLADirectoryStatisticsReply_t)	SLADirectoryStatisticsReply_t
totalSize (defined in SLADirectoryStatisticsReply_t)	SLADirectoryStatisticsReply_t
totalSizeUpper (defined in SLADirectoryStatisticsReply_t)	SLADirectoryStatisticsReply_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLADisplayAngle_t Member List

This is the complete list of members for [SLADisplayAngle_t](#), including all inherited members.

cameraIndex (defined in SLADisplayAngle_t)	SLADisplayAngle_t
decayRate (defined in SLADisplayAngle_t)	SLADisplayAngle_t
rotationDegrees (defined in SLADisplayAngle_t)	SLADisplayAngle_t
rotationLimit (defined in SLADisplayAngle_t)	SLADisplayAngle_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLADoDetectSnapShot_t Member List

This is the complete list of members for [SLADoDetectSnapShot_t](#), including all inherited members.

baseName (defined in SLADoDetectSnapShot_t)	SLADoDetectSnapShot_t
cameraIndex (defined in SLADoDetectSnapShot_t)	SLADoDetectSnapShot_t
flags (defined in SLADoDetectSnapShot_t)	SLADoDetectSnapShot_t
maxPerFrame (defined in SLADoDetectSnapShot_t)	SLADoDetectSnapShot_t
minFramesBetween (defined in SLADoDetectSnapShot_t)	SLADoDetectSnapShot_t
mode (defined in SLADoDetectSnapShot_t)	SLADoDetectSnapShot_t
reserved1 (defined in SLADoDetectSnapShot_t)	SLADoDetectSnapShot_t
sizeType (defined in SLADoDetectSnapShot_t)	SLADoDetectSnapShot_t
snapSize (defined in SLADoDetectSnapShot_t)	SLADoDetectSnapShot_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLADoSnapShot_t Member List

This is the complete list of members for [SLADoSnapShot_t](#), including all inherited members.

autoFolder (defined in SLADoSnapShot_t)	SLADoSnapShot_t
fileName (defined in SLADoSnapShot_t)	SLADoSnapShot_t
flags (defined in SLADoSnapShot_t)	SLADoSnapShot_t
frameStep (defined in SLADoSnapShot_t)	SLADoSnapShot_t
maxFiles (defined in SLADoSnapShot_t)	SLADoSnapShot_t
numFrames (defined in SLADoSnapShot_t)	SLADoSnapShot_t
shouldScan (defined in SLADoSnapShot_t)	SLADoSnapShot_t
snapAllCameras (defined in SLADoSnapShot_t)	SLADoSnapShot_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLADrawObject_t Member List

This is the complete list of members for [SLADrawObject_t](#), including all inherited members.

a (defined in SLADrawObject_t)	SLADrawObject_t
action (defined in SLADrawObject_t)	SLADrawObject_t
b (defined in SLADrawObject_t)	SLADrawObject_t
backgroundColor (defined in SLADrawObject_t)	SLADrawObject_t
c (defined in SLADrawObject_t)	SLADrawObject_t
d (defined in SLADrawObject_t)	SLADrawObject_t
objId (defined in SLADrawObject_t)	SLADrawObject_t
propertyFlags (defined in SLADrawObject_t)	SLADrawObject_t
text (defined in SLADrawObject_t)	SLADrawObject_t
type (defined in SLADrawObject_t)	SLADrawObject_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLADrawOverlay_t Member List

This is the complete list of members for [SLADrawOverlay_t](#), including all inherited members.

a (defined in SLADrawOverlay_t)	SLADrawOverlay_t
action (defined in SLADrawOverlay_t)	SLADrawOverlay_t
b (defined in SLADrawOverlay_t)	SLADrawOverlay_t
backgroundColor (defined in SLADrawOverlay_t)	SLADrawOverlay_t
c (defined in SLADrawOverlay_t)	SLADrawOverlay_t
cameraIndex (defined in SLADrawOverlay_t)	SLADrawOverlay_t
d (defined in SLADrawOverlay_t)	SLADrawOverlay_t
e (defined in SLADrawOverlay_t)	SLADrawOverlay_t
f (defined in SLADrawOverlay_t)	SLADrawOverlay_t
objId (defined in SLADrawOverlay_t)	SLADrawOverlay_t
propertyFlags (defined in SLADrawOverlay_t)	SLADrawOverlay_t
text (defined in SLADrawOverlay_t)	SLADrawOverlay_t
type (defined in SLADrawOverlay_t)	SLADrawOverlay_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAExternalProgram_t Member List

This is the complete list of members for [SLAExternalProgram_t](#), including all inherited members.

filename0 (defined in SLAExternalProgram_t)	SLAExternalProgram_t
filename1 (defined in SLAExternalProgram_t)	SLAExternalProgram_t
filename2 (defined in SLAExternalProgram_t)	SLAExternalProgram_t
programType (defined in SLAExternalProgram_t)	SLAExternalProgram_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAFocusParameters_t Member List

This is the complete list of members for [SLAFocusParameters_t](#), including all inherited members.

cameraIndex (defined in SLAFocusParameters_t)	SLAFocusParameters_t
colRoiFrac (defined in SLAFocusParameters_t)	SLAFocusParameters_t
focusDivider (defined in SLAFocusParameters_t)	SLAFocusParameters_t
focusMetric (defined in SLAFocusParameters_t)	SLAFocusParameters_t
highRoiFrac (defined in SLAFocusParameters_t)	SLAFocusParameters_t
rowRoiFrac (defined in SLAFocusParameters_t)	SLAFocusParameters_t
varianceDivider (defined in SLAFocusParameters_t)	SLAFocusParameters_t
wideRoiFrac (defined in SLAFocusParameters_t)	SLAFocusParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAFocusStats_t Member List

This is the complete list of members for [SLAFocusStats_t](#), including all inherited members.

cameraIndex (defined in SLAFocusStats_t)	SLAFocusStats_t
frameId (defined in SLAFocusStats_t)	SLAFocusStats_t
mean (defined in SLAFocusStats_t)	SLAFocusStats_t
metric (defined in SLAFocusStats_t)	SLAFocusStats_t
timeStamp (defined in SLAFocusStats_t)	SLAFocusStats_t
variance (defined in SLAFocusStats_t)	SLAFocusStats_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAFourAlignPoints_t Member List

This is the complete list of members for [SLAFourAlignPoints_t](#), including all inherited members.

index (defined in SLAFourAlignPoints_t)	SLAFourAlignPoints_t
LeftCameraPointACol (defined in SLAFourAlignPoints_t)	SLAFourAlignPoints_t
LeftCameraPointARow (defined in SLAFourAlignPoints_t)	SLAFourAlignPoints_t
LeftCameraPointBCol (defined in SLAFourAlignPoints_t)	SLAFourAlignPoints_t
LeftCameraPointBRow (defined in SLAFourAlignPoints_t)	SLAFourAlignPoints_t
LeftCameraPointCCol (defined in SLAFourAlignPoints_t)	SLAFourAlignPoints_t
LeftCameraPointCRow (defined in SLAFourAlignPoints_t)	SLAFourAlignPoints_t
LeftCameraPointDCol (defined in SLAFourAlignPoints_t)	SLAFourAlignPoints_t
LeftCameraPointDRow (defined in SLAFourAlignPoints_t)	SLAFourAlignPoints_t
RightCameraPointACol (defined in SLAFourAlignPoints_t)	SLAFourAlignPoints_t
RightCameraPointARow (defined in SLAFourAlignPoints_t)	SLAFourAlignPoints_t
RightCameraPointBCol (defined in SLAFourAlignPoints_t)	SLAFourAlignPoints_t
RightCameraPointBRow (defined in SLAFourAlignPoints_t)	SLAFourAlignPoints_t
RightCameraPointCCol (defined in SLAFourAlignPoints_t)	SLAFourAlignPoints_t
RightCameraPointCRow (defined in SLAFourAlignPoints_t)	SLAFourAlignPoints_t
RightCameraPointDCol (defined in SLAFourAlignPoints_t)	SLAFourAlignPoints_t
RightCameraPointDRow (defined in SLAFourAlignPoints_t)	SLAFourAlignPoints_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAFrameIndex_t Member List

This is the complete list of members for [SLAFrameIndex_t](#), including all inherited members.

cameraIndex (defined in SLAFrameIndex_t)	SLAFrameIndex_t
index (defined in SLAFrameIndex_t)	SLAFrameIndex_t

Sightline Intelligence

Command, Control, and Script

API

v3.11

SLAGetAcquisitionParameters_t Member List

This is the complete list of members for [SLAGetAcquisitionParameters_t](#), including all inherited members.

cameraIndex (defined in [SLAGetAcquisitionParameters_t](#)) [SLAGetAcquisitionParameters_t](#)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAGetADCParameters_t Member List

This is the complete list of members for [SLAGetADCParameters_t](#), including all inherited members.

cameraIndex (defined in [SLAGetADCParameters_t](#)) [SLAGetADCParameters_t](#)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAGetDetectionParameters_t Member List

This is the complete list of members for [SLAGetDetectionParameters_t](#), including all inherited members.

cameraIndex (defined in SLAGetDetectionParameters_t)	SLAGetDetectionParameters_t
detIdx (defined in SLAGetDetectionParameters_t)	SLAGetDetectionParameters_t

Sightline Intelligence

Command, Control, and Script

API

v3.11

SLAGetDisplayParameters_t Member List

This is the complete list of members for [SLAGetDisplayParameters_t](#), including all inherited members.

cameraIndex (defined in [SLAGetDisplayParameters_t](#)) [SLAGetDisplayParameters_t](#)

Sightline Intelligence

Command, Control, and Script

API

v3.11

SLAGetEthernetDisplayParameters_t Member List

This is the complete list of members for [SLAGetEthernetDisplayParameters_t](#), including all inherited members.

displayId (defined in [SLAGetEthernetDisplayParameters_t](#)) [SLAGetEthernetDisplayParameters_t](#)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAGetEthernetVideoParameters_t Member List

This is the complete list of members for [SLAGetEthernetVideoParameters_t](#), including all inherited members.

displayId (defined in [SLAGetEthernetVideoParameters_t](#)) [SLAGetEthernetVideoParameters_t](#)

Sightline Intelligence

Command, Control, and Script

API

v3.11

SLAGetH264Parameters_t Member List

This is the complete list of members for [SLAGetH264Parameters_t](#), including all inherited members.

displayId (defined in [SLAGetH264Parameters_t](#)) [SLAGetH264Parameters_t](#)

Sightline Intelligence

Command, Control, and Script

API

v3.11

SLAGetHardwareID_t Member List

This is the complete list of members for [SLAGetHardwareID_t](#), including all inherited members.

hwID (defined in [SLAGetHardwareID_t](#)) [SLAGetHardwareID_t](#)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAGetImageSize_t Member List

This is the complete list of members for [SLAGetImageSize_t](#), including all inherited members.

cameraIndex (defined in [SLAGetImageSize_t](#)) [SLAGetImageSize_t](#)

Sightline Intelligence

Command, Control, and Script

API

v3.11

SLAGetNetworkParameters_t Member List

This is the complete list of members for [SLAGetNetworkParameters_t](#), including all inherited members.

index (defined in [SLAGetNetworkParameters_t](#)) [SLAGetNetworkParameters_t](#)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAGetOverlayMode_t Member List

This is the complete list of members for [SLAGetOverlayMode_t](#), including all inherited members.

cameraIndex (defined in [SLAGetOverlayMode_t](#)) [SLAGetOverlayMode_t](#)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAGetPortConfiguration_t Member List

This is the complete list of members for [SLAGetPortConfiguration_t](#), including all inherited members.

port (defined in [SLAGetPortConfiguration_t](#)) [SLAGetPortConfiguration_t](#)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAGetRegistrationParameters_t Member List

This is the complete list of members for [SLAGetRegistrationParameters_t](#), including all inherited members.

cameraIndex (defined in [SLAGetRegistrationParameters_t](#)) [SLAGetRegistrationParameters_t](#)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAGetSnapShot_t Member List

This is the complete list of members for [SLAGetSnapShot_t](#), including all inherited members.

cameraIndex (defined in [SLAGetSnapShot_t](#)) [SLAGetSnapShot_t](#)

Sightline Intelligence

Command, Control, and Script

API

v3.11

SLAGetStabilizationParameters_t Member List

This is the complete list of members for [SLAGetStabilizationParameters_t](#), including all inherited members.

cameraIndex (defined in [SLAGetStabilizationParameters_t](#)) [SLAGetStabilizationParameters_t](#)

Sightline Intelligence

Command, Control, and Script

API

v3.11

SLAGetTrackingParameters_t Member List

This is the complete list of members for [SLAGetTrackingParameters_t](#), including all inherited members.

cameraIndex (defined in [SLAGetTrackingParameters_t](#)) [SLAGetTrackingParameters_t](#)

Sightline Intelligence

Command, Control, and Script

API

v3.11

SLAGetVideoEnhancementParameters_t Member List

This is the complete list of members for [SLAGetVideoEnhancementParameters_t](#), including all inherited members.

cameraIndex (defined in [SLAGetVideoEnhancementParameters_t](#)) [SLAGetVideoEnhancementParameters_t](#)

Sightline Intelligence

Command, Control, and Script

API

v3.11

SLAGetVideoParameters_t Member List

This is the complete list of members for [SLAGetVideoParameters_t](#), including all inherited members.

cameraIndex (defined in [SLAGetVideoParameters_t](#)) [SLAGetVideoParameters_t](#)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAGPIO_t Member List

This is the complete list of members for [SLAGPIO_t](#), including all inherited members.

Action (defined in SLAGPIO_t)	SLAGPIO_t
Field (defined in SLAGPIO_t)	SLAGPIO_t
GpioNum (defined in SLAGPIO_t)	SLAGPIO_t
Value (defined in SLAGPIO_t)	SLAGPIO_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAI2CCommand_t Member List

This is the complete list of members for [SLAI2CCommand_t](#), including all inherited members.

addr (defined in SLAI2CCommand_t)	SLAI2CCommand_t
bus (defined in SLAI2CCommand_t)	SLAI2CCommand_t
data (defined in SLAI2CCommand_t)	SLAI2CCommand_t
nread (defined in SLAI2CCommand_t)	SLAI2CCommand_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAKlvClassFilters_t Member List

This is the complete list of members for [SLAKlvClassFilters_t](#), including all inherited members.

cameraIndex (defined in SLAKlvClassFilters_t)	SLAKlvClassFilters_t
horizonFilter (defined in SLAKlvClassFilters_t)	SLAKlvClassFilters_t
maxLat (defined in SLAKlvClassFilters_t)	SLAKlvClassFilters_t
maxLon (defined in SLAKlvClassFilters_t)	SLAKlvClassFilters_t
maxTargetHeight (defined in SLAKlvClassFilters_t)	SLAKlvClassFilters_t
maxTargetWidth (defined in SLAKlvClassFilters_t)	SLAKlvClassFilters_t
minLat (defined in SLAKlvClassFilters_t)	SLAKlvClassFilters_t
minLon (defined in SLAKlvClassFilters_t)	SLAKlvClassFilters_t
minTargetHeight (defined in SLAKlvClassFilters_t)	SLAKlvClassFilters_t
minTargetWidth (defined in SLAKlvClassFilters_t)	SLAKlvClassFilters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLALandingAid_t Member List

This is the complete list of members for [SLALandingAid_t](#), including all inherited members.

blackTargetSize16 (defined in SLALandingAid_t)	SLALandingAid_t
cameraIndex (defined in SLALandingAid_t)	SLALandingAid_t
camHFovDeg8 (defined in SLALandingAid_t)	SLALandingAid_t
ctrlParam0 (defined in SLALandingAid_t)	SLALandingAid_t
ctrlParam1 (defined in SLALandingAid_t)	SLALandingAid_t
ctrlParam2 (defined in SLALandingAid_t)	SLALandingAid_t
ctrlParam3 (defined in SLALandingAid_t)	SLALandingAid_t
keepOutMode (defined in SLALandingAid_t)	SLALandingAid_t
keepOutRadius16 (defined in SLALandingAid_t)	SLALandingAid_t
matchThresh (defined in SLALandingAid_t)	SLALandingAid_t
mode (defined in SLALandingAid_t)	SLALandingAid_t
reserved0 (defined in SLALandingAid_t)	SLALandingAid_t
reserved1 (defined in SLALandingAid_t)	SLALandingAid_t
reserved2 (defined in SLALandingAid_t)	SLALandingAid_t
whiteTargetSize16 (defined in SLALandingAid_t)	SLALandingAid_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLALandingPosition_t Member List

This is the complete list of members for [SLALandingPosition_t](#), including all inherited members.

angleDeg7 (defined in SLALandingPosition_t)	SLALandingPosition_t
cameraIndex (defined in SLALandingPosition_t)	SLALandingPosition_t
camHFovDeg8 (defined in SLALandingPosition_t)	SLALandingPosition_t
capHigh (defined in SLALandingPosition_t)	SLALandingPosition_t
capWide (defined in SLALandingPosition_t)	SLALandingPosition_t
col (defined in SLALandingPosition_t)	SLALandingPosition_t
confidence (defined in SLALandingPosition_t)	SLALandingPosition_t
ctrlParam0 (defined in SLALandingPosition_t)	SLALandingPosition_t
ctrlParam1 (defined in SLALandingPosition_t)	SLALandingPosition_t
ctrlParam2 (defined in SLALandingPosition_t)	SLALandingPosition_t
ctrlParam3 (defined in SLALandingPosition_t)	SLALandingPosition_t
distance16 (defined in SLALandingPosition_t)	SLALandingPosition_t
frameId (defined in SLALandingPosition_t)	SLALandingPosition_t
keepOutConfidence (defined in SLALandingPosition_t)	SLALandingPosition_t
keepOutDist16 (defined in SLALandingPosition_t)	SLALandingPosition_t
keepOutState (defined in SLALandingPosition_t)	SLALandingPosition_t
keepOutSz (defined in SLALandingPosition_t)	SLALandingPosition_t
row (defined in SLALandingPosition_t)	SLALandingPosition_t
timeStamp (defined in SLALandingPosition_t)	SLALandingPosition_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLALensCommand_t Member List

This is the complete list of members for [SLALensCommand_t](#), including all inherited members.

cameraIndex (defined in SLALensCommand_t)	SLALensCommand_t
commandId (defined in SLALensCommand_t)	SLALensCommand_t
data (defined in SLALensCommand_t)	SLALensCommand_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLALensRanges_t Member List

This is the complete list of members for [SLALensRanges_t](#), including all inherited members.

camIdx (defined in SLALensRanges_t)	SLALensRanges_t
focusMax (defined in SLALensRanges_t)	SLALensRanges_t
focusMin (defined in SLALensRanges_t)	SLALensRanges_t
zoomMax (defined in SLALensRanges_t)	SLALensRanges_t
zoomMin (defined in SLALensRanges_t)	SLALensRanges_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLALogoParameters_t Member List

This is the complete list of members for [SLALogoParameters_t](#), including all inherited members.

cameraIndex (defined in SLALogoParameters_t)	SLALogoParameters_t
logoOpacity (defined in SLALogoParameters_t)	SLALogoParameters_t
OffsetX (defined in SLALogoParameters_t)	SLALogoParameters_t
OffsetY (defined in SLALogoParameters_t)	SLALogoParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAMetadataStaticValues_t Member List

This is the complete list of members for [SLAMetadataStaticValues_t](#), including all inherited members.

displayId (defined in SLAMetadataStaticValues_t)	SLAMetadataStaticValues_t
type (defined in SLAMetadataStaticValues_t)	SLAMetadataStaticValues_t
value (defined in SLAMetadataStaticValues_t)	SLAMetadataStaticValues_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAModifyTrackIndex_t Member List

This is the complete list of members for [SLAModifyTrackIndex_t](#), including all inherited members.

cameraIndex (defined in SLAModifyTrackIndex_t)	SLAModifyTrackIndex_t
flags (defined in SLAModifyTrackIndex_t)	SLAModifyTrackIndex_t
height (defined in SLAModifyTrackIndex_t)	SLAModifyTrackIndex_t
trackIndex (defined in SLAModifyTrackIndex_t)	SLAModifyTrackIndex_t
width (defined in SLAModifyTrackIndex_t)	SLAModifyTrackIndex_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAModifyTracking_t Member List

This is the complete list of members for [SLAModifyTracking_t](#), including all inherited members.

cameraIndex (defined in SLAModifyTracking_t)	SLAModifyTracking_t
col (defined in SLAModifyTracking_t)	SLAModifyTracking_t
flags (defined in SLAModifyTracking_t)	SLAModifyTracking_t
height (defined in SLAModifyTracking_t)	SLAModifyTracking_t
row (defined in SLAModifyTracking_t)	SLAModifyTracking_t
width (defined in SLAModifyTracking_t)	SLAModifyTracking_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAStartTracking_t Member List

This is the complete list of members for [SLAStartTracking_t](#), including all inherited members.

cameraIndex (defined in SLAStartTracking_t)	SLAStartTracking_t
col (defined in SLAStartTracking_t)	SLAStartTracking_t
flags (defined in SLAStartTracking_t)	SLAStartTracking_t
framePTS (defined in SLAStartTracking_t)	SLAStartTracking_t
height (defined in SLAStartTracking_t)	SLAStartTracking_t
nearVal (defined in SLAStartTracking_t)	SLAStartTracking_t
row (defined in SLAStartTracking_t)	SLAStartTracking_t
userTrackId (defined in SLAStartTracking_t)	SLAStartTracking_t
width (defined in SLAStartTracking_t)	SLAStartTracking_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLANoise3D_t Member List

This is the complete list of members for [SLANoise3D_t](#), including all inherited members.

cameraIndex (defined in SLANoise3D_t)	SLANoise3D_t
Noise_temporal_8 (defined in SLANoise3D_t)	SLANoise3D_t
SIG_h_8 (defined in SLANoise3D_t)	SLANoise3D_t
SIG_t_8 (defined in SLANoise3D_t)	SLANoise3D_t
SIG_th_8 (defined in SLANoise3D_t)	SLANoise3D_t
SIG_tv_8 (defined in SLANoise3D_t)	SLANoise3D_t
SIG_tvh_8 (defined in SLANoise3D_t)	SLANoise3D_t
SIG_v_8 (defined in SLANoise3D_t)	SLANoise3D_t
SIG_vh_8 (defined in SLANoise3D_t)	SLANoise3D_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLANucParameters_t Member List

This is the complete list of members for [SLANucParameters_t](#), including all inherited members.

cameraIndex (defined in SLANucParameters_t)	SLANucParameters_t
deadFilter (defined in SLANucParameters_t)	SLANucParameters_t
deadFilterThresh (defined in SLANucParameters_t)	SLANucParameters_t
deadReplace (defined in SLANucParameters_t)	SLANucParameters_t
destripeAmount (defined in SLANucParameters_t)	SLANucParameters_t
destripeSections (defined in SLANucParameters_t)	SLANucParameters_t
maxDeadGain (defined in SLANucParameters_t)	SLANucParameters_t
maxDeadOff (defined in SLANucParameters_t)	SLANucParameters_t
maxDeadVal (defined in SLANucParameters_t)	SLANucParameters_t
maxNumDead (defined in SLANucParameters_t)	SLANucParameters_t
maxStdDevDead (defined in SLANucParameters_t)	SLANucParameters_t
minDeadGain (defined in SLANucParameters_t)	SLANucParameters_t
minDeadOff (defined in SLANucParameters_t)	SLANucParameters_t
minDeadVal (defined in SLANucParameters_t)	SLANucParameters_t
nucName (defined in SLANucParameters_t)	SLANucParameters_t
nucRunMode (defined in SLANucParameters_t)	SLANucParameters_t
nucShow (defined in SLANucParameters_t)	SLANucParameters_t
numFrames (defined in SLANucParameters_t)	SLANucParameters_t
numReplace (defined in SLANucParameters_t)	SLANucParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLANudgeTrackingCoordinate_t Member List

This is the complete list of members for [SLANudgeTrackingCoordinate_t](#), including all inherited members.

cameraIndex (defined in SLANudgeTrackingCoordinate_t)	SLANudgeTrackingCoordinate_t
offsetCol (defined in SLANudgeTrackingCoordinate_t)	SLANudgeTrackingCoordinate_t
offsetColExt (defined in SLANudgeTrackingCoordinate_t)	SLANudgeTrackingCoordinate_t
offsetRow (defined in SLANudgeTrackingCoordinate_t)	SLANudgeTrackingCoordinate_t
offsetRowExt (defined in SLANudgeTrackingCoordinate_t)	SLANudgeTrackingCoordinate_t
offsetZoom10 (defined in SLANudgeTrackingCoordinate_t)	SLANudgeTrackingCoordinate_t
rotate (defined in SLANudgeTrackingCoordinate_t)	SLANudgeTrackingCoordinate_t
trackIndex (defined in SLANudgeTrackingCoordinate_t)	SLANudgeTrackingCoordinate_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAReadWriteNuc_t Member List

This is the complete list of members for [SLAReadWriteNuc_t](#), including all inherited members.

cameraIndex (defined in SLAReadWriteNuc_t)	SLAReadWriteNuc_t
fileName (defined in SLAReadWriteNuc_t)	SLAReadWriteNuc_t
interpolationRatio (defined in SLAReadWriteNuc_t)	SLAReadWriteNuc_t
nucReadWriteMode (defined in SLAReadWriteNuc_t)	SLAReadWriteNuc_t
reserved (defined in SLAReadWriteNuc_t)	SLAReadWriteNuc_t
secondaryFileName (defined in SLAReadWriteNuc_t)	SLAReadWriteNuc_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAResetStabilizationParameters_t Member List

This is the complete list of members for [SLAResetStabilizationParameters_t](#), including all inherited members.

cameraIndex	(defined in SLAResetStabilizationParameters_t)	SLAResetStabilizationParameters_t
resetType	(defined in SLAResetStabilizationParameters_t)	SLAResetStabilizationParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASendToBTS_t Member List

This is the complete list of members for [SLASendToBTS_t](#), including all inherited members.

displayId (defined in SLASendToBTS_t)	SLASendToBTS_t
ipAddr (defined in SLASendToBTS_t)	SLASendToBTS_t
mode (defined in SLASendToBTS_t)	SLASendToBTS_t
port (defined in SLASendToBTS_t)	SLASendToBTS_t
rate (defined in SLASendToBTS_t)	SLASendToBTS_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASendTraceStr_t Member List

This is the complete list of members for [SLASendTraceStr_t](#), including all inherited members.

str (defined in [SLASendTraceStr_t](#)) [SLASendTraceStr_t](#)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetDisplayAdjustments_t Member List

This is the complete list of members for [SLASetDisplayAdjustments_t](#), including all inherited members.

cameraIndex (defined in SLASetDisplayAdjustments_t)	SLASetDisplayAdjustments_t
pan (defined in SLASetDisplayAdjustments_t)	SLASetDisplayAdjustments_t
ratio8 (defined in SLASetDisplayAdjustments_t)	SLASetDisplayAdjustments_t
tilt (defined in SLASetDisplayAdjustments_t)	SLASetDisplayAdjustments_t

Sightline Intelligence

Command, Control, and Script

API

v3.11

SLASetKlvData_t Member List

This is the complete list of members for [SLASetKlvData_t](#), including all inherited members.

displayId	(defined in SLASetKlvData_t)	SLASetKlvData_t
klvData	(defined in SLASetKlvData_t)	SLASetKlvData_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetLensMode_t Member List

This is the complete list of members for [SLASetLensMode_t](#), including all inherited members.

data (defined in SLASetLensMode_t)	SLASetLensMode_t
lensMode (defined in SLASetLensMode_t)	SLASetLensMode_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetLensParameters_t Member List

This is the complete list of members for [SLASetLensParameters_t](#), including all inherited members.

afMetricRegionSize (defined in SLASetLensParameters_t)	SLASetLensParameters_t
autofocusChangePct (defined in SLASetLensParameters_t)	SLASetLensParameters_t
autofocusMethod (defined in SLASetLensParameters_t)	SLASetLensParameters_t
autofocusRateAdjust (defined in SLASetLensParameters_t)	SLASetLensParameters_t
colRoiFrac (defined in SLASetLensParameters_t)	SLASetLensParameters_t
comPortNum (defined in SLASetLensParameters_t)	SLASetLensParameters_t
focusSpeed (defined in SLASetLensParameters_t)	SLASetLensParameters_t
highRoiFrac (defined in SLASetLensParameters_t)	SLASetLensParameters_t
lensType (defined in SLASetLensParameters_t)	SLASetLensParameters_t
rowRoiFrac (defined in SLASetLensParameters_t)	SLASetLensParameters_t
wideRoiFrac (defined in SLASetLensParameters_t)	SLASetLensParameters_t
zoomSpeed (defined in SLASetLensParameters_t)	SLASetLensParameters_t
zoomTrackFocus (defined in SLASetLensParameters_t)	SLASetLensParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLATagDataRate_t Member List

This is the complete list of members for [SLATagDataRate_t](#), including all inherited members.

displayId (defined in SLATagDataRate_t)	SLATagDataRate_t
frameStep (defined in SLATagDataRate_t)	SLATagDataRate_t
mode (defined in SLATagDataRate_t)	SLATagDataRate_t
reserved1 (defined in SLATagDataRate_t)	SLATagDataRate_t
reserved2 (defined in SLATagDataRate_t)	SLATagDataRate_t
tagId1 (defined in SLATagDataRate_t)	SLATagDataRate_t
tagId2 (defined in SLATagDataRate_t)	SLATagDataRate_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetMultipleAlignment_t Member List

This is the complete list of members for [SLASetMultipleAlignment_t](#), including all inherited members.

alignment (defined in SLASetMultipleAlignment_t)	SLASetMultipleAlignment_t
horizontal (defined in SLASetMultipleAlignment_t)	SLASetMultipleAlignment_t
hzoom (defined in SLASetMultipleAlignment_t)	SLASetMultipleAlignment_t
nAlignments (defined in SLASetMultipleAlignment_t)	SLASetMultipleAlignment_t
rotate (defined in SLASetMultipleAlignment_t)	SLASetMultipleAlignment_t
vertical (defined in SLASetMultipleAlignment_t)	SLASetMultipleAlignment_t
zoom (defined in SLASetMultipleAlignment_t)	SLASetMultipleAlignment_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetRegistrationParameters_t Member List

This is the complete list of members for [SLASetRegistrationParameters_t](#), including all inherited members.

bottom (defined in SLASetRegistrationParameters_t)	SLASetRegistrationParameters_t
cameraIndex (defined in SLASetRegistrationParameters_t)	SLASetRegistrationParameters_t
left (defined in SLASetRegistrationParameters_t)	SLASetRegistrationParameters_t
maxRotation (defined in SLASetRegistrationParameters_t)	SLASetRegistrationParameters_t
maxTranslation (defined in SLASetRegistrationParameters_t)	SLASetRegistrationParameters_t
right (defined in SLASetRegistrationParameters_t)	SLASetRegistrationParameters_t
top (defined in SLASetRegistrationParameters_t)	SLASetRegistrationParameters_t
zoomRange (defined in SLASetRegistrationParameters_t)	SLASetRegistrationParameters_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetSnapShot_t Member List

This is the complete list of members for [SLASetSnapShot_t](#), including all inherited members.

cameraIndex (defined in SLASetSnapShot_t)	SLASetSnapShot_t
downsample (defined in SLASetSnapShot_t)	SLASetSnapShot_t
format (defined in SLASetSnapShot_t)	SLASetSnapShot_t
ipAddr (defined in SLASetSnapShot_t)	SLASetSnapShot_t
mode (defined in SLASetSnapShot_t)	SLASetSnapShot_t
password (defined in SLASetSnapShot_t)	SLASetSnapShot_t
port (defined in SLASetSnapShot_t)	SLASetSnapShot_t
quality (defined in SLASetSnapShot_t)	SLASetSnapShot_t
secondaryFormat (defined in SLASetSnapShot_t)	SLASetSnapShot_t
source (defined in SLASetSnapShot_t)	SLASetSnapShot_t
userName (defined in SLASetSnapShot_t)	SLASetSnapShot_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetStabilizationBias_t Member List

This is the complete list of members for [SLASetStabilizationBias_t](#), including all inherited members.

autoBias (defined in SLASetStabilizationBias_t)	SLASetStabilizationBias_t
biasCol (defined in SLASetStabilizationBias_t)	SLASetStabilizationBias_t
biasRow (defined in SLASetStabilizationBias_t)	SLASetStabilizationBias_t
cameraIndex (defined in SLASetStabilizationBias_t)	SLASetStabilizationBias_t
updateRate (defined in SLASetStabilizationBias_t)	SLASetStabilizationBias_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetVideoMode_t Member List

This is the complete list of members for [SLASetVideoMode_t](#), including all inherited members.

analogDest (defined in SLASetVideoMode_t)	SLASetVideoMode_t
analogRes (defined in SLASetVideoMode_t)	SLASetVideoMode_t
cameraIndex0 (defined in SLASetVideoMode_t)	SLASetVideoMode_t
cameraIndex1 (defined in SLASetVideoMode_t)	SLASetVideoMode_t
cameraIndex2 (defined in SLASetVideoMode_t)	SLASetVideoMode_t
displayDest (defined in SLASetVideoMode_t)	SLASetVideoMode_t
displayIndex0 (defined in SLASetVideoMode_t)	SLASetVideoMode_t
displayIndex1 (defined in SLASetVideoMode_t)	SLASetVideoMode_t
displayIndex2 (defined in SLASetVideoMode_t)	SLASetVideoMode_t
displayIndex3 (defined in SLASetVideoMode_t)	SLASetVideoMode_t
displayMode (defined in SLASetVideoMode_t)	SLASetVideoMode_t
hdmiDest (defined in SLASetVideoMode_t)	SLASetVideoMode_t
hdmiRes (defined in SLASetVideoMode_t)	SLASetVideoMode_t
HDSDIDest (defined in SLASetVideoMode_t)	SLASetVideoMode_t
hdsdiRes (defined in SLASetVideoMode_t)	SLASetVideoMode_t
nCameras (defined in SLASetVideoMode_t)	SLASetVideoMode_t
nDisplays (defined in SLASetVideoMode_t)	SLASetVideoMode_t
net0Dest (defined in SLASetVideoMode_t)	SLASetVideoMode_t
net1Dest (defined in SLASetVideoMode_t)	SLASetVideoMode_t
pipColor (defined in SLASetVideoMode_t)	SLASetVideoMode_t
pipQuadrant (defined in SLASetVideoMode_t)	SLASetVideoMode_t
pipScale (defined in SLASetVideoMode_t)	SLASetVideoMode_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASetVMTI_t Member List

This is the complete list of members for [SLASetVMTI_t](#), including all inherited members.

col (defined in SLASetVMTI_t)	SLASetVMTI_t
confidence (defined in SLASetVMTI_t)	SLASetVMTI_t
displayId (defined in SLASetVMTI_t)	SLASetVMTI_t
high (defined in SLASetVMTI_t)	SLASetVMTI_t
newTargetDetectionFlag (defined in SLASetVMTI_t)	SLASetVMTI_t
numTargets (defined in SLASetVMTI_t)	SLASetVMTI_t
row (defined in SLASetVMTI_t)	SLASetVMTI_t
targetID (defined in SLASetVMTI_t)	SLASetVMTI_t
targetPack (defined in SLASetVMTI_t)	SLASetVMTI_t
wide (defined in SLASetVMTI_t)	SLASetVMTI_t

Sightline Intelligence

Command, Control, and Script

API

v3.11

SLAShiftSelectedTrack_t Member List

This is the complete list of members for [SLAShiftSelectedTrack_t](#), including all inherited members.

cameraIndex (defined in [SLAShiftSelectedTrack_t](#)) [SLAShiftSelectedTrack_t](#)

Sightline Intelligence

Command, Control, and Script

API

v3.11

SLAStopSelectedTrack_t Member List

This is the complete list of members for [SLAStopSelectedTrack_t](#), including all inherited members.

cameraIndex (defined in [SLAStopSelectedTrack_t](#)) [SLAStopSelectedTrack_t](#)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAStopTracking_t Member List

This is the complete list of members for [SLAStopTracking_t](#), including all inherited members.

cameraIndex (defined in SLAStopTracking_t)	SLAStopTracking_t
reserved1 (defined in SLAStopTracking_t)	SLAStopTracking_t
reserved2 (defined in SLAStopTracking_t)	SLAStopTracking_t
reserved3 (defined in SLAStopTracking_t)	SLAStopTracking_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAStreamingControl_t Member List

This is the complete list of members for [SLAStreamingControl_t](#), including all inherited members.

streamingControlMask (defined in [SLAStreamingControl_t](#)) [SLAStreamingControl_t](#)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLASystemStatusMessage_t Member List

This is the complete list of members for [SLASystemStatusMessage_t](#), including all inherited members.

errorFlags (defined in SLASystemStatusMessage_t)	SLASystemStatusMessage_t
load0 (defined in SLASystemStatusMessage_t)	SLASystemStatusMessage_t
load1 (defined in SLASystemStatusMessage_t)	SLASystemStatusMessage_t
load2 (defined in SLASystemStatusMessage_t)	SLASystemStatusMessage_t
load3 (defined in SLASystemStatusMessage_t)	SLASystemStatusMessage_t
missedFrames0 (defined in SLASystemStatusMessage_t)	SLASystemStatusMessage_t
missedFrames1 (defined in SLASystemStatusMessage_t)	SLASystemStatusMessage_t
missedFrames2 (defined in SLASystemStatusMessage_t)	SLASystemStatusMessage_t
missedFrames3 (defined in SLASystemStatusMessage_t)	SLASystemStatusMessage_t
temperatureC (defined in SLASystemStatusMessage_t)	SLASystemStatusMessage_t
temperatureF (defined in SLASystemStatusMessage_t)	SLASystemStatusMessage_t

Sightline Intelligence

Command, Control, and Script

API

v3.11

SLASystemStatusMode_t Member List

This is the complete list of members for [SLASystemStatusMode_t](#), including all inherited members.

systemDebugBits	(defined in SLASystemStatusMode_t)	SLASystemStatusMode_t
systemStatusBits	(defined in SLASystemStatusMode_t)	SLASystemStatusMode_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLATagData_t Member List

This is the complete list of members for [SLATagData_t](#), including all inherited members.

displayId (defined in SLATagData_t)	SLATagData_t
klvData (defined in SLATagData_t)	SLATagData_t
reserved1 (defined in SLATagData_t)	SLATagData_t
reserved2 (defined in SLATagData_t)	SLATagData_t
reservedInternal (defined in SLATagData_t)	SLATagData_t
tagId (defined in SLATagData_t)	SLATagData_t
tagSubId (defined in SLATagData_t)	SLATagData_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLATagSourceSelector_t Member List

This is the complete list of members for [SLATagSourceSelector_t](#), including all inherited members.

displayId (defined in SLATagSourceSelector_t)	SLATagSourceSelector_t
mode (defined in SLATagSourceSelector_t)	SLATagSourceSelector_t
reserved1 (defined in SLATagSourceSelector_t)	SLATagSourceSelector_t
reserved2 (defined in SLATagSourceSelector_t)	SLATagSourceSelector_t
selector (defined in SLATagSourceSelector_t)	SLATagSourceSelector_t
tagId1 (defined in SLATagSourceSelector_t)	SLATagSourceSelector_t
tagId2 (defined in SLATagSourceSelector_t)	SLATagSourceSelector_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLATrackingBoxPixelStats_t Member List

This is the complete list of members for [SLATrackingBoxPixelStats_t](#), including all inherited members.

cameraIndex (defined in SLATrackingBoxPixelStats_t)	SLATrackingBoxPixelStats_t
frameId (defined in SLATrackingBoxPixelStats_t)	SLATrackingBoxPixelStats_t
index (defined in SLATrackingBoxPixelStats_t)	SLATrackingBoxPixelStats_t
max (defined in SLATrackingBoxPixelStats_t)	SLATrackingBoxPixelStats_t
mean (defined in SLATrackingBoxPixelStats_t)	SLATrackingBoxPixelStats_t
min (defined in SLATrackingBoxPixelStats_t)	SLATrackingBoxPixelStats_t
numTracks (defined in SLATrackingBoxPixelStats_t)	SLATrackingBoxPixelStats_t
stat (defined in SLATrackingBoxPixelStats_t)	SLATrackingBoxPixelStats_t
timeStamp (defined in SLATrackingBoxPixelStats_t)	SLATrackingBoxPixelStats_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLATrackingMultiClass_t Member List

This is the complete list of members for [SLATrackingMultiClass_t](#), including all inherited members.

baseClassConf (defined in SLATrackingMultiClass_t)	SLATrackingMultiClass_t
baseClassLabel (defined in SLATrackingMultiClass_t)	SLATrackingMultiClass_t
cameraIndex (defined in SLATrackingMultiClass_t)	SLATrackingMultiClass_t
frameId (defined in SLATrackingMultiClass_t)	SLATrackingMultiClass_t
index (defined in SLATrackingMultiClass_t)	SLATrackingMultiClass_t
multiClass (defined in SLATrackingMultiClass_t)	SLATrackingMultiClass_t
numSubClass (defined in SLATrackingMultiClass_t)	SLATrackingMultiClass_t
numTracks (defined in SLATrackingMultiClass_t)	SLATrackingMultiClass_t
reserved1 (defined in SLATrackingMultiClass_t)	SLATrackingMultiClass_t
reserved2 (defined in SLATrackingMultiClass_t)	SLATrackingMultiClass_t
reserved3 (defined in SLATrackingMultiClass_t)	SLATrackingMultiClass_t
reserved4 (defined in SLATrackingMultiClass_t)	SLATrackingMultiClass_t
reserved5 (defined in SLATrackingMultiClass_t)	SLATrackingMultiClass_t
reserved6 (defined in SLATrackingMultiClass_t)	SLATrackingMultiClass_t
subClassConf1 (defined in SLATrackingMultiClass_t)	SLATrackingMultiClass_t
subClassConf2 (defined in SLATrackingMultiClass_t)	SLATrackingMultiClass_t
subClassConf3 (defined in SLATrackingMultiClass_t)	SLATrackingMultiClass_t
subClassConf4 (defined in SLATrackingMultiClass_t)	SLATrackingMultiClass_t
subClassLabel1 (defined in SLATrackingMultiClass_t)	SLATrackingMultiClass_t
subClassLabel2 (defined in SLATrackingMultiClass_t)	SLATrackingMultiClass_t
subClassLabel3 (defined in SLATrackingMultiClass_t)	SLATrackingMultiClass_t
subClassLabel4 (defined in SLATrackingMultiClass_t)	SLATrackingMultiClass_t
timeStamp (defined in SLATrackingMultiClass_t)	SLATrackingMultiClass_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLATrackingPosition_t Member List

This is the complete list of members for [SLATrackingPosition_t](#), including all inherited members.

cameraIndex (defined in SLATrackingPosition_t)	SLATrackingPosition_t
frameId (defined in SLATrackingPosition_t)	SLATrackingPosition_t
offsetCol (defined in SLATrackingPosition_t)	SLATrackingPosition_t
offsetRow (defined in SLATrackingPosition_t)	SLATrackingPosition_t
rotation (defined in SLATrackingPosition_t)	SLATrackingPosition_t
sceneAngle7 (defined in SLATrackingPosition_t)	SLATrackingPosition_t
sceneCol (defined in SLATrackingPosition_t)	SLATrackingPosition_t
sceneColFrac8 (defined in SLATrackingPosition_t)	SLATrackingPosition_t
sceneConfidence (defined in SLATrackingPosition_t)	SLATrackingPosition_t
sceneRow (defined in SLATrackingPosition_t)	SLATrackingPosition_t
sceneRowFrac8 (defined in SLATrackingPosition_t)	SLATrackingPosition_t
sceneScale8 (defined in SLATrackingPosition_t)	SLATrackingPosition_t
timeStamp (defined in SLATrackingPosition_t)	SLATrackingPosition_t
trackingCol (defined in SLATrackingPosition_t)	SLATrackingPosition_t
trackingConfidence (defined in SLATrackingPosition_t)	SLATrackingPosition_t
trackingRow (defined in SLATrackingPosition_t)	SLATrackingPosition_t
userTrackId (defined in SLATrackingPosition_t)	SLATrackingPosition_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLATrackingPositions_t Member List

This is the complete list of members for [SLATrackingPositions_t](#), including all inherited members.

cameraIndex (defined in SLATrackingPositions_t)	SLATrackingPositions_t
col (defined in SLATrackingPositions_t)	SLATrackingPositions_t
confidence (defined in SLATrackingPositions_t)	SLATrackingPositions_t
flags (defined in SLATrackingPositions_t)	SLATrackingPositions_t
frameId (defined in SLATrackingPositions_t)	SLATrackingPositions_t
high (defined in SLATrackingPositions_t)	SLATrackingPositions_t
index (defined in SLATrackingPositions_t)	SLATrackingPositions_t
numTracks (defined in SLATrackingPositions_t)	SLATrackingPositions_t
row (defined in SLATrackingPositions_t)	SLATrackingPositions_t
timeStamp (defined in SLATrackingPositions_t)	SLATrackingPositions_t
track (defined in SLATrackingPositions_t)	SLATrackingPositions_t
velCol8 (defined in SLATrackingPositions_t)	SLATrackingPositions_t
velRow8 (defined in SLATrackingPositions_t)	SLATrackingPositions_t
wide (defined in SLATrackingPositions_t)	SLATrackingPositions_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLATrackingPositionsExtended_t Member List

This is the complete list of members for [SLATrackingPositionsExtended_t](#), including all inherited members.

cameraIndex (defined in SLATrackingPositionsExtended_t)	SLATrackingPositionsExtended_t
classifierConf (defined in SLATrackingPositionsExtended_t)	SLATrackingPositionsExtended_t
classifierLabel (defined in SLATrackingPositionsExtended_t)	SLATrackingPositionsExtended_t
col (defined in SLATrackingPositionsExtended_t)	SLATrackingPositionsExtended_t
confidence (defined in SLATrackingPositionsExtended_t)	SLATrackingPositionsExtended_t
flags (defined in SLATrackingPositionsExtended_t)	SLATrackingPositionsExtended_t
flags1 (defined in SLATrackingPositionsExtended_t)	SLATrackingPositionsExtended_t
frameId (defined in SLATrackingPositionsExtended_t)	SLATrackingPositionsExtended_t
high (defined in SLATrackingPositionsExtended_t)	SLATrackingPositionsExtended_t
index (defined in SLATrackingPositionsExtended_t)	SLATrackingPositionsExtended_t
numTracks (defined in SLATrackingPositionsExtended_t)	SLATrackingPositionsExtended_t
row (defined in SLATrackingPositionsExtended_t)	SLATrackingPositionsExtended_t
timeStamp (defined in SLATrackingPositionsExtended_t)	SLATrackingPositionsExtended_t
track (defined in SLATrackingPositionsExtended_t)	SLATrackingPositionsExtended_t
trackColFrac8 (defined in SLATrackingPositionsExtended_t)	SLATrackingPositionsExtended_t
trackRowFrac8 (defined in SLATrackingPositionsExtended_t)	SLATrackingPositionsExtended_t
userTrackId (defined in SLATrackingPositionsExtended_t)	SLATrackingPositionsExtended_t
velCol8 (defined in SLATrackingPositionsExtended_t)	SLATrackingPositionsExtended_t
velRow8 (defined in SLATrackingPositionsExtended_t)	SLATrackingPositionsExtended_t
wide (defined in SLATrackingPositionsExtended_t)	SLATrackingPositionsExtended_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLATrackTrails_t Member List

This is the complete list of members for [SLATrackTrails_t](#), including all inherited members.

cameraIndex (defined in SLATrackTrails_t)	SLATrackTrails_t
detectionLen (defined in SLATrackTrails_t)	SLATrackTrails_t
flags (defined in SLATrackTrails_t)	SLATrackTrails_t
tracksLen (defined in SLATrackTrails_t)	SLATrackTrails_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAUsb3VisionFeature_t Member List

This is the complete list of members for [SLAUsb3VisionFeature_t](#), including all inherited members.

cameraIndex (defined in SLAUsb3VisionFeature_t)	SLAUsb3VisionFeature_t
name (defined in SLAUsb3VisionFeature_t)	SLAUsb3VisionFeature_t
type (defined in SLAUsb3VisionFeature_t)	SLAUsb3VisionFeature_t
value (defined in SLAUsb3VisionFeature_t)	SLAUsb3VisionFeature_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAUsbWebcamFeature_t Member List

This is the complete list of members for [SLAUsbWebcamFeature_t](#), including all inherited members.

cameraIndex (defined in SLAUsbWebcamFeature_t)	SLAUsbWebcamFeature_t
reset (defined in SLAUsbWebcamFeature_t)	SLAUsbWebcamFeature_t
type (defined in SLAUsbWebcamFeature_t)	SLAUsbWebcamFeature_t
value (defined in SLAUsbWebcamFeature_t)	SLAUsbWebcamFeature_t

Sightline Intelligence

Command, Control, and Script

API

v3.11

SLAUserFont_t Member List

This is the complete list of members for [SLAUserFont_t](#), including all inherited members.

fontFileName (defined in SLAUserFont_t)	SLAUserFont_t
userFontIndex (defined in SLAUserFont_t)	SLAUserFont_t

Sightline Intelligence Command, Control, and Script API v3.11

[Main Page](#) [Related Pages](#) [Modules](#) [Classes](#) ■ [Camera Configuration](#)

SLAUserWarningLevel_t Member List

This is the complete list of members for [SLAUserWarningLevel_t](#), including all inherited members.

warnLevel (defined in [SLAUserWarningLevel_t](#)) [SLAUserWarningLevel_t](#)

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAVMTIChips_t Member List

This is the complete list of members for [SLAVMTIChips_t](#), including all inherited members.

displayId (defined in SLAVMTIChips_t)	SLAVMTIChips_t
format (defined in SLAVMTIChips_t)	SLAVMTIChips_t
maxPerFrame (defined in SLAVMTIChips_t)	SLAVMTIChips_t
minFramesBetween (defined in SLAVMTIChips_t)	SLAVMTIChips_t
mode (defined in SLAVMTIChips_t)	SLAVMTIChips_t
reserved0 (defined in SLAVMTIChips_t)	SLAVMTIChips_t
sizeHint (defined in SLAVMTIChips_t)	SLAVMTIChips_t
sizeType (defined in SLAVMTIChips_t)	SLAVMTIChips_t

Sightline Intelligence

Command, Control, and Script

API v3.11

SLAVMTIFields_t Member List

This is the complete list of members for [SLAVMTIFields_t](#), including all inherited members.

displayId (defined in SLAVMTIFields_t)	SLAVMTIFields_t
fields (defined in SLAVMTIFields_t)	SLAVMTIFields_t
ontologySeriesRate (defined in SLAVMTIFields_t)	SLAVMTIFields_t

Sightline Intelligence Command, Control, and Script API v3.11

SVPEndData_t Member List

This is the complete list of members for [SVPEndData_t](#), including all inherited members.

[buf](#) [SVPEndData_t](#)

Sightline Intelligence

Command, Control, and Script

API v3.11

Modules

Here is a list of all modules:

Blending	Two channel video blending
Capture	Camera acquisition controls
Compression	Compression, encoding and network streaming of video
Detection	Automatic object detection - moving target, color anomaly, radiometric, marine
Display	Video output display controls
Enhancement	Enhancement of moving video
Focus	Lens auto focus and brightness control
General	General system configuration - command camera, save parameters, user warnings, system profile etc
KLV Metadata	Metadata embedding into network stream
Landing Aid	Visual landing aid detection and tracking
Lua Script Functions	Lua script interface for system control
Network	Network communications and setup
Non Uniformity Correction	Non Uniformity Correction and Dead Pixel Replacement
Overlays	Graphics overlays
Recording	Video recording and snapshots to FTP server, microSD card, or other media
Serial Port	Serial port communications and setup
Stabilization	Stabilize moving video
Telemetry	Telemetry result messages and configuration
Tracking	User designated tracking of objects
Classification	AI Classifier
Message Identifiers	List of SightLine command & reply message ID (type) constants